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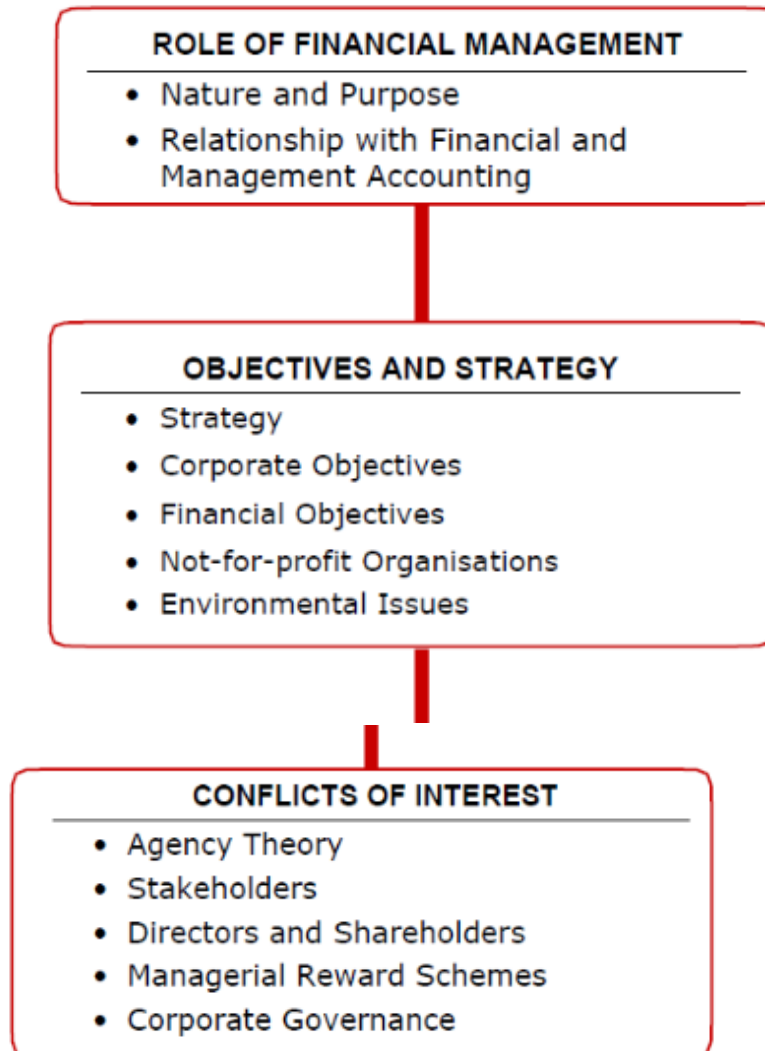
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CHAPTER 1: Visual Overview

Objective: To understand the nature of financial management and appreciate the various organisational stakeholders and their respective objectives.



1.1 Nature and Purpose

i Definition

Financial management – the management of activities associated with acquiring and using short- and long-term financial resources.

The three key financial management decisions are investing, financing and dividend policy. More specifically:

- What types of funds should be raised – equity or debt?
- How should the funds be raised?
- On which proposed investments should the funds be spent?
- How much dividend should be paid to the shareholders?
- How much working capital should the organisation have and how should it be financed?
- How should risk be managed?

These decisions are inter-related:

- An increase in dividends (the dividend decision) will reduce the level of retained cash and increase the need for external finance (the financing decision) to fund capital investment projects (the investment decision).
- An increase in asset ("capital") expenditure (the investment decision) would also increase the need for finance (the financing decision). One possible source of finance is internal, by reducing dividends (the dividend decision).

1.2 Relationship with Financial and Management Accounting

1.2.1 Financial Accounting

Financial accounting may influence financial management in various ways. For example:

- Directors of quoted companies need to consider the effect of financing decisions on the published financial statements. For example, leases longer than 12 months would increase reported financial gearing (i.e. the amount of debt relative to equity finance.)
- Directors of quoted companies need to consider the effect of investment decisions on key financial ratios such as return on capital employed.

1.2.2 Management Accounting

Management accounting information can often assist the financial manager. For example:

- The budgeting process may identify potential cash deficits/surpluses which the financial manager must plan to finance/invest.
- Analysis of costs into fixed and variable elements may assist financial management decisions.
- Variance analysis may help to control the costs of new projects.

2.1 Strategy

A strategy is a course of action to achieve an objective. A strategy might be short-term or long-term, depending on the objective.

Corporate strategies relate to the general direction of the entire organisation.

For example, a car manufacturer may have a strategy over the next 10 years focusing on sustainable energy, collectively owned transportation, and driving automation. Its investments and actions would be directed towards achieving objectives in these areas.

2.2 Corporate Objectives

Corporate objectives are relevant to the entirety of an organisation.

In practice, companies are likely to have a variety of different corporate objectives which may include several of the following:

- Profit targets
- Market share targets
- Share price growth
- Local and environmental concerns
- Contented workforce
- Meeting short-term targets
- Long-term plans.

These objectives can be classified as follows:

- Profit goals – objectives which lead directly to increased profits (e.g. cost reduction measures).
- Surrogate profit goals – objectives which lead indirectly to increased profits (e.g. maintaining a contented workforce).
- Constraints on profit – objectives restricting profit (e.g. ensuring that the company's operations do not harm the environment).
- Dysfunctional goals – objectives which do not provide a benefit even in the long run (e.g. the pursuit of market leadership at all costs).

A company may aim at either maximising or satisficing these objectives.

- Maximising involves seeking the best possible outcome.
- Satisficing involves finding an adequate outcome.

2.3 Financial Objectives

Financial objectives include targets for earnings per share (EPS), dividend per share, gearing level and operating profitability.

2.3.1 Maximisation of Shareholder Wealth

In theoretical terms, a single corporate (financial) objective is assumed, and this is *the maximisation of shareholder wealth*.

Shareholder wealth is the combination of dividend and share price growth, referred to as total shareholder returns (TSR).

$$\text{TSR} = \frac{\text{Year-end share price} - \text{Share price at start of year} + \text{Dividend per share}}{\text{Share price at start of year}} \times 100$$

If a company's shares are traded on a stock market, shareholder wealth increases when the share price rises. This happens if the company makes good profits, which it either pays out as dividends or reinvests to support future growth in both the business and in dividends (assuming the profits have been achieved without undue business or financial risk, which would concern shareholders).

Therefore, the objective of maximising share prices is often used as a substitute for maximising shareholder wealth.

Example 1 TSR and Changes in Dividends and Share Prices

An investor purchased 100 shares in ABC Co at the beginning of 20X0 at the market price of \$10 per share. During the year, the investor received a dividend of \$1.20 per share. At the end of 20X0, the shares had an ex-div market value of \$12.90 per share.

$$\text{TSR} = (\$12.90 - \$10 + \$1.20) / \$10 = 0.41 \text{ or } 41\%$$

This is made up of two components:

1. Capital gain: $(\$12.90 - \$10) / \$10 = 0.29$ or 29%
2. Dividend yield: $\text{Dividend/price at the beginning of the period} = \$1.20 / \$10 = 0.12$ or 12%

The objective of maximising shareholder wealth can be justified in the following ways:

- The company that provides the highest returns for its investors will find it easiest to raise new finance and grow. This should ensure optimal resource allocation within the economy and maximise the overall wealth of the nation.
- Companies that provide customers with what they require will achieve the highest returns for their investors, consistent with the nation's economic health.
- Companies that fail to provide adequate returns may become targets for hostile takeovers.
- Directors have a legal duty to run the company on behalf of the shareholders. It is generally considered a reasonable assumption that the shareholders of listed companies (mainly institutional investors) seek to

- Directors have a legal duty to run the company on behalf of the shareholders. It is generally considered a reasonable assumption that the shareholders of *listed* companies (mainly institutional investors) seek to maximise wealth.

Criticisms of the above include the following:

- ✗ It ignores market imperfections – it might not be in the public interest to allow monopolies to maximise returns as this may cause high consumer prices.
- ✗ It ignores social needs like health, education, police and so on.
- ✗ Maximising TSR ignores the interests of other stakeholders (e.g. employees, customers and arguably, society as a whole).
- ✗ In the case of *unlisted* companies, even the shareholders may not require maximised returns (e.g. some closely held companies are run as "lifestyle" companies whose main objective is to create prestige for the owners).

2.3.2 Profit Maximisation

In practice, many companies find the theoretical objective of maximising TSR hard to follow because:

- A listed company's share price is often influenced more by overall stock market conditions than its own performance.
- In the case of *unlisted* companies, there is no quoted share price to be maximised.

Therefore, profit maximisation is often used as a proxy for shareholder wealth maximisation. Unfortunately, profit is not necessarily a reliable proxy:

- The value of the company's equity is more closely connected with its cash generation than accrual-based accounting profits.
- Management may become "myopic", aiming to boost short-term profit at the expense of long-term growth (e.g. by cutting investment in research or training).
- Excessive pressure to maximise profits can lead to data manipulation and, in extreme cases, fraudulent financial reporting.
- Accounting profit reports debt financing costs (i.e. interest expense) but not equity financing costs (i.e. required return of shareholders). Therefore, reported profit will be maximised when there is no debt in the company's capital structure. However, the company would pay more tax as it will lose the benefit of tax-allowable interest expense (the "tax shield").

2.3.3 Earnings per Share Growth

Earnings per share (EPS) is calculated as:

$$\text{EPS} = \frac{\text{Profit after tax} - \text{preference dividends}}{\text{Weighted average number of ordinary shares in issue}}$$

EPS growth is a key indicator for financial analysts who advise institutional investors in listed companies. Analysts often publish quarterly forecasts for a company's EPS.

Exam advice

If preference shares have been classified as non-current liabilities, preference dividends would be included in finance costs and, therefore, already deducted from profit.

This leads to pressure on management to meet the analysts' expectations. Because it is based on accounting profits, EPS shares all of the problems listed above. Additionally:

- EPS can be cosmetically boosted by undertaking a share consolidation (i.e. replacing several existing shares with one new share).
- EPS can be legitimately increased through a share buyback scheme (i.e. using surplus cash to repurchase and cancel part of the share capital). Research shows, however, that company directors often perform a buyback for personal gain; the share price usually rises on the announcement of a buyback scheme, increasing the value of any shares issued to directors through executive share option schemes (see s.3.4).

Example 2 Earning per Share (EPS)

During the year ended 31 March 20X5, George Co reported EPS of 10.3 cents. The company had 400,000 shares in issue at 31 March 20X5.

On 1 October 20X5 George Co issued 60,000 shares at the market price. Profit after tax for the year ended 31 March 20X6 was \$57,000. There were no preference dividends.

For the year ended 31 March 20X6, $\text{EPS} = \$57,000 / 430,000 \text{ (W)} = 13.3 \text{ cents}$.

EPS growth was $(13.3 \div 10.3 - 1) \times 100 = 29.1\%$

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Weighted average number of shares:

400,000 × 6/12	200,000
460,000 × 6/12	<u>230,000</u>
	<u>430,000</u>

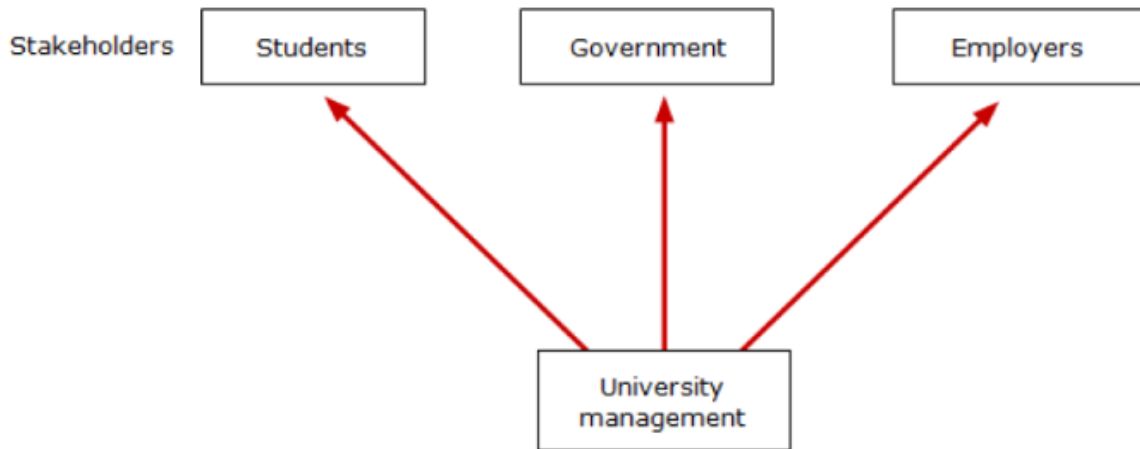
2.4 Not-for-profit Organisations (NPOs)

2.4.1 Objectives

Not-for-profit organisations (NPOs) include different types of organisation, for example charities and the public sector. Such organisations are not constrained by cost/profit objectives to the same extent as companies. However, they are often constrained by having multiple stakeholders with potentially conflicting objectives.

Consider a state-funded university in *Example 3*:

Example 3 Conflicting Objectives



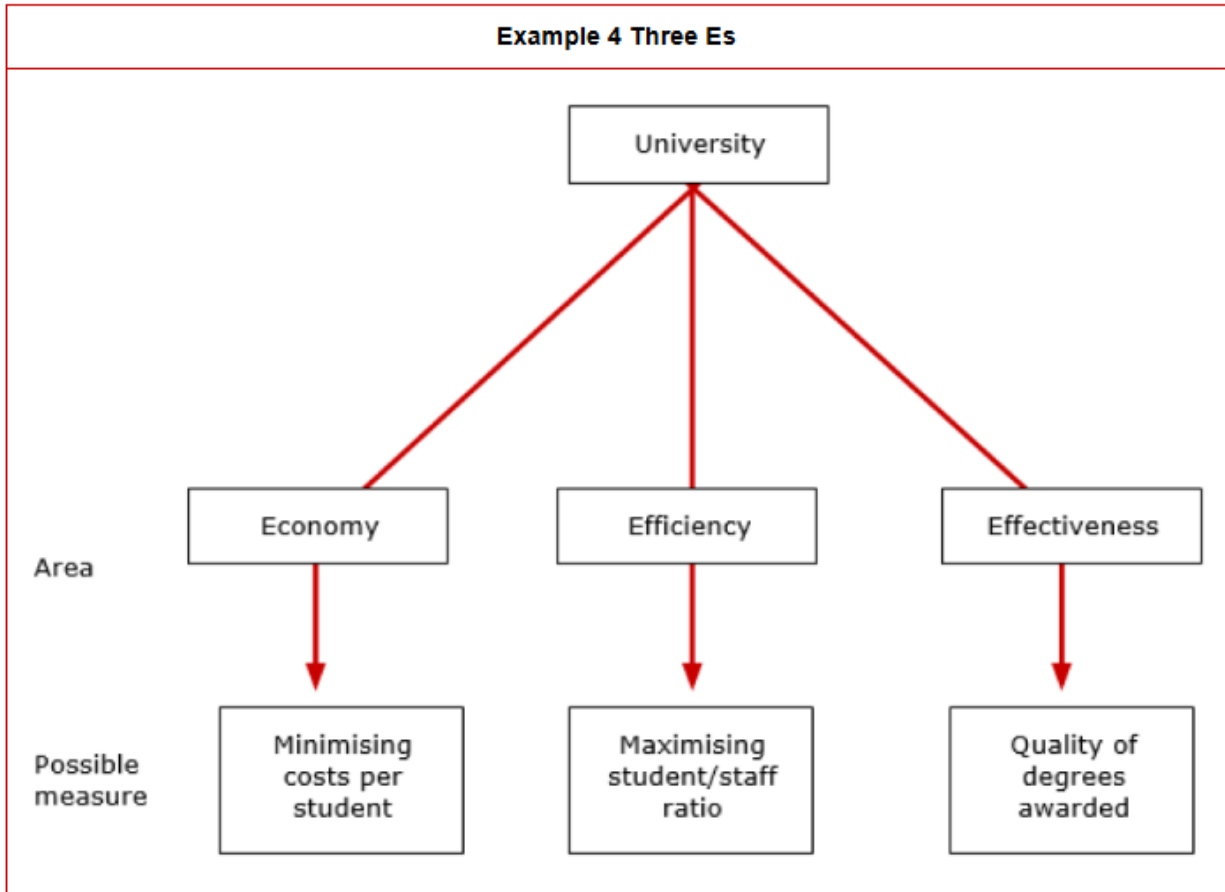
- The university management is accountable to multiple stakeholders (i.e. students, government and potential employers).
- However, the requirements of the stakeholders may *conflict*.
 - Students may desire small class sizes, a large library and courses that meet their interests.
 - The government may wish to minimise costs per student.
 - Potential employers may want graduates with relevant skills for industry (e.g. engineering graduates as opposed to anthropology graduates).
- Designing a performance measurement system to deal with multiple and conflicting objectives is very difficult.
- Management must try to rank its principals/stakeholders and prioritise objectives.

2.4.2 Value for Money

NPOs aim to maximise the difference between the benefits generated and the costs of their operations. The maximisation of this difference is commonly known as Value for Money and is similar to profit maximisation apart from society's interests being maximised rather than profit.

There is an increasing emphasis on Value for Money and achieving the "3 Es", which are:

1. Economy – securing resources as economically as possible, that is minimising the input costs of the organisation.
2. Efficiency – employing resources as efficiently as possible within the organisation, that is maximising the output/input ratio.
3. Effectiveness – using resources as effectively as possible to meet the organisation's objectives.



Activity 1 Three Es

[Hide](#)

Categorise the following measures of value for money for an international charity as economy, efficiency or effectiveness:

- a) Service delivery per dollar spent
- b) Beneficiary satisfaction
- c) Cost per project activity
- d) Change in beneficiaries lives
- e) Cost per beneficiary
- f) Utilisation of volunteer hours
- g) Resource consumption in programs
- h) Procurement savings

**Please use the notes feature in the toolbar to help formulate your answer.*

Measure	Category	Comment
(a) Service delivery per dollar spent	Efficiency	Conversion of financial resource into output
(b) Beneficiary satisfaction	Effectiveness	High satisfaction indicates effective service delivery
(c) Cost per project activity	Economy	The cost associated with specific project activities
(d) Change in beneficiaries lives	Effectiveness	Achieving intended outcomes
(e) Cost per beneficiary	Economy	The cost incurred for each individual beneficiary
(f) Utilisation of volunteer hours	Efficiency	Converting volunteer resource into output
(g) Resource consumption in programs	Efficiency	Use of energy, water, etc in program implementation
(h) Procurement savings	Economy	Comparing actual costs to budgeted amounts

2.4.3 Measuring Performance

As well as having multiple, conflicting objectives, it is often difficult to measure the output of NPOs in a meaningful way. For example, is a high proportion of first-class degrees awarded alone an adequate measure of the quality of a university course?

To overcome these issues performance can be measured by:

- Using judgments by experts, accepting that measurement must, to some extent, be subjective;
- Where possible, using comparisons to other similar bodies (e.g. for the public sector) or against historical results;
- Using inputs (e.g. teaching staff hours paid or costs), bearing in mind the apparent disadvantage of such an approach – that high inputs do not necessarily guarantee an objective is more likely to be achieved. For example, teaching staff might only be teaching for 75% of the hours they are paid for.

However, it is not impossible to measure the output of NPOs in a meaningful way. Continuing with the university example, outputs might be measured in terms of:

- Broader performance measures – number of students finding employment after graduation, number of publications by teaching staff, proportion of students graduating;
- Operational performance measures – number of courses offered, unit costs for each operating “unit”, class sizes, etc.

The following activity assumes knowledge of ratios from *Management Accounting*.

Activity 2 Not-for-profit Performance

[Show](#)

Lewisville is a town with a population of 100,000 people. The town council of Lewisville operates a bus service (Lewisville Bus Company, or LBC) which links all parts of the town with the town centre. The service is non-profit seeking, and its mission statement is "to provide efficient, reliable and affordable public transport to all the citizens of Lewisville". Attempting to achieve this mission requires operating services considered uneconomic by private-sector bus companies, due to the small number of passengers travelling on some routes or the low fares charged.

However, one council member has recently criticised the performance of the Lewisville bus service compared with those operated by private-sector bus companies in other towns. She has produced the following information for the most recent financial year:

Summarised Income and Expenditure Account

Summarised Income and Expenditure Account		
	\$000	\$000
Passenger fares		1,200
Staff wages	600	
Fuel	300	
Depreciation	<u>280</u>	
		(1,180)
Surplus		<u>20</u>
Capital employed \$2,210,000		
Operating statistics for Lewisville bus service		
Total passengers carried	2,400,000 passengers	
Total passenger miles travelled	4,320,000 passenger miles	
Industry average ratios for private sector bus companies		
Return on capital employed	10%	
Return on sales (net margin)	30%	
Asset turnover	0.33 times	
Average cost per passenger mile	\$0.37	

Required:

- Using the information provided above, compare the performance of the LBC to the industry average for private sector companies.
- Discuss the validity of comparing the performance of the LBC with private sector bus companies.
- Explain the meaning of the following terms in the context of performance measurement and suggest a measure of each one appropriate to a public sector bus service:
 - Economy;
 - Effectiveness; and
 - Efficiency.

*Please use the notes feature in the toolbar to help formulate your answer.

(a) Performance comparison

Regarding **profitability**, LBC has performed poorly compared to the industry averages for the private sector.

- Return on capital employed is 0.9% compared to 10% for private companies.
- Profit margins are low at 1.7% net profit margin compared to 30% for private companies.

This is probably because, LBC has to provide services on less profitable routes as a public sector provider.

It also charges lower fares than the private sector.

However, given that LBC is not a commercial enterprise and its mission is to provide efficient, affordable transport to the residents of Lewisville, such profitability measures are not relevant.

Asset turnover is higher for LBC than for the private sector.

LBC's revenue is 0.54 times capital employed, compared to 0.33 for the private sector.

This suggests that the LBC buses achieve higher utilisation than private sector buses. This may be because LBC fares are lower compared with the private sector. The buses are utilised to a greater extent, with more daily journeys than private sector buses.

This is a good sign, as it shows that LBC is achieving greater utilisation of its capital than the private sector bus companies (for revenue generation).

Cost per passenger mile is only 73% of the industry average (\$0.27/\$0.37). This is positive because it means that the company is managing to provide more journey distance for a lower cost, representing better resource use. This may be due to a higher number of passengers or longer bus routes, especially if LBC covers routes to outlying suburbs far from the city centre.

The fares per passenger mile charged by LBC are only 53% (\$0.28/\$0.53) of the fares charged by the private sector bus companies. As already discussed, this reduces the profitability of LBC.

However, low fares are likely to be appreciated by the bus users. The lower fares are part of the policy of providing affordable public transport.

Calculation of ratios

- Return on capital employed = $\frac{\text{Operating profit}}{\text{Capital employed}} \times 100 = \frac{\$20,000}{\$2,210,000} \times 100 = 0.9\%$
- Net profit margin = $\frac{\text{Operating profit}}{\text{Sales}} \times 100 = \frac{\$20,000}{\$1,200,000} \times 100 = 1.7\%$
- Asset turnover = $\frac{\text{Sales}}{\text{Capital employed}} \times 100 = \frac{\$1,200,000}{\$2,210,000} \times 100 = 0.54 \text{ times}$
- Average cost per passenger mile = $\frac{\text{Operating costs}}{\text{Passenger miles}} = \frac{\$1,180,000}{4,320,000} = \0.27
- Fares charged by bus service:

$$\text{Lewisville fare per passenger mile} = \frac{\text{Passenger fares}}{\text{Passenger miles}} = \frac{\$1,200,000}{4,320,000} = \$0.28$$

$$\text{Private sector fare per passenger mile} = \frac{\text{Average cost}}{1 - \text{Net margin}} = \frac{\$0.37}{1 - 0.30} = \$0.53$$

(b) Validity of comparison

There are several reasons why comparing the performance of LBC with the average ratios for the private sector is not valid. First, the objectives of LBC, as stated in its mission, are to provide efficient, reliable and affordable public transport to the citizens of Lewisville. Private sector companies are more likely to wish to maximise shareholders' wealth. This means that direct comparisons become less meaningful for the following reasons:

- Return on capital and profit margin are irrelevant for LBC, as the organisation does not exist to make a profit.
- Private sector bus companies may "cherry pick" the most profitable routes, and avoid providing services on less populated routes. A public sector company such as LBC may provide services on less populated routes as part of its mission of providing services to all citizens.
- The public sector bus company may aim to charge lower fares – particularly to specific groups such as pensioners and children. Private sector companies may not feel obliged to do this and will use a pricing policy to maximise profits.

Despite these limitations, comparing the costs of running the services, quality and utilisation could lead to some benefits.

(c) Economy, effectiveness and efficiency

It is sometimes suggested that measuring performance of public sector organisations should be assessed based on the 3 Es – economy, effectiveness and efficiency.

Economy is an input measure based on the organisation's expenditure. In the case of the LBC, it could be measured by total expenditure compared to budget.

Effectiveness is an output measure that looks at what the organisation achieves in terms of its objectives. LBC can measure its effectiveness by the number of passengers carried or passenger miles travelled.

Efficiency is a combination of the above two measures. It considers output from a given input. For LBC, it could be measured by cost per passenger mile travelled.

2.5 Environmental Issues

Environmental or "green" issues are an area of growing concern to all parties, companies included.

Managers must understand how the organisation's operations affect the environment to satisfy public concerns and, increasingly, avoid any penalties or costs due to environmental regulations.

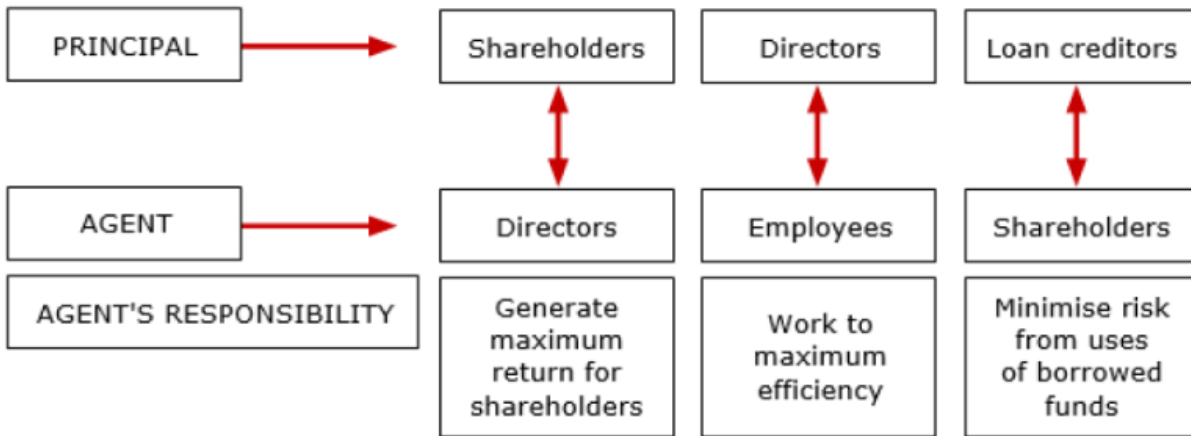
For these reasons, environmental reporting is becoming more common in company financial reporting.

Some organisations have adopted the triple bottom line approach: a method of *true cost accounting*. This approach considers the effect of production decisions in terms of environmental and social value, as well as economic value. Those companies that create environmental and social value alongside economic value are often considered to have a sustainable triple bottom line.

Green policies and sustainability issues are considered further in *Chapter 2*.

3.1 Agency Theory

Agency theory examines the duties and conflicts between the parties within an agency relationship. In a company, these include the following:



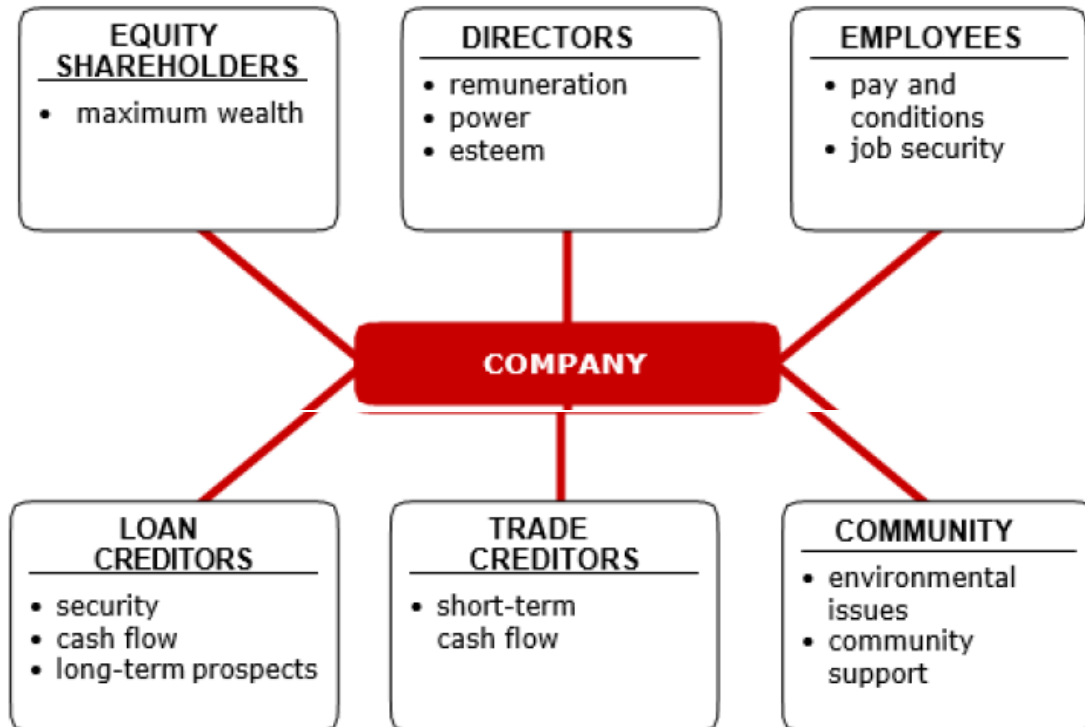
A company can be viewed as a set of contracts between these various interest groups. The company will not succeed unless all groups work towards the same objectives. Although most research has focused on the potential conflicts between directors and shareholders, there are other possible sources of tension within a company.

For example, when the amount of a company's assets are close to the amount of its liabilities, the debt investors will pressure the directors to undertake only low-risk projects. This is because the debt investors have nothing to gain from risky projects (they receive fixed returns) but everything to lose (if assets fall below liabilities, the company may become unable to service its debts).

Equity investors, on the other hand, may encourage the directors to take on risky projects. This is because equity investors participate in any excess returns, but due to the protection provided by their limited liability status, they cannot be forced to cover any losses.

3.2 Stakeholders

Companies are made up of various interest groups or stakeholders, all of whom are likely to have different interests and objectives for the company.



A company's stakeholders can be grouped as follows:

- Internal – e.g. employees, directors;
- Connected – e.g. shareholders, customers, suppliers, competitors; and
- External – e.g. local and national communities, government, pressure groups.

Conflict between these various stakeholders is possible. For example, efforts to reduce environmental effects (a community objective) will likely reduce profit and, therefore, shareholder wealth.

While shareholders are the key stakeholders, modern corporate governance suggests that directors should take account of the objectives of a broader range of interested parties. Directors are therefore expected to show responsibility to creditors (e.g. reasonable payment terms), employees (e.g. health and safety) and ultimately to society as a whole through corporate social responsibility (e.g. minimising pollution, investing in social projects, etc).

Therefore, the overall corporate objective may become satisficing (i.e. producing satisfactory rather than maximum returns for shareholders). With the rise of the ethical investor on world stock markets, many shareholders are willing to accept slightly lower returns in exchange for their companies following a wide range of financial and nonfinancial objectives.

3.3 Directors and Shareholders

In larger companies, the shareholders entrust its management to the directors. This is referred to as the separation of ownership from control. The directors manage the company on behalf of the shareholders. They should always act in the shareholders' best interests while also taking into account the objectives of other stakeholder groups.

This may not always be the case, as the directors may have other personal objectives, such as:

- increasing personal remuneration levels;
- maximising bonus payments;
- empire building; and/or
- job security.

In addition to the personal aspects shown above, a small number of directors have been guilty of not fulfilling their fiduciary duties by encouraging an organisation to:

- use creative accounting, which can flatter the accounts (also known as window dressing);
- use off-balance sheet financing (e.g. keeping liabilities off the statement of financial position by using special purpose vehicles);
- reject takeover offers, with the directors' protecting their jobs rather than acting in the shareholders' interests; and
- disregard environmental issues, with directors allowing pollution emitting processes and product testing on animals.

If directors follow personal objectives which conflict with those of their shareholders, this leads to *agency costs* (i.e. lost potential returns for shareholders). This is often referred to as an agency problem.

Good corporate governance procedures (see s.3.5) should be implemented to minimize the effect of agency costs. Unfortunately, implementing corporate governance brings additional costs (particularly in the case of the Sarbanes-Oxley Act [2002] in the United States). Therefore, a cost-benefit approach should be taken to determine an appropriate level of control over directors.

It can be argued that the actual return to shareholders = Maximum potential return – agency costs – the cost of engaging in corporate social responsibility.

To some degree, shareholders should be more active in monitoring the behavior of directors. Most shares in listed companies are held by institutional investors (e.g. pension funds). Fund managers have often been guilty of operating in a very passive way. For example, proxy voting rights are not always exercised by institutional investors. Until there is a rise in shareholder activism, it remains likely that some directors will continue to work in their own best interest.

3.4 Encouraging the Achievement of Stakeholder Objectives– Managerial Reward Schemes

Goal congruence occurs when the objectives of agents acting within an organisation align with the organisation's objectives. For example, managers should be encouraged to aim for long-term growth and prosperity rather than short-term reported profitability.

Methods of encouraging goal congruence between directors and shareholders include:

- Performance-related pay – however, relating pay or bonuses to profits include short-termism (e.g. not investing for future growth to maintain high profits in the short term).
- Executive share option scheme – the evidence is mixed regarding the success of such schemes in motivating directors to improve performance (e.g. a company's share price may rise due to a general rise in the stock market rather than the quality of its management).
- Long-term incentive plans (LTIPs) – paying a bonus to directors if the company's performance over several years is good when benchmarked against competitors.
- Transparency in corporate reporting.
- Increased shareholder activism (e.g. using voting rights).
- Improved corporate governance (e.g. appointing genuinely independent non-executive directors).

3.5 Encouraging the Achievement of Stakeholder Objectives– Corporate Governance

Definition

Corporate governance – the system by which companies are directed and controlled.

An objective of corporate governance is to reduced agency costs to a level acceptable to shareholders.

3.5.1 Principles of Good Governance

Various countries have developed codes of **corporate governance**. Although detailed knowledge of specific requirements is not required, candidates should have an awareness of the main principles that underlie these codes:

- Every company should be headed by an effective board which should lead and control the company.
- There should be a clear division of responsibilities at the head of the company between running the board (chair) and running the business (CEO); no single individual should dominate.
- The board should have a balance of executive and *independent* non-executive directors.
- All directors should be required to submit themselves for re-election regularly.
- Remuneration committees should be comprised of independent non-executive directors.
- Remuneration committees should provide the packages needed to attract, retain and motivate executive directors and avoid paying more.
- No director should be involved in setting their remuneration.
- The board should maintain a solid system of internal control to safeguard shareholders' investment and the company's assets.

3.5.2 Government Regulation

In the UK, the Listing Rules require companies:

- to apply the Main Principles of the UK Corporate Governance Code ("the Code"); and
- report to shareholders on how they have done so.

Although compliance with the provisions of the Code is not obligatory, any listed company which chooses not to comply with one or more provisions must explain its reasons for doing so.

The US Sarbanes–Oxley Act applies to all companies listed on a US stock market, including their foreign subsidiaries. Compliance is mandatory.

In South Africa, the King IV™ Code provides a set of voluntary principles and leading practices on ethical leadership, corporate citizenship, risk management, and the governance of information and technology.

In Australia, the Australian Securities Exchange (ASX) corporate Governance *Council's Corporate Governance Principles and Recommendations* set out recommended corporate governance practices for entities listed on the ASX that, in the Council's view, are likely to achieve good governance outcomes and meet the reasonable expectations of most investors in most situations.

3.5.3 Listing Regulations

Stock exchanges have rules and regulations to ensure that the stock market operates fairly and efficiently for all parties involved. To become listed on a stock exchange, a company must meet the listing requirements set out in the listing rules as part of its approval process.

Syllabus Coverage

This chapter covers the following Learning Outcomes.

A. Financial Management Function

1. The nature and purpose of financial management

- a) Explain the nature and purpose of financial management.
- b) Explain the relationship between financial management and financial and management accounting.

2. Financial objectives and the relationship with corporate strategy

- a) Discuss the relationship between financial objectives, corporate objectives and corporate strategy.
- b) Identify and describe a variety of financial objectives, including:
 - i) shareholder wealth maximisation
 - ii) profit maximisation
 - iii) earnings per share growth.

3. Stakeholders and impact on corporate objectives

- a) Identify the range of stakeholders and their objectives.
- b) Discuss the possible conflict between stakeholder objectives.
- c) Discuss the role of management in meeting stakeholder objectives, including the application of agency theory.
- d) Describe and apply ways of measuring achievement of corporate objectives including:
 - i) changes in dividends and share prices as part of total shareholder return.
- e) Explain ways to encourage the achievement of stakeholder objectives, including:
 - i) managerial reward schemes such as share options and performance-related pay
 - ii) regulatory requirements such as corporate governance codes of best practice and stock exchange listing regulations.

4. Financial and other objectives in not-for-profit organisations

- a) Discuss the impact of not-for-profit status on financial and other objectives.
- b) Discuss the nature and importance of Value for Money as an objective in not-for-profit organisations.
- c) Discuss ways of measuring the achievement of objectives in not-for-profit organisations.

Summary and Quiz

- The first step in developing the objectives of an organisation is to identify the relevant stakeholders.
- In the corporate sector, the principal stakeholders are the shareholders. Therefore, most traditional finance theory is built on the assumption that a company's objective is to maximise the wealth of its shareholders.
- Shareholder wealth is measured by total shareholder returns, which combines dividend and share price growth.
- Often profit maximisation and EPS growth are used as proxies for shareholder wealth maximisation.
- Corporate social responsibility suggests that directors should take account of other stakeholders and follow a range of non-financial objectives (e.g. employee satisfaction and reducing environmental effects).
- Non-financial objectives may conflict with maximising shareholder wealth. Therefore, the overall goal may be to produce satisfactory returns for shareholders while attempting to meet the demands of other interest groups.
- Not-for-profit organisations (NPOs) are often constrained by multiple stakeholders with potentially conflicting objectives other than profit.
- The objective of many NPOs is to maximise Value for Money (the "3 Es").
- Agency theory examines the duties and conflicts between the parties in an agency relationship.
- A company's managers may have personal objectives which conflict with their responsibilities as agents of the shareholders. These conflicts create agency costs for the shareholders. Managerial reward schemes can be used to encourage goal congruence, but these can be manipulated.
- Good corporate governance systems should reduce agency costs to an acceptable level.

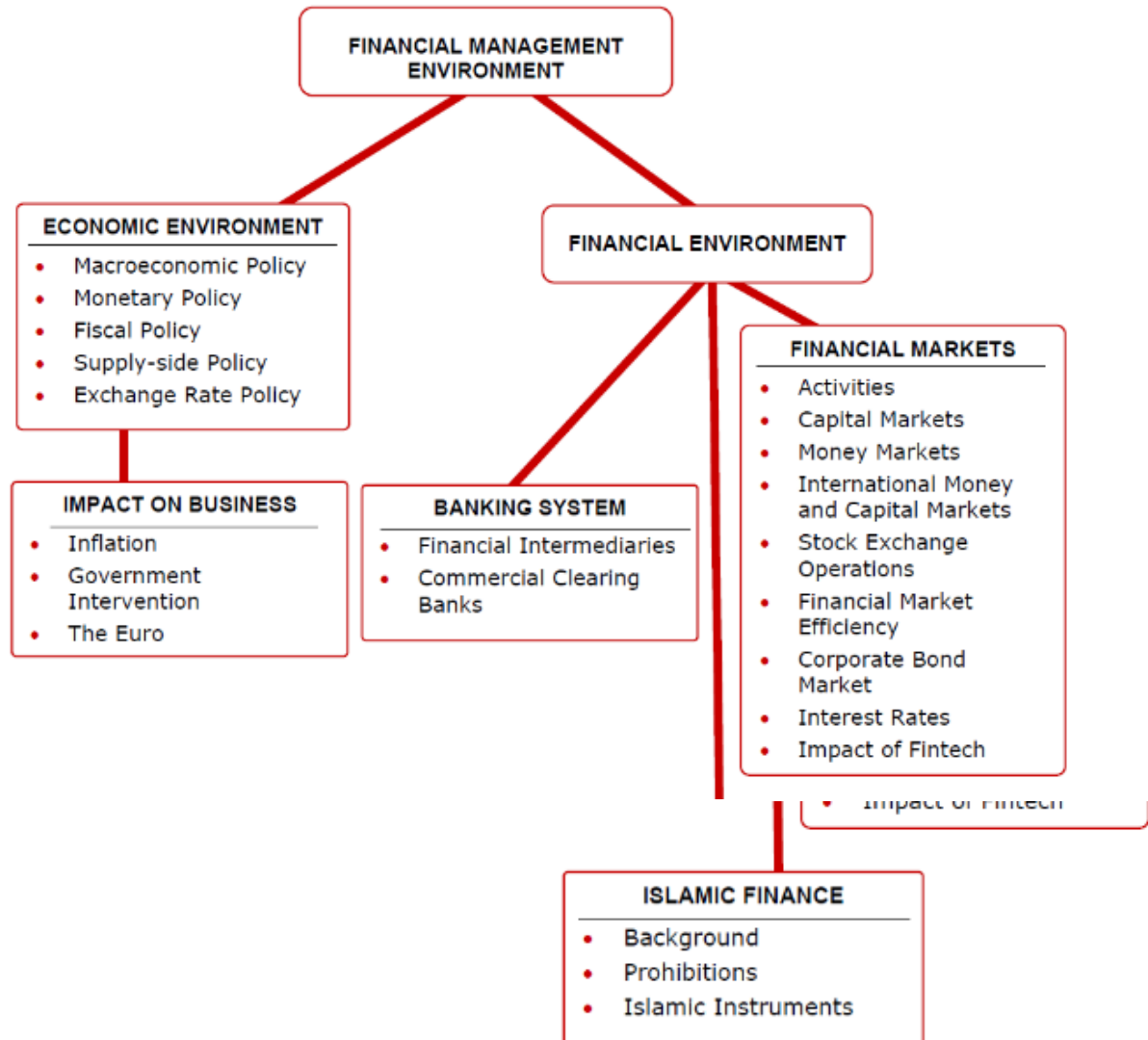
Chapter 1 Quiz

Chapter 2: The Financial Management Environment

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CHAPTER 2: Visual Overview

Objective: To understand the economic and financial environments in which organisations practise financial management.



1.1 Macroeconomic Policy

i Definition

Macroeconomic policy– the setting of economic objectives by the government (e.g. full employment, economic growth, the avoidance of inflation) and the use of control instruments to achieve those objectives (e.g. fiscal policy and monetary policy).

1.1.1 Objectives

Macroeconomic policies are adopted to achieve the following objectives:

- full employment;
- economic growth and thereby improved living standards;
- an acceptable distribution of wealth;
- price stability and therefore limited inflation; and
- solid balance of payments (because a continual external deficit, where a country imports more goods and services than it exports, is unsustainable and is likely to lead to an exchange rate crisis).

The above objectives can often conflict (e.g. economic growth can lead to excess resource demand and an increase in inflation).

1.1.2 Global Economic Events

Financial managers must understand not only national government economic policy but also world macroeconomic trends, due to the increasing interdependence of economies through the movement of trade and capital.

Recent world economic events include the following:

- The growing importance of emerging economies (e.g. the BRIC countries of Brazil, India and China).
- Using technology to raise funds (e.g. in crowdfunding and peer-to-peer lending) allows small businesses to raise capital without using banks or traditional financial markets.
- The coronavirus (COVID-19) pandemic (2020-2023) considerably impacted economic growth and global business, affecting, for example, tourism, international travel, supply chains and oil prices.
- Already weakened by the pandemic, higher-than-expected inflation worldwide has triggered tighter monetary policies.

1.2 Monetary Policy

1 Definition

Monetary policy– the set actions taken by the government or the central bank to achieve economic objectives using monetary instruments.

Monetary policy actions may either:

- *directly* control the amount of money in circulation (the *money supply*); or
- attempt to reduce the demand for money through its price (interest rates). For example, increasing interest rates increases the cost of borrowing, decreasing the demand for goods and the inflation rate.

By exercising control in these ways, governments can regulate the level of demand in the economy. Those who see monetary policies as crucial to managing macroeconomic activity are called *monetarists*.

A central bank's interest rate policy is just one of the factors that influences interest rates on financial instruments (see s.4.8).

1 Definition

Financial instrument – a contract for a monetary asset.

1.2.1 Direct Control of the Money Supply

Governments or central banks can *directly* control the money supply in the following ways:

1. Open market operations

- If the central bank sells government securities, the money supply is contracted, as some of the funds available in the market are "soaked up" by purchasing the government securities.
 - The sale of government securities will reduce bank deposits due to the level of funds that have been soaked up.
 - This in turn can lead to a further reduction in the money supply, as the banks' ability to lend is reduced. This is known as the *multiplier effect*.
- Equally, if the central bank were to buy back securities, funds would be released into the market.

2. **Reserve asset requirements (cash reserve ratio):** the central bank can set a minimum level of liquid assets which banks must maintain. This limits their ability to lend and thereby reduces the money supply.
3. **Special deposits:** the central bank can have the power to call for *special deposits*. These deposits do not count as part of the bank's reserve base against which it can lend. They have the effect, therefore, of reducing the bank's ability to lend and thereby reducing the money supply.
4. **Direct control:** the central bank may set specific limits on the amount banks lend. Credit controls are challenging to impose as, with fairly free international movement of funds, they can easily be circumvented.

1.2.2 Indirect Control of the Money Supply

Governments can reduce the demand for money, and indirectly reduce the money supply by encouraging an increase in short-term interest rates.

1.2.3 Problems of Monetary Policy

Various problems arise with monetary policy.

- ✗ A significant time lag often exists between implementing a policy and its effects.
- ✗ Credit control is ineffective in the modern global economy.
- ✗ The relationship between interest rates, level of investment and consumer expenditure is not stable and predictable.
- ✗ Increasing interest rates produces undesirable side effects, including:
 - Less investment, reduced industrial capacity, and increased unemployment (as higher interest rates increase the cost of capital for a company using debt finance).
 - A downward pressure on share prices, making it more difficult for companies to raise funds from new share issues.
 - Decreases in consumer demand.
 - An overvalued currency, which reduces demand for exports.

1.3 Fiscal Policy

1 Definition

Fiscal policy – government actions to achieve economic objectives through the fiscal instruments of taxation, public spending and the budget deficit or surplus. Governments can use public expenditure and taxation to regulate the level of demand within the economy.

1.3.1 Keynesian Approach

Those who view **fiscal policy** as crucial in controlling macroeconomic activity are called *Keynesians* (after the economist John Maynard Keynes).

If the economy is in recession, fiscal policy can be used to reflate the economy by taking the following actions:

- ✓ Increase government spending to increase the level of demand in the economy directly (e.g. if a government initiates large road-building projects or establishes training schemes for sections of the population, the demand for goods and services in the economy is increased).
- ✓ Reduce taxation to boost both consumption and investment.

However, problems can occur in reflation the economy due to the following:

- ✗ Government spending is an intervention in a free market, which can lead to the misallocation of resources (e.g. support for inefficient industries).
- ✗ There is often a significant time lag between the authorisation of additional spending and its actual occurrence.
- ✗ Tax cuts are not efficient at boosting domestic demand as, in times of recession, some of the extra disposable income made available will be saved, and some of the extra monies spent will be on imports.
- ✗ A large budget deficit is likely to occur, which will lead to a large Public Sector Net Cash Requirement (PSNCR).
- ✗ The inflation rate is expected to rise as demand may increase for resources in limited supply, and their prices may rise.

Formerly known as the Public Sector Borrowing Requirement (PSBR), the PSNCR is the difference between the expenditure of the public sector and its income. A deficit is financed by borrowing – principally via the sale of government gilt-edged stocks (“gilts”).

If there is too much demand in the economy (it is “overheating”), fiscal policy can be used to depress demand or deflate the economy by possibly taking the following actions:

- Reduce government spending to decrease directly the level of demand in the economy.
- Increase taxation to reduce consumption and assist with the redistribution of wealth.

However, problems can arise in deflating the economy due to the following:

- ✗ It is not possible to cut government spending dramatically in sectors such as health care or education.
- ✗ Increasing taxation discourages enterprise and innovation.

Keynesians prefer adjusting government spending to changing tax rates, as they believe it has a quicker and more significant impact on the level of demand in the economy.

1.3.2 Relationship Between Fiscal and Monetary Policy

Fiscal and monetary policies are interdependent and governments will use both fiscal and monetary policies to achieve their monetary and budgetary targets. Which policies dominate depends on the economic theory preferred by the government of the day.

For example, a Keynesian approach to managing the economy has been implemented in many countries at some time. For example:

- In Australia in response to economic challenges. For instance, during the global financial crisis, implementing stimulus packages such as infrastructure spending and cash transfers to households, to support domestic demand and mitigate the impact of the economic downturn.
- In Canada in response to economic downturns. For instance, during the 2008 financial crisis, implementing infrastructure spending and tax cuts, to stimulate economic growth and job creation.
- In the UK from the 1930s to the late 1970s. However, as this was believed to have contributed to the “boom-bust” economic cycles experienced, later governments have followed a more monetarist approach.
- In the US during economic recessions. For instance, implementing the Troubled Asset Relief Program (TARP) and the American Recovery and Reinvestment Act (ARRA) during the 2008 financial crisis, to boost economic activity.

1.4 Supply-side Policy

i Definition

Supply-side policies – policies which focus on creating the right conditions in which private enterprise can grow and raise the capacity of the economy to provide the output demanded.

1.4.1 Supply-side Policy Examples

For **supply-side policy** adherents, the private sector (being driven by the profit motive) is deemed more efficient at providing the output required than the public sector.

Supply-side policies include:

- low corporate tax rates to encourage private enterprise;
- promoting a stable, low inflation economy with minimal government intervention;
- limited government spending;
- a balanced fiscal budget;
- deregulation of industries;
- a reduction in the power of trade unions;
- an increase in the training and education of the workforce;
- an increase in infrastructure required by businesses (e.g. business parks); and
- a reduction in planning legislation.

1.4.2 Supply-side Policies and Fiscal Policy

Supporters of supply-side policies believe that for business to flourish, the economy must be stable and fiscal policy should be in equilibrium. In other words, government spending should not exceed government receipts from taxation.

Additionally, if the private sector is to be encouraged, tax rates should be kept to a minimum and government expenditure also should be kept to a minimum.

1.4.3 Supply-side Policies and Monetary Policy

Monetary policy is used to control inflation, and this provides the stable economy required for business to flourish.

1.4.4 Problems with the Supply-side Approach

Problems with using supply-side policies include the following:

- ✗ There is a time delay before the policies have any impact.
- ✗ The private sector will not provide all the goods and services society requires.

1.5 Exchange Rate Policy

i Definition

Exchange rate policy – how a government manages its currency in relation to foreign currencies.

Exchange rate policy is closely related to monetary policy and generally depends on many of the same factors.

1.5.1 Reasons for Controlling Exchange Rates

- To rectify a balance of trade deficit. If inflation in one country is higher than in others, its export prices will become uncompetitive in overseas markets so its trade deficit will grow. Therefore the government may try to bring about a fall in the exchange rates to make exports less expensive.
- To prevent a balance of trade surplus. A government may try to bring about a limited rise in exchange rates to make imports less expensive.
- To stabilise the exchange rate. If importers and exporters face less exchange rate risk, confidence in the currency will improve, facilitating international trade.

1.5.2 Floating Exchange Rate

A "freely floating" exchange rate means that the currency's value is allowed to move freely with supply and demand market forces.

The main sources of demand are:

- exports of goods;
- exports of services;
- inflows of foreign investment;
- speculative demand.

The main sources of supply are:

- imports of goods;
- imports of services;
- outflows of foreign investment;
- speculative selling.

Floating exchange rates tend to correct a trade surplus or deficit. If imports exceed exports, the supply of the home currency will exceed demand and the home currency will depreciate, boosting demand for exports and correcting the trade imbalance.

A "managed float" means that the exchange rate is allowed to float but the central bank intervenes to prevent major fluctuations.

1.5.3 Fixed Exchange Rate

A fixed exchange rate ("fixed peg") regime is one in which the rate is kept fixed against that of another currency, or basket of currencies. **No** fluctuations are permitted; the central bank intervenes to maintain the exchange rate.

Occasional revaluation or devaluation may be required when economic fundamentals demand it (e.g. during the 2008 global financial crisis, many governments devalued their currencies to boost growth through higher exports).

1.5.3 Crawling Peg

In a crawling peg system, a currency is allowed to fluctuate but only within a relatively narrow range around a target rate. The target rate may have to be reset due to inflation.

Example 1 Crawling Peg

In the 1990s, Mexico fixed its peso against the US dollar. However, due to the significant inflation in Mexico compared to the US, it was evident that the peso would need to be significantly devalued. However, because a rapid devaluation would create instability, Mexico adopted a crawling peg exchange-rate adjustment system. This allowed the peso to transition to a more appropriate exchange rate without a sudden devaluation.

2.1 Inflation

i Definition

Inflation – the rate of increase of the general level of prices in the economy.

2.1.1 Measurement

Inflation is usually measured by the Consumer Price Index (CPI).

i Definition

Consumer Price Index (CPI) – a measure of the average change over time in the prices paid by consumers for a market basket of consumer goods and services.

The CPI relates to retail goods and services and indicates how fast average prices are rising across the economy generally. (The inflation of specific goods and services may be higher or lower than the CPI.)

The maintenance of a low level of inflation is a critical economic objective.

Alternative measures of inflation also exist. For instance, look at the increases in the costs of the manufacturing industry.

2.1.2 Causes of Inflation

Keynesians consider there to be two major causes of inflation:

1. **Demand-pull inflation:** inflation arises due to demand exceeding the maximum output of the economy with full employment.
2. **Cost-push inflation:** increases in the cost of raw materials or the cost of labour lead to increases in the unit costs of production, which inevitably lead to price increases as higher costs are passed on to the consumer. The term "imported cost-push inflation" is used if the increased costs arise on imported goods.

Conversely, monetarists believe that inflation arises due to "too much money chasing too few goods". Therefore, in the short term, inflation should be controlled by controlling the money supply; in the long term, inflation should be managed by improving the economy's ability to produce goods and services.

Inflation is also thought to be brought about by people's expectations, as anticipation of future price increases is built into wage negotiations to protect future real incomes. In turn, expected cost increases, such as wage costs, are built into output prices. The term "wage-price spiral" suggests that inflation is ongoing and inevitable.

2.1.3 General Economic Consequences of Inflation

The general economic consequences of inflation include the following:

- The redistribution of income from those in a weak bargaining position to those in a strong bargaining position: those in a strong position are able to maintain the real value of their income.
- A disincentive to save, as the purchasing power of investments may be reduced.
- When inflation reaches very high levels, money can no longer carry out its primary functions as medium of exchange and a store of value.
- A fall in the exchange rate.
- A need for higher nominal interest rates (i.e. interest rates incorporating inflation).

2.1.4 Consequences of Inflation for Businesses

Inflation affects business operations in numerous ways, including:

- Entrepreneurial activity is reduced, as it is harder to estimate the likely returns from a new venture, and higher interest rates make borrowing more expensive.
- International competitiveness suffers where prices rise faster than those of foreign competitors.
- Increased uncertainty reduces new investment by existing businesses.
- Higher interest rates reduce the number of profitable investment opportunities, reducing the level of investment.
- In periods of rapid inflation, significant costs are added to industry because of the need to:
 - search for the best price currently available for the goods and services required; and
 - be constantly updating selling prices.
- Inflation will also affect business performance and the evaluation of this performance.
- Conventional historical cost accounts have the following problems during periods of significant inflation:
 - The historical cost of assets understates the value of the assets.
 - Changes in asset values are ignored until they are realised.
 - Gains arising from holding assets are treated as being fully distributable.
- The result is that profits become overstated (current revenues are charged with a measure of historical cost) and capital becomes understated. Therefore, the ROCE (return on capital employed) is overstated.

Alternative approaches which have been suggested include the following:

- The valuation of assets at their deprival value, which is the lower of replacement cost and recoverable amount (i.e. the higher of value-in-use and selling price);
- The use of current purchasing power or current cost accounting to eliminate the above distortions and ensure the maintenance of the shareholders' funds in real terms.

2.2 Government Intervention

2.2.1 Reasons Governments Intervene in the Free Market

Governments will intervene in the operation of the free market if:

- monopolies, mergers or restrictive practices operate against the public interest;
- an industry is of national strategic importance;
- the free market creates social injustice;
- companies fail to take into account how their actions affect others outside of the company (such effects are known as externalities, and a typical example is pollution caused by a company);
- the free market fails to provide sufficient public goods, such as health care or education; or
- the free market cannot provide the required capital (e.g. when a large infrastructure project, such as constructing a new tunnel or bridge, is undertaken).

Exam advice

You must be able to explain the effect of government policy (e.g. competition policy) on a business if asked in the exam.

2.2.2 Competition Policy

Governments develop policies to stimulate competition to increase the efficiency of the economy. Competition policy can vary across countries based on their economic structures, legal systems, and policy priorities. The main components of competition policy in the UK have been as follows:

- Monopolies and mergers legislation prevents the development of monopolies, which would have the power to act against the public interest.
- Restrictive practices legislation has eliminated practices such as the setting of retail prices by manufacturers;
- Deregulation in specific industries has removed regulations that restrict competition. For example, deregulation of the UK stock market reduced dealing costs, which increased the volume of trading considerably.
- The creation of internal markets within certain areas of the public sector (e.g. hospitals or schools). The providers of public sector services must compete for the resources they require based on the services they provide to their users.
- The Competition and Markets Authority (CMA), a government department, was established to investigate situations which may be against the public interest (e.g. excessive market power). The CMA uses a 25% market share to indicate potential unfair influence.

While the UK has a well-established competition policy framework, other key components can be found in other countries. For example:

- **Industrial policy integration:** Integrating competition policy with industrial policy to achieve broader economic development goals might involve promoting specific industries or protecting domestic companies in strategic sectors. For example, in China, this supports and promotes key industries that are strategically important for economic development (e.g. semiconductor industry).
- **Oversight of State-owned Enterprises (SOEs):** Regulations seek to ensure that SOEs operate on a “level playing field” with private entities and do not engage in anticompetitive practices. For example, in Singapore, competition policies and regulatory frameworks address the conduct of private entities and SOEs to ensure fair competition.
- **Trade practices and export controls integration:** Integrating trade practices and export control policies with competition policy addresses issues such as export cartels, anti-competitive practices in international trade and restrictions on certain exports. For example, Australia's competition policy is complemented by trade practices regulations that address anti-competitive conduct and unfair trading practices. Australia also has export control measures to regulate certain exports.
- **Regional integration agreements:** Countries participating in regional economic blocs may harmonise or coordinate their competition policies to promote fair competition within the region. For example, the regional trade bloc in South America (MERCOSUR) aims to promote economic cooperation and integration among its member states (Argentina, Brazil, Paraguay, Uruguay and Venezuela).
- **Public procurement regulations:** Some countries have procurement regulations governing the awarding of public (i.e. government) contracts to prevent collusion and promote transparency. For example, the US seeks to ensure fair competition in government contracting through rules and oversight by agencies like the Federal Trade Commission and the Department of Justice.

2.2.3 Privatisation

Privatisation of state-owned organisations has been a global trend and many countries have transferred public enterprises to the private hands to the private sector through sale to the general public, direct sale to another company or management buyout. Examples include:

- **Brazil (mining):** Vale do Rio Doce (Vale), a major global mining company created by the government in the 1940s, was privatised 1998 to increase efficiency, attract investment and enhance competitiveness in the global market.
- **China (telecommunications):** China Telecom and China Unicom have undergone partial privatisation, with the introduction of private capital and the listing of shares on stock exchanges. This has allowed for increased efficiency and competition in the telecommunications sector.
- **India (national airline):** In 2021, the government sold its ownership stake in Air India to the Tata Group, aiming to improve the airline's performance and reduce the financial burden on the government.
- **Argentina (public utilities):** A series of privatisations of public utilities in the 1990s, including water, gas and electricity distribution, transferred ownership and operational control to private companies as part of broader economic reforms.
- **Russia (oil industry):** Large-scale privatisation in the 1990s included state-owned assets in the oil industry. Yukos was formed through the transfer of assets from the state-owned company, Yuganskneftegaz.
- **UK:** British Telecom, British Gas and electricity distribution companies.

Arguments in favour of privatisation include:

- ✓ An increase in competition where a state monopoly is split into several operating companies before sale or where the monopoly position is removed.
- ✓ A short-term boost to government revenues, which has a favourable impact on the PSNCR.
- ✓ Widening share ownership, which increases individuals' stake in the economy as a whole.
- ✓ A reduction in the PSNCR in the future, as the newly privatised industry's borrowings are no longer government funded.

Arguments against privatisation include:

- ✗ The creation of private sector monopolies, which have then required regulation to ensure that their monopoly position is not abused.
- ✗ Breaking up large organisations into smaller companies results in the loss of economies of scale.
- ✗ The quality of service may deteriorate.

2.2.4 Government Assistance for Business

Governments also use official aid schemes to intervene in the economy. Such schemes include cash grants, consultancy advice and tax incentives to encourage investment in high technology or investment in areas of particularly high unemployment.

Grants may also be available from transnational institutions. For example, the European Regional Development Fund has assisted with many infrastructure projects in the more remote regions of EU Member States.

Other examples of government assistance programs include:

- Enterprise Ireland: The Irish government agency responsible for supporting Irish businesses, particularly small and medium-sized enterprises (SMEs), provides a range of financial supports, including grants, funding for research and development and assistance for market expansion. It also offers consultancy services to help businesses improve their competitiveness and navigate international markets.
- Singapore Productivity and Innovation Credit (PIC) Scheme: This tax incentive program aims to encourage businesses to invest in productivity and innovation. Under the scheme, businesses can claim tax deductions or cash payouts for qualifying expenditures in areas such as research and development, automation, employee training and the acquisition of intellectual property.
- Canada Small Business Financing Program: This federal government initiative aims to help small businesses access finance for acquiring or improving assets. While not a direct grant program, it provides a loan guarantee to financial institutions, making it easier for small businesses to secure loans for eligible purposes. It also provides consulting services and advice to assist businesses in accessing financing.

2.2.5 Green Policies and Sustainability Issues

Governments are increasingly taking active steps to improve the environmental performance of organisation. Measures include:

- Carbon credits where the government allocates each polluting organisation a quota of the number of tonnes of greenhouse gasses that it can emit. If an organisation then switches to *greener* production techniques, it may have a surplus of carbon credits which can then be sold to an organisation that is exceeding its quota (i.e. carbon trading).
- Formation of government environment agencies, such as the UK Environment Agency. The Agency was established "to protect and improve the environment". One of its priorities is to "support sustainable growth".

i Definition

Sustainability – meeting the needs of the present without compromising the ability of future generations to meet their own needs.

– *United Nations*

For investors, sustainable investing means investing in progress and recognising that companies solving the world's biggest challenges can be best positioned to grow. Investors from global institutions to individuals now take a sustainable approach to pursuing their investment goals based on environmental, social and governance (ESG) insights.

However, increasing demand for eco-friendly products and environmentally ethical business practices has led to increased greenwashing (i.e. making people believe that a company is doing more to protect the environment than it is).

In 2020, the EU introduced regulations to establish a list of environmentally sustainable economic activities. An activity is environmentally sustainable only if it:

In 2020, the EU introduced regulations to establish a list of environmentally sustainable economic activities. An activity is environmentally sustainable only if it:

- "substantially contributes" to at least one of six environmental objectives (see opposite);
- does no significant harm to any of the other objectives;
- meets minimum social safeguards.

Six environmental objectives are:

- climate change mitigation;
- climate change adaptation;
- sustainable use and protection of water/marine resources;
- transition to a circular economy (i.e. one that keeps products, etc in circulation for as long as possible);
- pollution prevention and control;
- protection and restoration of biodiversity and ecosystems.

The regulation requires certain sustainability-related disclosures and applies to “financial market participants” selling financial products in the EU and large companies. For example, websites should disclose which products/investments contribute to which objectives, and make a disclaimer for all other products.

For **financial managers**, as well as seeking to meet the increasing demand for sustainable investments, ESG factors help to identify risks and opportunities to earn returns. ESG factors represent significant financial risks for businesses and their providers of finance (e.g. the risk of environmental damage due to climate change can increase the cost of capital, insurance costs, procurement costs and regulatory costs).

2.2.6 Corporate Governance

As previously noted in *Chapter 1*, the UK Corporate Governance Code is included in the Listing Rules of the London Stock Exchange. Although compliance is not obligatory, any listed company which does not comply with the Corporate Governance Code must explain its reasons for non-compliance.

Exam advice

Detailed knowledge of specific corporate governance codes is not required for the examination. Refer to *Chapter 1* for the main principles.

The US Sarbanes-Oxley Act (“SOX”) was introduced in 2002 and represented the most significant review of US corporate governance since the Securities Exchange Act of 1934. SOX applies to all companies listed on a US stock market, including their foreign subsidiaries. Compliance is mandatory.

One of the main provisions of SOX is that the chief executive officer (CEO) and chief financial officer (CFO) should sign off personally on company accounts. Fraudulent certification (i.e. signing accounts known to be inaccurate) leads to criminal penalties, including fines of up to \$5m and up to 20 years in prison.

However, some CEOs and CFOs have tried to avoid their responsibilities under SOX by asking divisional heads to certify their division's accounts before they are sent to head office.

Furthermore, the detailed reporting required to comply with SOX is very high. Such high compliance costs have discouraged many companies from listing their shares in New York; instead they choose London, where corporate governance codes are based more on principles than detail.

Therefore organisations need to consider the consequences of a failure to adhere to or comply with corporate governance regulations, which range from criminal and financial penalties to bad publicity and a negative impact on share price, depending on whether compliance is mandatory.

2.3 The Euro

The Euro is the common currency of most countries that are members of the European Union. Member states of the European Union (EU), exception for Denmark, must introduce the Euro once they fulfil the convergence criteria. Therefore, businesses in the "eurozone" face no foreign exchange risk (or transaction costs) when trading within the eurozone. However, the rise/fall in the euro against other currencies affects their global competitiveness.

- Businesses within Europe, but outside of the Eurozone (e.g. businesses in the UK) may trade with businesses within the eurozone. These businesses are exposed to exchange rate risk between their domestic currencies and the euro.
- The euro allows easy comparison of prices between member countries, leading to increased competition.
- Within the eurozone, there is also a move towards tax harmonisation (e.g. introducing the same corporation tax rates across the area). This has implications for financial management (e.g. project appraisal).

3.0 Introduction

Having looked at the economic environment and its impact on organisations, **the remainder of this chapter considers** the financial environment.

3.1 Financial Intermediaries

i Definition

Financial intermediaries – organisations which bring together potential lenders and potential borrowers.

3.1.1 Financial Institutions

The following financial institutions act as **financial intermediaries**:

- Commercial banks (see discussion in s.3.2 below).
- Merchant/investment banks, which provide banking services for business customers. These services include advice on items such as share issues and mergers.
- Building societies, which take deposits from the domestic sector and lend to those buying their own house.
- Insurance companies, which can invest much of their premium income in long-term assets, as their outgoings are reasonably easy to predict.
- Investment trusts and unit trusts/mutual funds, which attract investors and then reinvest the funds raised in other companies.
- Pension funds, which receive regular premiums and therefore have predictable cash outflows, can invest for the long term.
- Finance companies, which provide business and domestic credit, leasing finance and factoring/invoice discounting services. These companies are often a subsidiary of another financial institution.
- Discount houses, which trade in investments such as bills of exchange.

3.1.2 Role of Financial Intermediaries

Financial intermediaries are important as they carry out the following roles:

- Aggregation: small deposits are combined and lent to large borrowers.
- Maturity transformation: a continuing stream of short-term deposits can be used to lend monies in the long term.
- Risk diversification: the risk of each particular borrower is effectively spread across many lenders.
- Liquidity: providing a liquid market with flexibility and choice for lenders and borrowers.
- Hedging: providing businesses with instruments for hedging risk (e.g. forward contracts, options and swaps).

The role of financial institutions as intermediaries that facilitate the flow of funds between two parties is threatened by the emergence of Fintech (see s.4.9).

3.2 Commercial Clearing Banks

The commercial clearing banks carry out the following roles:

- They accept deposits from their customers into current or deposit accounts.
- They issue certificates of deposit, which may then be traded. These relate to large deposits with a term to maturity of at least three months.
- They lend money in several ways, ensuring adequate returns are made. At the same time, some cash must be held to ensure that sufficient liquidity is maintained. Therefore, banks have to find the right balance between profitability and liquidity.
- They provide a money transmission service through the clearing system.

Bank lending takes the following forms:

- Overdraft facilities and term loans to individuals and business customers.
- Investments in other financial intermediaries, such as leasing companies.
- The purchase of short-term government securities.
- The purchase of trade or commercial bills.
- The lending of money in the very short term to discount houses, which will re-lend in the longer term, as not all of their borrowings are likely to be called in at any one time.

4.1 Activities

The financial markets include the capital markets (for medium- and long-term capital) and the money markets (for short-term capital). The following activities take place in these markets:

- Primary market activity – selling new securities to raise new funds.
- Secondary market activity – trading existing securities.

4.2 Capital Markets

The main UK capital markets are:

- The Official List at the London Stock Exchange.
- The Alternative Investment Market (AIM), which has fewer regulations and less cost than the Official List and is therefore attractive to smaller companies.
- The Eurobond market, where bonds are generally issued by large international companies with a 10- to 15-year term. These bonds are denominated in any currency other than that of the national currency of the issuer.

Capital markets exist in many other countries, and large international companies may trade and raise funds in multiple capital markets.

- In the US, in addition to the New York Stock Exchange (NYSE), the NASDAQ (National Association of Securities Dealers Automated Quotations) market was set up for technology focussed companies.
- The largest stock exchange in Japan, the Tokyo Stock Exchange (TSE), plays a crucial role in the global financial landscape, especially in the Asia-Pacific region.
- The Hong Kong Stock Exchange (HKEX) is known for international listings and provides a gateway for companies looking to access global capital markets, particularly from mainland China.

These markets provide long-term capital: equity capital, ordinary and preference shares, or debt capital such as loan notes. Companies requiring funds for five years or more will use the capital markets.

See s.4.5 for further details on the operation of a stock exchange.

4.3 Money Markets

The money market is not a physical market; it is the term used to describe trading between banks and other financial institutions. Although the money markets principally involve borrowing and lending by banks, some large companies and the government are engaged in money market operations.

4.3.1 Principal Roles

The principal roles of the money markets are to:

- transfer money from parties with surplus funds to parties with a deficit;
- allow governments and businesses to raise short-term funds;
- help governments implement monetary policy;
- determine short-term interest rates; and
- allow businesses to manage their exposures to risk.

The money markets fulfil these roles by providing:

- Interest-bearing instruments: debt issued at nominal value, paying "coupon" interest on this nominal value (see s.4.3.2);
- Discount instruments: debt issued at a discount to nominal value paying no interest (i.e. zero coupon) but redeemed at nominal value (see 4.3.3);
- Derivative instruments: financial institutions provide their clients with customised "over-the-counter" (OTC) instruments for hedging currency risk or interest rate risk (see 4.3.4).

4.3.2 Interest-bearing Instruments

Certificate of deposit (CD): a savings certificate issued by a commercial bank entitling the holder to receive interest. A CD bears a maturity date and a specified fixed interest rate and can be issued in any denomination. CDs are generally issued for terms from one month to five years. The holder can:

- keep the CD until its maturity date; or
- resell it on the secondary market (i.e. it is a negotiable instrument).

Repurchase agreements (Repo): short-term loans (normally for less than two weeks and often for one day) arranged by selling securities to an investor with an agreement to repurchase them at a fixed price on a specified date.

Municipal notes: short-term debt issued by cities in anticipation of future tax receipts or other revenues.

4.3.3 Discount Instruments

Bill of exchange: a short-term financial instrument consisting of a written order addressed by the seller of goods to the buyer requiring the latter to pay a certain sum of money on demand or at a future time. Bills of exchange are often used in international transactions, and the holder of such a bill may convert it immediately into cash by selling it to a bank at a discount.

Commercial paper: unsecured, but high-quality, corporate debt with a fixed maturity of one to 270 days. It is sold at a discount to nominal value and pays a zero coupon. Commercial paper allows companies to raise large amounts of finance quickly but is only available to those companies with very high credit ratings. This is another example of a negotiable instrument.

Banker's acceptance: a short-term debt issued by a company that is guaranteed by a commercial bank. Banker's acceptances are issued at a discount to nominal value, pay zero coupon and are traded on a secondary market, which allows the investor to sell the instrument rather than hold it to maturity. They are regularly used in international trade and the maturity date typically ranges between 30 and 180 days from the issue date. They are considered relatively safe investments because the bank and the borrower are liable for the amount due when the instrument matures.

Treasury bills: short-term debt obligations of a national government that are issued to mature in three to 12 months. These are issued at a discount to nominal value and pay a zero coupon.

4.3.4 Derivative Products

Definition

Derivative – a financial instrument whose value or price depends on an underlying asset.

Money market derivatives are derivative products whose value derives from instruments such as CDs and Treasury bills (see above). Businesses can use derivatives to reduce the risk of adverse fluctuations in the value of an asset through “hedging”.

Details of examples, including futures, forward contracts, options and swaps, are described in *Chapter 17*.

4.4 International Money and Capital Markets

Larger companies wishing to raise significant amounts of finance can use international money and capital markets.

Eurocurrency markets are the international money markets, whereas Eurobond markets are the international capital markets.

Eurocurrency is currency held by individuals and institutions outside the country of issue of that currency (e.g. US dollars on deposit in a UK bank). The Eurocurrency markets involve depositing funds with a bank outside the country of origin of the currency and relending these funds, usually for three months.

Instead of borrowing from a bank, a large company can issue bonds and borrow directly from investors, with banks arranging the transaction. Eurobonds are raised by international companies and sold to investors in several countries simultaneously, and typically have a term of 10 to 15 years. Borrowers must consider the exchange rate risk associated with Eurobonds.

4.5 Stock Exchange Operations

4.5.1 Functions and Purpose

The primary function of a stock exchange is to ensure a fair, orderly and efficient market for the transfer of securities and the raising of new capital through the issue of new securities. To do this, a stock exchange sets stringent regulations which are designed to ensure that:

- only suitable companies are allowed to trade their securities on the stock exchange;
- all relevant information is made publicly available as soon as possible – in this way, investors can make informed decisions and so funds will be attracted to the most successful companies; and
- all investors deal on the same terms and at the same prices.

The more efficient and fairer a stock exchange is seen to be, the more willing people will be to invest their money in the exchange and the more successful it will become.

4.5.2 Who Owns Shares?

A decline in the importance of private investors and a rise in the significance of institutional investors, particularly pension funds, is a trend that has been observed in many other developed economies. For example:

- There has been a notable shift toward institutional ownership of shares and other financial assets in the US. This trend has been influenced by the growth of retirement savings vehicles, which are often managed by institutional investors.
- Canada has experienced a similar trend. Institutional investors such as the Canada Pension Plan Investment Board and other large pension funds hold substantial portfolios of equities and other financial assets.
- Australia has seen a shift in the dominance of institutional investors, particularly superannuation (pension) funds. The compulsory superannuation system, in which employers contribute to employees' retirement savings, has led to the growth of large institutional investors that manage significant pools of capital.
- Nordic Countries (e.g. Sweden, Norway, Denmark) have also experienced a shift toward institutional ownership in their financial markets. Pension funds in these countries manage significant assets and their investment decisions have a substantial impact on the capital markets.

4.5.3 How Are Shares Bought and Sold?

In general, a broker is needed to buy and sell shares. The broker will either act as an agent and deal through a market maker or deal himself, in which case he is a broker-dealer. The broker will charge a fee for his services; a market maker will generate a profit through the bid-offer spread, which is the difference between the price he is willing to pay for a share and the price at which he is willing to sell it.

4.5.4 How Are Shares Valued?

Shares are valued using market forces at the price at which there are as many willing sellers as willing buyers. For instance, if a share is overvalued, more people will be keen to sell their shares than willing buyers, which will inevitably depress the market price.

Some trading is done for speculative reasons:

- A **bull** or bullish investor believes that prices will rise. Shares are bought in the hope of selling them for a profit.
- A **bear** or bearish investor believes that prices will fall. Shares are sold in the hope that they can be repurchased for less.

When there are more bulls than bears, prices will rise; when there are more bears than bulls, prices will fall.

Such speculative dealing has two consequences:

1. It reduces fluctuations in the market. For instance, as the market falls and prices fall, more speculators become bullish and start to buy again, and so arrest the fall in the market.
2. It ensures that there is always a ready market in all shares. In other words, someone will always be willing to buy or sell at the right price.

4.6 Financial Market Efficiency

1 Definition

An efficient market – one in which the market price of all securities traded on it reflects all the available information.

A perfect market – one which responds immediately to the information made available to it.

An efficient and perfect market will ensure that quoted share prices are as fair as possible as they accurately and quickly reflect a company's financial performance, prospects and any relevant news or events.

Efficiency can be looked at in several ways:

- **Allocative efficiency:** does the market attract funds to the best companies?
- **Operational efficiency:** does the market have low transaction costs and a convenient trading platform? These features promote a "deep" market with high liquidity (i.e. a high volume of transactions with low transaction costs).
- **Informational efficiency:** is all relevant information available to all investors at low cost?
- **Pricing efficiency:** do share prices quickly and accurately reflect all known information about the company? This is also referred to as information processing efficiency.

4.6.1 The Efficient Market Hypothesis

The Efficient Market Hypothesis (EMH) considers information processing efficiency/pricing efficiency. To test this hypothesis, three potential levels of efficiency are considered.

1. **Weak-form efficiency**

Share prices reflect all the information contained in the record of past prices. Share prices will follow a *random walk*.

If this efficiency level has been achieved, it should not be possible to forecast price movements by reference to past trends. This means chartists (also called technical analysts), who look at statistical trends such as movements in a share's price and volume, should not be able to outperform the market consistently. On the other hand, fundamental analysts, who evaluate shares by studying everything from the overall economy and industry conditions to the financial strength and management of individual companies, would be able to predict future share price movements.

2. **Semi-strong form efficiency**

Share prices reflect all information currently publicly available. Therefore, the price will change only when new information is published.

If this efficiency level has been reached, price movements can only be forecast using inside information (i.e. material non-public information). This is known as insider trading, which is illegal in most markets and unethical in all markets.

3. **Strong-form efficiency**

Share prices reflect all information, published and unpublished, that is relevant to the company.

If this efficiency level has been reached, share prices cannot be predicted, and gains through insider dealing are impossible as the market already knows everything.

Major markets (e.g. the London and New York Stock Exchanges), have strict rules to prohibit insider dealing. Therefore, such markets are regarded as **semi-strong efficient**.

The three forms of efficiency describe how share prices react to information from the company. If prices do not react to new information, there is no efficiency at all.

 Exam advice

Ensure that you learn and understand the Efficient Markets Hypothesis.

Activity 1 Market Efficiency

[Show](#)

Omar believes that share prices reflect all relevant information at all times. Griselda plots the historic movements of share prices and uses this analysis to make her investment decisions.

Required:

Identify the level of capital market efficiency that best reflects each of Omar and Griseldas beliefs.

**Please use the notes feature in the toolbar to help formulate your answer.*

Oliver believes markets the market to be **strong-form** efficient.

Griselda is basing her investment decisions on technical analysis, which means that she believes the stock market is **not efficient** at all, not even weak-form efficient.

Activity 2 Strong-form Efficiency

[Show](#)

Explain whether the existence of each of the following contradicts the belief that stock markets are strong-form efficient:

- (1) Projects with positive expected net present values; and
- (2) Information content in dividends.

**Please use the notes feature in the toolbar to help formulate your answer.*

- (1) The existence of projects with positive expected net present values (NPVs) **does not contradict** the idea that the stock market is strong-form efficient because a strongform efficient market will have taken into account the fact that there are projects with positive NPVs.
- (2) The existence of information content in dividends **contradicts** the idea that the stock market is strong-form efficient because shareholders will already know all relevant information (so the dividends themselves do not provide any additional information).

4.6.2 Implications for Financial Managers

The level of efficiency of the stock market has implications for financial managers:

- Timing of new issues: unless the market is fully efficient, the timing of new issues remains important. This is because the market does not reflect all the relevant information. Therefore, an advantage could be obtained by issuing shares just before or after additional information becomes available to the market.
- Project evaluation: if the market is not fully efficient, the share price is not fair; therefore, the rate of return required from that company by the market cannot be accurately known. If this is the case, it is not easy to decide what rate of return to use to evaluate new projects.
- Creative accounting: unless a market is fully efficient, creative accounting can still be used to mislead investors.
- Mergers and takeovers: where a market is fully efficient, the price of all shares is fair. Therefore, if a company is taken over at its current share value, the purchaser cannot hope to make any gain unless economies can be made through scale or rationalisation when operations are merged. Unless these economies are significant, an acquirer should not be willing to pay a substantial premium over the current share price.
- Validity of current market price: the share price is fair if the market is fully efficient. In other words, an investor receives an appropriate risk/return combination for his investment, and the company can raise funds at a reasonable cost. If this is the case, there should be no need to discount new issues to attract investors.

4.6.3 The Paradox of Efficient Markets

Market efficiency relies on investors actively buying/selling shares to create a liquid market where fair prices emerge. Investors will actively trade shares if they believe that bargains are available (a signal to buy) and other shares are overvalued (a signal to sell).

However, if the market is efficient, such mispriced shares will not exist. This is the paradox of efficient markets – investors must believe that the market is inefficient so that it becomes efficient.

4.7 Corporate Bond Market

4.7.1 Features

Corporate bonds (also called “loan notes”) are often used to raise funding for large-scale projects (e.g. business expansion, takeovers, new premises or product development). They can also be used to replace bank finance or to provide long-term working capital.

The main features of a corporate bond are:

- the nominal value – the price at which the bonds are initially issued;
- the “coupon” interest rate paid to the bond owner – this is usually a fixed percentage of the nominal value; and
- the redemption date – when the nominal value must be repaid to the bondholder.

4.7.2 Market Structure

4.7.2 Methods of Issue

Public issue – corporate bonds can be issued through the stock exchange, similar to a public share issue. Bonds issued on the public market can be traded and reach a market price (similar to how a share price is established). Minimum public issue sizes for corporate bonds can vary by country and market conditions, for example:

- In the Eurozone (e.g. Germany): €100m–€500m or more, depending on the issuer's size and credit rating;
 - In Japan: ¥ 10bn–¥ 100bn or more, depends on the issuer's funding needs, creditworthiness and prevailing market conditions;
 - In the UK: typically £100–200m;
 - In the US: More than \$500m for large, well-established corporations (\$100– \$500m for less creditworthy issuers).
- **Private placement** – a bond issue to a relatively small number of selected investors such as banks, insurance companies and pension funds. Private placements avoid the compliance costs of a public issue and can be used to accommodate smaller transactions.

4.8 Interest Rates

4.8.1 Influences

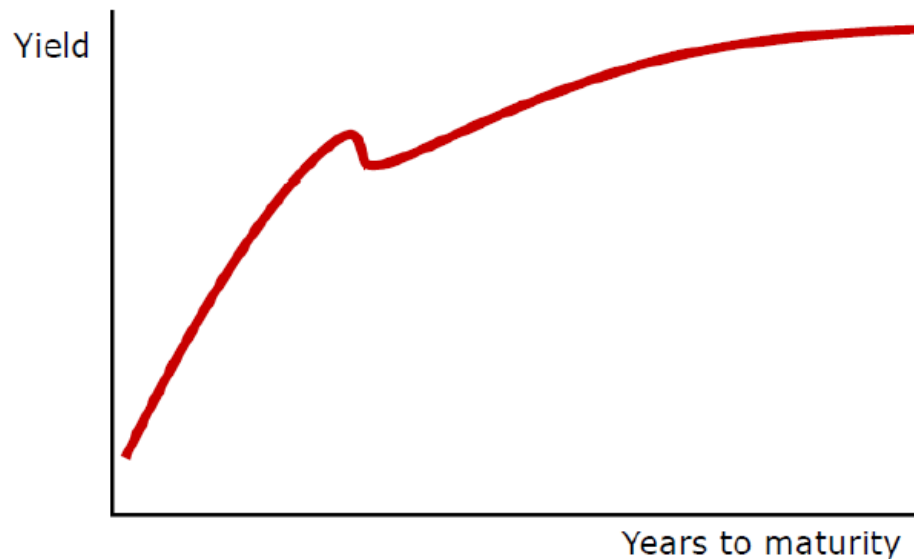
Different financial instruments offer different interest rates. To understand why, it is necessary to appreciate the factors determining the appropriate interest rate for a particular financial instrument. These factors are:

- General level of interest rates in the economy: These rates are affected by:
 - Inflation;
 - Government monetary policy;
 - The demand for borrowing;
 - Investors' preference for cash;
 - International factors (e.g. interest rates overseas and exchange rate movements).
- Level of risk: the higher the level of risk, the greater return an investor will expect.
- Duration of the loan: if it is assumed that, in the long-term, interest rates are expected to remain stable, then the longer the length of the loan, the higher the interest rate will be. This is because lending money in the longer term exposes the lender to additional risk (e.g. the risk of default increases).
- The need for the financial intermediaries to make a profit: for instance, a depositor at a building society will receive a lower rate of interest than a borrower will be charged. The building society earns the difference in the rates.
- Size: if a large sum of money is lent or borrowed, there are administrative savings. Therefore, a higher interest rate can be paid to a lender, and a lower interest rate can be charged to a borrower than would normally be the case.
- Term structure of interest rates: the return provided by a security will vary according to the length of time before the security matures (see below).

4.8.2 The Term Structure of Interest Rates and Yield Curves

The term structure of interest rates refers to how the yield on a security varies according to the borrowing term.

If a graph shows the yield to maturity (gross redemption yield) of various government securities against the number of years to maturity, the following "yield curve" might result.



Financial managers need to be aware of the shape of the yield curve, as it indicates the likely future movements in interest rates; therefore, it assists in the choice of finance for the company.

Although yield curves are usually constructed by reference to government interest rates, the yield curve for a corporate borrower will follow a similar trend. The shape of the yield curve can be explained by various theories and factors:

- **Liquidity Preference Theory** – Yields will need to rise as the term to maturity increases, as investors require higher returns as compensation for increased illiquidity of long-dated bonds. Therefore, a "normal" yield curve slopes upwards, as shown above.
- **Expectations Theory** – If interest rates are expected to increase in the future, the curve may rise even more steeply. On the other hand, the curve may fall (i.e. invert) if interest rates are expected to decline.
- **Market Segmentation Theory ("Preferred Habitat Theory")** - Different investors have a preference for and are interested in different segments of the yield curve.
 - Short-term yields, for example, are of interest to financial intermediaries such as banks. Therefore, the shape of the yield curve in that segment reflects the attitudes of the investors active in that sector.
 - Pension fund managers often prefer investing in long-dated bonds to match against the long-term liabilities of the fund. This can drive up the price of longdated bonds, lowering their yield and possibly resulting in an inverted (falling) yield curve.
 - Where two sectors meet, there is often a disturbance or apparent discontinuity in the yield curve, as shown in the above diagram.
- **Risk** - On high-quality government/sovereign debt (e.g. UK Gilt-edged Securities ("Gilts") and US Treasury Securities), the risk of default is not significant even for long-dated bonds. Default risk may, however, be more significant on corporate debt. Therefore, the yield curve for corporate debt may rise more steeply than for government debt.

4.9 Impact of Fintech

4.9.1 Introduction

Definition

Fintech – technologically enabled innovation in financial services that could result in new business models, applications, processes or products with an associated material effect on financial markets and institutions and the provision of financial services.

Financial innovation in Fintech has focussed on two main areas: better use of data and a “frictionless” customer experience. It also includes developing and using cryptocurrencies (“digital money”) such as bitcoin. One of the main features of Fintech is that it may reduce or eliminate the costs of financial intermediation, especially in the three broad areas of finance: raising, allocating and transferring capital.

Fintechs now affect all the primary functions of banks:

- maturity transformation (through competition in lending);
- allocation (through robo advisors and crowd investing platforms);
- payment services (through new payment platforms and interfaces); and
- information processing (big data, machine learning and artificial intelligence).

Many organisations and individuals now use Fintech for a wide range of financial activities without the assistance of a bank or financial advisor:

- depositing money and making money transfers using a smartphone;
- applying for credit;
- raising money for a business start-up (e.g. through crowd funding);
- managing an investment portfolio.

The impacts of Fintechs on financial markets and institutions include:

- Disintermediation;
- Availability of credit;
- Potential source of finance (security token offering).

4.9.2 Disintermediation

The main advantage of the role of financial institutions as intermediaries (see s.3.1) is their ability to provide safety, liquidity and economies of scale. Financial **disintermediation** involves transferring of funds between parties directly, without a financial intermediary. Fintechs can provide efficient banking services; directly matching lenders and borrowers, investors and investment opportunities and, therefore, disintermediate the flow of funds.

4.9.3 Availability of Credit

Fintech has expanded the availability of credit by allowing borrowers rejected by the traditional banks to access the funding they need.

i Definition

Peer-to-peer (P2P) lending – a method of debt financing that enables individuals to lend money to small businesses without using an official financial institution as an intermediary.

P2P loans give businesses access to financing when banks refuse credit or would charge very high interest rates. P2P loans can offer relatively attractive interest rates to lenders and borrowers because the costs of the intermediary, the formal banking system, are eliminated.

However, in some countries where P2P lending is unregulated, investors have lost significant sums. And P2P lending is illegal in some countries.

4.9.4 Security Token Offerings

Security Token Offerings (STOs) combine blockchain technology with the requirements of regulated securities. The blockchain environment:

- enhances the regulatory objectives of disclosure, fairness and market integrity and;
- supports innovation and efficiency through automation and “smart contracts” (i.e. self-executing contracts with agreed terms written into their source code).

i Definition

Security token offering (STO) – the first issue of tokens (or crypto coins) that represent an investment contract in an underlying asset.

A security token is a digital representation of ownership of a share in an asset (e.g. gold or property) or economic rights (e.g. a share of profits or revenue). Whereas a traditional share in a company is issued in a physical document (a share certificate), a security token is a digital certificate (and may be stored in a digital wallet).

The STO process can be generalised in five phases:

Phase		Principal activities
1	Preparation	• Prepare a “deck” of investor information including: - business plan; - profit forecast; - capital requirement; and - independent valuation.
2	Designing the offering	• Decide the structure of the token including: - quantity - value - embedded rights - duration of offering.
3	Selection of service providers	• Select a blockchain platform and assess its listing rules. • Create the security token. • Appoint a reputable services provider as a custodian of the underlying asset(s). • The distributors of the token (i.e. the brokers) should conduct due diligence on: - the issuer; and - the deck, to ensure it is complete and accurate.
4	Raising capital	• Identify target investors, hold roadshows and meetings and distribute the deck. (Marketing activities may require pre- approval from the relevant authority in certain jurisdictions.) • The brokers then: - collect funds from investors; and - distribute tokens to investors.
5	Listing of security on the trading venue	• Depending on the trading platform, the tokens will usually: - be sold to a “special purpose vehicle”; and - reissued on the trading platform.

5.1 Background

1 Definition

Islamic finance – a system of banking consistent with the principles of Islamic law (Sharia). Sharia prohibits:

- the payment or acceptance of interest (riba); and
- investing in businesses that provide goods or services considered contrary to its principles.

Islamic finance is based on Islamic law, or Sharia, whose primary sources are the Qur'an and the sayings of the Prophet Muhammad. Sharia emphasises justice and partnership.

5.1.1 Main Principles

The main principles of Islamic finance are that:

- Wealth must be generated from legitimate trade (i.e. making money from money is forbidden).
- Investment should also have a social and ethical benefit to broader society beyond pure return.
- Risk should be shared.
- All harmful activities (*haram*) should be avoided.
- Islamic banks must obtain their earnings through profit-sharing investments or fee-based returns. When loans are given, the lender should take part in the risk. Otherwise, the receipt of any gain over the amount loaned is regarded as interest.

5.1.2 The Sharia Board

The Sharia board is responsible for ensuring that all products and services offered by an institution comply with the principles of Sharia law; it is usually made up of a committee of Islamic scholars.

An institution's Sharia board will review and oversee all new product offerings before they are launched. It can also be asked to deliver judgments on individual cases referred to it, such as whether a specific customer's business proposals are Sharia-compliant.

5.1.3 Developments

The main problem is the absence of a single, worldwide body to set standards for Sharia compliance. Some financial aspects of Sharia law can be open to interpretation; therefore, a contract might be unexpectedly declared incompatible with Sharia law and rendered invalid.

In Malaysia, the world's biggest sukuk (debt finance) market, the Sharia Advisory Council ensures consistency to help create certainty across the market. Some industry bodies, notably the Accounting and Auditing Organisation for Islamic Financial Institutions (AAOIFI) in Bahrain, also work towards common standards.

However, despite these movements towards consistency, some differences between national jurisdictions will likely remain.

5.2 Prohibitions

- ✗ Charging and receiving interest (riba). A lender making a straight interest charge, irrespective of how the underlying assets perform, violates the concepts of risk sharing, partnership and justice. The definition of riba in classical Islamic law was "to ensure equivalency in real value". During this period, gold and silver currencies were the benchmark metals defining the value of all other traded materials. Applying interest to the benchmark made no sense, as its value remained constant relative to all other materials. Therefore, Islamic banking is based on profit sharing rather than interest.
- ✗ Investment in companies with too much borrowing (debt totalling more than 33% of the company's stock market value).
- ✗ Investments in businesses dealing with alcohol, gambling, drugs, pork, pornography or anything else that the Sharia considers unlawful or undesirable (haram).
- ✗ Transactions which involve speculation or extreme risk, which is seen as gambling. This prohibits speculating on the futures and options markets. On the other hand, mutual insurance is permitted as members contribute to a fund, not for profit, but in case one of the members suffers misfortune. Advocates of Islamic finance claim that it avoided much of the financial crisis in 2008 because of its prohibitions on speculation and its emphasis on risk sharing and justice.
- ✗ Uncertainty about the subject matter and terms of contracts. This includes a prohibition on selling something that one does not own. Therefore, complex derivative instruments and short selling are prohibited under Islamic finance.

5.3 Islamic Instruments

- **Murabaha (trade credit):** trade credit for asset acquisition that is structured to avoid the payment of interest. A bank buys the asset and then sells it to the customer on a deferred basis at a price that includes an agreed mark-up. The markup cannot be increased, even if the client does not take the asset within the time agreed in the contract.
- **Ijara (lease finance):** lease finance whereby the bank buys an item for a customer and then leases it back over a specific period at an agreed amount. In 2003, HSBC was the first bank to offer UK mortgages designed to comply with Sharia. HSBC's Islamic mortgage involves the bank purchasing a house and then leasing it to the customer. The customer's payments include a contribution to the purchase price, rent for use of the property and insurance charges. At the end of the finance term, the customer can exercise an option to have the property transferred into their name.
- **Mudaraba (equity finance):** a bank provides all the capital, and its customer provides expertise and knowledge and invests the capital. Profits generated are distributed in a predetermined ratio. Any losses are borne by the capital provider, thereby exposing the bank to considerable investment risk.
- **Musharaka (venture capital):** a joint venture or partnership between two parties provides capital for financing new or established projects. Profits are shared on a pre-agreed ratio, with losses shared based on the relative amounts of equity invested.

- **Sukuk (debt finance):** this involves Islamic bonds where the sukuk holders' return for providing finance is a share of the income generated by the project's assets. Typically, a sukuk issuer would acquire property to be leased to tenants to generate income. The issuer collects the income and distributes it to the sukuk holders. This entitlement to a share of the income generated by the assets (rather than a return based on interest on money loaned) can make the arrangement Sharia-compliant. However, the rental income is often linked to market interest rates e.g. the Secured Overnight Financing Rate ("SOFR" on USD-denominated loans) as opposed to the returns from the underlying asset, and hence such bonds may contravene the prohibition on *riba*. Indeed, it does appear that Sukuk bonds are often structured to make *money from money* rather than making returns from a physical asset.
- **Hibah (gift):** this is where Islamic banks voluntarily pay their customers a "gift" on savings account balances. This gift represents a portion of the profit made using those balances in other activities.
- **Qard hassan/Qardul hassan (good loan/benevolent loan):** this involves a loan extended on a goodwill basis, and the debtor is only required to repay the amount borrowed. However, the debtor may, at their discretion, pay an extra amount to the creditor.

Syllabus Coverage

This chapter covers the following Learning Outcomes.

B. Financial Management Environment

1. The economic environment for business

- Identify and explain the main macroeconomic policy targets.
- Define and discuss the role of fiscal, monetary, interest rate and exchange rate policies in achieving macroeconomic policy targets.
- Explain how government economic policy interacts with planning and decision-making in business.
- Explain the need for, and the interaction with, planning and decision-making in business of:
 - competition policy
 - government assistance for business
 - green policies and sustainability issues
 - corporate governance regulation.

2. The nature and role of financial markets and institutions

- Identify the nature and role of money and capital markets, both nationally and internationally.
- Explain the role of financial intermediaries.
- Explain the functions of a stock market and corporate bond market.
- Explain the impact of Fintech in changing the nature and role of financial markets and institutions.

3. The nature and role of the money market

- a) Describe the role of the money markets in:
 - i) Providing short-term liquidity to the private sector and the public sector
 - ii) Providing short-term trade finance
 - iii) Allowing an organisation to manage its exposure to foreign currency risk and interest rate risk.
- b) Explain the role of banks and other financial institutions in the operation of the money markets.
- c) Explain and apply the characteristics and role of the principal money market instruments:
 - i) Interest-bearing instruments
 - ii) Discount instruments
 - iii) Derivative products

E. Business Finance

1. Sources of, and raising, business finance

- d) Identify and discuss methods of raising short- and long-term Islamic finance.

F. Business Valuations

4. Efficient Market Hypothesis (EMH) and practical considerations in the valuation of shares

- a) Distinguish between and discuss weak form efficiency, semi-strong form efficiency and strong form efficiency.

G. Risk Management

2. Causes of exchange rate differences and interest rate fluctuations

- c) Describe the causes of interest rate fluctuations, including:
 - i) structure of interest rates and yield curves
 - ii) expectations theory
 - iii) liquidity preference theory
 - iv) market segmentation.

Summary and Quiz

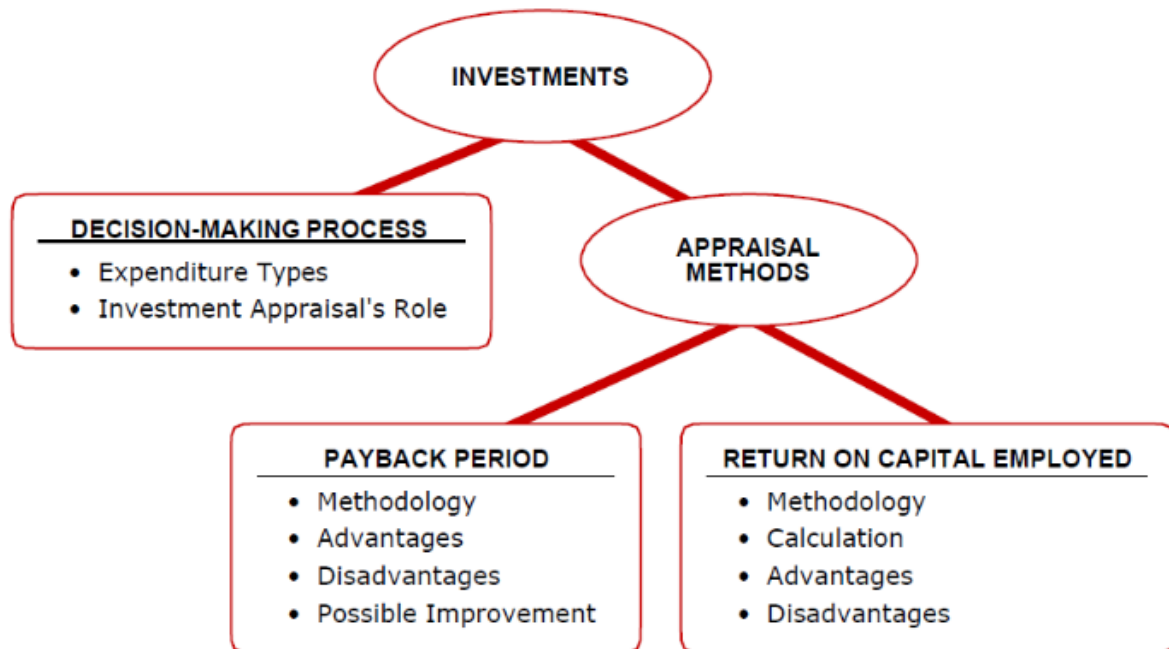
- Under macroeconomic policy:
 - the government sets economic objectives and uses control instruments to achieve those objectives.
 - targets include full employment, economic growth, rising living standards, acceptable distribution of wealth, price stability and a solid balance of payments.
- Under monetary policy:
 - the government or the central bank uses monetary instruments to achieve economic objectives.
 - actions may either directly control the amount of money in circulation or attempt to reduce the demand for money through its price.
- Under fiscal policy: the government uses the fiscal instruments of taxation, public spending and the budget deficit or surplus to achieve economic objectives.
- Keynesians identify two significant causes of inflation: demand-pull and cost-push inflation.
- Monetarists believe that inflation results from too much money chasing too few goods.
- Financial intermediaries bring together potential lenders and potential borrowers.
- The financial markets include both the capital markets and the money markets.
- The stock exchange's primary function is to ensure a fair, orderly and efficient market for the transfer of securities and the raising of new capital through the issue of new securities.
- An efficient market is one in which the market price of all securities traded reflects all the available information. A perfect market responds immediately to the information made available to it.
- The term structure of interest rates may cause interest rate fluctuations. The shape of the yield curve is influenced by liquidity preference theory, expectations theory, market segmentation and risk.
- Some of the main impacts of Fintechs, which integrate technology (e.g. blockchain) and financial services, are financial disintermediation, expanding the availability of credit and security token offerings.
- Islamic finance is a system of banking consistent with the principles of Islamic law (Sharia).
- The major difference between Islamic finance and conventional finance is that, under Islamic law, interest (riba) cannot be charged.

Chapter 3: Investment Decisions

Overview	CHAPTER 3: Visual Overview
3.1 Decision-making Process	3.1.1 Expenditure Types
	3.1.2 Investment Appraisal's Role
3.2 Payback Period	3.2.1 Methodology
	3.2.2 Advantages
	3.2.3 Possible Improvement
3.3 Return on Capital Employed (ROCE)	3.3.1 Methodology
	3.3.2 Calculation
	3.3.3 Advantages and Disadvantages of ROCE
3 Conclusion	3 Syllabus Coverage
	3 Summary and Quiz
	3 Technical Articles

CHAPTER 3: Visual Overview

Objective: To assess an investment using the payback period and the return on capital employed methods.



1.1 Expenditure Types

Asset ("capital") expenditure – incurred in the acquisition or improvement of non-current assets.

Expenses ("revenue expenditure") – incurred to maintain non-current assets (e.g. repairs).

1.2 Investment Appraisal's Role

Investment appraisal plays a key role in analysing and evaluating investment proposals.

- If the company's main financial objective is to maximise (or at least produce satisfactory) shareholder wealth, the superior investment appraisal technique is net present value (NPV). This is because NPV shows the theoretical absolute change in shareholder wealth due to a project.
- Managers may also require other measures as part of their decision-making process (e.g. payback as a liquidity measure and return on capital employed (ROCE) to judge the effect on published financial statements).
- Providers of finance may wish to know the project's internal rate of return (IRR). In particular, banks compare project IRR to the interest rate on proposed loans in order to measure the "headroom" on the project and the consequential risk of default on the debt: the higher the project IRR over the proposed loan interest rate, the lower the risk of default.

2.1 Methodology

Payback period is the time it takes for the undiscounted operating cash flows from a project to pay back the initial investment.

Key Point

The decision rule for payback period is:

- If payback period is less than target ACCEPT
- If payback period is greater than target REJECT

Example 1 Payback Period

Investment	\$1.4m
Annual cash flows (before depreciation but after tax)	\$0.3m
Project life	10 years

Solution

Payback period = $\$1.4\text{m} / \$0.3\text{m} = 4.7$ years

2.2 Advantages and Disadvantages of Payback Period

Advantages

- ✓ It is simple to calculate.
- ✓ It is easy to understand.
- ✓ It concentrates on earlier flows, which are:
 - more certain; and
 - more critical if the organisation has liquidity concerns.
- ✓ It focuses on recovering the original capital as soon as possible to exploit new investment options.

Disadvantages

- ✗ It ignores cash flows after the payback period.
- ✗ The target period is subjective.
- ✗ It ignores the time value of money.
- ✗ It gives no information about the change in shareholder wealth.

2.3 Possible Improvement

Discounted payback: this method requires cash flows to be discounted to present value first, and then calculating a discounted payback period. This approach allows for the time value of money (see *Chapter 7*).

3.1 Methodology

1 Definition

Return on capital employed (ROCE) – the average annual operating profit expressed as a percentage of the initial (or average) investment.

ROCE is also referred to as accounting rate of return (ARR) or return on investment (ROI).

Key Point

The decision rule for ROCE is:

- If ROCE is greater than target ACCEPT
- If ROCE is less than target REJECT

3.2 Calculation

ROCE is a financial accounting measure based on amounts reported in the statement of profit or loss and the statement of financial position. Therefore, it includes:

- Sunk costs (money already spent);
- Carrying amounts ("book values") of assets;
- Depreciation and amortisation; and
- Allocated fixed overheads.

It is calculated as:

$$\text{ROCE} = \frac{\text{Average annual operating profit}}{\text{Initial investment}} \times 100$$

OR

$$\frac{\text{Average annual operating profit}}{\text{Average investment}} \times 100$$

Where:

$$\text{Average investment} = \frac{\text{Initial investment} + \text{scrap value}}{2}$$

Activity 1 ROCE

[Show](#)

Initial investment	\$200m
Scrap value	\$20m
Operating cash flows:	
Year 1	\$100m
Years 2, 3 and 4	\$50m a year

Required:

Calculate ROCE on:

- (i) initial investment; and
- (ii) average investment.

**Please use the notes feature in the toolbar to help formulate your answer.*

**Please use the notes feature in the toolbar to help formulate your answer.*

$$\begin{aligned}\text{Average annual profit} &= \frac{\text{Total cash flows} - \text{Total depreciation}}{\text{Number of project years}} \\ &= \frac{\$250\text{m} - \$180\text{m}}{4} = \$17.5\text{m}\end{aligned}$$

(i) ROCE using initial investment

$$\text{ROCE} = \frac{\$17.5\text{m}}{\$200\text{m}} \times 100 = 8.75\%$$

(ii) ROCE using average investment

$$\begin{aligned}\text{Average investment} &= \frac{\text{Initial investment} + \text{Scrap value}}{2} \\ &= \frac{\$200\text{m} + \$20\text{m}}{2} = \$110\text{m}\end{aligned}$$

$$\text{ROCE} = \frac{\$17.5\text{m}}{\$110\text{m}} \times 100 = 15.9\%$$

3.3 Advantages and Disadvantages of ROCE

Advantages	Disadvantages
✓ It uses readily available accounting information. ✓ It is simple to calculate and understand. ✓ It is often used by financial analysts to appraise performance.	✗ Different methods of calculation may confuse. ✗ It is based on profits rather than cashflow. Profits are affected by accounting policy choices. It ignores the time value of money. ✗ The target rate is subjective. ✗ As a relative (percentage) measure, it gives no information about the absolute change in shareholder wealth.

Activity 2 ROCE and Payback

[Show](#)

A project being considered would require a machine costing \$80,000. Market research of \$ 8,000 has already been carried out and capitalised as an asset. The project is expected to last for six years and generate net cash inflows of \$20,000 for each of the first three years and \$15,000 for each of the last three years. Assume that cash flows arise evenly during the year.

Required:

Evaluate the project using:

(i) ROCE;

(ii) ROCE using the average investment approach;

(iii) Payback period.

Hint: For (iii), draw up a table showing cumulative cash flows.

**Please use the notes feature in the toolbar to help formulate your answer.*

**Please use the notes feature in the toolbar to help formulate your answer.*

(i) ROCE

$$\text{Average annual cash inflow} = \frac{(3 \times \$20,000 + 3 \times \$15,000)}{6} = \$17,500$$

$$\text{Average annual depreciation} = \frac{\$80,000 - \$4,000 + \$8,000}{6} = \$14,000$$

$$\text{ROCE} = \frac{\$17,500 - \$14,000}{\$88,000} \times 100 = 4.0\%$$

(ii) ROCE using the average investment

$$\text{Average investment} = \frac{\$88,000 + \$4,000}{2} = \$46,000$$

$$\text{ROCE} = \frac{\$17,500 - \$14,000}{\$46,000} \times 100 = 7.6\%$$

(iii) Payback period

Time	Cash flow	Cumulative cash flow
0	(88,000)	(88,000)
1	20,000	(68,000)
2	20,000	(48,000)
3	20,000	(28,000)
4	15,000	(13,000)
5	15,000	2,000
6	15,000	17,000

$$\text{Payback period} = 4 \text{ years} + \frac{\$13,000}{\$15,000} = 4.9 \text{ years}$$

Syllabus Coverage

This chapter covers the following Learning Outcomes.

D. Investment Appraisal

1. Investment appraisal techniques

- b) Calculate payback period and discuss its usefulness as an investment appraisal method.
- d) Calculate return on capital employed (accounting rate of return) and discuss its usefulness as an investment appraisal method.

Summary and Quiz

- Payback period and return on capital employed (ROCE) are commonly used in practice. However, neither method informs management of the absolute change in shareholder wealth due to a particular project.
- Decision rule for payback period: Accept the project if payback period is shorter than the target. Reject the project if payback period is longer than the target.
- Discounted payback period (see *Chapter 7*) offers a potential improvement over payback period.
- ROCE is also called the accounting rate of return (ARR) or return on investment (ROI).
- Decision rule for ROCE: Accept the project if ROCE exceeds the target. Reject the project if ROCE is less than the target.
- Because ROCE is a financial accounting measure based on amounts reported in financial statements, it includes sunk costs, "book values" of assets, depreciation and amortisation and allocated fixed overheads.

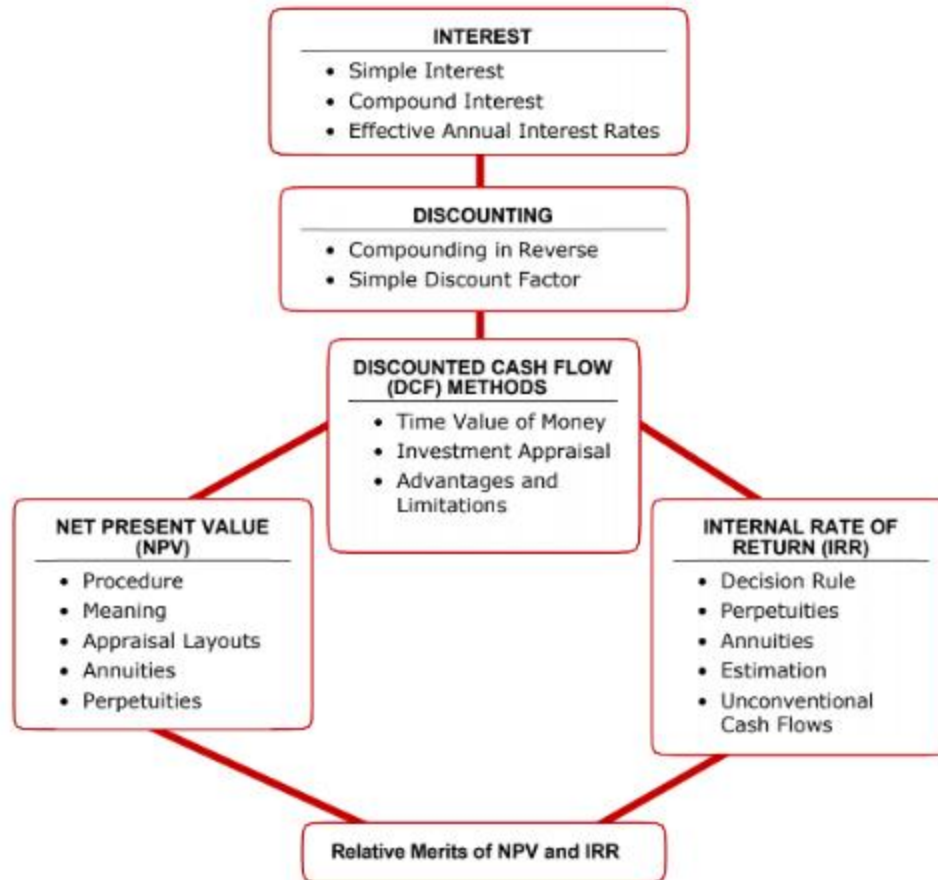
Chapter 3 Quiz

Chapter 4: Discounted Cash Flow Methods

Overview	CHAPTER 4: Visual Overview
4.1 Interest	4.1.1 Simple Interest
	4.1.2 Compound Interest
	4.1.3 Effective Annual Interest Rates (EAIR)
4.2 Discounting	4.2.1 Compounding in Reverse
	4.2.2 Simple Discount Factor
4.3 Discounted Cash Flow (DCF) Techniques	4.3.1 Time Value of Money
	4.3.2 Investment Appraisal
	4.3.3 Advantages and Limitations of DCF Techniques
4.4 Net Present Value (NPV)	4.4.1 Procedure
	4.4.2 Meaning
	4.4.3 DCF Appraisal Layouts
	4.4.4 Annuities
	4.4.5 Perpetuities
4.5 Internal Rate of Return (IRR)	4.5.1 Decision Rule
	4.5.2 Perpetuities
	4.5.3 Annuities
	4.5.4 Estimation
	4.5.5 Unconventional Cash Flows
	4.5.6 Relative Merits of NPV and IRR

CHAPTER 4: Visual Overview

Objective: To apply the time value of money to investment decisions.



1.1 Simple Interest

Key Point

Interest accrues only on the initial amount invested.

Example 1 Simple Interest

If \$100 is invested at 10% a year, simple interest is calculated as follows:

Year	Amount on deposit (beginning of year)	Interest	Amount on deposit (year-end)
1	\$100	$0.1 \times \$100 = \10	\$110
2	\$110	$0.1 \times \$100 = \10	\$120
3	\$120	$0.1 \times \$100 = \10	\$130

For simple interest, a single principal sum, P , invested for n years at an annual interest rate, r (expressed as a decimal), will amount to a future value (FV).

Where: $FV = P(1 + nr)$

1.2 Compound Interest

Key Point

Interest is reinvested alongside the principal.

When a single sum is invested to earn compound interest, the formula to find the FV is:

$$FV = P(1 + r)^n$$

where: P = initial principal

r = annual rate of interest (as a decimal)

n = number of years for which the principal is invested

Example 2 Compound Interest

If Zarosa placed \$100 in the bank today (t_0) earning 10% interest annually, what would this sum amount to after three years?

Solution

After 1 year, \$100 would have increased by 10% to \$110

After 2 years, \$110 would have increased by 10% to \$121

After 3 years, \$121 would have increased by 10% to \$133.10

Activity 1 Simple and Compound Interest

[Show](#)

\$500 is invested in a fund on 1.1.X1.

Required:

Calculate the amount on deposit by 31.12.X4 if interest is:

- a) 7% a year, simple;
- b) 7% a year, compounded.

**Please use the notes feature in the toolbar to help formulate your answer*

(a) 7% simple interest

$$FV = P (1 + nr)$$

$$FV = \$500 [1 + (4 \times 0.07)] = \$500 \times 1.28$$

$$FV = \$640$$

(b) 7% compound interest

$$FV = P (1 + r)^n$$

$$FV = \$500 (1 + 0.07)^4 = \$500 \times 1.3108$$

$$FV = \$655.40$$

Activity 2 Compound Interest

[Hide](#)

\$1,000 is invested on 1.1.X0 in a fund earning 5% a year. \$500 is added to the fund on 1.1.X1 and a further \$700 is added on 1.1.X2.

Required:

Calculate the amount on deposit by 31.12.X2 if compound interest is 5% a year.

**Please use the notes feature in the toolbar to help formulate your answer*

Date	Amount invested \$	×	Compound Interest Factor	=	Compounded Cash Flow \$
1.1.X0	1,000	×	$(1 + 0.05)^3 = 1.157625$		1,157.63
1.1.X1	500	×	$(1 + 0.05)^2 = 1.1025$		551.25
1.1.X2	700	×	$(1 + 0.05)^1 = 1.05$		<u>735.00</u>
	Amount on deposit			=	<u>2,443.88</u>

1.3 Effective Annual Interest Rates (EAIR)

Where interest is charged on a non-annual basis, it is useful to know the effective annual interest rate (EAIR). For example, interest on bank overdrafts (and credit cards) is often charged monthly. To compare the cost of finance to other sources, it is necessary to know the EAIR (also called the annual percentage rate or APR), which is calculated as follows:

$$1 + R = (1 + r)^n$$

Where: R = annual rate (EAIR)

r = interest rate for a period (month/quarter)

n = number of periods in a year

Example 3 EAIR

Sarah borrows \$100 at 2% compound interest a month.

How much (principal + interest) will be owed after a year?

Solution

$$\begin{aligned}\text{Using FV} &= P (1 + r)^n \\ &= \$100 \times (1.02)^{12} \\ &= \$100 \times 1.2682 = \$126.82\end{aligned}$$

EAIR is 26.82%, which can also be found using the formula:

$$1 + R = (1 + r)^n = (1 + 0.02)^{12} = 1.2682$$

$$R = 0.2682 = 26.82\%$$

The EAIR is an important concept that gives the true interest rate associated with an investment or loan. The more frequent the compounding periods, the higher the EAIR. This has implications for the cost of capital (see *Chapter 10*).

2.1 Compounding in Reverse

Discounting calculates the *sum which must be invested now* (at a fixed interest rate) to receive a given sum in the future.

Example 4 Discounting

If Zarosa needed to receive \$251.94 in three years (t_3), what sum would she have to invest today (t_0) at an interest rate of 8% a year?

Solution

The formula for compounding is: $FV = P(1 + r)^n$

Rearranging this: $P = FV \times \frac{1}{(1+r)^n}$

Alternatively, $PV = CF \times \frac{1}{(1+r)^n} = CF(1 + r)^{-n}$

where:	PV	=	the present value of a future cash flow (CF)
	r	=	annual rate of interest/discount rate
	n	=	number of years before the cash flow arises

In this case: $PV = \$251.94 \times \frac{1}{(1.08)^3} = \200

The present value of \$251.94 receivable in three years is \$200.

2.2 Simple Discount Factor

The "simple discount factor" formula $(1 + r)^{-n}$ gives the present value of \$1 receivable at the end of n years at a discount rate, r .

- For a cash flow arising now (at t_0), the discount factor will **always** be 1.
- t_1 is the point in time exactly one year after t_0 .

Exam advice

The formula for simple discount factors is provided at the top of the present value table in the exam as $(1 + r)^{-n}$.

Activity 3 Present Value

[Hide](#)

See Tables in the Formulae and Tables section of the Introduction. Find the present value of:

- a) **\$250 received or paid in 5 years, $r = 6\%$ a year.**
- b) **\$30,000 received or paid in 15 years, $r = 9\%$ a year.**

**Please use the notes feature in the toolbar to help formulate your answer*

- (a) From the tables: $r = 6\%$, $n = 5$, discount factor = 0.747
Present value = $\$250 \times 0.747 = \186.75
- (b) From the tables: $r = 9\%$, $n = 15$, discount factor = 0.275
Present value = $\$30,000 \times 0.275 = \$8,250$

Exam advice

Assume that cash flows arise at the *end* of the year to which they relate (unless told otherwise).

3.1 Time Value of Money

The "time value of money" concept is fundamental to discounted cash flow (DCF) methods. It is based on the assumption that investors prefer to receive \$1 today rather than \$1 in one year.

Several possible reasons underlie this assumption:

- **Liquidity preference:** money received today can be spent or reinvested to earn more. Therefore, investors have a preference for having cash/liquidity today.
- **Risk:** cash received today is safe; future cash receipts may be uncertain.
- **Inflation:** cash today can be spent at today's prices, but the value of future cash flows may be eroded by inflation.

Key Point

DCF methods account for the time value of money by restating each future cash flow at its equivalent value today.

3.2 Investment Appraisal

DCF methods can be used to evaluate business projects (i.e. for investment appraisal). Two methods are examinable:

1. Net present value (NPV) – an absolute measure; and
2. Internal rate of return (IRR) – a relative measure.

3.3 Advantages and Limitations of DCF Methods

DCF methods of appraisal are superior to non-DCF appraisal methods because they:

- ✓ Take into account:
 - the time value of money;
 - all of a project's cash flows over its entire duration; and
 - the timing of cash flows.
- ✓ Provide a more objective basis for evaluating and selecting investment projects as they are unaffected by financial accounting policies (e.g. capitalisation and depreciation versus expense).

As an absolute measure, NPV is regarded as the best single measure of the value of an investment (see s.4.2).

Despite the theoretical superiority of DCF methods, many company managers prefer to use non-DCF methods such as payback or ROCE (see *Chapter 3*).

Possible reasons for this reluctance to use DCF methods include:

- ✗ The potentially complex and time-consuming process of calculating NPV or IRR (see later in this chapter).
- ✗ Difficulty in explaining DCF methods to non-financial managers.
- ✗ The complexity of estimating an appropriate discount rate (i.e. the applicable interest rate), particularly for unquoted companies.
- ✗ Managers may feel little connection between DCF methods and their reported performance and bonus systems.

4.1 Procedure

The NPV method is applied as follows:

1. Forecast the relevant cash flows from the project.
2. Estimate the required return of investors (i.e. the discount rate). For a project with risk equal to the company's risk, investors' required return represents its overall cost of finance (also referred to as its cost of capital).
3. Discount each cash flow (receipt or payment) to its present value (PV).
4. Sum present values to give the NPV of the project.
5. If NPV is positive, accept the project as it provides a higher return than investors require (i.e. the company's cost of capital).

Key point

The cost of capital is an important concept in the NPV method of investment appraisal; it is the minimum return needed to justify an investment.

4.2 Meaning

A project's NPV shows the theoretical change in the dollar value of the company due to the project.

Therefore, it indicates the change in shareholder wealth due to the project. In other words, if the NPV of a project is \$1m, the company's market value will increase by \$1m, and shareholder wealth will increase by \$1m.

- The main objective of financial management is to maximise shareholder wealth. Shareholder wealth is increased if projects with positive NPVs are accepted, and shareholder wealth will be maximised if a company invests in all such projects.
- Therefore, NPV is considered the superior technique in business decision making.

4.3 DCF Appraisal Layouts

There are two possible layouts for presenting a DCF calculation.

4.3.1 Layout 1

The "cash flow" layout is most appropriate for more complex questions, especially when the complications of tax and inflation are incorporated (see *Chapter 5*).

Time	0	1	2	3
	\$000	\$000	\$000	\$000
Assets ("capital") expenditure	(x)	—	—	x
Cash from sales	—	x	x	x
Materials	(x)	(x)	(x)	—
Labour	—	(x)	(x)	(x)
Overheads	—	(x)	(x)	(x)
Advertising	(x)	—	(x)	—
Grant	—	x	—	—
Net cash flow	<u>(x)</u>	<u>x</u>	<u>x</u>	<u>x</u>
		$\frac{1}{1+r}$	$\frac{1}{(1+r)^2}$	$\frac{1}{(1+r)^3}$
r% discount factor	1			
Present value	(x)	x	x	x
NPV = $\sum x$ (i.e. the sum of the xs)				

Exam advice

The details of each question scenario must be read **very carefully** to ascertain all relevant cash flows (see *Chapter 5*) and their exact timing.

4.3.2 Layout 2

The following layout may be preferable when cash flows are relatively few and involve annuities (see s.4.4)

Time		Cash flow	Discount factor	Present value
		\$000	@ r%	\$000
0	Asset expenditure	(x)	1	(x)
1–10	Contribution	x	x	x
1–10	Fixed costs	(x)	x	(x)
10	Scrap value	x	x	<u>x</u>
Net present value				<u>x</u>

Exam advice

This layout becomes time consuming, cumbersome and error-prone when there are more than a few cash flows or annuities. Therefore, it is unsuitable for constructed response questions in Section C of the exam.

Activity 4 Net Present Value

[Hide](#)

Elgar has \$10,000 to invest for five years. He could deposit it in a bank, earning compound interest at 8%, annually.

He has been offered an alternative investment in a low-risk project: it is expected to produce net cash inflows of \$3,000 for each of the first three years, \$5,000 in the fourth year and \$1,000 in the fifth year.

Required:

Calculate the net present value of the project.

**Please use the notes feature in the toolbar to help formulate your answer*

Time	Description	Cash Flow	8% DF	PV
		\$		\$
0	Investment	(10,000)	1	(10,000)
1	Net inflow	3,000	0.926	2,778
2	Net inflow	3,000	0.857	2,571
3	Net inflow	3,000	0.794	2,382
4	Net inflow	5,000	0.735	3,675
5	Net inflow	1,000	0.681	<u>681</u>
			NPV =	<u>2,087</u>

4.3.3 Spreadsheet Function

The spreadsheet in the computer-based exam (CBE) includes an NPV function to calculate NPV based on a supplied discount rate and series of future cash flows:

$$NPV = NPV(\text{rate}, \text{value1}, [\text{value2}], \dots)$$

Where:

rate = discount rate (as a decimal)

value1 = the item, cell reference or range within which cash flows are to be discounted

value2,... = additional numbers, cell references or ranges to be discounted

For example, =NPV(0.1, B12:E12) discounts values in cells B12-E12 at a discount rate of 10%, where B12 is **one year** from now. This means that any cash flows now (e.g. the initial investment) will still require adjustment.

Activity 5 NPV Spreadsheet Function

[Hide](#)

As for Activity 4.

Required:

Calculate the net present value of the project using the NPV spreadsheet function.

**Please use the notes feature in the toolbar to help formulate your answer*

B8					
	A	B	C	D	E
1	Time	Cash flow			
2		\$			
3	1	3,000			
4	2	3,000			
5	3	3,000			
6	4	5,000			
7	5	1,000			
8	NPV	12087			
9	0	-10,000			
10		2,087			

Using the NPV function to discount the future cashflows and deducting the initial investment gives NPV **\$2,087** (as for Solution 4).

4.4 Annuities

1 Definition

Annuity– a stream of identical cash flows arising *each* year for a finite period.

The present value of an **annuity** is given as

$$CF \times \frac{1}{r} \left[1 - \frac{1}{(1+r)^n} \right]$$

Where: CF is the cash flow received each year commencing at t_1 .

$\frac{1}{r} \left[1 - \frac{1}{(1+r)^n} \right]$ is known as the "annuity factor" or cumulative discount factor".

Exam advice

The formula is given in the exam as $\frac{1 - (1+r)^{-n}}{r}$. Annuity tables are also provided in the exam.

Key Point

The exam formula and tables are based on the *assumption that the cash flow starts after one year*.

Example 5 PV of Annuity

Calculate the present value of \$1,000 receivable each year for three years if interest rates are 10%.

Time	Description	Cash flow	10% Annuity factor	PV
		\$		\$
t_{1-3}	Annuity	1,000	$\frac{1}{0.1} \left[1 - \frac{1}{1.1^3} \right] = 2.487$	2,487

(An annuity received for the next three years is written as t_{1-3})

Note that the annuity tables save time in the exam, compared to using the formula:

8%	9%	10%	
0.926	0.917	0.909	1
1.783	1.759	1.736	2
2.577	2.531	2.487	3
3.312	3.240	3.170	4
3.993	3.890	3.791	5

Activity 6 Annuity

[Hide](#)

Calculate the present value of \$2,000 receivable for each of 10 years commencing three years from now. Assume interest at 7%.

**Please use the notes feature in the toolbar to help formulate your answer*

Time	Description	Cash flow	7% Annuity Factor	PV
		\$		\$
t_{3-12}	Annuity	2,000	6.135 (W)	12,270

WORKING

Annuity factor $_{3-12}$ @ 7% = Annuity factor $_{1-12}$ @ 7% – Annuity factor $_{1-2}$ @ 7%

= 7.943 – 1.808 (per tables) = 6.135

7%	8%	9%	10%	
0.935	0.926	0.917	0.909	1
1.808	1.783	1.759	1.736	2
2.624	2.577	2.531	2.487	3
3.387	3.312	3.240	3.170	4
4.100	3.993	3.890	3.791	5
4.767	4.623	4.486	4.355	6
5.389	5.206	5.033	4.868	7
5.971	5.747	5.535	5.335	8
6.515	6.247	5.995	5.759	9
7.024	6.710	6.418	6.145	10
7.499	7.139	6.805	6.495	11
7.943	7.536	7.161	6.814	12

4.5 Perpetuities

Definition

Perpetuity – a stream of identical cash flows arising each year to infinity.

The present value of a perpetuity = $CF \times \frac{1}{r}$ where CF is the cash flow received each year.

The “perpetuity factor”, $\frac{1}{r}$, is not an exam formula but is easily remembered. Its derivation, which is not examinable, is as follows:

- As $n \rightarrow \infty$ (i.e. as n gets nearer and nearer to infinity), so $(1 + r)^n \rightarrow \infty$ (because anything to the power n , if n is near infinity, will also tend to infinity).
- This means $\frac{1}{(1 + r)^n} \rightarrow 0$ (because 1 divided by a big number gets closer to 0 as the number gets larger).
- Therefore, $\frac{1}{r} \left[1 - \frac{1}{(1 + r)^n} \right] \rightarrow \frac{1}{r} (1 - 0) \rightarrow \frac{1}{r}$

Key Point

The present value of a perpetuity formula is based on the assumption that the cash flow starts after one year.

Example 6 PV of Perpetuity

Calculate the present value of \$1,000 receivable each year in perpetuity if interest rates are 10%.

Time	Description	Cash flow	10% Annuity Factor	PV
		\$		\$
$t_1 - \infty$	Perpetuity	1,000	$\frac{1}{0.1} = 10$	10,000

Activity 7 Perpetuity

[Hide](#)

See Tables in the Formulae and Tables section of the Introduction chapter.

Required:

Calculate the present value of \$2,000 receivable in perpetuity commencing 10 years from now.
Assume interest at 7%.

**Please use the notes feature in the toolbar to help formulate your answer*

Time	Description	Cash flow	7% Annuity Factor	PV
		\$		\$
$t_{10} - \infty$	Perpetuity	2,000	7.771 (W)	15,542

WORKING

$$\text{Annuity factor}_{10-\infty} @ 7\% = \text{Annuity factor}_{1-\infty} @ 7\% - \text{Annuity factor}_{1-9} @ 7\%$$
$$= \frac{1}{0.07} - 6.515 \text{ (per tables)} = 14.286 - 6.515$$

Annuity factor to use = 7.771

5.1 Decision Rule

Definition

Internal rate of return – the discount rate at which the present value of a project's cash flows is zero.

IRR represents a project's average annual cash return as a percentage and the highest acceptable finance cost.

Key Point

The **decision rule** for IRR is:

- If $IRR > \text{cost of capital}$, accept the project.
- If $IRR < \text{cost of capital}$, reject the project.

5.2 Perpetuities

If a project has equal annual cash flows receivable in perpetuity:

$$\text{IRR} = \frac{\text{Annual cash inflows}}{\text{Initial investment}} \times 100$$

Example 7 Perpetuity IRR

An investment of \$1,000 gives an annual income of \$140 indefinitely. The return on the investment is given by

$$\text{IRR} = \frac{\text{Annual cash inflows}}{\text{Initial investment}} \times 100 = \frac{\$140}{\$1,000} = 14\%$$

Activity 8 IRR of a Perpetuity

[Hide](#)

An investment of \$15,000 today will provide \$2,400 each year to perpetuity.

Required:

Calculate the internal rate of return for the investment.

**Please use the notes feature in the toolbar to help formulate your answer*

$$\text{IRR} = \frac{2,400}{15,000} \times 100 = 16\%$$

5.3 Annuities

To give an NPV of zero, the *present value* of the cash inflows must equal the *initial cash outflow*.

- That is: Annual cash inflow $\times$ Annuity factor = Cash outflow. This equation can be rearranged to solve for the annuity factor:

$$\text{Annuity factor} = \frac{\text{Cash outflow}}{\text{Annual cash inflow}}$$

- Once the annuity factor is known, the discount rate can be established from the appropriate table.

Example 8 Annuity IRR

An investment of \$6,340 will yield an income of \$2,000 for four years.

Required:

Calculate the internal rate of return of the investment.

Solution

Year	Description	CF(\$)	DF	PV(\$)
0	Initial investment	(6,340)	1	(6,340)
1-4	Annuity	2,000	AF ₁₋₄ years	<u>6,340</u>
	NPV			<u>0</u>

$$AF_{1-4 \text{ years}} = \frac{\$6,340}{\$2,000} = 3.17$$

From the annuity table, the rate with a four-year annuity factor closest to 3.17 is 10% which is the approximate IRR for this investment.

Activity 9 IRR of an Annuity

[Hide](#)

See Tables in the Formulae and Tables section of the Introduction.

An immediate investment of \$10,000 will give an annuity of \$1,000 for the next 15 years.

Required:

Calculate the internal rate of return of the investment.

**Please use the notes feature in the toolbar to help formulate your answer*

Time	Description	Cash Flow	Discount Factor	PV
		\$		\$
0	Investment	(10,000)	= 1	(10,000)
1-15	Annuity	1,000	Annuity factor ₁₋₁₅ = 10 (6al)	<u>10,000</u>
				<u>0</u>

From the annuity table, the rate with a 15-year annuity factor of 10 lies between 5% and 6%. The IRR is between 5% and 6%.

Note that this annuity is not worthwhile if \$10,000 could be otherwise invested for a return of 6% or more.

5.4 Estimation

5.4.1 Linear Interpolation Formula

When cash flows are not perpetuities or annuities, the IRR is estimated as follows:

1. Calculate the NPV of the project at a chosen discount rate.
2. If NPV is positive, recalculate NPV at a higher discount rate (i.e. to get an NPV closer to zero).
3. If NPV is negative, recalculate at a lower discount rate (again to get an NPV closer to zero).
4. Use the following formula to estimate the IRR by “linear interpolation”:

$$IRR \sim A + \frac{N_A}{N_A - N_B} (B - A)$$

Where:

A = Lower discount rate

B = Higher discount rate

N_A = NPV at rate A

N_B = NPV at rate B

The linear interpolation formula always “works”, although care must be taken with + and – signs.

Exam advice

This is **not** an exam formulae and needs to be learnt.

Example 9 IRR Formula

The NPVs of a project with uneven cash flows are as follows:

NPV	
Discount rate	\$
10%	64,237
20%	(5,213)

Required:

Estimate the IRR of the investment using linear interpolation.

Solution

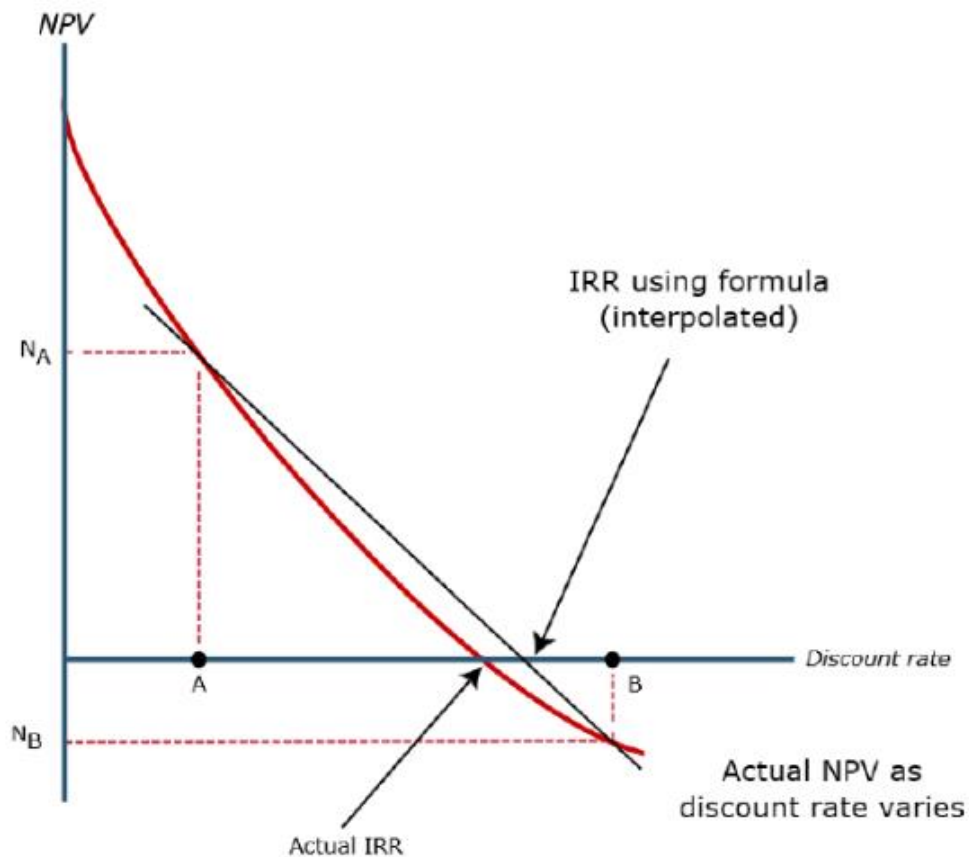
$$IRR \sim A + \frac{N_A}{N_A - N_B} (B - A)$$

$$IRR \sim 10\% + \frac{\$64,237}{\$64,237 - (-\$5,213)} (20\% - 10\%)$$

$$IRR \sim 19\%$$

Exam advice

When using linear interpolation, the answer should always be appropriately rounded due to the inherent inaccuracy of this method. The IRR calculated is only approximate, based on the simplifying assumption that there is a linear relationship between the NPV of a project and the discount rate. However, this is not so, and the following diagram illustrates the actual situation.



Activity 10 Linear Interpolation

[Hide](#)

An investment opportunity with uneven cash flows has the following net present values:

	\$
at 10%	71,530
at 15%	4,370

Required:

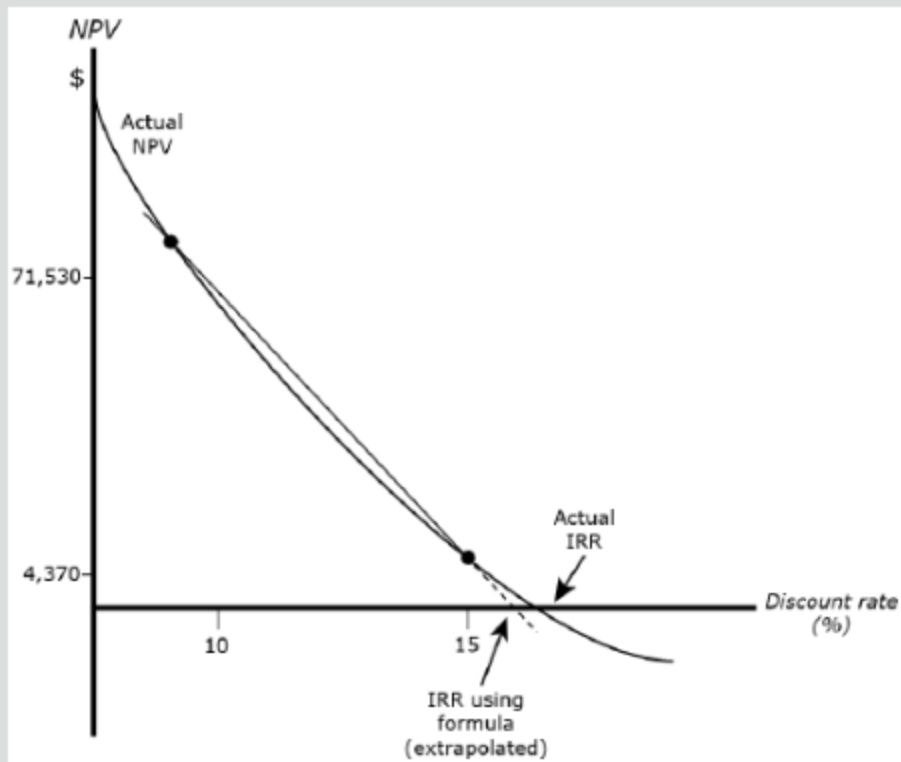
Estimate the IRR of the investment using linear interpolation.

**Please use the notes feature in the toolbar to help formulate your answer*

$$IRR \sim A + \frac{N_A}{N_A - N_B} (B - A)$$

$$IRR \sim 10\% + \frac{\$71,530}{\$71,530 - \$4,370} (15 - 10) = 10 + 5.325$$

IRR ~ 15.4% (rounded up to one decimal place). The graph below shows the results:



5.4.2 Spreadsheet Function

The example and activity in s.5.3.1 have been simplified to require only the estimate of the IRR for given NPVs. However, in exam questions, the IRR will likely need to be calculated from given cash flows, not the NPVs. In this case, the calculations are simplified using a function in the CBE spreadsheet:

$IRR = IRR(values;[guess])$, where:

- values (required) – is the range of cash flows
- guess (optional) – is a rate close to the result of the IRR, as a decimal.

Example 10 IRR Spreadsheet Function

A company is considering investment in a project requiring an outlay of \$1.6m. Annual net cash inflows of \$0.55m are expected for five years, add disposal proceeds of \$0.1m at the end of the five years. The cost of capital is 12% a year

Tabulating the cash flows:

	A	B	C
1	Year	CF	
2	0	-1.6	
3	1	0.55	
4	2	0.55	
5	3	0.55	
6	4	0.55	
7	5	0.65	
8	IRR	22.3%	

Using the IRR function $=IRR(B2:B7)$ gives **22.3%**

Activity 11 IRR of a Project

[Show](#)

Barcombe is considering a project X for which the following cash inflows have been estimated:

End of year	\$000
1	15
2	17
3	22
4	4

Project X requires an initial investment of \$45,000 in non-current assets. The assets would be expected to be sold at the end of the project for \$2,000.

Required:

Calculate the IRR of the project using the spreadsheet IRR function.

Calculate the IRR of the project using the spreadsheet IRR function.

**Please use the notes feature in the toolbar to help formulate your answer.*

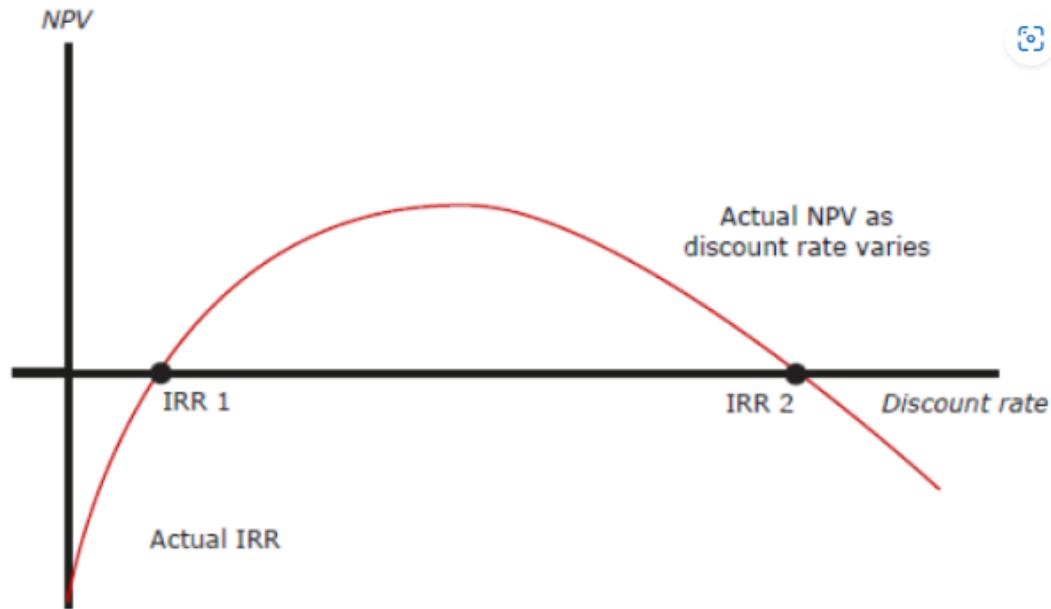
Tabulating the cash flows:

	A	B	C
1	Year	CF	
2	0	-45	
3	1	15	
4	2	17	
5	3	22	
6	4	6	(4+2)
7	IRR	13.6%	

Using the IRR function =IRR(B2:B6) gives **13.6%**

5.5 Unconventional Cash Flows

If cash outflows are followed by cash inflows and then *more cash outflows* (e.g. at the end of the project, a site has to be decontaminated), the situation of "multiple yields" may arise (i.e. more than one IRR). This outcome is shown graphically:



Suppose IRR 1 is 6% and IRR 2 is 30%. Using the NPV rule, the project will be acceptable as long as the cost of capital is between 6% and 30% (which is likely). However, if the target rate of return was 15%, say, the 6% IRR gives an incorrect reject decision.

5.6 Relative Merits of NPV and IRR

First, a summary:

NPV	IRR
• An absolute measure (\$) • If $NPV > 0$, accept • If $NPV < 0$, reject • Shows \$ change in shareholder wealth • A unique solution (i.e. a project has only one NPV) • Always reliable for decision making	• A relative measure (%) • If $IRR > \text{target \%}$, accept • If $IRR < \text{target \%}$, reject • Does not show an absolute change in wealth • May result in two answers (multiple IRRs) • Not always reliable

The NPV method is superior; it provides a single correct answer for decision-making in all circumstances. For example, if a choice has to be made between two or more projects, the higher the NPV, the greater the increase in shareholder wealth. Considering *Activity 11*, if Barcombe's cost of capital and target rate of return is 10%, Project X would be accepted using the IRR rule ($12.2\% > 10\%$). The NPV at 10% must be positive and is, in fact, \$1,931 (calculation not shown). If Project Y had ten times all the cash flows, its NPV would be \$19,310, but its IRR would still be 12.2%.

As a relative measure, IRR takes NO account of the relative size of a project; therefore, it cannot be relied on when making choices (see capital rationing in *Chapter 6*).

Finally, the perceived limitation of the "reinvestment assumption" in IRR is worth mentioning. Suppose that Project X (*Activity 11*) is to be funded with a loan carrying an interest rate of 10%. The NPV calculated at 10% is the value of the project after the cash inflows (e.g. \$15,000 at the end of year 1) have been applied to repay the loan. Alternatively, if Barcombe did not need to borrow, the cash inflows would be invested in other worthwhile projects (i.e. with a positive NPV).

Extending this idea suggests that there is an assumption in the IRR model that the interim cash flow receipts can be reinvested at the IRR. Therefore, an IRR that is significantly higher than a realistic reinvestment rate will be an overestimate of the project's actual return.

Syllabus Coverage

This chapter covers the following Learning Outcomes.

D. Investment Appraisal

1. Investment appraisal techniques

- e) Calculate net present value and discuss its usefulness as an investment appraisal method.
- f) Calculate internal rate of return and discuss its usefulness as an investment appraisal method.
- g) Discuss the superiority of discounted cash flow (DCF) methods over non-DCF methods.
- h) Discuss the relative merits of NPV and IRR.

Summary and Quiz

- The time value of money concept assumes that investors prefer to receive \$1 today rather than \$1 in one year.
- DCF methods have two significant advantages:
 - they focus on cash flow, which is more relevant than the accounting concept of profit; and
 - they take into account the time value of money.
- Net present value (NPV) is a DCF method used to determine whether a project should be undertaken.
- A project should be undertaken if its NPV > \$0.
- NPV is considered the superior decision-making technique as it is an absolute measure that tells management the change in shareholder wealth expected from a project.
- An annuity is an equal annual cash flow for more than one year.
- A perpetuity is a stream of identical cash flows arising each year to infinity.
- Internal rate of return shows the highest finance cost that can be accepted for the project.
 - If IRR > cost of capital, accept the project.
 - If IRR < cost of capital, reject the project.
- If cash flows are unconventional (the cash flows change signs more than once), IRR will have two solutions (i.e. multiple IRRs will be found).

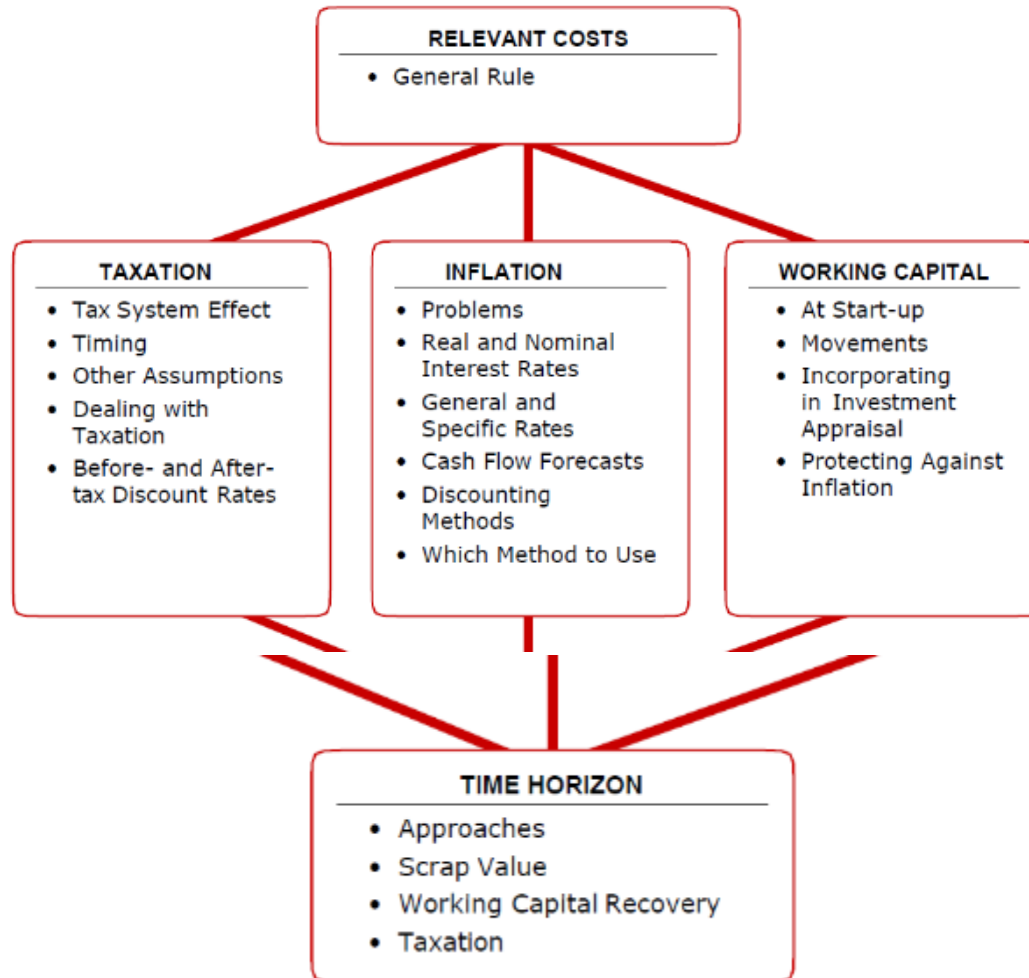
Chapter 4 Quiz

Chapter 5: Relevant Cash Flows

Overview	CHAPTER 5: Visual Overview
5.1 Relevant Costs	5.1.1 General Rule
5.2 Taxation	5.2.1 Tax System Effect
	5.2.2 Timing
	5.2.3 Other Assumptions
	5.2.4 Dealing with Taxation
	5.2.5 Before- and After-tax Discount Rates
5.3 Inflation	5.3.1 Problems Caused by Inflation
	5.3.2 Real and Nominal Interest Rates
	5.3.3 General and Specific Inflation Rates
	5.3.4 Cash Flow Forecasts
	5.3.5 Discounting Methods
	5.3.6 Which Method to Use
5.4 Working Capital	5.4.1 At Start-up
	5.4.2 Movements
	5.4.3 Incorporating in Investment Appraisal
	5.4.4 Protecting Against Inflation
5.5 Time Horizon	5.5.1 Approaches
	5.5.2 Scrap Value
	5.5.3 Working Capital Recovery
	5.5.3 Working Capital Recovery
	5.5.4 Taxation
5 Conclusion	5 Syllabus Coverage
	5 Summary and Quiz
	5 Technical Articles

CHAPTER 5: Visual Overview

Objective: To recognise relevant costs, determine the taxation effects of a new investment, deal with inflation, and deal with cash flows relating to working capital when completing discounted cash flow analysis.



1.1 General Rule

Key Point

Only cash inflows and outflows that are affected by a decision are relevant to the decision.

This means using only **future, incremental** cash flows.

Incremental cash flows are those that change because a project is undertaken (e.g. cash from sales, and operating costs such as materials and labour).

Do **not** include *financing cash flows* because the cost of finance is measured in the cost of capital/discount rate (i.e. finance costs are accounted for by the discounting process).

Exam advice

Exam questions may present large amounts of data; some will be relevant for discounting and some will not. Therefore, not all data provided will be utilised.

Specifically, *exclude*:

- sunk costs (i.e. money already spent);
- non-cash costs (e.g. depreciation);
- book values (e.g. FIFO inventory values);
- unavoidable costs (i.e. money already committed and *apportioned* fixed costs); and
- finance costs such as interest.

However, all **opportunity costs** and revenues must be included. For example, where there is "cannibalisation" (e.g. the launch of a new product will reduce the sales of an existing product), the lost contribution is an opportunity cost.

Key Point

Lost contribution is an opportunity cost and should be shown as a cash outflow.

Activity 1 Relevant Costs

[Show](#)

A research project, which has cost the company \$150,000 to date, is under review.

If the project proceeds, it will be completed in approximately one year, when the results will be sold to a government agency for \$300,000.

The additional expenses necessary to complete the work have been estimated as follows.

Materials

This material has just been purchased for \$60,000. It is toxic and must be disposed at a cost of \$5,000 if not used in this project.

Labour

Skilled labour is hard to recruit. The workers needed were transferred to the project from a production department; at a recent meeting, the production manager claimed that if the men were returned to him, they could generate sales of \$150,000 in the next year. The direct cost of these sales would be \$100,000, including \$40,000 for labour. The overhead absorbed in production costs would amount to \$20,000.

Research staff

It has already been decided that the research department will be closed when work on this project ceases. Research wages for the year are \$60,000, and redundancy and severance pay has been estimated at \$15,000 now or \$35,000 in one year.

Equipment

The project uses a special microscope, which cost \$18,000 three years ago.

It is expected to have a residual value of \$3,000 in another two years and has a current disposal value of \$8,000. If used in the project, the disposal value in one year is estimated at \$6,000. If the project is not undertaken, the microscope will be sold immediately.

Share of general building services

The project is charged \$35,000 a year to cover general building expenses. If the project is discontinued, the space occupied could be sublet immediately for an annual rent of \$7,000.

Required:

Advise the managing director whether the project should be allowed to proceed (ignore the time value of money).

Hint: List all the elements of cost and determine the relevant costs of each.

**Please use the notes feature in the toolbar to help formulate your answer*

Relevant costs and revenues of proceeding with the project.

	\$
(1) Costs to date of \$150,000 are sunk, so ignore.	–
(2) Materials – purchase price of \$60,000 is also sunk. The opportunity benefit is disposal costs saved if the project is undertaken.	5,000
(3) Labour cost – direct cost of \$40,000 will be incurred regardless of whether the project is undertaken, so is not relevant. Opportunity cost of lost contribution = \$150,000 – (\$100,000 – \$40,000)	(90,000)
Absorption of overheads – irrelevant as it is merely an apportionment of existing costs	–
(4) Research staff costs	
Wages for the year (closed down otherwise)	(60,000)
Redundancy pay increase (\$35,000 – \$15,000)	(20,000)
(5) Equipment	
Deprival value if used in the project (disposal value of \$8,000) less \$6,000 disposal proceeds in one year	(2,000)
(6) General building services	
Apportioned costs irrelevant	–
Opportunity costs rental forgone	<u>(7,000)</u>
	(174,000)
Sales value of the project	<u>300,000</u>
Increased cash flow contribution from the project	<u>126,000</u>

Advice: Proceed with the project.

2.1 Tax System Effect

Taxation has a negative effect and a positive effect on investment appraisal.

Negative Effect

Tax is charged on operating results.

- Operating results = revenues less operating costs (materials, labour and overheads)
- Any tax relief on *finance* costs is accounted for in the discount rate (cost of capital).

Positive Effect

Tax relief is given on non-current assets via *tax-allowable* depreciation (TAD).

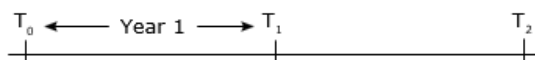
- *Financial accounting* depreciation is **not** a tax-allowable expense.
- Instead, businesses may claim TAD according to the applicable tax regime.
- Exam questions will state the depreciation method (reducing balance or straight-line) and rate (%).
- There is no TAD in the year of disposal. The difference between the tax written-down value and disposal proceeds (if any) is a tax allowable loss ("balancing allowance") or taxable gain ("balancing charge").

Exam advice

Exam questions will specify the tax rates/policies to use.

2.2 Timing

The timing of tax cash flows is complex. Some exam questions will specify that tax is paid *in* the year of taxable profits; others will state that tax is paid "one year in arrears" (i.e. in the following year), as follows:



- Assume the first net revenues (revenues minus operating costs) are received at the end of year 1 (T_1):
 - Tax is assessed at T_1 ;
 - Tax is paid at T_2 (as tax is paid one year in arrears)
- Assume an asset is bought at the *start* of year 1 (T_0):
 - First TAD claimed at T_1 (date of next tax assessment);
 - Reduces tax payment at T_2

Key Point

The tax benefit ("saving") is **not** the amount of the allowance; it is the allowance $\times$ tax rate.

Example 1 Tax Benefits

An asset is bought for \$5,000 at the start of an accounting period. It is sold at the end of the third accounting period for \$1,000.

The corporation tax rate is 30%, and tax is paid one year in arrears. Tax-allowable depreciation is available at 25% reducing balance, with a balancing adjustment in the year of disposal.

Required:

Calculate the tax benefits available and state when they arise.

Solution

		Tax benefit @ 30%	Timing
	\$	\$	
Cost	5,000		
Year 1 TAD @ 25%	(1,250)	375	T ₂
Tax written-down value c/f	3,750		
Year 2 TAD @25%	(938)	281	T ₃
Tax written-down value c/f	2,812		
Year 3 Disposal proceeds	(1,000)		
Tax allowable loss (balance)	1,812	544	T ₄

2.3 Other Assumptions

Other simplifying assumptions include the following:

- The tax rate is **constant**.
- **Sufficient taxable profits** are available to use **all tax deductions** in full.
- Working capital flows have **no tax effects** (e.g. an increase in accounts receivable does not affect tax, as tax is charged when revenues are recorded not when cash is received). (See the last section of this chapter.)

Key Point

All negative cash flows associated with a project are assumed to attract tax relief; asset ("capital") expenditure attracts tax-allowable depreciation and expenses ("revenue expenditure") reduce taxable profit. The sufficient taxable profits assumption means that even if the net cash flow for a year is negative (as is often the case in the first year), there will be a **positive** tax cash flow for the tax relief.

2.4 Dealing with Taxation

		T ₀	T ₁	T ₂	T ₃
Step 1	Set up table				
	↓				
REVENUE					
	(a) Put in revenues and				
	operating costs		x	x	
Step 2			(x)	(x)	
	↓				
	(b) Total columns for				
	net revenues		x	x	
	↓				
	(c) Calculate tax payable				
	on net revenues			(x)	(x)
	↓				
CAPITAL					
	Put in capital outlay and any	(x)			
	disposal value				
Step 3				x	
	disposal value			x	
Step 3					
	↓				
	Calculate tax saving on tax-				
	allowable depreciation			x	x
Step 4					
	↓				
	Total columns for net cash	(x)	x	x	(x)
	flows and discount				
Step 5					
	Discount factor r%	x	x	x	x
	Present value	(x)	x	x	x
<i>Note:</i> The schedule above assumes that the project has a two-year operating life and that tax is paid/saved one year in arrears.					

Activity 2 Tax Cash Flows

[Hide](#)

A company buys an asset for \$10,000 at the beginning of an accounting period (1 January 20X1) to undertake a two-year project.

Net cash inflows received at the end of year 1 and year 2 are \$5,000.

The company sells the asset on the last day of the second year for \$6,000.

The corporation tax rate is 33% and tax is paid one year in arrears. Tax-allowable depreciation is calculated on a reducing balance basis at 25%. The cost of capital is 10%.

Required:

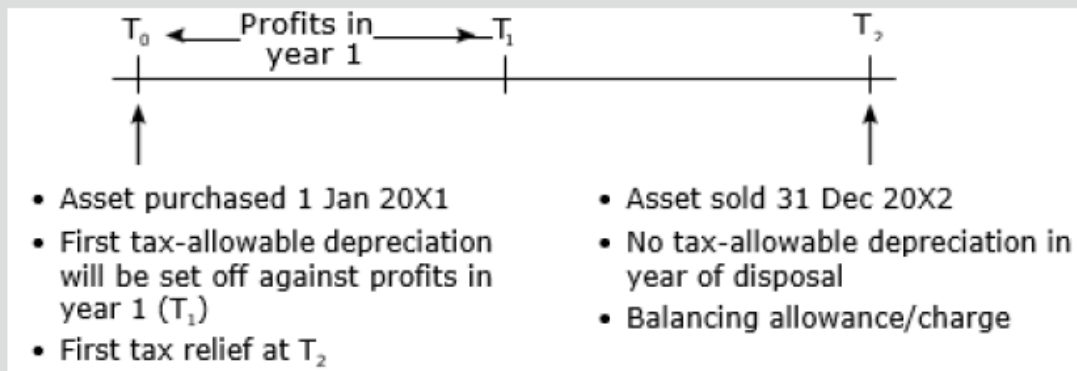
Calculate the project's NPV and recommend whether to accept or reject the project.

Hint: Draw a timeline to show the timing of the tax cash flows.

**Please use the notes feature in the toolbar to help formulate your answer*

	T ₀	T ₁	T ₂	T ₃
Net cash inflows		5,000	5,000	
Tax @ 33%			(1,650)	(1,650)
Asset purchase	(10,000)			
Scrap proceeds			6,000	
Tax benefits (W)			<u>825</u>	<u>495</u>
Net cash flow	(10,000)	5,000	10,175	(1,155)
10% discount factor	1	0.909	0.826	0.751
Present value	(10,000)	4,545	8,405	(867)
NPV = \$ 2,083				
Recommendation: Accept the project				
WORKING				

WORKING



		\$	Tax benefits at 33%	Timing
T ₀	Investment in asset	10,000		
Year 1	TAD @ 25%	(2,500)	825	T ₂
		7,500		
Year 2	Disposal proceeds	(6,000)		
	Tax allowable loss (balance)	(1,500)	495	T ₃

Activity 3 More Tax Cash Flows

[Show](#)

A company buys a new machine costing \$50,000 with a disposal value of \$25,000 at the end of four years. Tax-allowable depreciation is available on a straight-line basis at 20% on cost, with a balancing adjustment in the year of disposal. Corporate profits are taxed at 35% and tax is paid without delay.

Required:

Calculate the tax effects associated with the machine and state their timings.

**Please use the notes feature in the toolbar to help formulate your answer*

	\$	Timing
Tax-allowable depreciation ($\$50,000 \times 20\%$)	10,000	
Annual tax benefit ($\$10,000 \times 35\%$)	3,500	(Years 1–3)
Tax written-down value at disposal (end of third year – carried forward into fourth year) ($\$50,000 - (\$10,000 \times 3)$)	20,000	
Taxable gain on disposal ($\$25,000 - \$20,000$)	5,000	
Tax paid on gain on disposal ($\$5,000 \times 35\%$)	1,750	(Year 4)

2.5 Before- and After-tax Discount Rates

Before- and after-tax discount rates may also be called pre- and post-tax discount rates.

For examination purposes, the default approach to project appraisal is to:

- Forecast the *post-tax* cash flows from the project; and
- Discount at the stated discount rate, assuming this is the *post-tax* required return.

This approach is technically correct as a company's investors are only entitled to a project's cash flows *after* corporation tax has been paid.

An alternative approach is to discount the *pre-tax* cash flows at the *pre-tax* discount rate, where:

- Pre-tax cash flow = Post-tax cash flow/(1 – Corporate tax rate)
- Pre-tax discount rate = Post-tax discount rate/(1 – Corporate tax rate)

The two approaches will only give the same result in simplified situations because, in practice, the annual pre-tax cash flow will rarely equal the Post-tax cash flow/(1 – Corporate tax rate). This may be due to significant timing differences between cash flows and taxable profits, particularly in years of asset expenditure.

Example 2 Pre- and Post-tax Discount Rates

A project produces an annual post-tax cash flow of \$14,000. The post-tax required return is 14% and the corporate tax rate is 30%. Therefore, the pre-tax discount rate is 20% ($14\% / (1 - 0.30)$) and the pre-tax cash flow is \$20,000 ($\$14,000 / (1 - 0.30)$).

Consider two scenarios – cash flows in perpetuity and a single cash flow after one year:

	Cash flows in perpetuity	Single cash flow after one year
PV – post-tax method	$\$14,000 / 0.14 = \$100,000$	$\$14,000 / 1.14 = \$12,281$
PV – pre-tax method	$\$20,000 / 0.2 = \$100,000$	$\$20,000 / 1.2 = \$16,667$

The cash flow in perpetuity shows that the alternative approach may give the same result.

However, only the post-tax approach gives the correct result for the single post-tax cash flow of \$14,000 after one year.

The result of the pre-tax method is incorrect as it implies that an investor would be prepared to invest \$16,667 today but only receive \$14,000 after one year.

3.1 Problems Caused by Inflation

Inflation is a problem for project appraisal for many reasons:

- ✗ It is hard to estimate, especially when rates are high.
- ✗ It causes governments to take actions which may affect businesses (e.g. raising interest rates or cutting state spending).
- ✗ Different costs and revenues will inflate at different rates.
- ✗ It alters the cost of capital (in nominal terms).
- ✗ It makes historical costs irrelevant and causes return on capital employed (ROCE) to be overstated.
- ✗ It creates uncertainty for customers, which may lead to lower demand.
- ✗ It encourages managers to become short term in their outlook.

3.2 Real and Nominal Interest Rates

When dealing with inflation, it is important to distinguish between real and nominal ("money") interest rates:

- The real interest rate reflects the interest rate required in the absence of inflation. The market rarely quotes a real rate of interest.
- The nominal interest rate reflects the real interest rate adjusted for the effect of *general* inflation as measured, for example, by the Consumer Price Index (CPI) or the Eurozone Harmonised Index of Consumer Prices (HICP).

Example 3 Real and Nominal Interest Rates

Suppose you invest \$100 today for one year and, in the absence of inflation, you require a return of 5%. The inflation rate is expected to be 10% over the coming year.

In one year, in the absence of inflation, you require:

$$\$100 \times 1.05 = \$105$$

To maintain the purchasing power of your investment (i.e. to cover inflation) you require:

$$\$105 \times 1.1 = \$115.50$$

Therefore, you require a nominal return of $\frac{\$15.50}{\$100} = 15.5\%$ for the year.

The Fisher formula links nominal rates, real rates and general inflation:

$$(1 + i) = (1 + r)(1 + h)$$

Where:

i = nominal interest rate

r = real interest rate

h = general inflation rate

Applying this to *Example 3*: $(1 + 0.05)(1 + 0.10) = 1.155$. Therefore, $i = 15.5\%$.

Exam advice

The Fisher formula is provided in the exam.

3.3 General and Specific Inflation Rates

A specific inflation rate is the inflation rate on an individual item (e.g. wage inflation, materials price inflation). In comparison, the general inflation rate is a weighted average of many specific inflation rates.

3.4 Cash Flow Forecasts

If there is inflation in the economy, there are two possible approaches:

1. Discount **nominal** (money) cash flows at a **nominal** cost of capital; or
2. Discount **real** (today's money) cash flows at a **real** cost of capital.

3.5 Discounting Methods

3.5.1 Nominal Method

Cash flows are **inflated** to future price levels using the **specific** inflation rate for each type of revenue/cost. This produces a forecast of the actual cash amounts the company will receive and pay on future dates.

These nominal (or money) cash flows are discounted at a nominal (or money) cost of capital.

Example 4 Nominal Cash Flows

A new three-year project:

- requires an initial investment of \$5m at time 0, with a scrap value of \$1m at the end of year 3;
- is expected to generate revenues of \$10m at current prices for each of the next three years; these revenues are expected to rise at 5% a year;
- is expected to incur costs of \$5m in year 1, \$4m in year 2 and \$5m in year 3, at current prices; these costs are expected to rise at 3% in year 1, 6% in year 2 and 5% in year 3.

The general inflation rate is expected to be 4% for the next three years and the nominal weighted average cost of capital is 10%.

Calculate the NPV of the project by discounting nominal cash flows. Ignore tax.

Solution

Discounted nominal cash flows (\$m)

	T_0	T_1	T_2	T_3
Revenues (W1)		10.500	11.025	11.576
Costs (W2)		(5.150)	(4.367)	(5.732)
Capex	(5.000)			1.000
Relevant cash flows	(5.000)	5.350	6.658	6.844
DF at 10%	1	0.909	0.826	0.751
PV	(5.000)	4.863	5.500	5.140
NPV	10.50m			

WORKINGS

(1) Revenues	T_1	T_2	T_3
Current prices	\$10m	\$10m	\$10m
x inflation factor	$(1 + 5\%)$	$(1 + 5\%)^2$	$(1 + 5\%)^3$
= Nominal revenues	10.500	11.025	11.576
(2) Costs			
Current prices	\$5m	\$4m	\$5m
			$(1 + 3\%)(1 + 6\%)$
x inflation factor	$(1 + 3\%)$	$(1 + 3\%)(1 + 6\%)$	$(1 + 5\%)$
= Nominal costs	5.150	4.367	5.732

3.5.2 Real Method

Cash flows are expressed at today's prices (i.e. **before** the effects of inflation). These are then discounted at a real rate (i.e. *excluding* general inflation). Use the Fisher formula (see above) to calculate a real rate from a nominal (or money) rate and the general inflation rate.

Key Point

This method **only** works if **ALL** cash flows are inflating at the **same** general inflation rate.

Example 5 Real Cash Flows

A new three-year project:

- requires an initial investment of \$5m at time 0, with a scrap value of \$1m at the end of year 3 in today's money terms;
- is expected to generate sales revenues of \$10m a year for the next three years, growing at the general rate of inflation;
- is expected to incur costs of \$5m in year 1, \$4m in year 2 and \$5m in year 3, growing at the general inflation rate.

The real weighted average cost of capital is 3%.

Calculate the NPV of the project using the real discount rate. Ignore tax.

Solution

Discounted real cash flows (\$m)

Time	T ₀	T ₁	T ₂	T ₃
Revenues		10	10	10
Costs		(5)	(4)	(5)
Capex	(5)			1
Relevant cash flows	(5)	5	6	6
DF at 3%	<u>1</u>	<u>0.971</u>	<u>0.943</u>	<u>0.915</u>
PV	<u>(5.000)</u>	<u>4.855</u>	<u>5.658</u>	<u>5.490</u>
NPV	11.00m			

Activity 4 Discounting with Specific Inflation

[Show](#)

A company is considering a three-year project with an initial outlay of \$10,000.

Material costs at current prices will be \$1,500 a year for three years. Material costs inflate at 8% each year.

Labour savings at current prices will be \$4,000 a year for three years. Labour costs inflate at 5% each year.

Overhead savings at current prices will be \$2,000 a year for three years. Overhead costs inflate at 10% each year.

The nominal cost of capital is 15.5%.

General inflation is 7%.

Required:

Calculate the net present value of the project.

Ignore taxation.

Hint: Calculate discount factors to three decimal places.

**Please use the notes feature in the toolbar to help formulate your answer*

Using the nominal method to calculate cash flows:

	T ₀	T ₁	T ₂	T ₃
	\$	\$	\$	\$
Investment	(10,000)			
Materials (8%)		(1,620)	(1,750)	(1,890)
Labour savings (5%)		4,200	4,410	4,631
Overhead savings (10%)		<u>2,200</u>	<u>2,420</u>	<u>2,662</u>
Net cash flow	<u>(10,000)</u>	<u>4,780</u>	<u>5,080</u>	<u>5,403</u>
Discount factor @ 15.5%	<u>1</u>	<u>0.866</u>	<u>0.750</u>	<u>0.649</u>
Present value	<u>(10,000)</u>	<u>4,139</u>	<u>3,810</u>	<u>3,507</u>

NPV = \$1,456

Note that the cash outflows saved for labour and overhead increase each year per at their specific inflation rates.

3.6 Which Method to Use

If selling prices and costs have different inflation rates, the only way to calculate NPV accurately is:

- to forecast each cash flow in **nominal** terms (i.e. incorporating the specific inflation rate affecting that cash flow); and
- to discount the total nominal cash flow at the company's nominal cost of capital (i.e. incorporating the general inflation rate in the economy).

The only situation in which the real method is valid is when revenues and costs all increase at the general inflation rate. In this case cash flows before inflation can be discounted at the real cost of capital.

Activity 5 Tax and Inflation

[Show](#)

A company is considering a project which requires a machine costing \$250,000 on 1 January 20X4. Net inflows from the project are expected to be \$80,000 a year in current price terms for the next four years. At the end of the project, the machine will be sold for estimate cash proceeds of \$50,000. The project flows are expected to inflate at 5% and the company's nominal cost of capital is 15%.

The company has a December year end and pays tax one year in arrears at 33%. Tax-allowable depreciation is available at 25% reducing balance.

Required:

Determine whether the company should proceed with the project.

Hint: Show a separate working for the calculation of tax-allowable depreciation.

**Please use the notes feature in the toolbar to help formulate your answer*

		\$	@ 33%	Time
Tax allowable depreciation				
y/e 31 December 20X4				
Purchase		250,000		
TAD @ 25%		<u>(62,500)</u>	20,625	T ₂
		187,500		
y/e 31 December 20X5 TAD @ 25%		<u>(46,875)</u>	15,469	T ₃
		140,625		
y/e 31 December 20X6 TAD @ 25%		<u>(35,156)</u>	11,602	T ₄
		105,469		
y/e 31 December 20X7 Disposal proceeds		<u>(50,000)</u>		
Tax allowable loss (balance)		<u>55,469</u>	18,305	T ₅

Project appraisal	T ₀	T ₁	T ₂	T ₃	T ₄	T ₅
Inflows		84,000	88,200	92,610	97,241	
Tax @ 33% in arrears			(27,720)	(29,106)	(30,561)	(32,090)
Initial investment	(250,000)					
Scrap					50,000	
Tax savings			<u>20,625</u>	<u>15,469</u>	<u>11,602</u>	<u>18,305</u>
	<u>(250,000)</u>	<u>84,000</u>	<u>81,105</u>	<u>78,973</u>	<u>128,282</u>	<u>(13,785)</u>
DF @ 15%	1	0.870	0.756	0.658	0.572	0.497
PV	<u>(250,000)</u>	<u>73,080</u>	<u>61,315</u>	<u>51,964</u>	<u>73,377</u>	<u>(6,851)</u>

NPV = \$2,885. Therefore, **accept** the project.

Key Point

Even though inflation is only general it is not possible to use the real method because not all cash flows are affected by the same general inflation rate.

4.1 At Start-up

A project usually starts with a cash outflow for the investment in non-current assets (e.g. plant and equipment). However, many projects also will require an investment in net *current* assets (i.e. working capital). For example, the project may require the purchase of inventory.

Working capital is defined as inventory + accounts receivable – accounts payable for project appraisal. This definition *excludes* cash. Cash flow is calculated as the change in inventory + accounts receivable – accounts payable.

4.2 Movements

Over the project's life, accounts receivable may increase, resulting in cash inflows less than sales revenues.

On the other hand, the accounts payable may also increase, reducing the investment in working capital and improving cash flows because payments to suppliers are less than purchases.

At the end of the project, inventory may be reduced to zero and all receivables collected, creating a cash inflow.

4.3 Incorporating in Investment Appraisal

The *initial* working capital requirement and the *movements* in working capital are relevant cash flows for investment appraisal. Cash flows are derived as follows:

- Increase in net working capital = cash outflow
- Decrease in net working capital = cash inflow

Exam advice

Unless an exam question specifies otherwise, working capital is assumed to be "released" at the end of a project (i.e. the inflow).

Also, it is assumed that changes in the level of working capital have **no tax effects**. This is a realistic assumption because tax will be charged when net revenues accrue, not when cash is received.

Activity 6 Working Capital

[Show](#)

New product sales are forecast at \$100,000 in the first year, increasing by a 10% compound rate each year. The product has a four-year life cycle. Working capital equal to 15% of annual sales is required at the start of each year. The company's contribution margin is 40%, and no incremental fixed costs are expected.

Required:

Determine the relevant operating cash flows after allowing for working capital.

**Please use the notes feature in the toolbar to help formulate your answer*

	T ₀	T ₁	T ₂	T ₃	T ₄
	\$	\$	\$	\$	\$
Contribution (40% x Sales)		40,000	44,000	48,400	53,240
Cash in working capital (W)	<u>(15,000)</u>	<u>(1,500)</u>	<u>(1,650)</u>	<u>(1,815)</u>	<u>19,965</u>
Total cash flow	<u>(15,000)</u>	<u>38,500</u>	<u>42,350</u>	<u>46,585</u>	<u>73,205</u>
WORKING					
Sales		100,000	110,000	121,000	133,100
Level of working capital	15,000	16,500	18,150	19,965	0
Cash invested in working capital	<u>(15,000)</u>	<u>(1,500)</u>	<u>(1,650)</u>	<u>(1,815)</u>	<u>19,965</u>

4.4 Protecting Against Inflation

Over the life of a project, the effect of general inflation would be to erode the real value of any initial investment in working capital.

To protect against this requires subsequent nominal increases in the level of working capital, with these incremental investments creating related cash outflows.

Activity 7 Working Capital and Inflation

[Show](#)

A four-year project requires an initial investment in working capital of \$7.5m. The level of working capital will increase in line with general inflation, which is expected to be 3% each year.

Required:

Calculate the cash flows associated with working capital over the life of the project.

**Please use the notes feature in the toolbar to help formulate your answer*

	T ₀	T ₁	T ₂	T ₃	T ₄
	\$000	\$000	\$000	\$000	\$000
Level of working capital	7,500	7,725	7,957	8,195	0
Associated cash flows	(7,500)	(225)	(232)	(238)	8,195

5.1 Approaches

Ideally, management should determine the expected operating life of an investment project. The project should then be evaluated over its expected operating life – this should help avoid sub-optimal investment decisions.

However, making a reliable forecast of the project's life may not be possible. In this case, management may implement a policy of evaluating projects over a specified time horizon (e.g. four years). It could be argued that this approach reduces the uncertainty associated with investment appraisal since uncertainty increases with project life.

However, if an artificially short period is used, care is needed regarding scrap value, working capital recovery and tax to avoid sub-optimal decisions.

5.2 Scrap Value

Scrap value, salvage value or terminal value must be included in the evaluation of a project as a cash inflow at the end of the project life. This applies even if an artificially short time horizon is used. Ignoring scrap value would reduce the NPV and may lead to rejecting an otherwise acceptable investment project.

5.3 Working Capital Recovery

The recovery of working capital should be shown as a cash inflow at the end of the project. This follows the usual assumption that the investment in working capital is "released" on project completion.

5.4 Taxation

If, say, the project's duration is four years and tax is paid "one year in arrears", the tax charged on the fourth year's profit will be a cash outflow at the end of the fifth year.

Furthermore, the project's scrap value at the end of the fourth year should be compared to the tax written-down value of the asset at that point, and a tax allowable loss or taxable gain should be calculated as appropriate. The related tax benefit/expense should be shown as a cash inflow/outflow in the fifth year (if tax is paid one year in arrears).

Exam advice

The cash flow forecast is extended by a year if tax is paid in arrears.

Syllabus Coverage

This chapter covers the following Learning Outcomes.

D. Investment Appraisal

1. Investment appraisal techniques

a) Identify and calculate relevant cash flows for investment projects.

2. Allowing for inflation and taxation in DCF

- a) Apply and discuss the real-terms and nominal-terms approaches to investment appraisal.
- b) Calculate the taxation effects of relevant cash flows, including the tax benefits of tax-allowable depreciation and the tax liabilities of taxable profit.
- c) Calculate and apply before- and after-tax discount rates.

Summary and Quiz

- Relevant costs are only those costs which are affected by a decision, which means that they are (1) in the future, (2) incremental and (3) part of operating cash flows.
- Lost contribution is an opportunity cost and should be shown as a cash outflow.
- *Never* deduct depreciation because it is not a cash flow.
- *Never* deduct finance costs because the cost of finance is measured in the discount rate and has already been taken into account.
- Financial accounting depreciation is not a tax-allowable expense; however, the tax regime may allow a business to claim tax-allowable depreciation on non-current assets.
- Exam questions will specify the tax rates/policies to use.
- The timing of tax cash flows is complex. Questions will specify whether tax is paid *in* the year of taxable profits or the year after.
- Consistency is essential when discounting with inflation (i.e. if inflation is included in the cash flow forecast, it must be included in the discount rate).
- The Fisher formula relates nominal interest rates (i), real interest rates (r) and the general level of inflation (h). The exam formula is:
- $(1 + i) = (1 + r)(1 + h)$
- To deal with inflation, inflate cash flows and use a nominal discount rate (except when all cash flows are inflating at the same general inflation rate).
- Changes to working capital balances are relevant cash flows: inventory + accounts receivable – accounts payable.

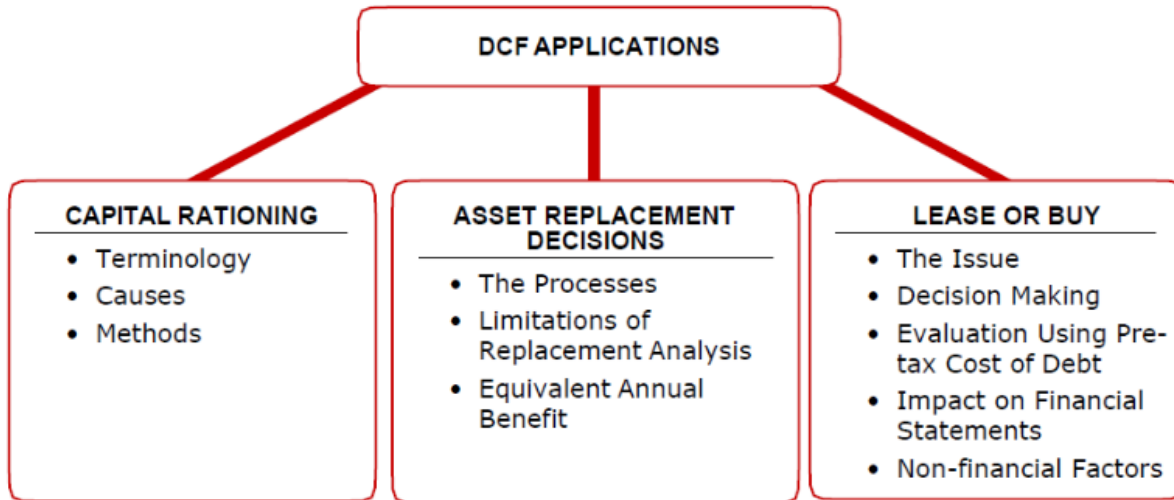
Chapter 5 Quiz

Chapter 6: Applications of Discounted Cash Flow Techniques

Overview	CHAPTER 6: Visual Overview
6.1 Capital Rationing	6.1.0 Introduction
	6.1.1 Terminology
	6.1.2 Causes of Capital Rationing
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6.2 Asset Replacement Decisions	6.2.1 The Process
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	6.3.3 Evaluation Using Pre-tax Cost of Debt
	6.3.4 Impact on Financial Statements
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6 Conclusion	6 Syllabus Coverage
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CHAPTER 6: Visual Overview

Objective: To apply discounted cash flow techniques to capital rationing, asset replacement and lease or buy decisions.



1.0 Introduction

The NPV decision rule, to accept all projects with a positive NPV, assumes the existence of a perfect capital market in which access to funds for capital investment is not restricted. In practice, companies are likely to find that funds available for capital investment are restricted or rationed.

1.1 Terminology

1 Definition

Capital rationing – a situation in which there is not enough finance (capital) available to undertake all available positive NPV projects. Therefore, capital has to be rationed.

Hard capital rationing – the capital markets limit the amount of finance available.

Soft capital rationing – the company sets internal limits on finance availability.

Single-period capital rationing – capital is in short supply in only one period.

Multi-period capital rationing – capital is rationed in two or more periods.

1.2 Causes of Capital Rationing

1.2.1 Hard Capital Rationing

Reasons why capital markets may restrict the funds available to a particular company include:

- High business risk (i.e. the company's operating cash flows are very sensitive to the economic cycle).
- High country/political risk (i.e. concerns about corporate governance or state appropriation of assets).
- High financial risk (i.e. the company already has high debt levels).
- Lack of reliable independent information available about the company. This is especially relevant to small- and medium-sized entities (SMEs).

1.2.2 Soft Capital Rationing

There are several reasons why company directors or managers might restrict available funds for capital investment:

- A preference for slower organic growth to a sudden increase in size arising from accepting several large investment projects. This particularly applies to a family-owned business that wishes to avoid hiring new managers.
- Managers may wish to avoid raising further equity finance if this will dilute the control of existing shareholders.
- Managers may wish to avoid issuing new debt if their expectations of future economic conditions are such that an increased commitment to fixed interest payments would be unwise.
- To create an internal market for investment funds. This can be seen as a way of reducing the risk and uncertainty associated with investment projects, as it leads to only accepting projects with greater margins of safety.

1.3 Methods

1.3.1 Divisible Projects

A project is divisible if any partial or proportionate investment can be made in it (i.e. between 0% and 100%). However, the project cannot be repeated.

The three steps required for rationing capital to divisible projects are as follows:

1. Calculate a *profitability index* for each project:

Profitability index = NPV of net cash inflows/Initial investment.

2. Next, rank projects according to their index.
3. Then, allocate funds starting with the highest ranking to maximise NPV.

Activity 1 Divisible Projects[Show](#)

Projects	A	B	C	D
	\$000	\$000	\$000	\$000
NPV	100	(50)	84	45
Cash flow at t_0	(50)	(10)	(10)	(15)

Additional information:

(1) Cash is rationed to \$50,000 at t_0 .

(2) The projects are divisible.

Required:**Determine the optimal investment plan.****Please use the notes feature in the toolbar to help formulate your answer.*

Projects	A	B	C	D
	\$000	\$000	\$000	\$000
NPV	100	(50)	84	45
Cash flow at t_0	(50)	(10)	(10)	(15)
NPV	100	(50)	84	45
/Investment	/50	/10	/10	/15
Profitability index	= 2	Reject	= 8.4	= 3
Rank	3	N/A	1	2
Plan			Cash	NPV
			50	
Accept	C		(10)	84
			40	
Accept	D		(15)	45
			25	
Accept	50% of A		25	50
			0	179

1.3.2 Non-divisible Projects

A non-divisible/indivisible project must be done 100% or not at all. The three steps for capital rationing then become:

1. Do not calculate a profitability index.
2. Simply list all possible combinations of projects which can be utilised given the capital budget (limit).
3. Choose the combination with the highest NPV.

Activity 2 Non-divisible Projects

[Show](#)

Details are as for *Activity 1*, but the projects are non-divisible.

Required:

Determine the optimal investment plan.

**Please use the notes feature in the toolbar to help formulate your answer.*

Possible Combinations	NPV \$000
A only (spend \$50,000)	100
C + D (spend \$25,000)	129

Choose the combination with the highest total NPV: C + D.

1.3.3 Mutually Exclusive Divisible Projects

Mutually exclusive projects are two or more projects that cannot be undertaken simultaneously (e.g. because they use the same land).

Assuming that projects are divisible (unless otherwise told), one approach is to rank and fund the projects, bearing in mind that any feasible combination of projects can include only one mutually exclusive project.

Activity 3 Mutually Exclusive Divisible Projects

[Show](#)

Details are as for *Activity 1*, but projects C and D are mutually exclusive.

Required:

Determine the optimal combination of projects.

**Please use the notes feature in the toolbar to help formulate your answer.*

Following on from Solution 1, Project B will be rejected. Project C is still the most profitable to be accepted, but investment in D is now precluded, leaving \$40,000 to be invested in A. The plan is therefore:

Plan		Cash	NPV
		50	
Accept	C	<u>(10)</u>	84
Accept	80% of A	<u>(40)</u>	<u>80</u>
		0	<u>164</u>

1.3.4 Multi-period Capital Rationing

If finance is limited in several periods, a *linear programming model* must be set up and solved to find the optimal investment strategy.

Exam advice

Multi-period capital rationing using linear programming is not examinable.

2.1 The Process

Assume that a company has already decided it requires a particular non-current asset. A secondary decision is how often to replace the asset. For example, how often should the company replace its fleet of motor vehicles or its computer equipment? This is referred to as an "asset replacement decision".

The three steps in the asset replacement decision process are:

1. Calculate the NPV of each possible replacement cycle.
2. Calculate the *equivalent annual cost* (EAC) of each cycle:

$$\text{EAC} = \text{NPV} / \text{Annuity factor}$$

3. Choose the cycle with the lowest EAC.

Definition

Equivalent annual cost – the annual cost of owning, operating and maintaining an asset over its entire life.

NPVs cannot be compared directly for projects or investments with different durations (e.g. in *Example 1*, the NPV of replacement after two years –\$3,418 cannot be compared directly with the NPV of replacement after three years –\$4,813). The calculation of EAC takes account of the different lifespans, allowing comparisons to be made.

The asset replacement decision is just one example of the application of the EAC method. Other applications include deciding when to replace an existing asset with a new asset and choosing between alternative assets (e.g. equipment with different capital outlays, annual maintenance expenses and expected useful lives).

Example 1 Equivalent Annual Cost

A machine has a cost of \$3,500. The annual maintenance costs of the machine are forecast to be \$900 in the first year, \$1,000 in the second year and \$1,200 in the third year of ownership.

The residual value of the machine is expected to be \$2,100 after two years and \$1,600 after three years.

The cost of capital is 11% per year.

Calculate the optimal replacement cycle for the machine.

Solution

Step 1 – Calculate the NPV of cost for each potential replacement cycle

As there is no amount for residual value after one year, it is not possible to calculate NPV for a one-year replacement cycle. The choice, therefore, is between a two- or three-year replacement cycle. The usual assumptions are made about the timing of cash flows (e.g. maintenance cost cash flows paid at the end of each year).

Two-year replacement cycle

	T_0	T_1	T_2
Initial cost	(3,500)		
Maintenance		(900)	(1,000)
Residual value			<u>2,100</u>
Net cash flows	(3,500)	(900)	1,100
11% Discount factors	1	0.901	0.812
Present values	(3,500)	(811)	893
Net present value	(3,418)		

Three-year replacement cycle

	T_0	T_1	T_2	T_3
Initial cost	(3,500)			
Maintenance		(900)	(1,000)	(1,200)
Residual value				<u>1,600</u>
Net cash flows	(3,500)	(900)	(1,000)	400
11% Discount factors	1	0.901	0.812	0.731
Present values	(3,500)	(811)	(812)	292
Net present value	(4,831)			

Exam advice

Although there is an element of repetition in these calculations, you are strongly advised to use a similar simple and logical format. Formats which try to combine the calculations tend to lead to mistakes being made.

Step 2 – Calculate the EAC for each potential replacement cycle

Replacement cycle	Two years	Three years
NPV	(\$3,418)	(\$4,831)
Annuity factor at 11%	1.713	2.444
EAC	(\$1,995)	(\$1,977)

Conclusion: The three-year replacement cycle is the optimal replacement cycle.

Activity 4 Machine Replacement

[Show](#)

The following information is available for a machine which costs \$20,000:

	Running costs	Scrap proceeds
Year 1	\$5,000	\$16,000
Year 2	\$5,500	\$13,000

The company's cost of capital is 10%.

Required:

Determine whether the machine should be replaced each year or every two years. Use a discount rate of 10%.

**Please use the notes feature in the toolbar to help formulate your answer.*

Exam advice

If an exam question states that maintenance is an annual overhaul required at the end of each year (rather than ongoing maintenance throughout the year), the logical assumption is that there will be no maintenance in the year of disposal.

Replace Every Year

Time	Narrative	Cash flow	Discount factor	Present value
0	Purchase	(20,000)	1	(20,000)
1	Running costs	(5,000)	0.909	(4,545)
1	Scrap proceeds	16,000	0.909	<u>14,544</u>
			NPV=	<u>(10,001)</u>

$$\text{Annual equivalent cost} = \frac{\text{NPV}}{\text{1-year annuity factor}} = \frac{10,001}{0.909} = (\$11,002)$$

Replace Every Two Years

Time	Narrative	Cash flow	Discount factor	Present value
0	Purchase	(20,000)	1	(20,000)
1	Running costs	(5,000)	0.909	(4,545)
2	Running costs	(5,500)	0.826	(4,543)
2	Scrap proceeds	13,000	0.826	<u>10,738</u>
			NPV=	<u>(18,350)</u>

$$\text{Annual equivalent cost} = \frac{\text{NPV}}{\text{2-year annuity factor}} = \frac{18,350}{1.736} = (\$10,570)$$

Decision: Replace every two years.

Exam advice

If an exam question states that maintenance is an annual overhaul required at the end of each year (rather than ongoing maintenance throughout the year), the logical assumption is that there will be no maintenance in the year of disposal.

2.2 Limitations of Replacement Analysis

- ✗ Like-with-like replacement (in perpetuity) is rarely possible, even if desirable. Asset requirements may change over time.
- ✗ Changing technology may also require earlier replacement (e.g. IT equipment) than the optimal solution suggests.
- ✗ Non-financial factors (e.g. employees may be more satisfied if their company cars are replaced more often.)

2.3 Equivalent Annual Benefit

If faced with mutually exclusive projects where only one can be accepted, the general rule is to choose the project that generates the highest NPV (to maximise shareholder wealth). However, if the same situation is faced repeatedly (in perpetuity) and the projects have different lives, the equivalent annual benefit (EAB) approach can be used.

Example 2 Equivalent Annual Benefit

Two mutually exclusive projects are being considered:

Project A has an NPV of \$47m and is expected to last three years.

Project B has an NPV of \$58m and is expected to last four years.

If either project is chosen it will be possible to repeat it for the foreseeable future.

The cost of capital is 13%.

Calculate which project the company should accept.

Solution

Project	A	B
NPV	\$47m	\$58m
Annuity factor at 13%	2.361	2.974
EAB	\$19.9m	\$19.5m

Decision: Although Project B has the higher NPV, Project A should be chosen if it can be repeated for the foreseeable future. This result arises because although the shorter project has a lower NPV, that NPV will be obtained more frequently.

A good practical example of this is where there is a choice between alternative technologies with different expected lives. The organisation should choose the technology with the higher equivalent annual benefit.

Activity 5 Equivalent Annual Benefit

[Show](#)
[Show](#)

Activity 5 Equivalent Annual Benefit

Parrott plans to install a new IT system, which it expects to use indefinitely. Two alternative technologies are available:

Option 1

Initial cost \$450,000 with a four-year economic life at the end of which there would be no residual value. Annual net operating cash inflows would be as follows (in \$000s):

Year 1	Year 2	Year 3	Year 4
200	170	130	100

Option 2

Initial cost \$650,000 with a five-year economic life at the end of which there would be no residual value. Annual net operating cash inflows would be as follows (in \$000s):

Year 1	Year 2	Year 3	Year 4	Year 5
220	210	180	160	140

Parrott's cost of capital is 10% and taxation may be ignored.

Required:

Using the annual equivalent benefit method, determine whether Parrott Co should choose Option 1 or Option 2.

**Please use the notes feature in the toolbar to help formulate your answer.*

Option 1

	Year 0	Year 1	Year 2	Year 3	Year 4
Cash flow (\$000)	(450)	200	170	130	100
Discount factor	1	0.909	0.826	0.751	0.683
Present value (\$000)	(450)	182	140	98	68
Net present value = \$38,000					

Annual equivalent benefit = NPV/annuity factor = $\$38,000/3.170 = \$11,987$

Option 2

	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Cash flow (\$000)	(650)	220	210	180	160	140
Discount factor	1	0.909	0.826	0.751	0.683	0.621
Present value (\$000)	(650)	200	173	135	109	87
Net present value = \$54,000						

Annual equivalent benefit = NPV/annuity factor = $\$54,000/3.791 = \$14,244$

Conclusion: Parrott should choose Option 2 as it is the technology with the highest annual equivalent benefit.

3.1 The Issue

Use of an asset may be obtained through:

- an outright purchase (e.g. by borrowing to buy); or
- a lease agreement.

The important issue for financial management is the *cash flows* created by a lease, compared to an outright asset purchase.

3.2 Decision Making

The "lease or buy" decision is a financing decision that is relevant to the investment decision. It assumes that the preferred financing decision is the one with the lower present value (PV) of cost and that the purchased asset is financed with a bank loan. Therefore, cash flows are discounted at an after-tax cost of borrowing (i.e. pre-tax cost of borrowing $\times (1 - \text{tax rate})$).

TWO DECISIONS?

1. Investment Decision

Does the asset give operational benefits?

Focus on the NPV of the operating cash flows.

Discount cash flows using a rate which reflects *operating risk* of investment (e.g. average cost of capital)

2. Financing Decision

Is it cheaper to buy or lease?

Focus on the relative benefits of tax-allowable depreciation from buying and the tax relief on the lease payments.

Discount these cash flows using *after-tax cost of borrowing*.

3.2.1 Investment Decision

For the investment decision (which is made first), discount the cash flows from using the asset (sales, materials, labour, overheads, tax on net cash flows, etc) at the company's weighted average cost of capital (WACC).

3.2.2 Financing Decision

For the financial decision, which is made after the investment decision, discount the cash flows specific to each financing option at the **after-tax cost of debt**. Assuming that shareholders view borrowing and leasing as equivalent in terms of financial risk, the after-tax cost of debt is an appropriate discount rate for both options.

The preferred financing option is that with the lowest PV of cost.

The typical relevant cash flows to consider include:

- Buy asset
 - Purchase cost
 - Tax benefit of TAD
 - Scrap proceeds.
- Lease asset
 - Lease payments (assume all to be tax-allowable deductions)
 - Tax benefit of lease payments.

Other costs may be relevant depending on the terms of the lease agreement. For example, operational costs associated with the asset (e.g. maintenance or insurance) may differ depending on whether the lessee or lessor incurs these costs directly. Such costs will be relevant to the financing decision if they are not the same as for the purchase option.

As before, interest payments are **not** included as a relevant cash flow because they are already accounted for in discounting at the cost of capital.

3.2.3 Decision Criterion

Key Point

If the PV of the cost of the best finance source is less than the PV of the operating cash flows, the project should be undertaken.

Activity 6 Lease or Buy

[Show](#)

A new project is being considered:

The asset costs \$200,000 on the first day of a new accounting period.

The scrap value is \$25,000 on the last day of the next accounting period.

Operating inflows are \$150,000 for two years.

The tax rate is 33% and taxes are paid one year in arrears.

The company's WACC is 10%.

Tax-allowable depreciation is at 25% on a reducing balance basis with a balancing adjustment in the year of disposal.

Finance options:

(1) using a bank loan at a 10.5% interest rate; or

(2) leasing for \$92,500 a year in advance for two years (lease payments are tax allowable).

Required:

- Determine the operational benefit of the project.
- Determine how the project should be financed. Hint: Prepare a separate TAD working.
- Decide whether the project is worthwhile.

**Please use the notes feature in the toolbar to help formulate your answer.*

(a) Operational Benefit

Time	Cash flow \$	Narrative	DF @ 10%	PV \$
1-2	150,000	Project returns	1.736	260,400
2-3	(49,500)	Taxes on above	1.578	<u>(78,111)</u>
	Present value			<u>182,289</u>

(b) Financing Decision

(1) Borrow and Buy Asset Cash Flows

Post-tax cost of debt = pre-tax cost of debt $\times$ (1 – tax rate) = 10.5% $\times$ (1 – 0.33) $\approx$ 7%

Time	Cash flow \$	Narrative	DF @ 7%	PV \$
0	(200,000)	Purchase cost	1	(200,000)
2	25,000	Sale proceeds	0.873	21,825
2	16,500	TAD (W)	0.873	14,405
3	41,250	TAD (W)	0.816	<u>33,660</u>
				<u>(130,110)</u>

WORKING

<i>Time</i>			Tax effect at 33%	Time
		\$	\$	
0	Purchase	200,000	16,500	2
1	TAD at 25%	<u>(50,000)</u>		
	Tax written-down value	150,000		
	Sale proceeds	<u>25,000</u>		
2	Tax allowable loss	<u>125,000</u>	41,250	3

(2) Leasing Flows

Time	Cash flow	Narrative	DF @ 7%	PV
	\$			\$
0–1	(92,500)	Lease payments	1.935	(178,988)
2–3	30,525	Tax relief thereon*	1.689	<u>51,557</u>
				<u>(127,431)</u>

* This is the cash effect of the tax relief ($\$92,500 \times 33\%$) one year in arrears for two years.

Conclusion: The project should be financed using a lease.

(c) Final Decision

	\$
PV of operating flows (a)	182,289
PV of lowest cost finance ((b) is leasing)	<u>(127,431)</u>
NPV	<u>54,858</u>

Decision: The asset should be acquired using a lease.

3.3 Evaluation Using Pre-tax Cost of Debt

If a business is not in a tax-paying position, there are implications for a lease-versus-buy decision process:

- There would be no tax benefit from TAD if the asset was bought.
- No tax benefit arises from the lease payments.
- There would be no tax shield on debt (i.e. interest expense on borrowings would not reduce tax expense).

The last point means that the discount rate used to evaluate the financing options should be the **pre-tax** cost of debt (i.e. the gross interest rate quoted on a bank loan or gross redemption yield if borrowing would be in the form of bonds).

Situations where a business is not in a tax-paying position include:

- losses in current year;
- losses are brought forward from prior years;
- incorporation in a tax-free special economic zone;
- tax holidays granted by the host government; and
- tax-exempt charitable status.

Activity 7 Evaluation Using Pre-tax Cost of Debt

[Show](#)

A machine costs \$500,000 and has a \$150,000 residual value at the end of three years. Annual repairs and maintenance costs are \$20,000 in the first two years and zero in the third year.

Finance options:

- (1) 8% bank loan; or
- (2) lease at \$160,000 a year for three years, payable in arrears.

The machine is required for a three-year project in a tax-free high-tech park. The project has a strong positive NPV.

Required:

Determine whether the machine should be bought or leased.

**Please use the notes feature in the toolbar to help formulate your answer.*

(1) Borrow and Buy Asset Cash Flows

Time	Cash flow	Narrative	DF @ 8%	PV \$
0	(500,000)	Purchase cost	1	(500,000)
1	(20,000)	Maintenance	0.926	(18,520)
2	(20,000)	Maintenance	0.857	(17,140)
3	150,000	Disposal	0.794	<u>119,100</u>
				<u>(416,560)</u>

(2) Leasing Flows

Time	Cash flow	Narrative	DF @ 8%	PV \$
1-3	(160,000)	Lease payments	2.577	<u>(412,320)</u>

Conclusion: The cheapest method of finance is to lease.

3.4 Impact on Financial Statements

The financial accounting implications of borrowing or leasing an asset may be a factor in management's final decision.

This is certainly relevant for managers of listed companies (i.e. whose shares are traded on the stock market) as key ratios may be influenced – particularly financial risk indicators such as the company's debt-to-equity ratio and interest cover.

The implications of each financing option can be summarised as follows:

3.4.1 Borrow to buy

- The bank loan (or loan notes) will be recorded in non-current liabilities, increasing the reported level of financial gearing (debt to equity ratio).
- Interest on the debt will reduce interest cover (i.e. earnings before interest and tax (EBIT) divided by interest expense).
- However, the overall effect also depends on the profits generated by the asset as these will increase EBIT, and any retained profit would increase the level of equity.

3.4.2 Lease

- Both the asset and a related liability would be recognised in the statement of financial position (as for the borrow-to-buy option). Therefore, reported financial gearing would rise initially.
- However, as lease payments would be allocated to interest expense and principal repayments, the liability would amortise (i.e. reduce) over time and ultimately fall to zero. Interest expense in early years would be relatively high, tending to reduce interest cover, but lower in later years.

The accounting treatment above applies to any lease longer than 12 months. For leases less than 12 months, payments are simply a rental expense, which does not affect the statement of financial position.

Exam advice

Detailed knowledge of IFRS 16 *Leases* is not required for this exam.

3.5 Non-financial Factors

For any lease-or-buy decision, several non-financial or qualitative factors will be considered before making a final decision.

To buy an asset may be the preferred option if, for example:

- The asset is of fundamental importance to the business and is in constant use.
- Management wants to have absolute control over the asset.
- To update or upgrade the asset to scale up capacity is relatively easy.
- The asset is easily maintained (e.g. under a supplier's warranty).
- There is an active second-hand market for the asset (so it is not only financially more affordable, but can be disposed of easily).

To lease an asset may be the preferred option if, for example:

- The asset may only be needed for a short time or there is significant uncertainty about how long it may be used.
- The asset requires specialist support or maintenance that is provided by the lessor.
- The asset needs to be regularly updated or upgraded for any reason (e.g. due to technical obsolescence or rapid expansion).
- The lease agreement includes options to upgrade to newer models or other flexibility.
- End-of-life disposal of the asset is subject to strict regulations.

Syllabus Coverage

This chapter covers the following Learning Outcomes.

D. Investment Appraisal

4. Specific investment decisions (Lease or buy, asset replacement, capital rationing)

- a) Evaluate leasing and borrowing to buy using the before- and after-tax costs of debt.
- b) Evaluate asset replacement decisions using equivalent annual cost and equivalent annual benefit.
- c) Evaluate investment decisions under single-period capital rationing, including:
 - i) the calculation of profitability indexes for divisible investment projects
 - ii) the calculation of the NPV of combinations of non-divisible investment projects
 - iii) a discussion of the reasons for capital rationing.

E. Business Finance

3. Sources of finance and their relative costs

- d) Assess the impact of sources of finance on financial position, financial risk and shareholder wealth using appropriate measures, including:
 - iii) leasing or borrowing to buy.

Summary and Quiz

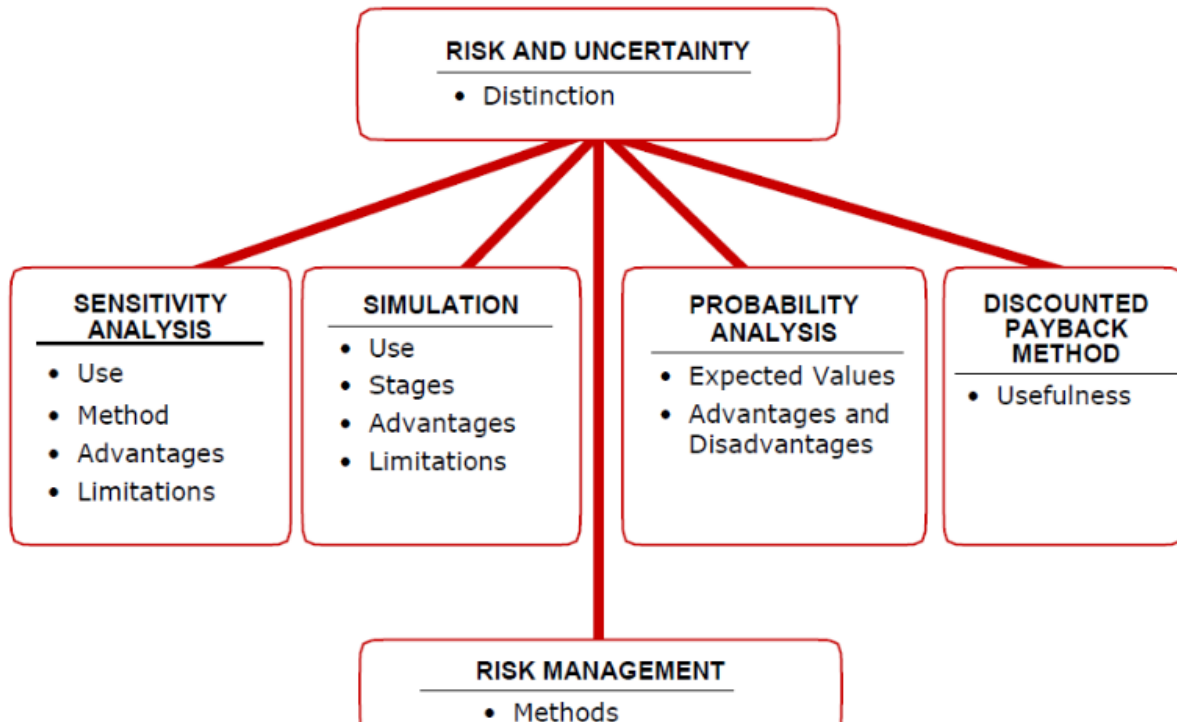
- Capital rationing occurs when insufficient finance (capital) is available to undertake all positive NPV projects available.
 - With capital rationing, it is essential to identify the nature of the projects (i.e. divisible or non-divisible, mutually exclusive or not).
 - A divisible project is one in which the company can undertake between 0% and 100% of the project. A non-divisible/indivisible project must be done 100% or not at all.
 - The profitability index for a project is its NPV divided by its initial investment.
 - Profitability indexes can be used to rank divisible projects.
 - For asset replacement decisions, use the equivalent cost (EAC) method to compare cycles of different lengths.
 - The EAC is calculated as $\text{NPV}/\text{Annuity factor}$.
 - The lowest EAC is chosen in making the replacement decision.
 - For lease or buy decisions, separate the financing decision from the investment decision and analyse each at a discount rate reflecting the risk of the cash flows.
 - In assessing a lease cost, assume that all lease payments are tax-deductible expenses.
 - If the PV of the cost of the best finance source is less than the PV of the operating cash flows, the project should be undertaken.
 - If a business pays no taxes, the discount rate used to evaluate the financing options should be the pre-tax cost of debt.
-

Chapter 7: Project Appraisal Under Risk

Overview	CHAPTER 7: Visual Overview
7.1 Risk and Uncertainty	7.1.1 Distinction
7.2 Sensitivity Analysis	7.2.1 Use
	7.2.2 Method
	7.2.3 Advantages of Sensitivity Analysis
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7.5 Discounted Payback Method	7.5.1 Usefulness
7.6 Risk Management	7.6.1 Methods
7 Conclusion	7 Syllabus Coverage
	7 Summary and Quiz

CHAPTER 7: Visual Overview

Objective: To appraise investment projects where there is more than one possible outcome.



1.1 Distinction

Definitions

Risk - a condition in which several possible outcomes exist, the probabilities of which can be quantified from historical data.

Uncertainty - the inability to predict possible outcomes due to a lack of historical data (i.e. information) being available for quantification.

Key Points

Although “risk” and “uncertainty” are often used interchangeably, only **risk** is measurable.

The main risk to the success of an investment project will be the variability of future cash flows. This could be the variability of income streams, the variability of cost cash flows, or a combination of both.

In general, risky projects are those whose predicted future cash flows (which affect project returns) are more variable. The greater the variability, the greater the risk. The risk associated with capital investments is increased because expenditure and benefit estimates might be for several years; costs might escalate and long-term benefits can only be estimated.

Note that uncertainty increases as a project's life increases.

2.1 Use

Definition

Sensitivity analysis – the analysis of changes made to significant variables to determine their effect on a planned course of action.

In project appraisal, sensitivity (“what if?”) analysis is used to analyse the effect of changes in the value of an input variable (occasionally more than one input variable), assuming other inputs are kept constant.

The cash flows, probabilities or cost of capital are varied until the decision changes (i.e. NPV becomes zero). This will show the sensitivity of the decision to changes in those variables.

Therefore, the estimation of IRR is a measure of the sensitivity of the project's NPV to changes in the cost of capital.

2.2 Method

Sensitivity analysis has TWO steps:

- Step 1** Calculate the NPV of the project based on best estimates.
- Step 2** For each element of the decision (cash flows, cost of capital), calculate the change necessary for the NPV to fall to zero.

The sensitivity can be expressed as a percentage change for an individual cash flow:

$$\text{Sensitivity} = \frac{\text{NPV}}{\text{PV of relevant cash flow}} \times 100$$

The lower the percentage, the more sensitive the NPV is to that project variable, as the variable would need to change by a smaller amount to make the project non-viable.

Key Point

For a change in sales volume, the relevant cash flow is contribution. This may involve combining several cash flows (i.e. revenue and variable costs).

Activity 1 Sensitivity Analysis

[Show](#)

Williams has just set up a company, JPR Manufacturing Company, and estimates its cost of capital to be 15%. His first project is to invest \$150,000 in equipment, which has a useful life of 15 years and a final scrap value of \$15,000.

The equipment will produce 15,000 deluxe pairs of rugby boots annually, generating a contribution of \$2.75 each pair. He estimates annual fixed costs at \$15,000.

Required:

- a) Determine, whether the project is worthwhile.
- b) Calculate what percentage changes in the following factors would cause your decision in (a) to change:
 - i. Initial investment;
 - ii. Sales volume;
 - iii. Fixed costs;
 - iv. Scrap value;
 - v. Cost of capital.

Comment on each result.

**Please use the notes feature in the toolbar to help formulate your answer*

a)	Time	Cash Flow	DF @ 15% PV	
		\$		\$
0	Initial cost	(150,000)	1	(150,000)
1 – 15	Contribution			
	(\$2.75 × 15,000)	41,250	5.847	241,189
1 – 15	Fixed costs	(15,000)	5.847	(87,705)
15	Scrap value	15,000	0.123	<u>1,845</u>
				<u>5,329</u>

Decision: The project is worthwhile as the NPV is positive.

(b) Sensitivity of the decision in (a)

(i) Initial investment

$$\text{Sensitivity} = \frac{\$5,329}{\$150,000} \times 100 = 3.6\%$$

Comment: If the initial investment increases by more than \$5,329 (3.6%), the project will be rejected.

(ii) Sales volume

$$\text{Sensitivity} = \frac{\$5,329}{\$241,189} \times 100 = 2.2\%$$

Comment: The \$241,189 contribution is directly proportional to volume. If this PV is reduced by more than \$5,329 (2.2%), the project will be rejected.

(iii) Fixed costs

$$\text{Sensitivity} = \frac{\$5,329}{\$87,705} \times 100 = 6\%$$

Comment: If this PV rises by more than \$5,329 or (6%), the project will be rejected. It can be seen, for example, that the project breaks even if fixed costs become $\$15,000 \times 1.06 = \$15,900$.

(iv) Scrap value

$$\text{Sensitivity} = \frac{\$5,329}{\$1,845} \times 100 = 289\%$$

Comment: The scrap value is relatively irrelevant to the investment decision – the company would have to pay to have the plant removed for the project to be rejected.

From the above calculations, the decision to accept the project is extremely sensitive to most of the variables. The project will be rejected in the event of small rises in the initial investment or fixed cost figures or falls in contribution or volume.

(v) Cost of Capital

This can be found by calculating the project's IRR, which is probably only marginally above 15% because the NPV was small at a 15% cost of capital. Therefore, a 16% cost of capital can be utilised, and then the IRR can be estimated.

Year	Cash flow	DF @ 16%	Present value
	\$		\$
0	(150,000)	1	(150,000)
1 – 15*	26,250	5.575	146,344
15	15,000	0.108	<u>1,620</u>
NPV at 16%			<u>(2,036)</u>

$$\begin{aligned}
 \text{IRR} &= A + \frac{\text{NPV}_A}{\text{NPV}_A - \text{NPV}_B} (B - A) \\
 &= 15\% + \frac{5,329}{5,329 + 2,036} (16\% - 15\%) \\
 &= 15.7\%
 \end{aligned}$$

* Contribution less fixed costs

Comment: If the cost of capital rises from 15% to more than 15.7%, the project will be rejected.

2.3 Advantages of Sensitivity Analysis

- ✓ It gives an idea of how sensitive the project is to changes in each of the original estimates.
- ✓ It prompts management to check the quality of data for the most sensitive variables.
- ✓ It identifies the *critical success factors* for the project and directs project management.
- ✓ It is easily adapted for use in spreadsheet packages.

2.4 Limitations of Sensitivity Analysis

- ✗ Although it can be adapted to deal with multi-variable changes, sensitivity analysis is typically used to examine what happens when one variable changes and others remain constant. However, variables are often interdependent.
- ✗ It assumes data for all other variables is accurate.
- ✗ Without a computer, it can be time-consuming.
- ✗ It does not consider the probability of change in a variable.
- ✗ It does not provide a decision rule. Management must decide the level of sensitivity that is acceptable.

3.1 Use of Simulation

Definition

Simulation – a technique that allows more than one variable to change simultaneously.

An example of simulation is the Monte Carlo simulation method. The output estimates not only a project's outcomes but also their associated probabilities.

3.2 Stages in a Monte Carlo Simulation

1. Specify the major variables (e.g. revenue and costs).
2. Specify the relationship between the variables (e.g. revenue minus costs equals profit).
3. Attach probability distributions to each variable and assign random numbers to reflect the distribution (e.g. a 10% probability that costs are \$1,000, a 50% probability that they are \$10,000 and a 40% probability that they are \$12,000).
4. Simulate the environment by generating random numbers for revenue and costs. For example, using random numbers from 00 to 99:
 - assign 00 to 09 to \$1,000 cost with 10% probability;
 - assign 10 to 59 to \$10,000 with 50% probability assigned to it, and so on.
5. Record the outcome of each simulation (e.g. the profit based on the revenue and costs randomly selected).
6. Repeat the simulation many times to obtain a probability distribution of the possible outcomes (e.g. the probabilities of achieving various profit levels).

Exam advice

Calculations will **not** be required in the exam; an awareness of the stages will be sufficient.

3.3 Advantages of Simulation

- ✓ Monte Carlo simulation provides more information about the possible outcomes and their relative probabilities.
- ✓ This data can be used to calculate an expected NPV (and the standard deviation of the expected NPV).

Exam advice

The calculation of standard deviation (a measure of the spread of values around a mean value) is **not** examinable.

3.4 Limitations

- ✗ Monte Carlo simulation is not a technique for making a decision, only for obtaining more information about the possible outcomes.
- ✗ It can be very time-consuming without a computer.
- ✗ It could prove expensive in designing and running the simulation, even on a computer.
- ✗ Simulations are only as good as the probabilities, assumptions and estimates used.

4.0 Introduction

A probability analysis of expected cash flows can be used to measure risk.

4.1 Expected Values

i Definition

Expected value – the quantitative result of weighting uncertain events by the probability of their occurrence.

An **expected value** is calculated as follows:

Expected value	=	weighted arithmetic mean of possible outcomes= $\sum xp(x)$
Where x	=	value of an outcome
$P(x)$	=	the probability of outcome x
$\sum$	=	sum

Activity 2 Expected Values

[Show](#)

The values in the table below show the NPV of three projects given three different states of the market.

State of market	Contracting	Static	Expanding
Probability	0.4	0.3	0.3
Project 1	\$100	\$200	\$1,000
Project 2	\$0	\$500	\$600
Project 3	\$180	\$190	\$200

Required:

Determine which is the best project, based on expected values.

**Please use the notes feature in the toolbar to help formulate your answer*

Project 1	Expected value	=	$100 \times 0.4 + 200 \times 0.3 + 1,000 \times 0.3 =$	400
Project 2	Expected value	=	$0 \times 0.4 + 500 \times 0.3 + 600 \times 0.3 =$	330
Project 3	Expected value	=	$180 \times 0.4 + 190 \times 0.3 + 200 \times 0.3 =$	189

The best project based on expected values is Project 1.

Expected values can be discounted to calculate an estimated value of an NPV and to measure risk.

Example 1 Probability Analysis

An organisation is considering making a \$40,000 investment in a project which is estimated will generate cash inflows over the project's two-year life as follows:

Year 1	Cash flow	Probability
	\$20,000	0.4
	\$30,000	0.6

Year 2	If cash flow in Year 1 is:	there is a probability of:	that cash flow in Year 2 will be:
	\$20,000	0.25	\$10,000
		0.75	\$20,000
	\$30,000	0.30	\$20,000
		0.70	\$25,000

The organisation's cost of capital is 10%.

Required:

Calculate the EV of the project's NPV and the probability that the NPV will be negative.

Solution

Calculate the PV of the cash inflows:

Year	Cash flow	Discount factor	PV
	\$000		\$000
1	20	0.909	18,180
1	30	0.909	27,270
2	10	0.826	8,260
2	20	0.826	16,520
2	25	0.826	20,650

Calculate the EV of the PVs of the cash inflows.

Year 1 PV \$000	Probability	Year 2 PV \$000	Probability	Joint probability	Total PV \$000	EV of PV \$000
(a)	(b)	(c)	(d)	(b) × (d) = (e)	(a) + (c) = (f)	(e) × (f)
18,180	0.4	8,260	0.25	0.10	26,440	2,644
18,180	0.4	16,520	0.75	0.30	34,700	10,410
27,270	0.6	16,520	0.30	0.18	43,790	7,882
27,270	0.6	20,650	0.70	0.42	47,920	20,126
				1.00		41,062
Less project cost						40,000
EV of the project's NPV						1,062

The probability that the NPV will be negative (a measure of the risk) is the probability that the total PV of cash inflows is less than \$40,000. From the Total PV column, this corresponds to the joint probabilities 0.1 + 0.3 = 0.4. This might be considered an unacceptably high risk.

4.2 Advantages and Disadvantages

Advantages	Disadvantages
✓ Expected value reduces the information to one value for each choice. ✓ The idea of an average is readily understood	✗ The probabilities of possible outcomes may be difficult to estimate. ✗ The average may not correspond to any of the possible outcomes. ✗ Unless the same decision has to be made many times, the average will not be achieved; therefore, it is not appropriate in "one-off" situations. ✗ The average does not indicate the spread of possible results (i.e. it <i>ignores risk</i>).

5.1 Usefulness

i Definition

Discounted payback period – the time it takes for the *discounted* returns (i.e. present values of the operating cash flows) from a project to recover the initial investment. It is also called the *adjusted* payback period.

Discounted payback resolves one of the limitations of the standard payback method: it allows for the time value of money. (The usefulness of unadjusted payback as a measure of project risk is strictly limited as it gives equal weighting to cash flows irrespective of when they are received.) Discounted payback also considers more of the project's cash flows because the amounts are discounted; therefore, the payback period is longer.

However, it does not overcome all the limitations of the payback method as it still ignores cash flows after the payback period.

Example 2 Limitation of Payback

Two alternative projects each require an initial investment of \$1,000,000 and have the following forecast operating cash flows (\$000):

	Project A	Project B
Year 1	600	100
Year 2	200	300
Year 3	200	600
Year 4	205	205
Year 5	150	150

Each project has a payback period of three years and, on this measure, would be ranked equally in terms of liquidity and risk.

However, Project A produces strong cash flows in the early years, which may be less uncertain than later returns.

This payback limitation can be dealt with using the discounted payback period (i.e. first discount the project returns at a rate reflecting the level of operating risk, and recalculate the payback period).

Activity 3 Discounted Payback Period[Hide](#)

The company's WACC of 10% is believed to reflect the risk attached to both Project A and Project B in Example 2.

Required:

Calculate the discounted payback period of each project.

**Please use the notes feature in the toolbar to help formulate your answer*

	Project A	10% DF	PV	Cumulative PV
Year 0	(1,000)	1	(1,000)	(1,000)
Year 1	600	0.909	545	(455)
Year 2	200	0.826	165	(290)
Year 3	200	0.751	150	(140)
Year 4	205	0.683	140	0
Year 5	150	0.621	93	93

Discounted payback = 4 years

Discounted payback = 4 years

	Project B	10% DF	PV	Cumulative PV
Year 0	(1,000)	1	(1,000)	(1,000)
Year 1	100	0.909	91	(909)
Year 2	300	0.826	248	(661)
Year 3	600	0.751	451	(210)
Year 4	205	0.683	140	(70)
Year 5	150	0.621	93	23

Discounted payback = $4 + 70/93 = 4.75$ years

6.1 Methods

Methods of keeping project risk within acceptable levels include:

- Setting a maximum adjusted payback period in the initial screening process of potential projects.
- Using risk-adjusted discount rates for both NPV and adjusted payback. A higher discount rate should be applied to higher-risk projects, reducing the influence of more distant cash flows. In addition, investors prefer cash now to later and require a higher return for long-term investments. Project-specific discount rates can be found using the capital asset pricing model (see *Chapter 12*).
- Using conservative forecasts, such as reducing the forecast returns downwards to reflect the guaranteed minimum inflows from a project (certainty equivalents). Then discount these lower cash flows at the risk-free interest rate (i.e. risk is removed from the cash flows rather than adjusted for in the discount rate).
- Selecting projects with a combination of an acceptable expected NPV and a relatively low standard deviation of NPV.
- Focusing attention on the critical success factors indicated by sensitivity analysis.

Exam advice

The calculation of reduced cash flows to reflect guaranteed minimum inflows is not examinable.

Syllabus Coverage

This chapter covers the following Learning Outcomes.

D. Investment Appraisal

1. Investment appraisal techniques

- c) Calculate discounted payback period and discuss its usefulness as an investment appraisal method.

3. Adjusting for risk and uncertainty in investment appraisal

- a) Describe and discuss the difference between risk and uncertainty in relation to probabilities and increasing project life.
- b) Apply sensitivity analysis to investment projects and discuss the usefulness of sensitivity analysis in assisting investment decisions.
- c) Apply probability analysis to investment projects and discuss the usefulness of probability analysis in assisting investment decisions.
- d) Apply and discuss other techniques of adjusting risk and uncertainty in investment appraisal, including:
 - i) simulation
 - ii) adjusted payback
 - iii) risk-adjusted discount rates.

Summary and Quiz

- Risk is a condition in which several possible outcomes exist, the probabilities of which can be quantified from historical data.
- Uncertainty is the inability to predict possible outcomes due to a lack of historical data for quantification.
- The major risks to the success of an investment project will be the variability of the future cash flows.
- Exam calculations on project risk are likely to focus on sensitivity analysis (i.e. finding the value of key variables at which NPV = 0).
- The sensitivity for individual cash flows can be expressed as a percentage using the following formula:

$$\text{Sensitivity} = \frac{\text{NPV}}{\text{PV of flow considered}} \times 100\%$$

- Monte Carlo simulation can provide more information about the possible outcomes and their relative probabilities; this data can be used to calculate a NPV (and its standard deviation).
- Monte Carlo simulations are only as good as the probabilities, assumptions and estimates used.
- An expected value is the quantitative result of weighting uncertain events by the probability of their occurrence.
- Discounted payback period ("adjusted payback period") is the time it takes for the discounted returns from a project to recover the initial investment.
- Adjusting the discount rate to reflect a project's risk is dealt with later in *Chapter 12* on the capital asset pricing model (CAPM).

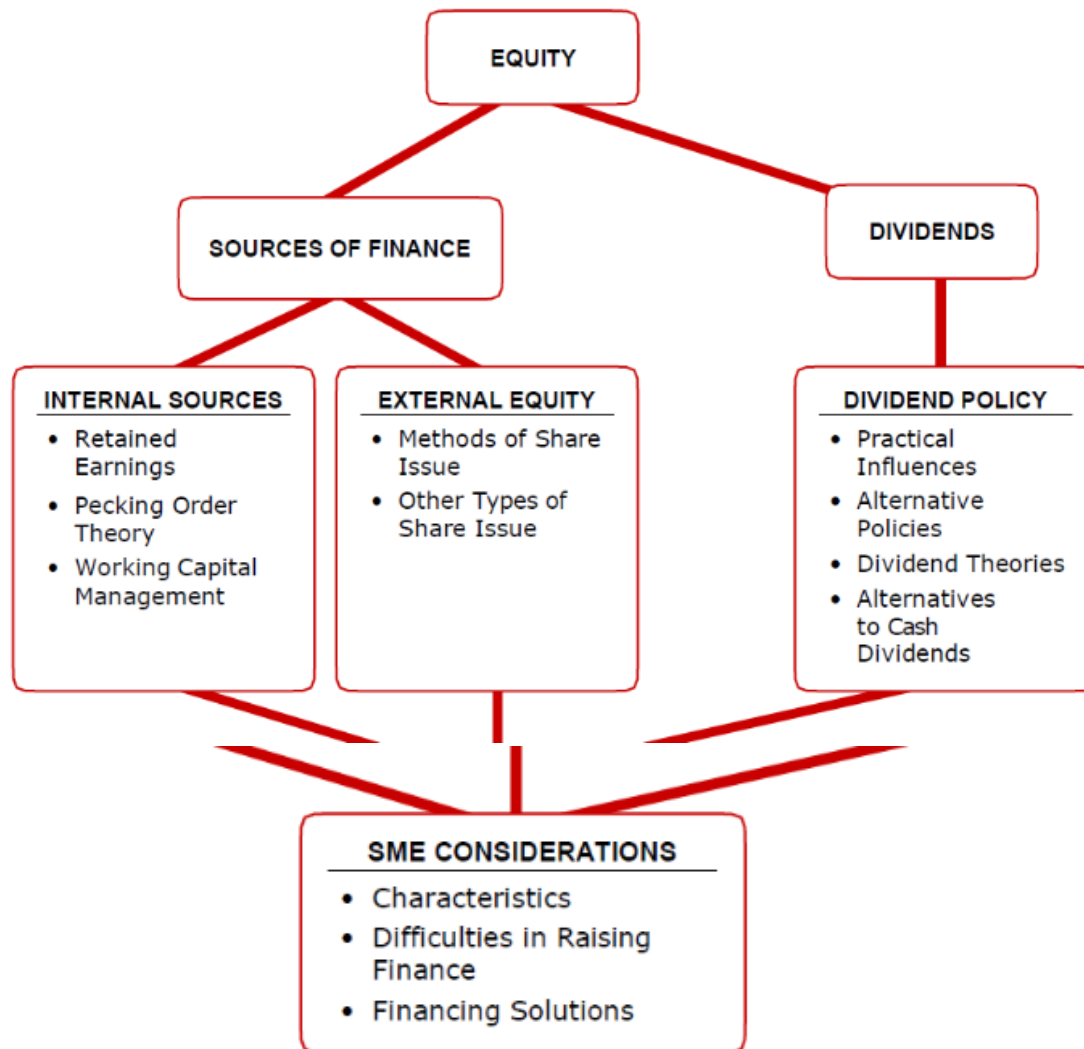
Chapter 7 Quiz

Chapter 8: Equity Finance

Overview	CHAPTER 8: Visual Overview
8.1 Internal Sources of Finance	8.1.1 Retained Earnings
	8.1.2 Pecking Order Theory
	8.1.3 Working Capital Management
8.2 External Equity	8.2.0 Introduction
	8.2.1 Methods of Share Issue
	8.2.2 Other Types of Share Issue
8.3 Dividend Policy	8.3.0 Introduction
	8.3.1 Practical Influences
	8.3.2 Alternative Dividend Policies
	8.3.3 Dividend Theories
	8.3.4 Alternatives to Cash Dividends
8.4 SME Considerations	8.4.1 Characteristics
	8.4.2 Difficulties in Raising Finance
	8.4.3 Financing Solutions
8 Conclusion	8 Syllabus Coverage
	8 Summary and Quiz
	8 Technical Articles

Visual Overview

Objective: To understand the issues relating to equity finance.



1.1 Retained Earnings

Key Point

The primary internal source of finance is **retained earnings** (accumulated profits).

Internal finance can also be generated by increasing working capital management efficiency.

However, a company may have substantial retained earnings in its statement of financial position but no cash in the bank; therefore, it will not be able to finance investment. Therefore, retained cash, rather than retained earnings, is the source of finance.

Microsoft, for example, did not pay any dividends for many years. It reinvested all cash to produce growth in the company and its share price. Any shareholder that required a dividend could sell some shares to take a capital gain and create a "home-made" dividend.

Retained cash as a source of finance has no issue costs, is flexible in terms of amounts used or repayment patterns and does not affect shareholdings. Shareholders may not appreciate the loss or reduction of dividends.

1.2 Pecking Order Theory

1.2.1 Issue Costs

According to the Pecking Order Theory, companies should follow an established pecking order to raise finance simply and efficiently. The first justification for this theory is that companies should want to minimise issue costs:

- The cheapest source of new finance is retained earnings, as it has no issue costs. This is a form of equity, since retained earnings would otherwise be distributable as a dividend to shareholders.
- A bond issue or bank loan is the next cheapest source of new finance. These are forms of debt.
- The most expensive source of new finance is a fresh equity issue. For example, an initial public offering (IPO) on a regulated exchange typically incurs issue costs of between 2% and 8% of the amount raised.

The second justification is that companies will want to minimise the time and expense involved in persuading outside investors of the merits of the project:

- If the company can use retained earnings, it does not have to spend time persuading outside investors.
- Otherwise, the time and expense associated with issuing debt is usually significantly less than that associated with a share issue.

1.2.2 Asymmetrical Information

A further justification is the existence of asymmetrical information and the presumed information transfer ("signal") that results from management actions. This recognises that management knows more about the companies' prospects than the outside investors/the markets. (Management knows all the detailed inside information, whilst the markets only have access to past and publicly available information.)

Because of this asymmetry in information, management's actions are closely scrutinised by the markets and often interpreted as the insiders' view of the company's prospects. For example, an unexpected dividend increase shows management's increased confidence in the company's prospects, which will typically increase the share price.

Management with favourable inside information would want to avoid issuing shares because they would be undervalued. Conversely, outside investors assume that managers with unfavourable inside information would like to issue shares as they would be overvalued. Therefore, an equity issue is interpreted as a "bad" sign; investors may start to sell their shares, causing the share price to fall. The issue of equity is a last resort.

1.2.3 Available Sources

The preference for internal finance is supported by research showing that directors often choose "the path of least resistance" when it comes to financing. Although the pecking order theory explains the real-world preference for internal finance, it does not imply that it is always optimal. Instead, the theory argues that companies will choose the cheapest **available** source:

- New start-ups have no retained earnings and cannot yet borrow. They are typically equity-financed by their owners or private equity.
- As they grow, companies will seek debt in preference to new equity as it is cheaper, quicker and easier to issue, up to the limit of their debt capacity.
- Well-established, stable companies can finance their expansion with retained earnings. They will become heavily equity financed (as retained earnings are a component of equity) without financial management.

1.3 Working Capital Management

Creating accounting profits does not guarantee the availability of internal equity finance, as the company must also convert profits into positive cash flows. Therefore, a company's ability to generate internal finance depends on generating operating cash flow exceeding interest and taxes.

Potential internal finance available = operating cash flow – interest – tax

As interest and tax are committed costs, the focus must be maximising operating cash flows.

Operating cash flows are calculated as follows:

Earnings before interest and tax, depreciation and amortisation (EBITDA)	x
Rise/fall in inventory	(x)/x
Rise/fall in receivables	(x)/x
Rise/fall in payables	<u>x/(x)</u>
Operating cash flow	<u>x</u>

Improved working capital management can help to release more internal equity finance. Potential areas for improvement (which are covered in *Chapters 13-16*) include:

- ✓ Reducing the time taken to receive payments from customers (e.g. offering discounts for quick payment or outsourcing debt collection to a factor).
- ✓ Reducing inventory (e.g. through improved supply chain management or even moving to Just-in-Time (JIT) production).
- ✓ Taking increased credit from suppliers. However, care must be taken not to lose settlement discounts or compromise relationships with key suppliers.

2.0 Introduction

As well as internal finance, a company may be able to issue new shares (or debt, which is covered in *Chapter 9*) to finance its investment projects (i.e. external finance).

2.1 Methods of Share Issue

2.1.1 Quoted Companies – New Shares

If a company is already listed, the following methods are available for issuing new shares:

Method	Explanation
Offer for subscription (public issue)	A <i>direct</i> sale to the general public. This is generally the most expensive method of issuing new shares.
Offer for sale	An <i>indirect</i> sale to the public is achieved by selling shares directly to an issuing house (merchant/investment bank), which then sells them to the public. (The issuing house guarantees to buy the shares.)
Placing	In a placing, the sponsor (typically an investment bank) places the shares with its clients (usually pension funds and insurance companies) rather than offer shares to the general public. This is generally the least expensive method of issuing new shares.
Rights issue	An offer to <i>existing</i> shareholders to buy new shares in proportion to their existing holdings.
Offer for sale or subscription by tender	Like an <i>auction</i> , with the public being invited to bid for shares. It is useful where setting a price for the shares is difficult.

2.1.2 Options for Unquoted Companies

Options for unquoted companies include:

- To become quoted (i.e. raise new equity finance at the same time as becoming listed). This is known as an initial public offering (IPO). The method could be an offer for subscription or sale, tender, or placing. It is an expensive process.
- To stay unquoted. Use a rights issue or private placing. However, sources of funds are limited to existing owners and new private investors.
- To engage in what is known as "introduction".
 - No shares (neither existing nor newly created) are made available to the market, so no new finance is raised.
 - Stock market grants a quotation, given that shares must already be widely held, so that a market can be seen to exist.

Useful to gain greater marketability of shares, a valuation for inheritance tax purposes or future easier access to additional capital.

Exam advice

The terms quoted, floated and listed all refer to the same thing (i.e. shares traded on a stock exchange).

2.1.3 IPO Factors to Consider

Ordinary shareholders take more risk than any other type of investor in a company because:

- ordinary dividends are *discretionary* (i.e. the company has no legal obligation to pay an ordinary dividend); and
- ordinary shareholders rank last in the event of bankruptcy/liquidation.

Shareholders require high returns to compensate for this risk therefore, issuing new shares is an expensive source of finance. However, sometimes, a new share issue is the only available source of finance.

There are several factors that the company should consider before doing an IPO:

- Legal restrictions.
- Costs including:
 - Underwriting costs (i.e. fees that must be paid to an investment bank to underwrite/guarantee the share issue) are typically 5% to 7% of the offering.
 - Stock market listing fee (initial charge) for the new shares.
 - Fees for the issuing house (investment bank), solicitors, auditors, PR.
 - Cost of printing and distributing the prospectus (the document in which the shares are offered for sale).
 - Advertising in national newspapers.
- Valuation: setting a price for the new shares to be issued.
- Stock exchange rules.
- Timing.

2.1.4 Official Listing Requirements

For a company to trade its shares on a stock exchange, it must be able to meet that exchange's listing requirements and pay both the exchange's entry and yearly listing fees.

Listing requirements vary by exchange but typically measure the size and market share of the security to be listed and the underlying financial viability of the issuing company. Exchanges establish these standards as a means of maintaining their reputation and visibility. For example:

Stock exchange	Trading record	Public float*	Minimum capitalisation
Frankfurt	2-3 years	No specific percentage	No fixed minimum
Hong Kong	3 years	25%	HK\$200m
London	3 years	25%	£700,000
New York	No minimum	\$40m	No fixed minimum
Tokyo	3 years	35%	No fixed minimum

*** The portion of outstanding shares held by public investors, as opposed to shareholders with controlling interests, directors or non-public entities.**

2.1.5 Alternative Markets

The costs of acquiring and maintaining an official listing mean it is not feasible for small or medium-sized entities (SMEs). In the UK, for example, these companies may find the London Stock Exchange's Alternative Investment Market (AIM) more attractive as it has fewer regulations. Investors recognise that investing in AIM companies carries additional risk due to the more limited regulation. Other examples of alternative markets that cater to smaller and growing companies include:

- Euronext Growth, available in many European countries;
- Hong Kong Growth Enterprise Market (GEM);
- Mothers Market (Market of the High-growth and Emerging Stocks) in Japan; and
- NASDAQ First North in several Nordic and Baltic countries.

2.2 Other Types of Share Issue

2.2.1 Rights Issue

In a rights issue, the existing shareholders are offered more shares (usually at a discount to the current market price) in proportion to their existing holding.

Company law in many jurisdictions (e.g. Australia, Canada, Singapore, South Africa, the US and UK) guarantees shareholders *pre-emptive rights* (i.e. the right to purchase new shares before they can be offered to other investors). This is to protect shareholders from dilution of their control.

i Definitions

Theoretical ex-rights price (TERP) – the expected share price after the rights issue.

Although the formula is not published in the exam, the TERP is simply the forecast total market value of the company's equity after the rights issue divided into the number of shares that will be in issue after the rights issue.

$$\text{TERP} = \frac{\text{Pre-existing value of equity} + \text{proceeds of rights issue} + \text{project NPV}}{\text{Number of shares after the rights issue}}$$

Points to note for the calculation of the TERP per share:

- Proceeds of the rights issue should be added net of any issue costs.
- If the project has already been announced, and if the market is operating at the semi-strong level of efficiency, the project's NPV will *already* be reflected in the existing share price and it should **not** be included again in the formula above.

Activity 1 TERP

[Show](#)

At the close of trading, a company has 100,000 shares issued with a current market price of \$2 each.

After trading closes for the day, the company announces it will take on an project with a NPV of \$25,000.

The project will be financed by a rights issue of one new share for every two existing shares. The rights price is \$1 per new share.

Ignore issue costs and assume that the equity market operates at a semi-strong level of pricing efficiency.

Required:

Calculate the theoretical ex-rights price of the company's shares.

**Please use the notes feature in the toolbar to help formulate your answer*

The theoretical ex-rights price of shares is \$1.83 (calculated below).

If the market price of the existing shares had been given after the announcement of the project, the NPV of \$25,000 would already be included in the MV of the old shares. However, because this is **not** the case, the project NPV has to be added.

$$\begin{aligned}\text{TERP} &= \frac{\text{Pre-existing value of equity} + \text{proceeds so rights issue} + \text{project NPV}}{\text{Number of shares after the rights issue}} \\ &= \frac{(100,000 \times \$2) + (50,000 \times \$1) + \$25,000}{100,000 + 50,000} = \$1.8333\end{aligned}$$

Notes/Additional Information (not required):

- The NPV represents the theoretical increase in market value produced by a project.
- Value of a right per new share = TERP – subscription (issue) price = \$1.8333 – \$1.00 = \$0.8333
- Value of a right per existing share = \$0.8333/2 = \$0.4167

Activity 2 Shareholder's Wealth

[Show](#)

Assume in *Activity 1* above that Mr X owns 1,000 shares in the company. A shareholder's wealth depends on whether he exercises his rights.

Required:

Show Mr X's position if he:

- (i) takes up his rights;**
- (ii) renounces and sells his rights;**
- (iii) does nothing.**

Hint: Calculate his wealth before and after the rights issue for each scenario.

**Please use the notes feature in the toolbar to help formulate your answer*

(i)	Takes up rights		\$
	Wealth before the rights issue	$1,000 \times \$2$	<u>2,000</u>
	Wealth post-rights issue	$1,500 \times \$1.8333$	2,750
	Less: rights cost	$500 \times \$1$	<u>(500)</u>
			<u>2,250</u>
	Therefore, is \$250 better off.		
(ii)	Sells rights		\$
	Wealth before the rights issue	$1,000 \times \$2$	<u>2,000</u>
	Wealth post-rights issue		
	Shares	$1,000 \times \$1.8333$	1,833
	Sale of rights	$500 \times \$0.8333$	<u>417</u>
			<u>2,250</u>
	Therefore, is \$250 better off.		
(iii)	Does nothing		\$
	Wealth before the rights issue	$1,000 \times \$2$	2,000
	Wealth post-rights issue	$1,000 \times \$1.8333$	1,833
	Therefore, a loss of \$167.		

2.2.2 Bonus Issue

In a bonus issue, reserves (e.g. revaluation surplus or share premium accounts) are converted into share capital, which is distributed as new shares to existing shareholders in proportion to their existing holdings).

Key Point

No finance is raised.

The purpose is to increase the marketability of the shares, as it increases the number of shares in existence and reduces their price. This creates a more active secondary market for the shares, which will help future issues to raise cash (e.g. rights issues).

Bonus issues (also called "scrip issues" or a "capitalisation of reserves") signal a company's strength to the market.

2.2.3 Stock Splits

Stock splits occur when ordinary shares are split in value (e.g. each \$1 share is converted into two 50-cent shares). This reduces the market price per share, increasing their marketability.

3.0 Introduction

As introduced in *Chapter 1*, the three key decision areas in financial management – investing, financing and dividend policy –are interrelated:

- An increase in dividends (the dividend decision), will reduce the level of retained cash available and increase the need for external finance (the financing decision) to fund capital investment projects (the investment decision).
- An increase in asset expenditure (the investment decision),would also increase the need for finance (the financing decision) which may be sourced internally by reducing dividends (the dividend decision).

Therefore, directors must decide whether to pay a dividend or retain earnings.

3.1 Practical Influences

Practical factors have a significant influence on the company's dividend policy.

3.1.1 Legal Constraints

In most countries, a dividend can only be paid if there is a credit balance in the retained earnings account in the statement of financial position. No distributable reserves means no dividends.

- If a company has brought forward losses from previous years, the balance on retained earnings may be a debit ("negative"), even if a profit was made in the current year.
- This restriction on paying dividends for the foreseeable future could damage the company's ability to attract new equity investors to finance its growth. Therefore, some countries allow a debit balance on retained earnings to be written-off against other reserves, allowing the company to resume paying dividends.

3.1.2 Liquidity Requirements

Irrespective of any potentially detrimental effect on share price that a change in dividend policy might cause, companies have to maintain adequate liquidity. Dividend restrictions may be imposed to stay solvent or meet statutory capital maintenance requirements.

Cash for dividend payments may also be unavailable because the company may wish to hold significant cash to meet both routine and any unexpected expenses, and also to be able to move quickly into investment opportunities.

Example 1 Liquidity Requirements

As of June 2011, Apple Computers held an astonishing \$76 billion in cash and cash equivalents. This cash was held partly to be able to buy in advance large quantities of key components and ensure continuous production, and partly as a "war chest" to finance the quick acquisition of other innovative high-tech companies.

3.1.3 Shareholder Expectations

Most shares in quoted companies are held by influential institutional investors (e.g. pension funds). Therefore, the directors of quoted companies have to manage investor expectations of dividends carefully.

- If a large dividend is paid in the current year, this may create expectations of the same, or even higher, in future. If these expectations are not met, key investors may sell their shares.
- In smaller owner-managed companies, there is no such agency problem. The dividend decision is influenced more by the personal tax position of the owner-managers than sensitivity over expectations.

3.1.4 Signalling

In quoted companies, where there is a significant "divorce of ownership and control", investors do not have access to all information about the company's operations and prospects; they only have what is publicly available. Therefore, public announcements of the level of proposed dividends are signals of company' strength or weakness.

- A surprise cut in dividend may be interpreted as a signal of liquidity problems, even if the cut is actually to finance attractive projects.
- A surprise increase may be taken as a signal of company strength, although some investors may question why the directors have not found suitable strategies for reinvesting surplus cash.

3.1.5 Stability

Companies often strive for a stable level of dividends or a constant level of growth. This is done to avoid sharp movements in share price.

- Companies will try to maintain the level of dividends in the face of fluctuating earnings.
- This approach is widespread for quoted companies.

3.1.6 Borrowing Covenants

A loan agreement may include a clause that limits dividend payments (e.g. to a specified proportion of earnings) to help ensure the loan is more secure. If less cash is paid as dividends, liquidity might be better (though cash can still be consumed on the purchase of non-current assets).

3.2 Alternative Dividend Policies

3.2.1 Constant Dividend

A constant dividend policy avoids surprises and signals stability to shareholders, but shareholders might be dissatisfied if they receive relatively low dividends if earnings are rising. As the proportion of retained earnings increases, shareholders might question whether the company can find enough suitable investment opportunities.

3.2.2 Constant Growth

Constant growth is predictable and favoured by shareholders but again the dividend growth rate might not match the earnings growth rate.

3.2.3 Constant Payout Ratio

A constant payout ratio means paying a dividend that is a fixed proportion of each year's earnings. Although this approach sounds logical, the variability in the absolute amount of dividends creates uncertainty and is rarely adopted by quoted companies.

3.2.4 Residual Dividend Policy

Under the residual dividend policy, retained earnings fund all positive NPV projects. Any remaining earnings not needed to support such projects are paid out as a dividend.

This links to pecking order theory (i.e. retained earnings are the cheapest source of new finance, so they should be used to fund projects first).

A residual dividend policy is likely to lead to fluctuating dividends, so although it may excuse a cut in an otherwise stable dividend pattern, it is not common for quoted companies.

3.2.5 Zero Dividend Policy

A high-growth company may find that, in the early years, all surplus cash can be profitably reinvested back into the business – particularly if the company lacks access to external finance.

However, at some point, when fewer positive NPV projects are available, the company should start paying dividends.

3.3 Dividend Theories

3.3.1 Clientele Theory

The company's historical dividend policy may have attracted investors with specific needs (e.g. for current income or their tax situation). Therefore, management should view shareholders as the company's "clientele".

- For example, a pension fund must invest in a portfolio that produces income to pay pensioners. Therefore, it will therefore invest heavily in shares that pay regular, relatively predictable dividends.
- Similarly, tax can affect investment decisions if capital gains and investment income are taxed differently.
- An abrupt change in dividend policy will disturb investors' carefully constructed portfolios, and investors will have to adjust their mix of shares, incurring transaction costs. Although it may be argued that an investor can "manufacture" dividends by selling some shares, this would incur transaction costs and may attract different tax treatment.
- Therefore, the company should maintain a stable dividend policy or risk losing major investors such as financial intermediaries (e.g. insurance companies and pension funds).

3.3.2 "Bird-in-the-Hand" Theory

The "bird-in-the-hand" theory argues that shareholders prefer higher dividends (and therefore lower potential capital gains) because a cash dividend today is without risk, whereas future share price growth is uncertain.

Although persuasive, this is incorrect. Market forces should mean that a share price is correctly set for the level of risk and returns. If more dividends are paid out as cash the investor has to decide how to invest it. In another investment with higher returns and higher risk or lower returns and lower risks? According to the capital asset pricing model (see *Chapter 12*), diversified investors should be happy with either because extra returns correctly compensate for additional risk.

3.3.3 Dividend Irrelevance Theory

Modigliani and Miller (finance theorists) argue that shareholders are indifferent to dividend policy under certain conditions.

- If a company pays no dividend, the share price should rise due to reinvestment of earnings.
- Any shareholder who requires a dividend can sell part of their holding to create a capital gain (i.e. manufacture a "home-made dividend").

Under this theory, the pattern of dividends is irrelevant to shareholder wealth. However, Modigliani and Miller made a series of assumptions which may not hold in practice:

- No distortions from the personal tax system (i.e. dividends and capital gains are taxed at the same rate in the hands of investors).
- No transactions costs (i.e. investors can sell shares to create a home-made dividend without incurring any trading costs).
- Perfect markets (i.e. if the company defers dividend payments, the current share price will fully reflect its value).

3.4 Alternatives to Cash Dividends

Like the payment of dividends, other distribution methods also require the company to have sufficient distributable reserves and liquidity to fund the distribution.

3.4.1 Share Buyback Programmes

- The buyback can be performed by:
 - writing directly to all shareholders with an offer to buy shares at a fixed price (a tender offer) or:
 - purchasing shares via the stock market at the prevailing price.
- The shares are either cancelled (and a non-distributable reserve created) or held by the company as treasury shares for possible future reissue. If held by the company, the shares carry no voting rights and receive no dividends.
- Distributable reserves must be reduced by the **full value** of the buyback.
- As there are fewer shares in issue after the buyback, the share price should rise.
- Ratios such as earnings per share (EPS) and return on equity (ROE) should also improve.
- In many countries, a share buyback is treated as a capital gain in the hands of the investor rather than as income. This can have tax advantages if tax on capital gains is lower than tax on income.

3.4.2 Special Dividends

- If a quoted company announces a larger-than-expected dividend, this may raise market expectations that future dividends will also be higher.
- Therefore, to avoid creating expectations of an unsustainable level, a larger dividend may be announced as a "special" dividend - a **bonus dividend**.
- The directors are communicating to the markets that any exceptional cash surplus will be returned in this way from time to time but that this should not be built into dividend-per-share forecasts.
- Like a buyback, the company distributes cash. Also, **all** shareholders will receive cash (whereas, under a buyback, only those who sell receive cash).

3.4.3 Scrip Dividends

A scrip dividend is a choice between a cash dividend and shares instead of cash. Shareholders can choose to acquire new shares (if they do not need the cash dividend) without transaction costs. The company conserves cash for reinvestment (the preferred source of finance under the pecking order theory).

4.1 Characteristics

There is no official definition of a *small and medium-sized entity* (SME). However, McLaney (2000) identifies three characteristics of an SME:

1. the entity is likely to be unquoted;
2. ownership of the business is restricted to a few individuals, typically a family group; and
3. it is not a micro business (i.e. a very small business created for the self-employment of its owners).

SMEs contribute in a significant way to many economies in the world. Besides generating income (often a large proportion of the gross national product), they are frequently major employers and the sector most identified with new ideas and entrepreneurial spirit. These latter factors help sustain and support growth rates in many economies.

4.2 Difficulties in Raising Finance

SMEs are private companies with a limited number of shareholders and, unless those shareholders are wealthy, there is a limit to how much they can invest in the company.

Therefore, SMEs have to rely heavily on retained earnings for equity finance, but this source is limited, especially when profits are low. They may have to rely on debt if restricted in the amount of equity they can raise.

4.2.1 Perceptions of Risk and Uncertainty

SMEs are often viewed as unattractive investment opportunities because they have high levels of risk and uncertainty attached to them.

- Larger businesses have a track record, and a long-term relationship with their bankers. New businesses, typically SMEs, obviously do not have a track record.
- SMEs internal controls are often non-existent or very limited.
- Larger businesses conduct more of their activities in public than do SMEs. If information is public, there is less uncertainty. A company that is quoted on a stock exchange will be subject to press scrutiny and stock exchange rules that regulate its activities, including publishing accounts that have been audited. Many SMEs do not require audits (or even publish full accounts), and the press is not interested in them. Therefore, they have fewer external controls.
- The fact that potential investors in an SME have much less information about the business than its managers (i.e. *asymmetry of information*) contributes to their perception of high risk.
- Often, SMEs have one dominant owner-manager whose decisions are rarely questioned.
- They tend to have limited assets to offer as security (see below).

4.2.2 Absence of Security

If an SME seeks a bank loan, the bank will look to see what security (collateral) is available for any loan provided. This is likely to involve an audit of the entity's assets.

- Collateral is important because it can reduce the level of risk a bank is exposed to in granting a loan to a new business.
- Many SMEs are based in the service sector, where the main asset is likely human capital rather than physical assets. Therefore, the directors may be required to pledge personal assets (e.g. their homes) to secure business loans.

4.2.3 Shares Lack Marketability

The equity issued by small companies is unquoted and difficult to buy and sell.

- Sales are usually on a matched bargain basis, which means that a shareholder wishing to sell has to wait until an investor wishes to buy.
- This lack of marketability means that small companies may be unable to offer new equity to anyone other than family and friends.

4.2.4 Tax Considerations

The tax system may encourage individuals with cash to invest via large institutional investors rather than directly into small companies.

- In many countries, personal tax incentives are offered on contributions to pension funds.
- These institutional investors usually invest in larger companies (e.g. listed companies) to maintain an acceptable risk profile and ensure a steady stream of income to meet ongoing liabilities. This reduces the potential flow of funds to small companies, although the government may try to mitigate this effect by offering tax advantages for investment in SMEs.

4.2.5 Funding Gap and Maturity Gap

Initial owner finance is nearly always the first source of finance for an SME, whether from the owner or family connections. Many of the assets may be intangible at this stage, making sourcing external financing difficult. The "funding gap" is the difference between the finance available to SMEs and the funding they could productively use.

Bank loans may become available at a later stage. However, with small businesses, longer-term loans are often easier to obtain than medium-term loans because the longer-term loans are easily secured with mortgages against property. The fact that medium-term loans are hard to obtain is a well-known feature of SMEs, and many resort to financing medium-term assets with short-term finance such as an overdraft. This is known as the *maturity gap* as there is a mismatch of the maturity of the assets and liabilities within the business, which is not ideal.

Due to the gaps in funding and maturity, an SME may have to take an innovative approach to financing. Possible financing solutions are discussed below.

4.3 Financing Solutions

Government policies can influence the level of funds available. For example:

- Tax policy such as higher taxes on dividends (meaning less income for investors) or concessions to encourage companies to invest (e.g. tax-allowable depreciation).
- Interest rate policy, for example, low rates, meaning borrowing becomes cheaper, and the supply of funds increases as there is less incentive for investors to save.

4.3.1 Venture Capital

Venture capitalists are a potential source of financing for an SME. Venture capital is equity capital provided to small and growing businesses. It is usually offered by a wealthy individual, a venture capital firm (e.g. the 3i Group) that manages a venture capital fund, or a high-risk, potentially high-return subsidiary of an organisation with significant cash to invest (e.g. a bank). Typically, a \$1m minimum is involved for any one investment.

- Before investing, the providers look for:
 - a product or products with strong potential (e.g. an innovation);
 - solid management; and
 - potential for high returns.
- To invest, providers of funds would normally expect:
 - typically, 25% to 49% of the equity;
 - a business plan with medium-term cash flow and profit projections;
 - board representation (to monitor their investment and give advice);
 - a dividend policy which promotes growth (i.e. high reinvestment of earnings);
 - an "exit route" for selling their investment (e.g. proposed time-scale for seeking a market quotation);
 - and
 - the provision of regular management accounting information.

Venture capital trusts (VCTs) also serve as a potential financing source.

- VCTs are listed investment trust companies which invest at least 70% of their funds in a spread of small *unquoted trading* companies.
- Many governments give tax incentives to individual investors in VCTs (e.g. Australia, Canada, France, Singapore, the UK and the US).

4.3.2 Private Equity Funds

A private equity fund attempts to gain control over a company by putting it through a restructuring programme before either selling to another fund or listing the company on the stock market.

- The difference between private equity and venture capital is that private equity funds usually seek total control of the target company. In contrast, venture capitalists provide growth finance in return for partial control.
- Private equity funds do not only target SMEs; they also buy large quoted companies, take them off the stock market, restructure them, and then re-list them on the stock market.

4.3.3 Business Angels

Business angels are private individuals (or small groups of individuals) prepared to invest equity (or perhaps debt) into small businesses with considerable potential.

- Angels are often entrepreneurs who made their fortunes in the high-tech sector, were wise enough to sell before the "dot.com" bubble burst, and now invest in small business as a hobby (although they expect to make gains).
- Angels not only provide finance, they also offer advice, experience and business contacts. A typical business angel will hold a portfolio of investments and may, for example, add an investment in an entity that makes health drinks to complement existing investments in fitness clubs.
- Angels receive many applications for finance, and will only be prepared to invest in a business with an innovative product and talented management.

4.3.4 Government Assistance

Governments are often keen to support SMEs in raising finance for profitable investments:

- to avoid lost investment opportunities and national wealth being lower than it could be;
- to support innovation (an area in which SMEs often excel); and
- to boost employment.

Forms of government assistance include:

- Providing grants and guaranteeing loans (see *Chapter 9*).
- Providing tax breaks or incentives to those willing to take the risk of investing in SMEs.
- Providing advice. For example, in Scotland, "Business Gateway" is a government-funded organisation which assists those setting up and running a business, including advice on raising finance.
- Providing equity investment. Many countries have government-backed venture capital organisations willing to invest in the equity of SMEs. This is often done on a matching basis, where the organisation will match any equity investment raised from other sources. For example, "Enterprise Capital Funds" in the UK and the "Small Business Investment Company" programme in the US.

The UK Enterprise Investment Scheme (EIS) is an example of a scheme to help SMEs attract equity finance.

- The EIS scheme encourages private investors to buy shares in *unlisted* trading companies.
- Tax relief, at an income tax rate of 30%, is available for investors.
- Maximum annual tax relief is £1m (£2m if at least £1m is invested in knowledge-intensive companies).
- Shares must be held for at least three years (otherwise tax reliefs are withdrawn/ withheld).

4.3.5 Crowdfunding

1 Definitions

Crowdsourcing – using a large, evolving, relatively open group of people (usually assembled via the internet) to provide goods, services or finance.

Crowdfunding – is crowdsourcing, to fund a project. Under crowdfunding, there is no financial intermediary: each individual in the crowd has a direct relationship with the entity funded.

Crowdfunding may be appropriate for the early stages of "seed" finance for an SME.

The crowdfunding model is driven by:

- the project initiator who proposes the idea or project to be funded;
- individuals or groups who support the idea; and
- a moderating organisation (the "platform") that brings the parties together to launch the idea.

Crowdfunding lets people invest in projects or ideas that they believe in or are interested in, so they might be willing to take relatively higher risks and accept relatively lower returns. In addition, the "crowd" aspect means supporters often encourage others to invest.

Crowdfunding can be particularly beneficial for SMEs as it allows them to reach out directly to investors who may be happy with the risks associated with funding innovations and technologies often pioneered by SMEs.

The main types of crowdfunding are:

1. Donation-based—any crowdfunding campaign without financial returns to the investors or contributors (e.g. fundraising for disaster relief, charities, NPOs and medical bills). It may also be suitable to obtain support from a local community.

2. Reward-based— where entrepreneurs pre-sell a product or service to launch a business concept without incurring debt or issuing equity to outside investors. This is best for businesses with a marketable product and need funding before generating revenues.

3. Equity-based – where investors receive unlisted shares, usually in the company's early stages, in exchange for the money provided. This is typically best for businesses that can grow and scale quickly.

Example 2 Crowdfunding Platforms

Crowdfunding platforms for small businesses include:

Kickstarter: Rewards-based funding platform focuses on creative projects like art, music, film, etc. Funding is "all or nothing" (i.e. funds pledged will only be received if the funding goal is reached within the designated timeline). There are no fees if a campaign is unsuccessful.

Crowdfunder: A low-cost equity crowdfunding solution based on a subscription model.

Crowdfunding may also be debt-based (see peer-to-peer lending in *Chapter 9*).

Syllabus Coverage

This chapter covers the following Learning Outcomes.

E. Business Finance

1. Sources of, and raising, business finance

b) Identify and discuss the range of long-term sources of finance available to businesses, including:

- i) equity finance
- iv) venture capital.

c) Identify and discuss methods of raising equity finance, including:

- i) rights issue
- ii) placing
- iii) public offer
- iv) stock exchange listing.

e) Identify and discuss internal sources of finance, including:

- i) retained earnings
- ii) increasing working capital management efficiency
- iii) the relationship between dividend policy and the financing decision
- iv) the theoretical approaches to, and the practical influences on, the dividend decision, including legal constraints, liquidity, shareholder expectations and alternatives to cash dividends.

5. Finance for small and medium sized entities (SMEs)

a) Describe the financing needs of small businesses.

b) Describe the nature of the financing problem for small businesses in terms of the funding gap, the maturity gap and inadequate security.

c) Explain measures that may be taken to ease the financing problems of SMEs, including the responses of government departments and financial institutions.

d) Identify and evaluate the financial impact of sources of finance for SMEs, including sources already referred to in syllabus section E1 and also:

- i) Business angel financing
- ii) Government assistance
- iv) Crowdfunding

Summary and Quiz

- Retained earnings are the main internal source of finance.
- Internal finance can also be generated by increasing efficiency in working capital management.
- Company managers generally prefer to use internal finance rather than external finance. This preference is known as the pecking order theory.
- Issuing new shares in an IPO is an expensive source of finance.
- A company must meet several requirements before receiving an official listing on the full London Stock Exchange. These costly requirements preclude small and medium-sized entities (SMEs) from official listing.
- However, SMEs find the London Stock Exchange's AIM more attractive because the listing requirements are less onerous.
- In a rights issue, the existing shareholders are offered more shares, at a discounted price, in proportion to their existing holding.
- The theoretical ex-rights price (TERP) of a share is the expected share price following the rights issue.
- A company's dividend policy will be affected by legal constraints, liquidity requirements, shareholder expectations, the likely signal given, and the desire for share price stability.
- Under a residual dividend policy, any remaining earnings, after funding all attractive projects, are paid out as dividends.
- The dividend irrelevance theory states that investors are indifferent to whether a dividend is paid. This is because:
 - If a company pays no dividend, the share price should rise due to reinvestment of earnings.
 - Any shareholder requiring a dividend can sell part of their holding to create a capital gain (i.e. manufacture a home-made dividend).
- Alternatives to dividends include share buybacks, special dividends and scrip dividends.
- SMEs have difficulty raising finance because of business uncertainty, a lack of collateral, shares that are not marketable and tax incentives.
- Funding solutions for SMEs include venture capital, private equity, angel investors (business angels), crowdfunding and government schemes.

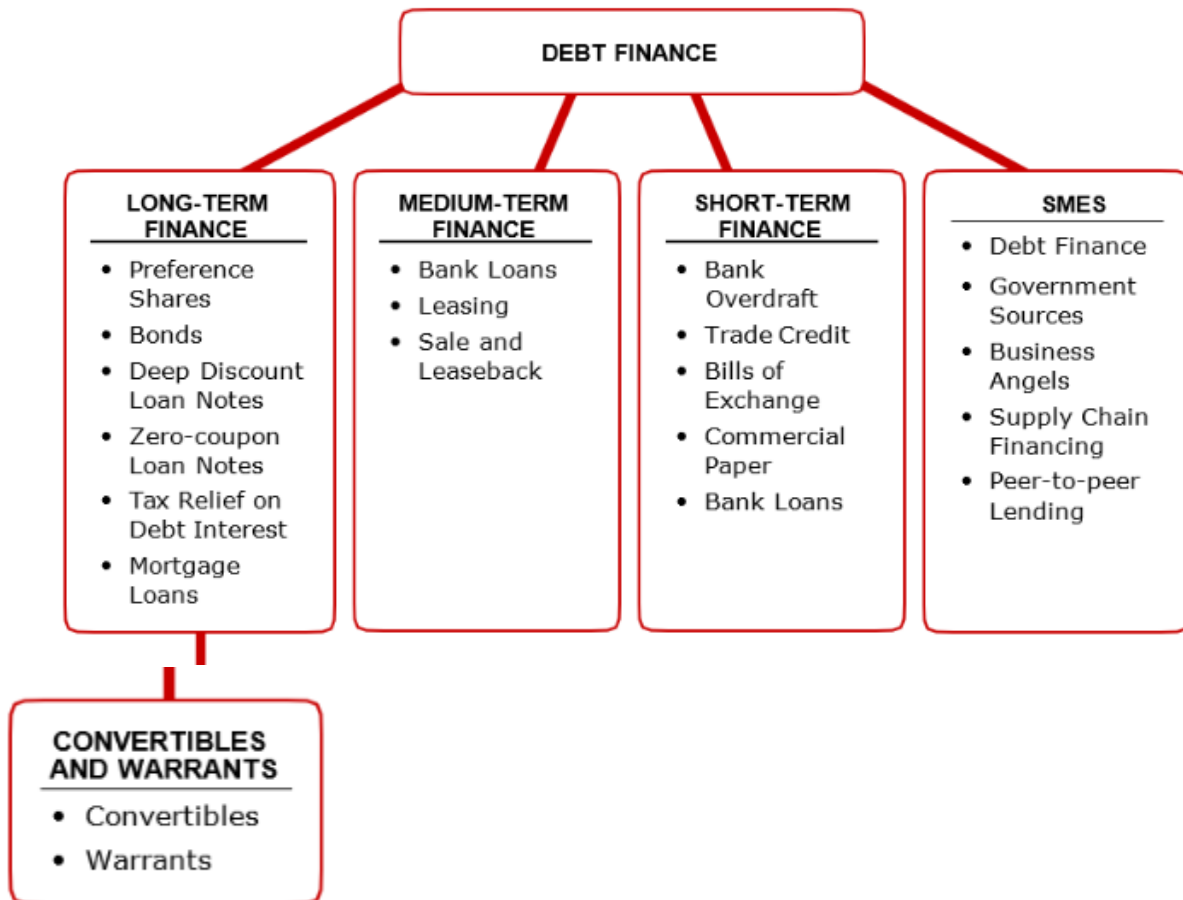
Chapter 8 Quiz

Chapter 9: Debt Finance

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	9.1.3 Deep Discount Loan Notes
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9.5 Small- and Medium-sized Entities (SMEs)	9.5.1 Debt Finance
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CHAPTER 9: Visual Overview

Objective: To appreciate the options available to a company for long-, medium- and short-term debt finance.



1.1 Preference Shares

i Definition

Preference shares – shares with a fixed rate of dividend that have a prior claim on profits available for distribution.

Preference shares ("preferred shares") are legally equity. They are often treated as debt (e.g. under IFRS Accounting Standards) as they are similar to debt.

Some of the common features of preference shares include:

- The shares have a fixed percentage ("coupon") dividend payable before ordinary dividends. This preference share dividend is expressed as a percentage of the share's nominal value.
- The dividend is only payable if there are sufficient distributable profits. However, if the shares are cumulative, the right to receive unpaid dividends is carried forward (i.e. cumulative preference dividends). Any dividend arrears are then payable before ordinary dividends.
- As for ordinary dividends, preference dividends are not deductible for corporate tax purposes. The preference dividends are considered a distribution of profit rather than an expense.
- On liquidation of the company, preference shareholders rank before ordinary shareholders and after debt holders.
- A participating preference share gives the holder the right to receive an additional dividend (if certain conditions are met) in addition to a fixed percentage of the share's nominal value.

Advantages	Disadvantages
✓ No voting rights; therefore no dilution of control. ✓ Compared to the issue of debt: - preferred dividends do not have to be paid in any specific year (e.g. if profits are poor); - preferred shares are not secured on company assets; and - non-payment of a dividend does not give holders the right to appoint a liquidator.	✗ Preferred dividends are not tax deductible (unlike interest on debt which is a tax allowable expense). Preference shares are a relatively rare source of finance in practice because of this. ✗ To attract investors to buy preferred shares, the company must pay a higher return than debt to compensate for the additional risk.

1.2 Bonds (“Loan Notes”)

i Definition

Bond – a negotiable security evidencing a debt governed by a contract which specifies, for example, the coupon rate, repayment schedule, security (if any), principal value, seniority (if subordinate to other debt) and other covenants.

The terms "debenture", "loan note" and "bond" all basically refer to the same thing (i.e. a written acknowledgement of a company's debt which can then be traded). Bonds may be **secured** by:

- a **fixed** charge over a specified asset (e.g. a specific building), which cannot be sold unless the debt is repaid; or
- a **floating** charge over a class of asset which changes (e.g. inventory). On default, the floating charge "crystallises" as a fixed charge, and the asset class can no longer be traded until the debt is repaid.

Bonds may be redeemable or irredeemable.

Holders of **unsecured** bonds have the same rights as unsecured trade creditors.

In the UK, bonds are usually issued with a "face" value ("nominal" or "par" value) of £100. They can then be traded on the bond market and reach a market price. A bond with a face value of £100 that is "selling at a premium" of 15%, sells for £115. Bonds in the US typically have a face value of \$1,000, and in the Eurozone €1,000 (or multiples thereof).

Interest is usually a fixed coupon expressed as a percentage of the bond's nominal value (e.g. an 8% loan note has a coupon of 8% and would pay interest of \$8 against a nominal value of \$100).

Variable (floating) interest rate loan notes automatically adjust their interest payment in line with an agreed market reference rate (e.g. Euro Short-Term Rate – "ESTER").

1.3 Deep Discount Loan Notes

i Definition

Deep discount loan notes – loan notes issued at a substantial discount to nominal value and redeemable at their nominal value (or above) on maturity.

Investors in **deep discount loan notes** receive a large capital gain on redemption but are paid a low coupon during the loan term.

These loan notes offer a cash flow advantage to the borrower. This is especially useful for financing projects which produce weak cash flows in the early years.

Example 1 Deep Discount Loan Notes

A five-year, \$100, 3% loan note issued at \$80 would generate the following cash inflows/(outflows) for the issuing company:

	t_0	t_1	t_2	t_3	t_4	t_5
Issue price	80					
Interest		(3)	(3)	(3)	(3)	(3)
Redemption						(100)

The coupon rate is **not** the cost of debt where the market price differs from the redemption price. In this example, the IRR of these cash flows (pre-tax cost of debt, k_d) is 8%.

1.4 Zero-coupon Loan Notes

i Definition

Zero-coupon loan notes – loan notes issued at a discount to nominal value and which pay no coupon.

Zero-coupon loan notes have the following advantages:

- ✓ The issuing company pays no interest; the only cash payout is at the loan note's maturity.
- ✓ The return to investors is wholly a capital gain (the difference between issue and redemption price).

1.5 Mortgage Loans

i Definition

Mortgage – legal agreement to lend money, secured by a legal charge over the borrower's property.

A mortgage loan is typically a long-term loan (15 to 30 years or more) secured by property.

1.5.1 Advantages

- ✓ Given the security, mortgage loans have a lower interest rate than shorter-term debt.
- ✓ Lower monthly repayments over a longer term help cash management, making it easier to budget and plan for the long term.
- ✓ The growth in the property's value accrues to the company.

1.5.2 Disadvantages

- ✗ There are likely to be restrictive covenants concerning the use of the property and its potential disposal.
- ✗ In the event of default on repayments, the lender may force the sale of the property to recover the loan. This is a significant risk to the business.
- ✗ A significant proportion of finance tied up in property may be unsuitable for a business that needs liquid assets for investment opportunities or operations.
- ✗ A lengthy approval process (including property valuations) make mortgage loans unsuitable for urgent finance.

1.6 Tax Relief on Debt Interest

Interest expense is tax deductible; it reduces corporation tax payments. Therefore, the issue of debt is preferable to the issue of shares, as dividends are not tax deductible.

Example 2 Debt Interest Tax Relief		
	Co A	Co B
Profit before tax	100	100
Interest	-	<u>(10)</u>
	100	90
Corporation tax 30%	<u>(30)</u>	<u>(27)</u>
	70	63
		
	\$7 difference	
Effective cost of debt in Co B		
Interest		10
Less: Tax relief		<u>(3)</u>
		<u>7</u>

Points to note:

- The after-tax **cost** of debt (i.e. to the company) = Pre-tax cost of debt $\times$ (1-tax rate).
- The pre-tax **cost** of debt to the company = Return required by the debtholders.
- The post-tax cost is said to be lower due to the "tax shield" on the coupon payments.

2.1 Convertibles

1 Definition

Convertibles – loan notes or preference shares which can be converted into a pre-determined number of ordinary shares.

Convertible loan notes and convertible preference shares:

- Pay a fixed coupon or dividend until converted.
- May be converted into ordinary shares:
 - on, or by, a pre-determined date;
 - at a pre-determined rate (the conversion ratio); and
 - at the holder's option.
- May have a conversion ratio which changes during the loan term to stimulate early conversion.

(Numerical examples showing how convertibles work will be covered in *Chapter 10*.)

Convertibles have the following advantages:

- ✓ For investors, they are a relatively low-risk investment with the opportunity to make high returns on conversion to ordinary shares.
- ✓ For the issuer, they can offer a lower coupon rate than would have to be paid on a non-convertible (straight) loan notes/preference shares (because the conversion option has value).
- ✓ For newer companies, investors may not want to risk investing in equity but may be prepared to invest in less risky loan notes. If the company does well, investors can opt to convert and benefit from capital growth or keep the safer loan notes.

Convertible loan notes require the calculation of "fully diluted" earnings per share (EPS) to indicate what the EPS might be if the debt is converted into equity (see *Chapter 19*).

2.2 Warrants

i Definition

Warrant – the investor's right, but not the obligation, to purchase new shares at a future date at a fixed price (the exercise or subscription price).

Warrants have the following features:

- They may be attached to loan notes to make them more attractive.
- They are an option to buy shares in the issuing company.
- They may be separated from the underlying debt so the holder of the warrants may sell them rather than keep them (i.e. they are traded independently).

Warrants offer several advantages for the issuing company:

- ✓ When initially attached to the loan note, the coupon rate on the loan note will be lower than for comparable straight debt. This is because the investor has the additional benefit of potentially purchasing equity shares at an attractive price.
- ✓ They may make an issue of unsecured debt possible when the company's assets are inadequate to secure the debt.
- ✓ They are a way of issuing equity (albeit with a delay) without the usual negative signal associated with an equity issue.

3.1 Bank Loans

Companies agree to borrow from the bank at a fixed rate for a specified period and with an agreed repayment schedule.

3.1.1 Advantages

- ✓ As the loan is for a fixed term, there is no risk of early recall (unlike overdrafts that are repayable on demand). This helps with financial stability.
- ✓ A predetermined repayment schedule makes it easier to budget and plan for loan repayments. This helps cash flow management.
- ✓ Interest will usually be tax deductible; reducing the company's overall tax liability is cost effective.

3.1.2 Disadvantages

- ✗ Terms and conditions tend to be inflexible; fixed repayment schedules may be non-negotiable. A bank may also restrict how a loan can be used; a loan obtained for a specific purpose cannot be redirected to another project without re-negotiation.
- ✗ Banks may require security ("collateral"). Valuable assets that are pledged as security will be at risk if the company defaults on the repayment schedule.
- ✗ It may require covenants, which are restrictions on the company (e.g. limits on dividend payments, limitations on further borrowing), designed to protect the debtholder and could reduce the interest rate on the debt.
- ✗ Penalties or fees for early repayment can be a disadvantage if a company want to repay a loan ahead of schedule (e.g. to reduce interest costs).

3.2 Leasing

Instead of buying an asset outright, using retained earnings or borrowed funds, a company may lease an asset. Under a lease contract, the lessee has the right to use an asset owned by the lessor in exchange for a series of payments.

3.2.1 Advantages

- ✓ There are many willing providers (often associated with asset manufacturers and therefore offering attractive terms).
- ✓ A lease “matches” finance to the right to use the asset. A company can use assets that might be too expensive to buy outright.
- ✓ Very flexible packages available, some of which include repairs and maintenance. This helps reduce the costs and operational burden of asset maintenance.

3.2.2 Disadvantages

- ✗ Over the long term, leasing can be more expensive than outright purchase; cumulative lease payments can exceed the cost of buying the asset.
- ✗ At the end of the lease term, the company typically does not own the asset unless it has an option to purchase it, at an additional cost.
- ✗ Leases can include restrictions on the use of the asset or penalties for early termination. These terms may restrict the company's operational decisions.

3.3 Sale and Leaseback

Under a sale and leaseback, a company sells property to an institution, such as a pension fund, and then leases it.

3.3.1 Advantages

- ✓ Immediate injection of cash into the business can be used for investment opportunities, debt reduction (interest savings) or working capital requirements.
- ✓ Operational continuity in using the asset without disruption.
- ✓ It may realise a profit on the sale (subject to IFRS requirements).

3.3.2 Disadvantages

- ✗ As the company no longer owns the property it will not benefit from any appreciation in its value.
- ✗ The future borrowing capacity of the company will be reduced, as there will be fewer assets to provide security for a loan.
- ✗ The net effect is equivalent to secured borrowing. A right-of-use asset and a lease liability will be recognised. If the right-of-use asset is greater than the carrying amount of the leased asset, gearing (the proportion of debt finance to equity finance) will increase.

4.1 Bank Overdraft

A bank overdraft is a borrowing facility associated with a current account. In jurisdictions where a bank overdraft is allowed, it is often the liquidity fallback of choice for small- and medium-sized entities (SMEs).

Advantages	Disadvantages
✓ Flexible terms of borrowing and repayment allow businesses to borrow funds as needed up to an approved limit, and repay when cash flow permits. ✓ Overdrafts can be established relatively quickly, providing immediate access to cash. ✓ Interest paid on the amount of the facility used is more cost effective.	✗ Potential uncertainty as banks can withdraw or reduce facilities with little notice: overdrafts are technically repayable on demand. ✗ Overdrafts are unsuitable for long-term financing needs and ongoing cash flow issues. ✗ Interest rates can be higher than other forms of short-term financing, making overdrafts expensive if used regularly.

4.2 Trade Credit

Businesses that offer trade credit invoice their customers with payment terms, for example, 30, 60 or 90 days. Customers can take advantage of trade credit by delaying payments to existing suppliers or using more of the credit offered.

Advantages	Disadvantages
✓ Trade credit allows a company to delay payment, providing interest-free finance for a period. ✓ Extending the time between receipt of goods/services and payment improves cash flow management. ✓ Used responsibly (i.e. adhering to terms) can build supplier relationship, potentially leading to more favourable terms and discounts.	✗ The loss of settlement (prompt payment) discounts is potentially costly (see <i>Chapter 16</i>). ✗ As it is specific to purchases from suppliers, it does not meet other short-term needs (e.g. meeting payroll obligations). ✗ Late payments may result in penalties and damage supplier relationships.

How to use trade credit in managing accounts payable is detailed in *Chapter 16*.

4.3 Bills of Exchange

1 Definition

Bill of exchange – an acknowledgement of a debt to be paid on a stated date.

In international trade, an exporter typically requires a customer to accept a bill before releasing documents of title to the goods.

The exporter may then:

- hold the bill to maturity (and receive payment from the customer); or
- "discount" the bill with a bank to receive cash earlier. (If the customer then fails to pay, the bank has recourse to the exporter for payment.)

Bills of exchange are not widely used today – having been replaced with alternatives (e.g. bank transfers and credit/debit card payments).

Example 3 Bill of Exchange

X sells \$2m worth of goods to Y. X writes out ("draws") a bill of exchange for \$2m payable in two months, which it sends to Y. Y signs the bill to acknowledge the debt and returns it to X.

X can hold the bill for two months until Y pays the debt or sell it at a discount (e.g. at 98% of nominal value). If Y then pays, the buyer of the bill receives the \$2m and makes a gain.

Advantages

- ✓ Bills of exchange provide short-term financing that can help bridge temporary gaps in cash flow.
- ✓ They facilitate payments between parties in different countries, promoting international trade.
- ✓ As negotiable instruments they can be bought, sold or discounted in secondary markets, providing liquidity.

Disadvantages

- ✗ There is a risk that the party obligated to pay the bill may not accept it or default on payment, leading to financial losses.
- ✗ The documentation and verification processes involved can be administratively complex.
- ✗ As their main use is trade-related they may not be suitable for all short-term financing needs

4.4 Commercial Paper

i Definition

Commercial paper – short-term unsecured debt issued by high-quality companies. Investors can then trade the paper on the secondary market.

Commercial paper is appropriate for financing short-term liabilities.

Advantages	Disadvantages
✓ It is a cost-effective option as large sums can be raised at lower interest rates than other short-term sources. ✓ No security is required; it relies on the creditworthiness of the company. ✓ Commercial paper can be issued in various denominations with different maturities to match specific needs.	✗ Only available to large companies with investment-grade credit ratings. ✗ Concerns about the issuing company can affect the marketability of the paper. ✗ Issuers of commercial paper must comply with regulatory requirements that add administrative and legal costs.

4.5 Short-term Bank Loans

A short-term cash injection can help companies in need of working capital. Short repayment terms are typically from three months to one year.

Advantages	Disadvantages
✓ Quick access to funds for immediate financial needs is available to most companies (assuming a wellfunctioning banking system). ✓ Typically unsecured (i.e. no need to provide collateral). ✓ Short-term interest rates are usually lower than long-term interest rates due to lower credit risk on short-term debt.	✗ Arrangement fees may be high when expressed as an annual effective cost. ✗ If security is required, this can tie up valuable assets. ✗ Refinancing risk (rollover risk). Every time a short-term loan matures, the borrower faces the risk that it cannot be easily replaced or refinanced or that interest rates have risen.

5.1 Debt Finance

Sources of debt finance available to SMEs include:

- Trade credit (i.e. obtaining goods or services from suppliers on credit terms).
- Factoring and invoice discounting (whereby a company effectively raises finance against the security of its outstanding receivables – see *Chapter 16*). These forms of short-term financing may be more expensive than a bank overdraft, but the finance available grows as the SME grows (based on the level of receivables).
- Leasing is useful for SMEs as it avoids raising capital to acquire assets. For SMEs, leasing can be cheaper than borrowing because:
 - Large leasing companies have bargaining power with suppliers, and so the reduced cost of assets can be partially passed on to lessees.
 - Leasing companies can usually dispose of old assets more effectively than SMEs.
 - The lease company can reclaim the asset if the lessee defaults on the lease payments, so the agreement has built in security.
 - The finance cost to a large established leasing company will likely be less than that for an SME.
- Bank finance, typically an overdraft or longer-term loans secured on major assets, is usually supported by business plans including cash flow forecasts and backed by personal guarantees of the owner-manager of the SME.

Exam advice

These options are particularly suitable for small- and medium-sized entities (SMEs), which often have difficulty finding debt finance. Such challenges may be caused by asymmetry of information (as covered in *Chapter 8*), such that banks fear making loans to entities which are not well-known and without published credit ratings.

5.2 Government Sources

5.2.1 Grants and Subsidies

Depending on the location and nature of the SME, regional, national or even international grants may be available, either to help start up the business or to contribute towards expansion costs. Subsidies may take the form of government loans offered to SMEs at interest rates below commercial levels.

5.2.2 Government Loan Guarantee Schemes

Government loan guarantee schemes exist in various countries to support businesses and stimulate economic activity. Under such schemes, the government act as a guarantor for commercial loans to SMEs. For example:

- Under the Enterprise Finance Guarantee scheme in the UK, the government will, for a small fee from the SME, guarantee a large proportion of any loan advanced by a bank. Therefore, banks are potentially more willing to lend, as most of their risk has been eliminated. The SME must, however, be able to demonstrate to the lender that it should be able to repay the loan in full.
- The Small Business Administration in the US offers various loan guarantee programs to support small businesses that provide guarantees to lenders, encouraging them to provide loans to small businesses that might not otherwise qualify for traditional financing.
- The German government-owned development bank, Kreditanstalt für Wiederaufbau (KfW) operates various loan guarantee programs to support different sectors, including SMEs.

5.3 Business Angels

As described in *Chapter 8*, business angels are wealthy individuals prepared to invest money and time in small companies if they see high growth potential.

If prepared to invest in debt, they also may want the opportunity for future equity participation. Therefore, convertible debt or debt with warrants may be appropriate.

5.4 Supply Chain Financing

Definition

Supply Chain Finance (SCF) – the use of financial instruments, practices and technologies to optimise the management of the working capital and liquidity tied up in supply chain processes for collaborating business partners.

SCF provides short-term credit that optimises working capital for both the buyer and the seller. It generally involves using a technology platform to automate transactions and track invoice approval and settlement processes from initiation to completion.

The growing popularity of SCF has been primarily driven by the increasing globalisation and complexity of the supply chain, especially in industries such as car manufacturing and the retail sector.

SCF can be particularly beneficial to SMEs with a poor credit rating (or lacking a credit history) that supply goods or services to a large company with a good credit rating. It allows the SME to take advantage of its customer's superior credit rating and receive immediate payment at a small discount to the sales invoice's face value. The customer may also benefit from its bank offering extended payment terms on the invoice.

However, a disadvantage for SME suppliers is that larger trading partners may be able to abuse their power by demanding lower prices from them in exchange for arranging SCF.

Example 4 Supply Chain Financing

Aplus Co, a large company with a good credit rating, buys goods from Beplus Co, an SME, on 30 days credit.

- Beplus Co issues a sales invoice to Aplus Co on standard terms (i.e. payment within 30 days).
- Aplus Co approves the invoice.
- Aplus Co is expected to pay the amount due to its financial institution, Bank C, in 30 days. Bank C would then make an immediate payment to Beplus Co.
- However, Beplus can request funds from Bank C any time during the 30 days. In this case Beplus will receive the amount due less a small discount which is Bank C's fee for the (minor) risk that the invoice will not be paid (by Aplus Co).
- If Aplus Co wants to delay payment to Bank C beyond 30 days it may ask for extended credit (e.g. an additional 30 days' credit). Aplus Co will be charged bank interest at a rate which reflects its credit rating.
- In the unlikely event that Aplus fails to pay Bank C, it is Bank C that suffers the bad debt.

5.5 Peer-to-peer (P2P) Lending

1 Definition

Peer-to-peer (P2P) lending – a method of debt financing that enables individuals to lend money to small businesses without using an official financial institution as an intermediary.

P2P lending is also referred to as "debt-based crowdfunding".

Advantages

- ✓ Loans generate interest income for *lenders*, often exceeding that earned on a bank deposit account.
- ✓ *Borrowers* have access to finance when banks refuse credit or would charge very high interest rates.
- ✓ By effectively cutting out the middleman (i.e. the formal banking system), interest rates can be relatively attractive to lenders and borrowers.

Example 5 P2P Lending

LendingClub is a debt-based crowdfunding platform that offers small business loans from 5,000 – \$500,000 with terms of up to five years with relatively low interest rates starting at 5.99% per year.

Exam advice

If asked to recommend a suitable financing method, consider the following factors (in so far as the information permits):

Availability. Finance may be limited if recent or forecast performance is poor and an SME will always find it difficult to raise equity.

Cash flow. Debt requires a company to pay out cash in the form of interest; therefore, the company's ability to generate cash will be relevant.

Control. Raising equity can lead to a change in control, but debt will not.

Cost. Debt finance is cheaper than equity finance, so if the company can take on more debt, there could be a cost advantage.

Ease and cost of issue. Raising equity is more difficult, time consuming and expensive than raising debt.

Maturity. In general, the term of the finance should match the term of the need (matching principle). However, the maturity dates of existing debt should also be considered.

Risk. Company directors must control the total risk (financial risk and business risk). If business risk rises, the company may seek to reduce financial risk and vice versa.

Security and covenants. Debt may require security (is this available?) or covenants (are these acceptable?) to maintain a certain liquidity level.

Yield curve. If, for example, the yield curve is getting steeper (see *Chapter 2*), there is an expectation that interest rates will rise. Therefore, fixed-rate debt may be appropriate if a company wishes to take on more debt.

Syllabus Coverage

This chapter covers the following Learning Outcomes.

E. Business Finance

1. Sources of, and raising, business finance

- a) Identify and discuss the range of short-term sources of finance available to businesses, including:
 - i) overdraft
 - ii) short-term loan
 - iii) trade credit
 - iv) lease finance.
- b) Identify and discuss the range of long-term sources of finance available to businesses, including:
 - ii) debt finance
 - iii) lease finance.

5. Finance for small- and medium-sized entities (SMEs)

- d) Identify and evaluate the financial impact of sources of finance for SMEs, including sources already referred to in syllabus section E1 and also:
 - i) Business angel financing
 - ii) Government assistance
 - iii) Supply chain financing
 - iv) Peer-to-peer funding.

Summary and Quiz

- Although legally they are equity, preference shares are in substance debt, as they pay a fixed committed dividend in priority to any ordinary dividend. They also rank ahead of ordinary shareholders on liquidation (although after "real" debt, such as bank loans and loan notes).
- Preference shareholders face lower risk than ordinary shareholders and require lower returns.
- Banks and holders of loan notes take lower risks as they rank ahead of preference shareholders on bankruptcy, and their debts may be secured by a fixed or floating charge over assets. Therefore, providers of loans require lower returns than other providers of finance.
- Loan notes can be secured or unsecured. The security can be specific assets (a fixed charge) or a class of assets (a floating charge). The assets are used as security and are sold to pay the loan note if default occurs.
- Deep discount loan notes are loan notes issued at a large discount to nominal value (issued well below nominal value) and redeemable at par on maturity.
- The ultimate deep discount loan note is a zero-coupon loan note, which is issued at a discount to nominal value and pays no coupon.
- Convertibles are loan notes or preference shares which can be converted into ordinary shares.
- A warrant offers the investor the right, but not the obligation, to purchase new shares at a future date at a fixed price.
- Medium-term finance includes bank loans, leasing, sale and leaseback and bank loans.
- Short-term finance includes bank overdrafts, trade credit, bills of exchange, shortterm bank loans and commercial paper. Commercial paper is short-term unsecured debt issued by high-quality companies, which investors can trade.
- SMEs may need to look at grants, subsidies, loan guarantees, business angels, supply chain financing and peer-to-peer lending to raise debt finance.

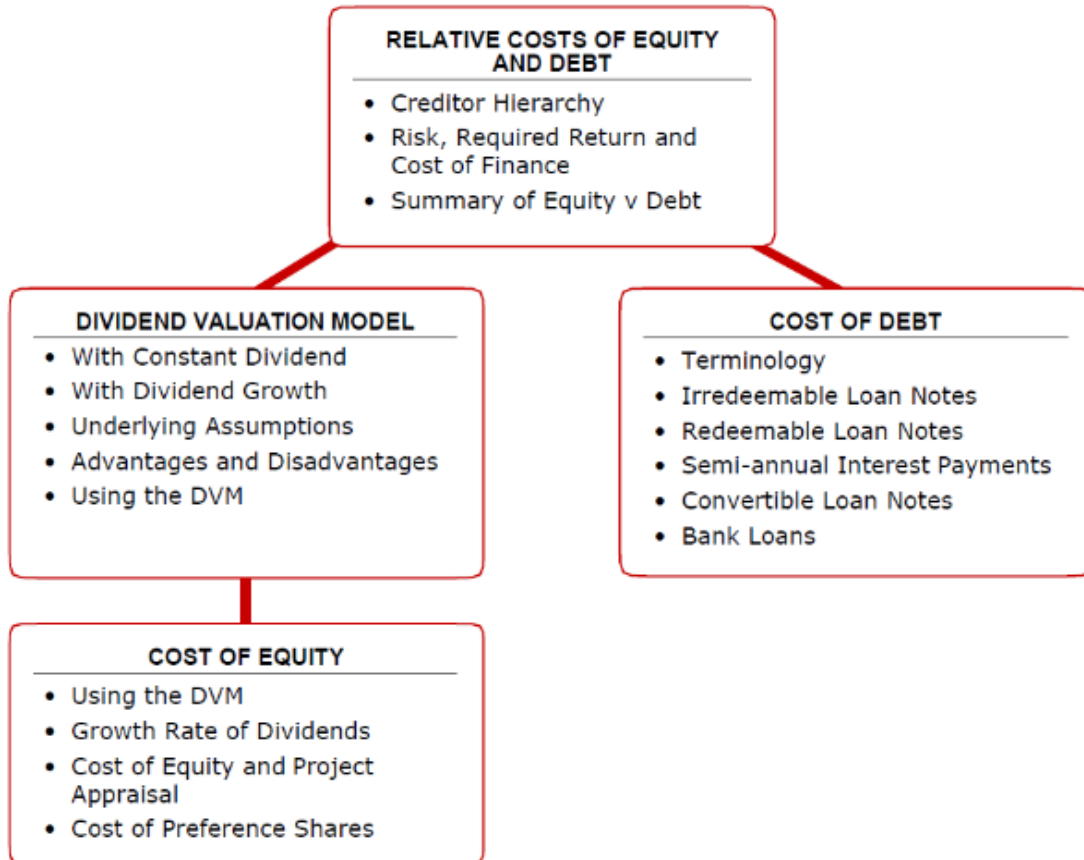
Chapter 9 Quiz

Chapter 10: Cost of Capital

Overview	CHAPTER 10: Visual Overview
10.1 Relative Costs of Equity and Debt	10.1.1 Creditor Hierarchy
	10.1.2 Risk, Required Return and Cost of Finance
	10.1.3 Summary of Equity v Debt
10.2 Dividend Valuation Model (DVM)	10.2.0 Introduction
	10.2.1 With Constant Dividends
	10.2.2 With Dividend Growth
	10.2.3 Underlying Assumptions
	10.2.4 Advantages and Disadvantages of the DVM
	10.2.5 Using the DVM
10.3 Cost of Equity	10.3.1 Using the DVM
	10.3.2 The Growth Rate of Dividends
	10.3.3 Cost of Equity and Project Appraisal
	10.3.4 Cost of Preference Shares
10.4 Cost of Debt	10.4.1 Terminology
	10.4.2 Irredeemable Loan Notes
	10.4.3 Redeemable Loan Notes
	10.4.4 Semi-annual Interest Payments
	10.4.5 Convertible Loan Notes
	10.4.6 Bank Loans

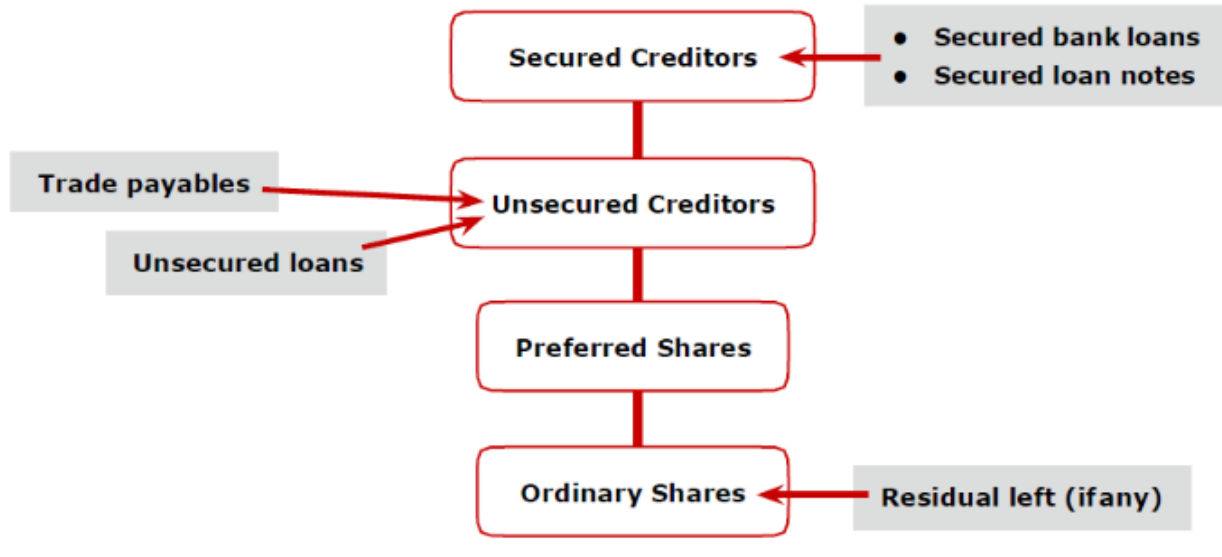
CHAPTER 10: Visual Overview

Objective: To estimate the cost of equity and the cost of debt.



1.1 Creditor Hierarchy

The relative risk/return relationship of equity and debt is based on their relative priority for repayment on liquidation – the creditor hierarchy:



On liquidation, the company's assets are sold and the cash raised is paid out according to the priority of creditors:

- Secured loans and secured loan notes are repaid first. Some may be secured by a "fixed charge" over a specific asset such as property, others by a "floating charge" over classes of assets such as working capital. Secured creditors expect to receive most, if not all, of what they are owed in the event of liquidation.
- Trade creditors and unsecured debt are repaid next. When a company is forced into liquidation, by definition the value of its assets is likely to be below the value of its liabilities. In this case, unsecured creditors may not receive everything they are owed.
- Preference shareholders would be repaid next, if there is any cash remaining after paying all creditors.
- Equity shareholders rank last on liquidation and are unlikely to receive anything.

In practice, other stakeholders may have claims on liquidation (e.g. employees, tax authorities and the liquidator). Exam questions would state where in the hierarchy these would rank, as it depends on the jurisdiction.

1.2 Risk, Required Return and Cost of Finance

Key Point

The risk faced by each class of investor drives the required return of that investor (there is a risk/return pay off) and the required return drives the company's cost of each source of finance.

1.2.1 Secured Loans and Secured Loan Notes

Debt investors have legally binding contracts with the company to receive interest and repayment of principal. While the company is trading, interest must be paid in priority to dividends. If the company goes into liquidation, secured creditors are repaid first. Therefore, secured debt can be considered a low-risk investment and investors in secured debt require relatively low returns, creating relatively cheap finance for the company.

1.2.2 Unsecured Debt

Unsecured debt is also legally binding, and its interest must be paid before dividends. However, as there is no guarantee of full repayment on liquidation, the required return will be higher than on secured debt, and hence the cost will be higher.

1.2.3 Preference Shares

Preference dividends are a fixed percentage of the share's nominal value paid after interest on debt and before dividends on equity shares ("ordinary dividends"). On liquidation, preference shares rank between creditors and equity shares, so the required return of preference shareholders (and hence the company's cost of preference capital) will be higher than on debt but lower than on equity shares.

1.2.4 Equity (Ordinary) Shareholders

Equity shareholders have no guarantee of receiving dividends (ordinary dividends are discretionary, whereas preference dividends are committed) and rank last on liquidation. Therefore, equity shareholders face high risk and expect high returns to compensate, making the company's cost of equity relatively high.

1.3 Summary of Equity v Debt

	Equity	Debt
Rank on liquidation	Last	Higher
Servicing of finance	Discretionary dividend	Committed interest
Risk to investor	High	Lower
Return required	High	Lower
Cost to company	High	Lower
	Cost of equity > cost of debt	
Servicing tax allowable?	No	Yes
	Post-tax cost of debt < pre-tax cost of debt	
Speed of issue	Slow	Faster
Issue costs	5–11% (IPO)	Loan notes 1–2% Bank loan – arrangement fees

2.0 Introduction

This section explains the dividend valuation model that can be used to provide the market value of a share.

The model can be rearranged to estimate the return required by shareholders.

The following section, builds on the model to determine the cost of equity.

2.1 With Constant Dividends

The formula for share valuation can be derived as follows:

$$\text{Ex-div market value at time 0} = \text{Present value of the future dividends discounted at the shareholder's required rate of return}$$

The ex-div market value assumes that a dividend has just been paid (or is just about to be paid).

Let:

P_0 = Current ex-div market value

D_n = Dividend at time n

r_e = Shareholders' required rate of return

The model then becomes:

$$P_0 = \frac{D_1}{(1+r_e)} + \frac{D_1}{(1+r_e)^2} + \frac{D_1}{(1+r_e)^3} \dots \frac{D_n}{(1+r_e)^n}$$

If the dividend is assumed to be *constant to infinity*, this becomes the present value of a perpetuity, which simplifies to:

$$P_0 = \frac{D}{r_e}$$

This version of the model can be used to determine the theoretical value of a share which pays a constant dividend (e.g. a preference share or an equity share in a zero growth company).

Exam advice

The formula for the present value of a perpetuity is assumed knowledge; it is not provided in the exam formula sheet.

This derivation of the DVM is **not** examinable but provided to emphasise that for this formula to “work” the current market value P_0 must be **ex-div** so the first dividend is receivable at time 1.

2.2 With Dividend Growth

If dividends are forecast to *grow* at a constant rate in perpetuity, where g = future annual growth rate:

$$P_0 = \frac{D_0(1+g)}{r_e - g} = \frac{D_1}{r_e - g}$$

Where D_0 = most recent dividend

D_1 = dividend in one year

r_e = required return of equity investors

The formula is published in the exam as:

$$P_0 = \frac{D_0(1+g)}{(r_e - g)}$$

2.3 Underlying Assumptions

Underlying assumptions of the DVM include:

- The same expectations for all investors; therefore, they all require the same rate of return.
- Perfect capital market assumptions:
 - rational investors;
 - no taxes or transaction costs;
 - large number of buyers and sellers of shares;
 - no individual can affect the share price; and
 - perfect information is freely available to all investors.
- Dividends are paid just once a year and one year apart.
- Dividends are either constant or are growing at a constant rate.

2.4 Advantages and Disadvantages of the DVM

2.4.1 Advantages

- ✓ It is easy to understand and can be applied to any share that offers a dividend.
- ✓ For minority shareholders without control over a company's policies, dividends are the only available metric.
- ✓ The model is not subjective (e.g. there is no ambiguity in determining amounts of dividends whereas "earnings" or "profit" are open to interpretation).
- ✓ It is particularly suitable for "mature" companies paying regular dividends.
- ✓ It is based solely on dividends, taking no account of market conditions, making comparisons across companies of different sizes and industries easier.

2.4.2 Disadvantages

- ✗ It cannot be applied to shares that do not pay dividends therefore, it cannot be used for smaller businesses and startups.
- ✗ It assumes shares have no issue costs.
- ✗ It does not explicitly incorporate risk.
- ✗ It is overly simplistic (e.g. can only be an approximation because dividends do not grow at a constant rate in reality).
- ✗ It makes no allowance for non-dividend factors that influence the value of a share (e.g. brand loyalty and other internally generated intangible assets).
- ✗ It makes no allowance for the effect of taxation.
- ✗ It ignores capital gains tax on investors (although it could be argued that the change of ownership of a share does not affect the PV of a dividend stream).

2.5 Using the DVM

If the company is listed and the share price is known, the model can be rearranged and used to estimate shareholders' required return, r_e .

Example 1 Investors' Required Return

A share has a current ex-div market value of \$0.80. Investors expect a dividend of \$0.10 per share to be paid each year, as in recent years.

Using the DVM, the investors' required return can be calculated as follows:

$$P_0 = \frac{D}{r_e}$$

$$\$0.80 = \frac{\$0.10}{r_e}$$

$$r_e = \frac{\$0.10}{\$0.80} = 12.5\%$$

All investors require this return, as the model assumes they are all rational and have the same information about the risk of this share.

If investors think the annual dividend will increase to \$0.15, the share gives a higher than 12.5% return at \$0.80 a share. Therefore, investors will buy the share, and the price will increase until, according to the model, the value will be:

$$P_0 = \frac{\$0.15}{0.125} = \$1.20$$

Alternatively, suppose that the investors' perceive that the dividend will remain at \$0.10 but require a 15% return because the risk of the share has increased. If the share only gives a return of 12.5% (on an \$0.80 share price), investors will sell, and the price will fall. The fair value of the share according to the model will be:

$$P_0 = \frac{\$0.10}{0.15} = \$0.67$$

3.1 Using the DVM

3.1.1 Constant Dividends

The basic DVM is:

$$P_0 = \frac{D}{r_e}$$

As seen in *Example 1*, this can be rearranged to find r_e :

$$r_e = \frac{D}{P_0}$$

If r_e is the return required by the shareholders so that the share value remains constant, it is also the return the company must pay its shareholders. Therefore, r_e also equates to the company's cost of equity, k_e .

Therefore, the cost of equity for a company with a constant annual dividend can be estimated as the dividend divided into the ex-div share price (i.e. the *dividend yield*).

The **ex-div** market value is the market value of the share, assuming that the current dividend has just been paid.

A **cum-div** market value includes the value of the dividend just about to be paid. If given a cum-div market value, this must be adjusted to an ex-div market value by deducting the current dividend.

Activity 1 Cost of Equity

[Show](#)

A company's shares have a market value of \$2.20 each. The company is about to pay a dividend of \$0.20 per share, as for the last ten years.

Required:

Calculate the company's cost of equity.

**Please use the notes feature in the toolbar to help formulate your answer.*

$$P_0 \text{ (cum-div)} = \$2.20$$

$$P_0 \text{ (ex-div)} = \$2.00$$

$$r_e = k_e = \frac{D}{P_0} = \frac{\$0.20}{\$2.00} \times 100\% = 10\%$$

3.1.2 Dividend with Constant Growth

The model can also deal with dividend growth at a constant annual rate, g .

The formula for valuing the share, as seen earlier, is:

$$P_0 = \frac{D_0(1+g)}{r_e - g} = \frac{D_1}{r_e - g}$$

Where D_0 = most recent dividend

D_1 = dividend in one year

Rearranged, this becomes:

$$r_e = \frac{D_0(1+g)}{P_0} + g$$

Where g = growth rate (assumed constant in perpetuity)

P_0 = ex-div market value

Therefore, if shareholders' required return is the same as the cost of equity, the cost of equity, k_e = Dividend yield + estimated growth rate.

Example 2 Dividend with Constant Growth

$$D_0 = \$0.12$$

$$P_0 \text{ (ex-div)} = \$1.75$$

$$g = 5\%$$

Calculate the value of k_e .

$$k_e = \frac{0.12(1.05)}{1.75} + 0.05 = 12.2\%$$

Activity 2 Dividend with Constant Growth

[Show](#)

The dividend just about to be paid by a company is \$0.24. The current market price of the share is \$2.76 cum-div. The historical dividend growth rate, which is expected to continue in the future, is 5%.

Required:

Calculate the estimated cost of equity.

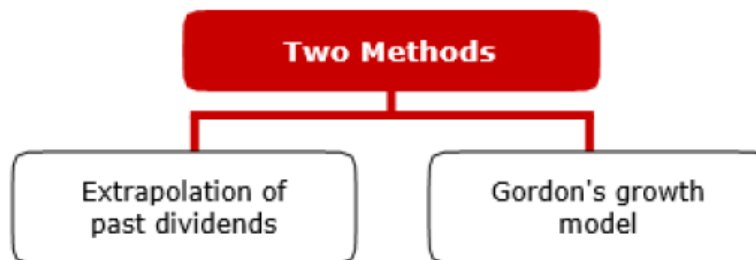
**Please use the notes feature in the toolbar to help formulate your answer.*

Ex-div market price, $P_0 = \text{Cum-div market price} - \text{dividend} = 2.76 - 0.24 = 2.52$

$$r_e = k_e = \frac{D_0(1+g)}{P_0} + g = \frac{0.24(1+0.05)}{2.52} + 0.05 = 15\%$$

3.2 The Growth Rate of Dividends

The growth rate of dividends can be estimated using either of two methods:



3.2.1 From Past Dividends

This method analyses historical growth to predict future growth:

- If dividends have grown 5% each year for the last ten years, the predicted future growth is 5%.
- For uneven but steady growth, take an average overall growth rate – the “geometric mean”.
- If there is a discontinuity in the growth rate, take the most recent evidence.
- For a new company with very high growth rates, consider that it is unlikely to produce such high growth in perpetuity.
- If there is no pattern (i.e. dividends up one year, down the next), do not use the DVM.

Activity 3 Growth Using Extrapolation

[Show](#)

A company has paid the following dividends over the last five years:

Dividend per share	
20X0	\$1.00
20X1	\$1.10
20X2	\$1.25
20X3	\$1.36
20X4	\$1.45

The current (20X4) ex-div market value is \$10.50 per share.

Required:

Estimate the growth rate and the cost of equity.

**Please use the notes feature in the toolbar to help formulate your answer.*

Estimate the growth rate and the cost of equity.

**Please use the notes feature in the toolbar to help formulate your answer.*

Over the period 20X0–20X4, there are four changes in dividend (i.e. four periods of growth). g can be estimated by considering the relationship between the earliest and latest dividends:

$$1.00 (1 + g)^4 = 1.45$$

Therefore:

$$1 + g = \sqrt[4]{\frac{1.45}{1.00}} = 1.097$$

$$g = 9.7\%$$

The cost of equity is then:

$$r_e = k_e = \frac{D_1}{P_0} + g = \frac{1.45(1.097)}{10.50} + 0.097 = 24.8\%.$$

Alternatively, g , can be calculated more accurately using the geometric mean function, GEOMEAN, available in the spreadsheet:

C7		=GEOMEAN(C3:C6)		
	A	B	C	D
1		Dividend	Growth	
2	20X0	1		
3	20X1	1.1	0.1	
4	20X2	1.25	0.15	
5	20X3	1.36	0.11	
6	20X4	1.45	0.09	
7			0.11	

The cost of equity is then $\frac{1.45(1.11)}{10.50} + 0.11 = 26.3\%$.

3.2.2 Gordon's Growth Model

According to Gordon's growth model, growth is achieved by retaining and reinvesting profits:

$$g = br_e$$

Where b = proportion of profits retained ("retention ratio")

r_e = rate of return on those retained profits (the return on capital employed)

Take an average of r and b over the preceding years to estimate future growth:

$$r_e = \frac{\text{Profit after tax}}{\text{Shareholder's funds}} = \frac{\text{Profit after tax}}{\text{Net assets}}$$

$$b = \frac{\text{Retained profit}}{\text{Profit after tax}}$$

These figures can be obtained from the statement of financial position and statement of profit or loss.

Activity 4 Gordon Growth Model

[Show](#)

A company has 300,000 equity shares in issue with an ex-div market value of \$2.70 per share. A dividend of \$40,000 has just been paid out of post-tax profits of \$100,000.

Net assets at the year end were valued at \$1.06m.

Required:

Estimate the cost of equity.

**Please use the notes feature in the toolbar to help formulate your answer.*

Growth rate, $g = br$

$b = \% \text{ profit retained}$

$$= \frac{60,000}{100,000} = 60\%$$

$r = \text{return on capital employed}$

$$= \frac{\text{Profit after tax}}{\text{Opening net assets}} = \frac{100,000}{1,060,000 - 60,000} \times 100 = 10\%$$

$$g = 0.6 \times 0.1 = 0.06 = 6\%$$

$$r_e = k_e = \frac{D_1}{P_0} + g$$

$$r_e = \frac{40,000(1.06)}{300,000 \times 2.70} + 0.06 = 11.2\%$$

Note: The return on capital employed could alternatively be calculated on average net assets instead of opening net assets.

Key Point

A company cannot alter its cost of equity by changing its dividends, which the equation for the DVM might suggest. A cut in dividends or the growth rate would cause the market value of the shares to fall to the level at which investors would obtain the return they require.

3.3 Cost of Equity and Project Appraisal

Example 3 Effect of a Project on Value of Equity

Axis Co, a listed company, is all equity financed and has one million shares quoted at \$2 each ex-div. It pays constant annual dividends of \$0.30 per share.

It is contemplating a project that will cost \$500,000 and has the same risk as its existing activities. A rights issue will finance the project. The project will produce inflows of \$90,000 per year in perpetuity. All inflows will be distributed as dividends.

Required:

Calculate the new value of the equity in Axis Co and the gain to the shareholders.

Ignore tax.

$$k_e = \frac{0.30}{2.00} = 15\%$$

	\$000
Existing total dividend	300
Dividends from the project	<u>90</u>
New total dividend	<u>390</u>

$$\begin{aligned}\text{Value of equity} &= \frac{\$390,000}{0.15} = \$2.6\text{m} \\ \text{Shareholders' gain} &= \$ (2.6\text{m} - 2\text{m}) - \$0.5\text{m} = \$100,000\end{aligned}$$

$$\text{Project NPV} = (\$500,000) + \frac{\$90,000}{0.15} = \$100,000$$

Therefore, new value of equity

$$= \text{Existing value} + \text{Equity outlay} + \text{NPV}$$

$$= \text{Existing value} + \text{PV of additional dividends}$$

As shown by *Example 3*, the NPV of a project serves to increase the value of the company's shares (i.e. the NPV of a project shows the increase in shareholders wealth).

This demonstrates that NPV is consistent with the assumed objective of maximising shareholder wealth: \$1 of NPV is \$1 increase in shareholders wealth.

3.4 Cost of Preference Shares

By definition, preference shares have a constant dividend:

$$k_e = \frac{D}{P_0}$$

Where D = constant annual dividend

Preference dividends are typically quoted as a percentage (e.g. 10% preference shares). This means the annual dividend will be 10% of the *nominal* value, **not** the market value.

Activity 5 Cost of Preference Shares

[Show](#)

A company has 100,000 12% preference shares in issue (nominal value \$1).

The current ex-div market value is \$1.15 per share.

Required:

Calculate the cost of the preference shares.

**Please use the notes feature in the toolbar to help formulate your answer.*

12% preference shares: dividend is $12\% \times \text{nominal value}$

$$r_e = k_e = \frac{D}{P_0} = \frac{\$0.12}{\$1.15} \times 100 = 10.4\%$$

4.1 Terminology

① Definition

Loan note ‐ a security instrument (so it is tradable) that acknowledges a company's debt.

Also called a **bond** or **debenture**, a loan note:

- Usually pays a fixed coupon.
- May be secured or unsecured.
- If quoted, it will trade on an exchange at a price determined by that market.

① Definition

Coupon rate ‐ the interest rate printed on the loan note certificate.

Therefore:

$$\text{Annual interest} = \text{coupon rate} \times \text{nominal value}$$

① Definition

Nominal value ‐ the value that is stated on an issued security.

Nominal value is also known as par value or face value.

💬 Exam advice

In the exam the nominal value of one loan note is usually \$100.

Market value (MV) is typically quoted as the MV of \$100 nominal value.

Example 4 Loan Note Terminology

10% loan notes quoted at \$95 means that \$100 is selling for \$95, and annual interest is \$10 per \$100.

Market value (ex-int) is when interest has just been paid.

Market value (cum-int) includes the value of accrued interest which is just about to be paid.

4.2 Irredeemable Loan Notes

Irredeemable loan notes entitle the holder to receive interest each year indefinitely. They are also referred to as undated loan notes or perpetual bonds.

The market value of undated debt can be calculated using the DVM:

Key Point

MV (ex-interest) = PV of future interest payments discounted at the debtholder's required rate of return.

For irredeemable loan notes the interest is a *perpetuity*.

$$P_0 = \frac{I}{k_d}$$

Where:

P_0 = Ex-interest market price

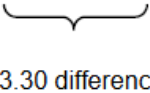
I = Annual interest payment

k_d = Bondholders' required return

The company's **pre-tax** cost of irredeemable bonds is the debtholder's required return $k_d = \frac{I}{P_0}$. This is the **interest yield**. If the company gets tax relief on interest expense, the cost to the company will be reduced by the "tax shield".

Example 5 Tax Shield of Debt Interest

Consider two companies with the same earnings before interest and tax (EBIT). The first company has some debt finance, the second has no debt.

	\$	\$
EBIT	100	100
Debt interest	<u>(10)</u>	
Profits before tax	90	100
Tax @ 33%	29.70	33
		

Therefore:

	\$
Debt interest	10.00
Less: tax shield	<u>(3.30)</u>
Effective cost of debt	<u>6.70</u>

Exam advice

An exam question will state whether the tax liability is settled in the year interest is paid (i.e. tax relief is “instant”) or at the end of the following year.

If the debt is **irredeemable**:

$$\begin{aligned}\text{Cost of debt to the company ("post-tax" cost of debt)} &= \text{Debt holders' required holders} \times (1 - T_c) \\ &= k_d \times (1 - T_c)\end{aligned}$$

Where T_c = corporate tax rate as a decimal

Key Point

k_d is used to denote the **pre-tax** cost of debt.

Activity 6 Cost of Irredeemable Debt

[Show](#)

12% undated loan notes with a nominal value of \$100 are quoted at \$92 cum-interest. The rate of corporation tax is 33%.

Required:

Calculate:

- a) **the return required by the debt investors; and**
- b) **the cost to the company.**

**Please use the notes feature in the toolbar to help formulate your answer.*

a) Return required by debt investors

$$r_d = \frac{\text{Int}}{\text{MV}_{\text{ex-int}}} = \frac{12}{92-12} \times 100 = 15\%$$

a) Cost to the company

$$k_d = \frac{\text{Int}(1-T_c)}{\text{MV}_{\text{ex-int}}} = \frac{12(1-0.33)}{92-12} = 10.05\%$$

4.3 Redeemable Loan Notes

Definition

Redeemable loan note – a loan note that can be redeemed by the issuer before its maturity date. Early redemption would usually be at a premium above the loan note's nominal value.

The cash flows are not a perpetuity because the principal will be repaid. However, the following rule is derived from the DVM:

Key Point

The cost of any source of funds is the IRR of the cash flows associated with that source.

- Looking at the return from an *investor's* point of view, interest payments are *gross*.
- Looking at the cost to the *company*, interest payments are included *net* of corporation tax.
- Assume instant tax relief and that the final redemption payment has no tax effects.
- To find the cost of debt for a company, find the IRR of the following cash flows:

Time		\$
0	Market value (ex-interest)	x
1–n	Post-tax interest	(x)
n	Redemption value	(x)

The IRR is found as usual using **linear interpolation**.

Exam advice

Usually, NPV calculations present future cash flows as inflows. However, when future cash flows are presented as **outflows** (as above), if the first NPV calculation is positive, NPV should be recalculated at a **lower** discount rate to get closer to zero.

Activity 7 Cost of Redeemable Debt

[Hide](#)

A company has in issue \$200,000 7% loan notes redeemable at a premium of 5% on 31 December 20X6. Interest is paid annually on 31 December. It is 1 January 20X3 and the loan notes are trading at \$98 ex-interest per \$100 nominal value. The corporation tax rate is 33%.

Required:

Calculate the company's cost of debt.

**Please use the notes feature in the toolbar to help formulate your answer.*

Using the IRR spreadsheet function:

	A	B	C	D	E
1	Year		\$		Working
2	0	Current market value (ex-interest)	-98		
3	1	Post-tax interest	4.69		$7 \times (1 - 0.33)$
4	2	Post-tax interest	4.69		
5	3	Post-tax interest	4.69		
6	4	Post-tax interest + redemption value	109.69		$4.69 + 105$
7	IRR		6.4%		

The cost of debt = **IRR(C2:C6) = 6.4%**

Note that the working column is only shown here for completeness. In the exam, these workings should be in the relevant cells (C3 and C6), where the marker can check them if necessary.

Required return of the redeemable debt holder	=	IRR of pre-tax cash flows from the bond	=	Gross redemption yield
---	---	---	---	------------------------

Gross redemption yield is also called **yield to maturity** (YTM).

4.4 Semi-annual Interest Payments

In practice, interest is usually paid every six months rather than annually. This practical aspect can be built into calculations for the cost of debt as follows:

- If interest is paid every six months, the IRR calculation must be based on cash flows at six-month intervals.
- The IRR, or cost of debt, will then be a six-month cost of debt and must be adjusted to determine the annual cost of debt.

$$\text{Effective annual cost} = (1 + \text{semi-annual cost})^2 - 1$$

Activity 8 Semi-annual Interest

[Show](#)

A company has 6% loan notes, the interest on which is paid on 30 June and 31 December each year. The loan notes are redeemable at \$100 nominal value on 31 December 20X9. It is now 1 January 20X7 and the loan notes are quoted at \$96.

Required:

Calculate the effective annual cost of debt, k_d . Ignore corporation tax.

**Please use the notes feature in the toolbar to help formulate your answer.*

Time 0 is 1 January 20X7. Interest payments are due:

30 June X7 Time 1

31 Dec X7 Time 2

30 June X8 Time 3

31 Dec X8 Time 4

30 June X9 Time 5

31 Dec X9 Time 6

Each interest payment will be just half the coupon rate, \$3 every 6 months.

Time	Cash flow	DF @ 3%	PV @ 3%	DF @ 5%	PV @ 5%
0	(96)	1	(96.00)	1	(96.00)
1-6	3	5.417	16.25	5.076	15.23
6	100	0.837	<u>83.70</u>	0.746	<u>74.60</u>
			<u>3.95</u>		<u>(6.17)</u>

$$\text{IRR} = 3 + \frac{3.95}{3.95 + 6.17} \times (5 - 3) = 3.78\%$$

This is the six-month cost of debt.

$$\text{IRR} = 3 + \frac{3.95}{3.95 + 6.17} \times (5 - 3) = 3.78\%$$

This is the six-month cost of debt.

The effective annual cost of debt is $(1.0378^2) - 1 = 7.7\%$

4.5 Convertible Loan Notes

Convertible loan notes allow the investor to choose between redemption at some future date or conversion into a predetermined number of equity shares (see *Chapter 9*).

To find the **post-tax cost of convertible debt**, find the IRR of the **after-tax** cash flows:

Time		\$
0	Market value (ex-interest)	x
1-n	Post-tax interest	(x)
n	Higher of redemption value/forecast conversion value	(x)

Activity 9 Convertible Loan Notes

[Show](#)

A company has in issue 8% convertible loan notes currently quoted at \$85 ex-interest. The loan notes are redeemable at a 5% premium in five years or can be converted into 40 equity shares at that date. The current ex-div market value of the shares is \$2 per share and dividend growth is expected at 7% per year. The corporation tax rate is 33%.

Required:

Calculate the cost to the company of the convertible loan notes.

**Please use the notes feature in the toolbar to help formulate your answer.*

First, decide whether the loan notes will be converted in five years by comparing the expected value of 40 shares in five years with the cash alternative.

Assuming that the MV of shares will grow at the same rate as the dividends: MV per share in five years = $2(1.07)^5 = \$2.81$

MV of 40 shares $\times \$2.81 = \112.40 , compared to cash alternative = \$105 (redeemed at 5% premium).

Therefore all debt holders will choose the share conversion.

To find the cost to the company, find the IRR of the post-tax flows.

	A	B	C	D
1	Year	\$		Working
2	0	-85		
3	1	5.36		$8*(1-0.33)$
4	2	5.36		
5	3	5.36		
6	4	5.36		
7	5	118		$5.36+112.4$
8	IRR	11.5%		

Therefore, cost to the company = $\text{IRR}(\text{B2:B7}) = 11.5\%$.

Activity 10 Different Conversion/Redemption Dates

[Show](#)

A company has 5% \$100 loan notes in issue, which are redeemable at nominal value in eight years. Alternatively, each loan note is convertible after seven years into 11 equity shares. The company's equity shares are currently trading at \$6.50 per share, which is expected to increase by 6% a year. The current market value of a loan note is \$88.70. The corporate tax rate is 30%.

Required:

Calculate the cost to the company of the convertible loan notes.

**Please use the notes feature in the toolbar to help formulate your answer.*

The company's after-tax cost of debt can be estimated as the IRR of the relevant cash flows:

	A	B	C	D	E
1	Year		\$		Working
2	0	Market price	-88.7		
3	1	Post-tax interest	3.5		$5 \times (1 - 0.3)$
4	2	Post-tax interest	3.5		
5	3	Post-tax interest	3.5		
6	4	Post-tax interest	3.5		
7	5	Post-tax interest	3.5		
8	6	Post-tax interest	3.5		
9	7	Post-tax interest + conversion	111		$3.5 + 107.5 (W)$
10	IRR		6.4%		

Therefore, cost to the company = $\text{IRR}(C2:C9) = 6.4\%$.

WORKING

The share price in seven years is expected to be \$9.77 per share (6.50×1.067), and the conversion value will be \$107.47 per loan note (9.77×11). Holders of the loan notes would prefer conversion after seven years rather than redemption after eight years for \$105 (redemption price + interest for year 8). Therefore, convert at \$107.50, say, as a close approximation.

4.6 Bank Loans

Unless the exam question states otherwise, the interest rate quoted on a bank loan can be assumed to be the pre-tax cost.

A profitable company should be able to claim tax relief on bank interest, therefore the post-tax cost of bank loans = Quoted interest rate $\times$ (1 – corporation tax rate).

As bank loans are **not** traded, their book value must be used as a proxy for market value.

Syllabus Coverage

This chapter covers the following Learning Outcomes.

B. Financial Management Environment

2. The nature and role of financial markets and institutions

- d) Explain the nature and features of different securities in relation to the risk/return trade-off.

E. Business Finance

2. Estimating the cost of capital

- a) Estimate the cost of equity including:
- i) application of the dividend growth model, its assumptions, advantages and disadvantages
- b) Estimating the cost of debt:
- i) irredeemable debt
 - ii) redeemable debt
 - iii) convertible debt
 - iv) preference shares
 - v) bank debt.

3. Sources of finance and their relative costs

- a) Describe the relative risk-return relationship and the relative costs of equity and debt.
b) Describe the creditor hierarchy and its connection with the relative costs of sources of finance.

Summary and Quiz

- On liquidation, the company's assets are sold, and the cash raised is paid out according to the priority of creditors in which the equity shareholders rank last. Therefore, equity shareholders require the highest returns.
- If capital markets are perfect and the market price of a share is known, the dividend valuation model (DVM) can be rearranged to find the required return of investors, which is the company's cost of equity.
- If dividends are growing, the growth rate can be determined by extrapolating past dividends or using Gordon's growth model.
- For loan notes, annual interest = coupon rate $\times$ nominal value (usually \$100 in the exam). Market value (ex-int) is when interest has just been paid. Market value (cum-int) includes accrued interest that is about to be paid.
- For irredeemable loan notes, pre-tax $k_d = \frac{I}{P_0}$
- For redeemable loan notes and convertible loan notes, post-tax k_d is the IRR of the post-tax cash flows associated with the source of funds.

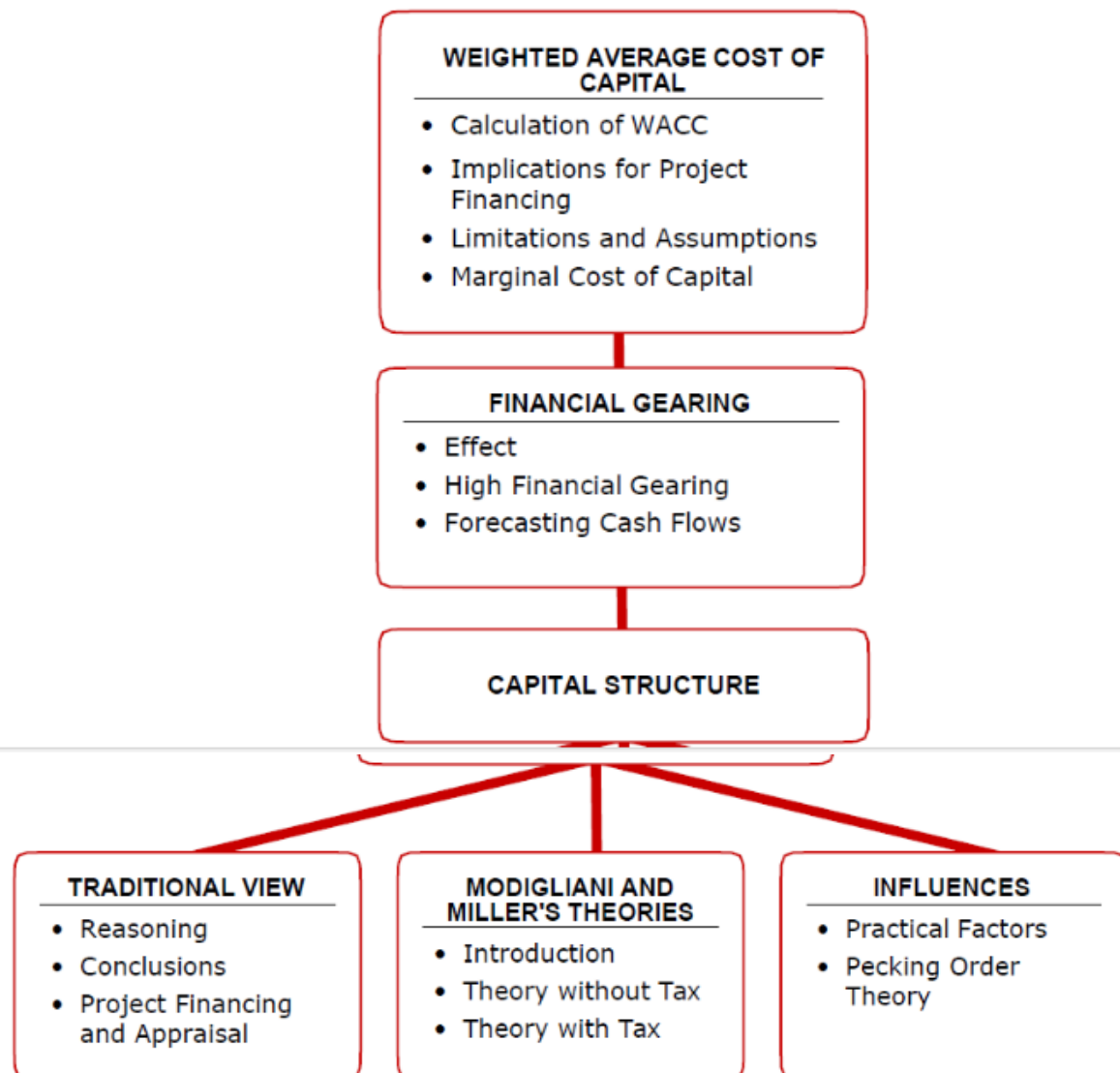
[Chapter 10 Quiz](#)

Chapter 11: Weighted Average Cost of Capital and Gearing

Overview	CHAPTER 11: Visual Overview
11.1 Weighted Average Cost of Capital (WACC)	11.1.1 Calculation of WACC
	11.1.2 Implications for Project Financing
	11.1.3 Limitations and Assumptions
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CHAPTER 11: Visual Overview

Objective: To understand the weighted average cost of capital (WACC), including how it is estimated and the effect of gearing on it; the traditional view of capital structure; Modigliani and Miller's models; and pecking order theory.



1.1 Calculation of WACC

Companies are usually financed by combination of debt and equity (i.e. they use some degree of financial/capital gearing). The WACC represents a company's average cost of long-term finance.

In the exam the formula given is:

$$\text{WACC} = \left[\frac{V_e}{V_e + V_d} \right] k_e + \left[\frac{V_d}{V_e + V_d} \right] k_d (1 - T)$$

Where:

V_e = Total market value of equity

V_d = Total market value of debt

k_e = Cost of equity "geared"

k_d = **Pre-tax** cost of debt

T = Corporation tax rate

Exam advice

$k_d(1 - T)$ in the exam formula represents the post-tax cost of debt. However, this is only correct for irredeemable loan notes.

For a **redeemable** loan note, the post-tax cost of debt must be calculated as the IRR of its post-tax cash flows (as already covered in *Chapter 10*).

Market values of equity/debt (where available) are used to weight the individual costs of capital. However:

- If the company's shares are not listed on the stock market, the book value of equity will have to be used;
- Similarly for debt (i.e. if the company has issued loan notes);
- Use book values for bank loans.

Activity 1 WACC

[Show](#)

A company has in issue:

45 million \$1 ordinary shares

10% irredeemable loan notes with a book value of \$55m

The loan notes are trading at par.

Share price	\$1.50
Dividend (just paid)	\$0.15
Dividend growth	5% p.a.
Corporation tax	33%

Required:

Estimate the WACC.

**Please use the notes feature in the toolbar to help formulate your answer.*

$$k_e = \frac{D_0(1+g)}{P_0} + g$$

$$= \frac{\$0.15(1+0.05)}{\$1.50} + 0.05 = 15.5\%$$

$$k_d = 10\% \text{ (pre-tax)} \times (1 - 0.33) = 6.7\% \text{ (after-tax)}$$

$$WACC = \left[\frac{V_e}{V_e + V_d} \right] k_e + \left[\frac{V_d}{V_e + V_d} \right] k_d (1 - T)$$

$$= 0.155 \left(\frac{45m \times \$1.50}{(45m \times \$1.50) + \$55m} \right) + 0.067 \left(\frac{\$55m}{(45m \times \$1.50) + \$55m} \right) = 11.5\%$$

WACC calculations can be set out in a spreadsheet using a tabular layout. In this example, the calculation of the WACC is D5/B5 = 11.5%, where B5 is the sum of the market values of the components of capital and D5 is the sum of their costs:

	A	B	C	D	E
1	Capital structure	MV	Cost of capital	MV * CoC	WACC
2		\$m			
3	Ordinary shares	67.5	15.5%	10.4625	
4	10% loan notes	55	6.7%	3.685	
5		122.5		14.1475	11.5%

Activity 2 WACC Using Book Values

[Show](#)

Lifestyle Co is a family-owned company whose shares are not listed on the stock market. However, the family members have stated that they require an annual return of 12% to compensate for the risk of their shareholdings.

The following is a summary of the most recent statement of financial position:

	\$m	\$m
Assets		
Non-current assets		300
Current assets		<u>211</u>
Total assets		<u>511</u>
Equity and liabilities		
Share capital		100
Retained earnings		<u>121</u>
Total equity		221
Non-current liabilities Long-term borrowings		200
Current liabilities		
Trade payables	80	
Accruals	<u>10</u>	
Total current liabilities		90
Total liabilities		<u>290</u>
Total equity and liabilities		<u>511</u>

Additional information

Additional information

Non-current assets include significant property acquired many years ago, recorded at historical cost.

The long-term borrowings consist of a bank loan at an annual interest rate of 8%. The company pays corporation tax at 25%.

Required:

- Estimate Lifestyle Co's WACC.
- Give FOUR reasons why it would be preferable to use market values rather than book values when estimating the WACC and suggest ONE improvement that Lifestyle could make to its estimate.

**Please use the notes feature in the toolbar to help formulate your answer.*

(a) WACC

As market values are not available book values have to be used as the weightings in the WACC:

$$\begin{aligned} \text{WACC} &= \left[\frac{V_e}{V_e + V_d} \right] k_e + \left[\frac{V_d}{V_e + V_d} \right] k_d (1 - T) \\ &= \left[\frac{221}{221 + 200} \right] 12\% + \left[\frac{200}{221 + 200} \right] 8\% (1 - 0.25) = 9.15\% \end{aligned}$$

Alternatively, using a spreadsheet:

	A	B	C	D	E
1	Capital structure	MV	Cost of capital	MV * CoC	WACC
2		\$m			
3	Equity	221	12.0%	26.52	
4	Debt	200	6.0%	12	
5		421		38.52	9.15%
-					

(b) Market value**Why preferable**

Although the market value of debt is often reasonably close to its book value (and in any case, bank loans are not traded on a secondary market), the market value of equity is often significantly above the carrying amount of net assets, as shown on the statement of financial position. In such cases, it would be preferable to weight the cost of equity using the market value of equity for the following reasons:

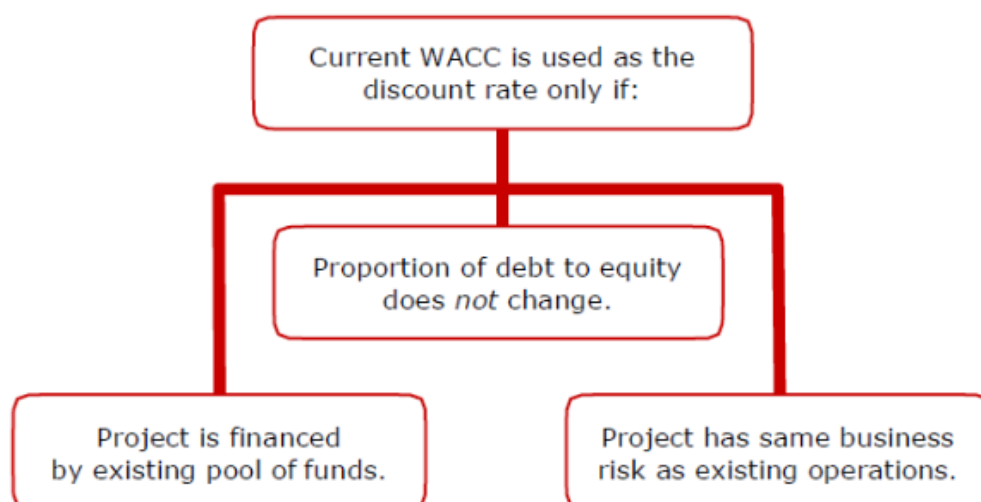
1. The book value of equity is subjective as it depends on accounting policies (e.g. revaluation or costs models for properties and depreciation methods).
2. The book value of equity fails to capture the value of significant intangible assets such as internally generated goodwill.
3. Market values are current and so reflect the most up-to-date information about the company's financial position, performance and prospects. The book value of equity is out of date due to the delay in publishing financial statements.
4. Weighting using the book value of equity will tend to understate the influence of the cost of equity, thereby understating the WACC, which could lead to incorrect investment decisions.

One Improvement

The company has significant property, which is currently recorded at historic cost. It may be advisable to restate this to market value (as permitted under IFRS), thereby creating a revaluation surplus within the equity section of the statement of financial position and moving the book value of equity closer to its market value.

1.2 Implications for Project Financing

WACC is a *potential* discount rate for project appraisal using NPV.



So, a company's existing WACC can be used as the discount rate *only* if:

- The project exposes investors to the same level of **business risk** as they currently face; and
- the finance for the project does not alter the **company's financial risk** (i.e. it does not change its gearing).

If the above conditions do not apply (which is likely to be the case, as they are very restrictive), investors will expect a different return as the project represents a change in risk; this changes the cost of capital. This is explored later in this chapter.

1.3 Limitations and Assumptions

The calculation of the WACC depends on the accuracy of the inputs to the formula (k_e , k_d , V_e , V_d and T). The accuracy of each of these will depend on the models (and the limitations and assumptions of those models) used to find them. For example:

- k_e – was this calculated using the DVM (*Chapter 10*) or CAPM (*Chapter 12*)? How was the growth rate found?
- k_d – is this likely to be constant over the project life?
- V_e – if a market price is used, is the market efficient? If the company is unquoted, how has this been estimated?
- T – will this be constant over the life of the loan?

1.4 Marginal Cost of Capital

Marginal cost of capital (MCC) is the cost of raising the most recent dollar of finance. At first, it may seem reasonable to use MCC as the discount rate for project appraisal. But this can lead to problems:

- ✗ Project finance may be drawn from the company's pool of funds and not from a specific source.
- ✗ Even if a project is financed from a specific source it may not be appropriate to evaluate it using the MCC. For example, if a project is financed by debt the discount rate for NPV should *not* be the cost of debt as this ignores the fact that the surplus cash flows belong to the equity investors and are exposed to business risk. Hence, the cost of debt understates the risk of the project and would lead to an overstatement of the project's NPV.

Therefore, the WACC is a more appropriate discount rate than the MCC, as the WACC is an average of the cost of equity (which measures business risk) and the cost of debt.

2.1 Effect

The current WACC reflects the company's *existing risk profile*: **business risk** and **financial risk**.

1 Definitions

Business risk – the variability in the company's *operating earnings* associated with the industry in which it operates. It is determined by general business and economic conditions.

Financial risk – the *additional variability* in returns due to introducing fixed-interest debt in the capital structure. As a committed fixed cost, interest on debt creates more volatile “bottom-line” profits for shareholders.

Example 1 Level of Gearing

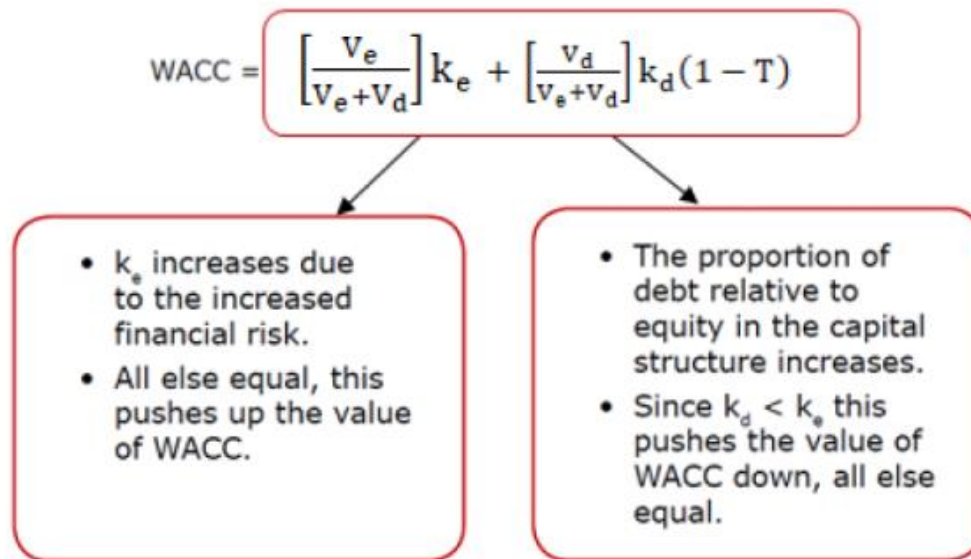
U Co (ungeared) and G Co (geared) operate in the same line of business. Between Years 1 and 2, their profits from operations halve as follows:

	U Co		G Co	
	Year 1	Year 2	Year 1	Year 2
Profits from operations	2,000	1,000	2,000	1,000
Interest	–	–	(400)	(400)
Profit after interest	2,000	1,000	1,600	600
Tax at 20%	(400)	(200)	(320)	(120)
Available to equity shareholders	1,600	800	1,280	480

The amount available to equity shareholders in U Co also halves as profits halve. Their risk or volatility/variability arises purely from the business operations (i.e. operating conditions are not as favourable). However, in G Co, the amount available to equity shareholders decreases by 62.5%. Their risk arises from two sources: business risk plus risk introduced by gearing.

The more highly geared a company is, the more fixed interest it has to pay regardless of profits; therefore, the greater the risk that there will be little or nothing available to distribute as a dividend to shareholders. High financial gearing makes a company more vulnerable to poor trading conditions.

Therefore, as a company *gears up*, shareholders require a higher return, so two things happen as gearing increases:



The effect of increased gearing on the WACC depends on the relative sizes of these two opposing effects.

There are two views on gearing and capital structure relevant to the Financial Management syllabus:

1. Traditional view (see s.3).
2. Modigliani and Miller's theories (see s.4).

A company's capital structure is the mixture of equity and debt finance that is used to finance its assets. The decision on this mixture is called the financing decision.

Key Point

If the gearing ratio does not change, the WACC will not change. If capital structure is changed (i.e. it varies the mixture of equity and debt finance), WACC will automatically change.

2.2 High Financial Gearing

A company's level of financial gearing may be considered relatively high when

- its debt to equity ratio is significantly above the industry average and/or
- its interest coverage ratio (EBIT divided by interest expense) is relatively low.

Potential problems of high levels of financial gearing include:

- ✗ High level of financial risk: The high levels of committed interest expense make earnings to equity investors more volatile, pushing up the cost of equity.
- ✗ Financial distress costs: The company may lose customers due to concerns about its survival and may lose key staff if employees become concerned about job security.
- ✗ Increased credit risk: The rating agencies may cut the company's credit rating, pushing up its cost of debt.
- ✗ Agency costs: Debt contracts include increasingly restrictive covenants (e.g. limiting dividend payments or forcing the company to stay within low-risk projects). In this way the directors become agents of debt holders rather than of the equity investors and shareholder returns are subsequently damaged.

2.3 Forecasting Cash Flows

To survive, all businesses must maintain an uninterrupted capacity to pay debts when they fall due. As debts are usually settled in cash, the cash flows of a business should be important to management.

A cash flow forecast helps managers to monitor future movements in cash. It sets out the anticipated cash inflows and outflows arising over a particular forecast period and can provide an early warning of problems. For example, it may identify potential breaches of an overdraft limit that has been agreed with the bank, and this may allow managers time to review their plans.

Where the forecast indicates a cash surplus, managers can consider whether this surplus should be reinvested, used to redeem debt early or distributed to shareholders.

Activity 3 Cash Flow Forecast

[Show](#)

Victor is a clothing wholesaler. At 1 June, it has a bank overdraft of \$75,000. Management is keen to keep within the \$90,000 overdraft limit agreed with the bank. The overdraft has grown significantly over the past three months and management is reviewing its plans for the next six months.

The following actual and forecast information for the period April to November is available:

	Actual		Forecast					
	April	May	June	July	August	Sept	Oct	Nov
	\$000	\$000	\$000	\$000	\$000	\$000	\$000	\$000
Sales	285	270	280	340	320	275	260	285
Purchases	188	164	175	204	183	188	167	194
Administration expenses	42	38	45	48	42	45	42	39
Selling expenses	36	74	45	43	47	42	46	48
Taxation payment				29				
Mortgage repayment					70			
Asset expenditure					38	59		

Notes:

- (1) All sales are on credit. Customers paying within one month receive a 2% discount for prompt payment.
- (2) 80% of credit customers pay within one month and 20% of customers pay two months after the sale.
- (3) Suppliers allow one month's credit.
- (4) Management has negotiated an early mortgage repayment on a property. The outstanding capital sum is due to be repaid in August.
- (5) Administration expenses are paid when incurred, including a monthly depreciation charge of \$12,000.
- (6) Selling expenses are payable one month after they are incurred.
- (7) Selling expenses are payable one month after they are incurred.

Required:

- a) Prepare a forecast cash flow statement for internal purposes for Victor for the six months ended 30 November, showing the cash balance at the end of each month.
- b) Comment on the information provided by the forecast cash flow statement provided in (a) above and suggest how the directors might deal with any problems which are revealed.

**Please use the notes feature in the toolbar to help formulate your answer.*

(a) Cash flow forecast for the six months to 30 November

	June	July	August	Sept	Oct	Nov
	\$000	\$000	\$000	\$000	\$000	\$000
Receipts						
Sales						
– within 1 month	212	220	267	251	216	204
– within 2 months	<u>57</u>	<u>54</u>	<u>56</u>	<u>68</u>	<u>64</u>	<u>55</u>
Total Receipts	<u>269</u>	<u>274</u>	<u>323</u>	<u>319</u>	<u>280</u>	<u>259</u>
Payments						
Purchases	164	175	204	183	188	167
Administration expenses	33	36	30	33	30	27
Selling expenses	74	45	43	47	42	46
Taxation due		29				
Mortgage repayment			70			
Asset expenditure			<u>38</u>	<u>59</u>		
Total Payments	<u>271</u>	<u>285</u>	<u>385</u>	<u>322</u>	<u>260</u>	<u>240</u>
Net cash flow	<u>(2)</u>	<u>(11)</u>	<u>(62)</u>	<u>(3)</u>	<u>20</u>	<u>19</u>
Opening balance	<u>(75)</u>	<u>(77)</u>	<u>(88)</u>	<u>(150)</u>	<u>(153)</u>	<u>(133)</u>
Closing balance	<u>(77)</u>	<u>(88)</u>	<u>(150)</u>	<u>(153)</u>	<u>(133)</u>	<u>(114)</u>

(b) Comment

The cash flow forecast indicates that the company will breach the overdraft limit agreed with the bank from August onwards. Although the size of the overdraft decreases in October and again in November, it will remain significantly above the agreed limit. Some revision of earlier plans is necessary.

Possible ways in which the directors might improve cash flows include the following:

- An injection of long-term capital, such as ordinary shares or debt.
- Delaying asset expenditure (assuming no binding commitments have been made).
- Postponing the early mortgage repayment.
- Chasing receivables that pay after two months rather than after one month.
- Delaying payments to suppliers.

The directors should be aware of the potential adverse repercussions of the last two suggestions. Although there may be cost savings, the forecast overdraft during the six months exceeds the agreed limit significantly. It is unlikely that cost savings alone would be sufficient to deal with the problem.

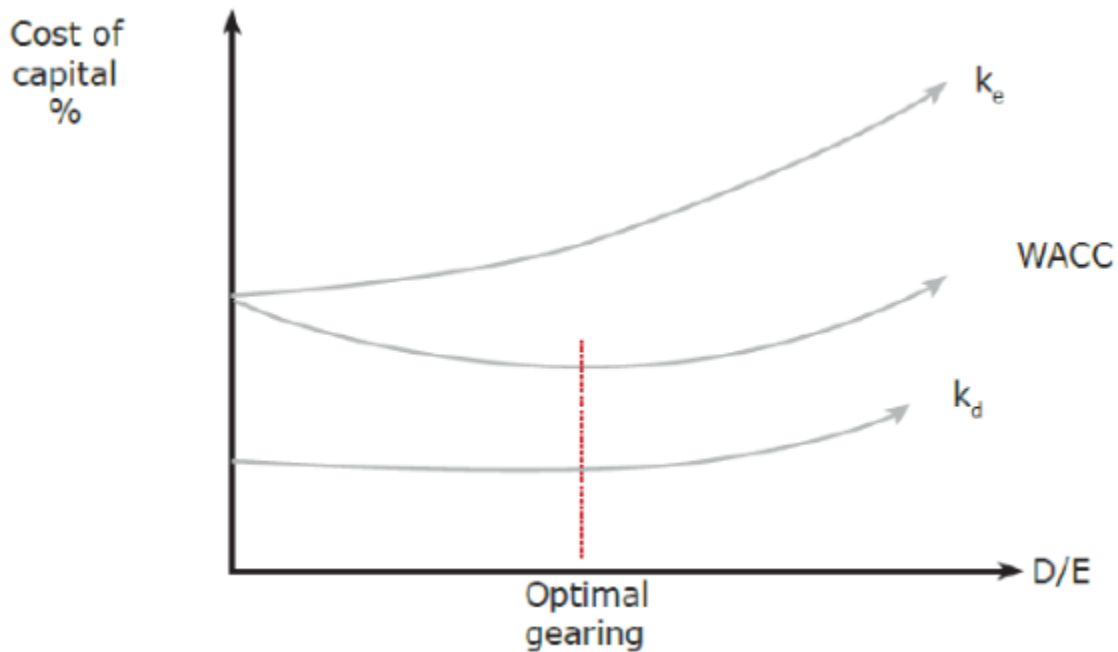
3.1 Reasoning

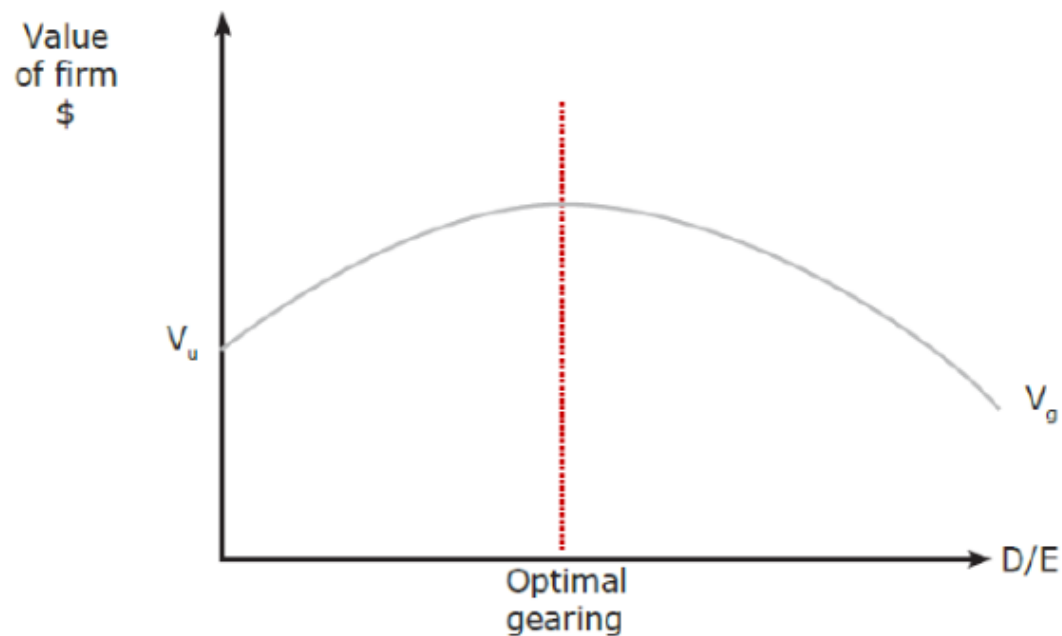
The traditional view has no theoretical foundation; it is often described as the "intuitive approach". It is based on the trade-off caused by gearing (i.e. using more, relatively cheap debt results in a rising cost of equity).

This model/view assumes the following:

- k_e rises slowly at low levels of gearing; therefore, the benefit of using lower-cost debt finance outweighs the cost of the rising k_e .
- At higher levels of gearing, the increased financial risk outweighs this benefit and WACC rises.
- At very high levels of gearing, the cost of debt rises due to the risk of default on debt payments (i.e. credit risk). This is referred to as **financial distress risk**; it should not be confused with *financial risk*, which occurs even at relatively safe debt levels.

These relationships are shown in the graph below:





Where:

V_u = Value of ungeared company

V_g = Value of geared company = Value of equity + Value of debt

Example 2 Market Value of a Company

As seen in *Chapter 8*, the market value of a company = Future cash flows/shareholders' required rate of return (WACC).

So, the lower the WACC, the higher the company's market value. For example:

$$100/0.15 = 667$$

$$100/0.10 = 1,000$$

When the WACC falls from 15% to 10%, the market value increases from 667 to 1,000. Therefore, if capital structure can be changed to lower the WACC, the market value will increase, increasing shareholder wealth.

3.2 Conclusions

As the graph shows, there is an *optimal* gearing level (an optimal capital structure) at which WACC is minimised. Hence, the NPV of all projects is maximised (i.e. the value of the company is maximised).

- Finance managers need to find the optimal capital structure resulting in the lowest WACC.
- However, there is no straightforward method of calculating k_e or WACC, much less the optimal capital structure.
- The optimal capital structure can only be found by trial and error.

3.3 Implications for Project Financing and Appraisal

When considering financing for project, management must determine whether the company is optimally geared.

- If it is optimally geared: raise finance to maintain the existing gearing ratio.
- If suboptimally geared: raise debt finance to increase the gearing ratio towards the optimal level.
- If supraoptimally geared: raise equity finance to reduce the gearing ratio to the optimal level.

Once optimal gearing has been achieved:

1. Appraise the **project** at the existing WACC. If the NPV of the project is positive, the project is worthwhile.
2. Appraise the additional **finance**. If the marginal cost of the finance exceeds the WACC, the finance is not appropriate and should be rejected.

4.1 Introduction

Modigliani and Miller (MM) constructed a mathematical model to provide a basis for company managers to make financing decisions.

- Mathematical models predict outcomes which would occur based on simplifying assumptions.
- Comparison of the model's conclusions to real-world observations allows researchers to understand the effect of the simplifying assumptions. By relaxing these assumptions, the model can be moved towards real life.

MM's assumptions include the following:

- All investors are rational, have the same expectations and are indifferent to personal and corporate borrowing.
- Capital markets are perfect: all investors have equal access to information, there are no transaction costs, and all securities can be bought or sold at fair market prices.
- There is no tax (either corporate or personal) in the initial MM theory. (But this assumption is relaxed in a later theory.)
- There is a single risk-free rate of borrowing (R_f).
- There is no financial distress risk (i.e. no risk of default) even at very high debt levels.
- Corporate debt is irredeemable.
- There are no agency costs associated with conflict of interest between shareholders and management (i.e. directors continue to put shareholders' interests first even at high debt levels).

4.2 Theory without Tax

4.2.1 Two Propositions

MM considered two companies of the same size and in the same line of business, therefore both having the same level of business risk:

1. One company, U, was ungeared.
2. One company, G, was geared.

Proposition 1: in the absence of corporation tax, the market values (V) and, hence, the WACCs of these two companies would be the same.

$$V_g = V_u$$

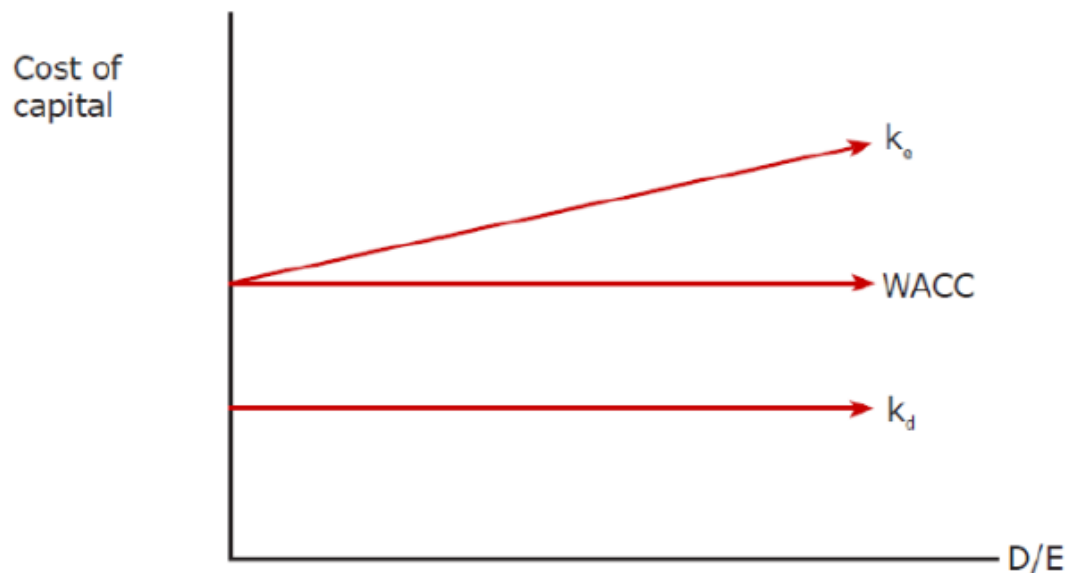
$$WACC_g = WACC_u$$

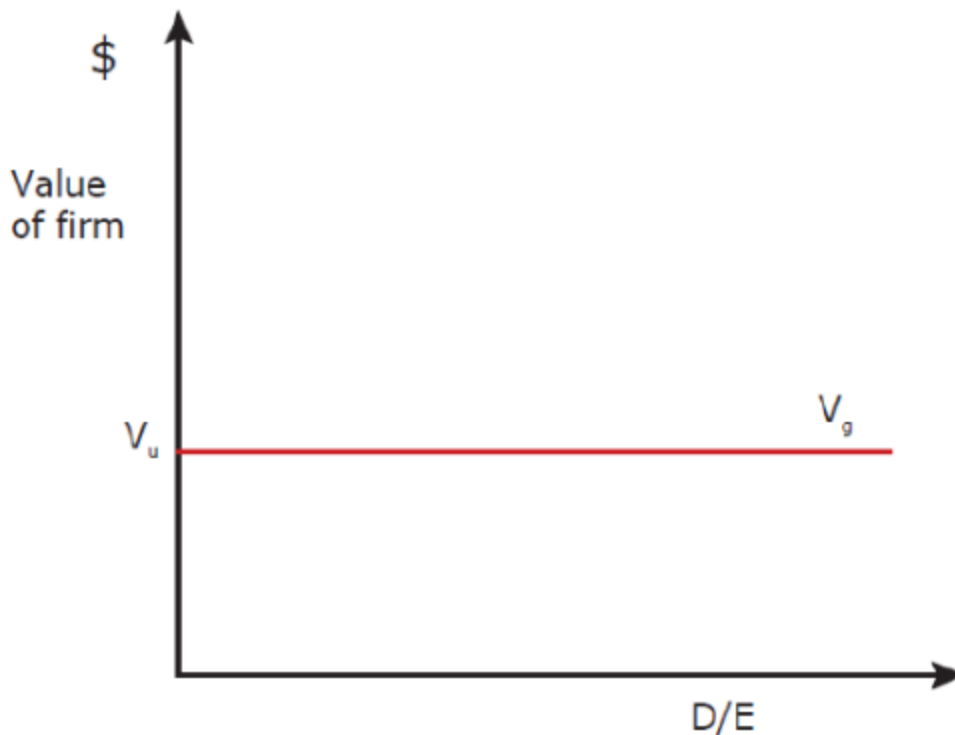
MM argued that the costs of capital would change as gearing changed in the following manner:

- k_e would increase *at a constant rate* as gearing increased due to the perceived increased financial risk –
- Proposition 2.**
- k_d would remain constant (at R_f) whatever the gearing level.
 - The rising k_e would offset the benefit of cheaper debt so the WACC remains constant.

4.2.2 Graphically

- This can be shown as a graph:





4.2.3 Conclusions

Conclusions that can be drawn from the without-tax theory include the following:

- Only *investment* decisions affect the value of the company.
- The value of the company is independent of the financing decision (i.e. capital structure is irrelevant).
- There is *no* optimal gearing level.

Of course, this is not true in practice because the assumptions are too simplistic. There are apparent differences between the real world and the model.

MM did not claim that gearing does not matter in the real world. They said that gearing should not matter in a world in which their assumptions hold true. They then relaxed these assumptions to see how the model's predictions would change.

The first assumption they relaxed was the no corporate tax assumption.

4.3 Theory with Tax

When MM allowed for corporation tax, their conclusions regarding capital structure were altered. This is because of the tax relief on debt interest (the tax shield).

4.3.1 Proposition

In the with-tax theory:

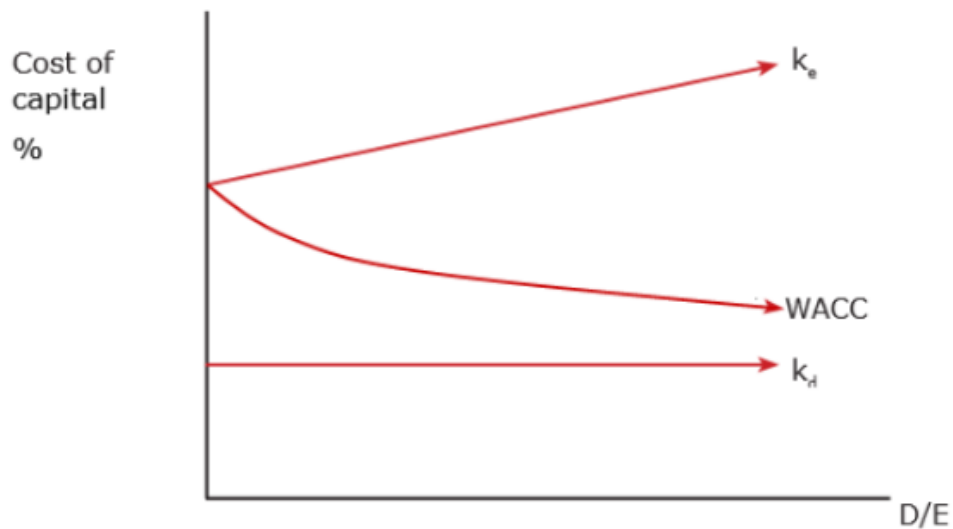
- k_e rises (as before) to reflect the increased uncertainty due to financial risk;
- k_d is constant (as MM ignored financial distress risks).

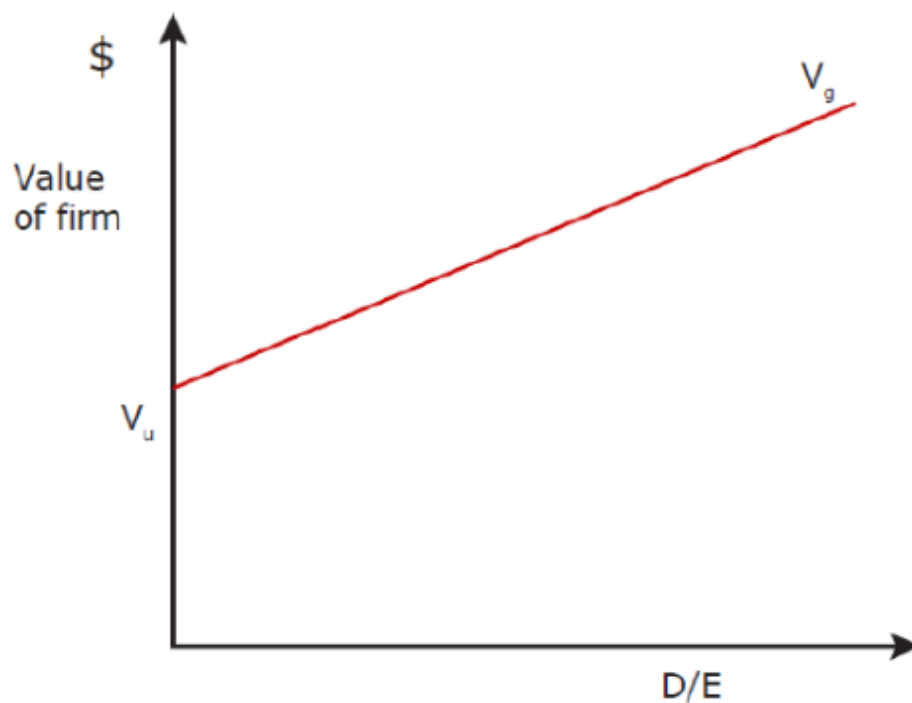
MM recognised that operating profits are distributed in three ways: equity, debt and tax.

- MM argued that taking on more debt reduced the tax payments.
- So the after-tax cash available to debt and equity rises with debt.
- As the company's risk has not changed (as the risk of financial distress is ignored), WACC must fall and the company's value must rise, as shown in the following graphs.

4.3.2 Graphically

- This can be shown as a graph:





4.3.3 Conclusions

MM's theory with tax implies that there **is** an optimal gearing level and that this is to maximise debt in the capital structure to generate maximum value for the shareholders.

This is **not true** in practice (because companies' capital structures are not geared as much as possible) because, for example:

- At high gearing levels, the risk of default on debt (bankruptcy) becomes significant; at very high levels, debt begins to take on the risk characteristic normally associated with equity. The cost of debt rises and the cost of equity increases even more (as shareholders rank last in the priority of claims in insolvency proceedings).
- Personal taxes exist, which may cause investors to prefer injecting equity rather than debt if dividends are taxed at lower rates than interest income.

5.1 Practical Factors

The following factors should be considered:

- The project's business risk. It is not advisable to finance high-risk projects with debt, as interest payment is a legally binding commitment.
- The existing level of financial gearing.
- The existing level of *operational* gearing (i.e. the proportion of fixed to variable operating costs). If already high, the company may prefer to avoid more debt as this increases fixed costs even further.
- The type and quality of assets that the company can offer as security.
- The personal tax positions of the shareholders and debtholders.
- Tax exhaustion (i.e. not enough profit to fully utilise the interest tax shield).
- Issue costs associated with new debt or equity finance, which links directly to *pecking order theory* (see s.5.2).
- Agency costs. At high levels of financial gearing, control of the company may move away from the shareholders towards the debt investors (e.g. restrictive debt covenants may restrict dividends or limit operations to low-risk areas). Limiting operations to low-risk areas may reduce returns to shareholders.
- Costs of financial distress. At dangerous levels of financial gearing, the company may find that operating costs start to rise (e.g. suppliers may ask for payment in advance, staff turnover may increase and customers may lose faith in the company's product warranties).

5.2 Pecking Order Theory

As discussed in Chapter 8, company managers usually prefer internal finance over external finance. This preference for internal finance is called the *pecking order theory*. It is supported by research, which found that company directors often choose "the path of least resistance" (i.e. the most convenient source) when financing a project.

Research has shown that the following hierarchy emerges:

5.2.1 Retained Earnings

Internal finance (i.e. reinvesting profit) is preferred to raising external finance. There are several practical advantages of using internal finance:

- ✓ Faster decision-making and management time is not consumed by paperwork.
- ✓ No issue costs make investments more cost-effective.
- ✓ No additional debt reduces financial risk.
- ✓ No requirements for collateral or personal guarantees.
- ✓ No change in the company's ownership, so no dilution of control.
- ✓ Privacy over financial decisions and project details (e.g. no need to publish a prospectus).
- ✓ External finance may not be available, particularly for SMEs, due to information asymmetry (a perceived high risk due to a lack of public information about the business).

5.2.2 Issue Debt

External finance must be raised if internal finance is insufficient (e.g. existing shareholders require a significant dividend). Here, the preference is for debt because:

- ✓ Debt is cheaper than equity.
- ✓ Interest is tax allowable, which results in a tax shield.
- ✓ Issue costs are lower on debt than equity.
- ✓ Debt can be raised more quickly than equity.

5.2.3 Issue Equity

If debt cannot be raised (e.g. due to lack of assets for security), a share issue is inevitable. A new share issue ranks last in pecking order theory because:

- ✗ The cost of equity is high because equity investors are exposed to high risk (business and operating risk).
- ✗ Dividends do not give a tax shield.
- ✗ Issue costs are high.
- ✗ Share issues take much time and effort to organise.

Syllabus Coverage

This chapter covers the following Learning Outcomes.

E. Business Finance

2. Estimating the cost of capital

- c) Estimating the overall cost of capital including:
 - i) distinguishing between average and marginal cost of capital.
 - ii) calculating the weighted average cost of capital (WACC) using book value and market value weightings.

3. Sources of finance and their relative costs

- c) Identify and discuss the problem of high levels of gearing.
- d) Assess the impact of sources of finance on financial position and financial risk and shareholder wealth using appropriate measures including:
 - ii) cash flow forecasting.
- e) Impact of cost of capital on investments including:
 - i) the relationship between company value and cost of capital
 - ii) the circumstances under which WACC can be used in investment appraisal.

4. Capital structure theories and practical considerations

- a) Describe the traditional view of capital structure and its assumptions.
- b) Describe the views of Miller and Modigliani on capital structure, both without and with corporate taxation, and their assumptions.
- c) Identify a range of capital market imperfections and describe their impact on the views of Miller and Modigliani on capital structure.
- d) Explain the relevance of pecking order theory to the selection of sources of finance.

Summary and Quiz

- WACC estimates the company's average cost of long-term finance.
- WACC is a potential discount rate to use for the calculation of the NPV of possible projects. However, the existing WACC should only be used if the project would not change the company's business risk or level of gearing (i.e. financial risk).
- Theories to explain how financial gearing affects the WACC include the traditional view Modigliani and Miller (MM) without and with corporate taxation. Each theory has useful elements, even if its conclusions lack practical relevance.
- According to the traditional view, there is an *optimal* gearing level (an optimal capital structure) at which WACC is minimised. Hence, the NPV of all projects is maximised (i.e. the company's value is maximised).
- MM without tax concludes: (1) there is no optimal capital structure and (2) a company's value is only affected by its investment decisions and not how it is financed.
- MM with tax concludes that a company should use as much debt financing as possible to maximise its overall market value.
- Under pecking order theory, managers choose the path of least resistance (e.g. the most convenient source) of financing for a project.

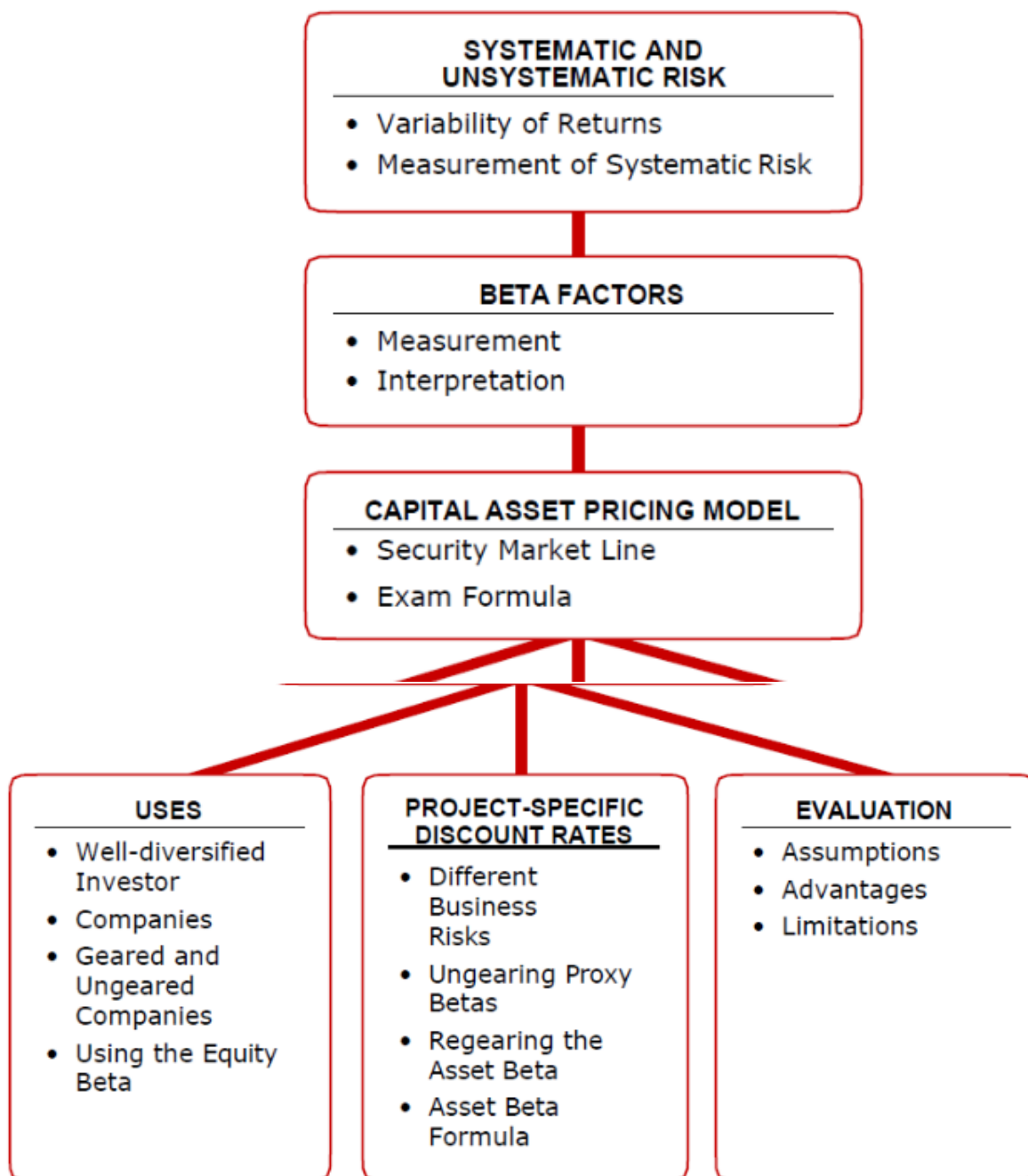
[Chapter 11 Quiz](#)

Chapter 12: Capital Asset Pricing Model

Overview	CHAPTER 12: Visual Overview
12.1 Systematic and Unsystematic Risk	12.1.1 Variability of Returns
	12.1.2 Measurement of Systematic Risk
12.2 Beta Factors	12.2.1 Measurement
	12.2.2 Interpretation
12.3 Capital Asset Pricing Model (CAPM)	12.3.0 Introduction
	12.3.1 Security Market Line
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12.4 Uses of the CAPM	12.4.1 Well-diversified Investors
	12.4.2 Companies
	12.4.3 Geared and Ungeared Companies
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12.5 Project-specific Discount Rates	12.5.1 Different Business Risks
	12.5.2 Ungearing Proxy Betas
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12.6 Evaluation of CAPM	12.6.1 Assumptions
	12.6.2 Advantages of CAPM
	12.6.3 Limitations of CAPM

CHAPTER 12: Visual Overview

Objective: To understand the Capital Asset Pricing Model and its uses in financial management.



1.1 Variability of Returns

If an investor has a portfolio of investments in shares of different companies, it might be supposed that the risk of the portfolio would be the average of the risks of the individual investments. However, this is not the case; the risk of the portfolio is less than the average due to diversification.

Investment risk, the uncertainty or variability of returns on investments, has two elements:

1. Unsystematic risk; and
2. Systematic risk.

i Definitions

Unsystematic risk – the risk which is unique to each company's shares. (Also called "unique", "diversifiable" or "industry-specific" risk.)

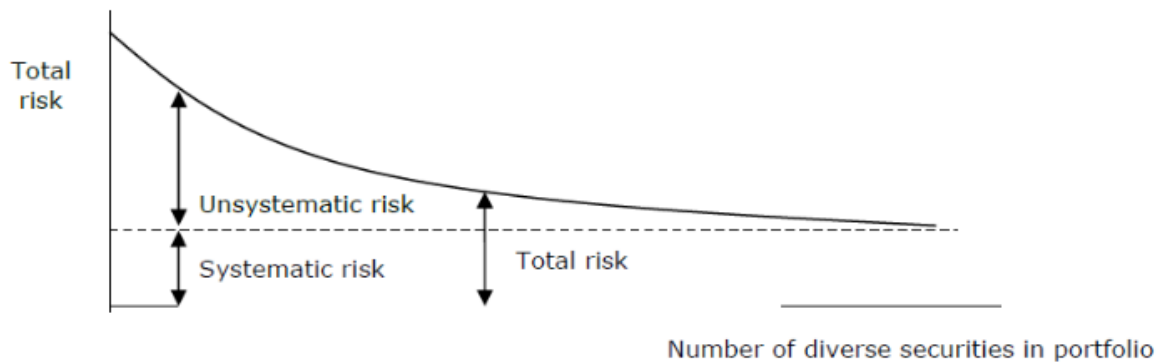
Systematic risk – the risk which affects the market as a whole rather than a specific company's shares. (Also known as "market" risk or "undiversifiable" risk.)

Unsystematic risk is the element that can potentially be eliminated ("diversified away") by investors building a diversified portfolio (i.e. positive and negative exposures cancel out). For example, if the price of copper rises, the share prices of copper mining companies will rise, but the share prices of manufacturing companies that use copper will fall.

Systematic risk *cannot* be diversified away. This risk is associated with the financial *system* and remains even in a well-diversified portfolio. For example, both copper mining and manufacturing companies are exposed to increases in interest rates and the general state of the economy.

The sum of systematic risk and unsystematic risk is called total risk.

Therefore, a portfolio of shares has a total risk comprising these two types of risk. As more different shares are added to the portfolio, the unsystematic risk is diversified away and the total risk reduces to approach systematic risk.



This means that investors should only be rewarded for risks they cannot avoid – systematic risk.

1.2 Measurement of Systematic Risk

A well-diversified portfolio of shares still has some degree of risk or variability (i.e. to macro-economic changes) because all shares are affected by market risk.

This systematic risk will affect the shares of all companies, although some will be affected to a greater or lesser degree than others. For example:

- A supermarket chain will have a relatively low risk because people will always buy food, regardless of the state of the economy.
- A tour operator will have a higher risk because, during a recession, demand for holidays will fall.

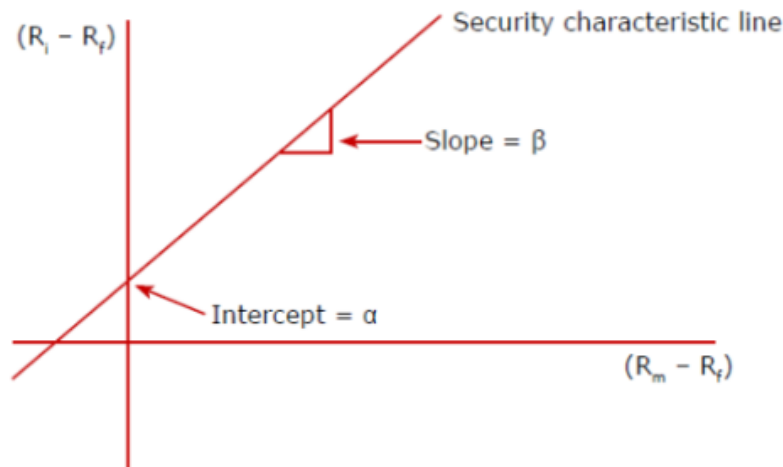
This sensitivity to systematic risk is measured by "an index of responsiveness of the returns on a company's shares compared to the returns on the market as a whole", called a *beta* value or *beta* factor, or simply *beta*.

2.1 Measurement

Beta factors for quoted shares are measured using historical data and published in "beta books". They are determined by comparing changes in a share's returns to changes in the stock market returns over many years (at least five years).

This can be illustrated by the "security characteristic line", which indicates the share's sensitivity to market changes.

Beta is estimated from these observations by determining the gradient or slope of the "line of best fit" through the observed points. The steeper the slope, the more volatile the share and the higher the beta.



Where $(R_i - R_f)$ = the excess return of the share over the risk-free return

$(R_m - R_f)$ = the excess return of the stock market over the risk-free return

R_f = the return on a risk-free investment

In the long run, the security characteristic line should pass through the point where the two axes meet.

In the short run, however, this may not always be the case, and any short-term difference or abnormal return is known as the *alpha factor*.

2.2 Interpretation

A beta describes a share's degree of sensitivity to changes in the market's returns caused by systematic risk. Typically, beta lies between 0.2 and 1.6:

- Beta = 1 – Indicates a “neutral” share that is as sensitive as the market to systematic risk. Such shares should earn the **market return**.
- Beta > 1 – Indicates an “aggressive” share that is more sensitive than the market. Therefore, if the market in general rises by 10%, the returns from this share are likely to be more than 10%.
- Beta < 1 – Indicates a “defensive” share that is less sensitive than the market and is likely to rise and fall in value less than the market in general. For example, if a share has a beta of 0.5, the expected will increase by only 5% if the return on the capital market increases by 10%.

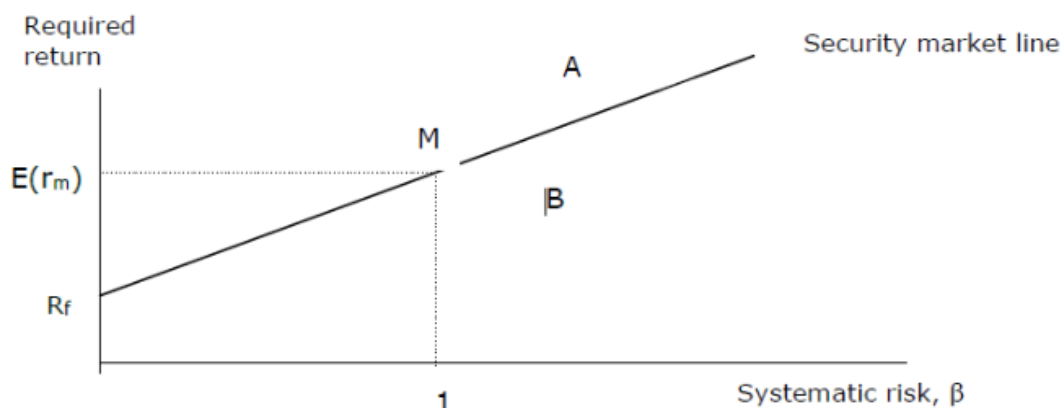
The market return is the return on the **market portfolio** (i.e. a fully diversified portfolio with a representative sample of all traded shares).

3.0 Introduction

The CAPM is a method of calculating the return required on an investment based on its risk. It was developed from portfolio theory, which is based on the idea that an investor who puts all their funds into one investment risks everything on the performance of that individual investment. A less risky policy would be to spread the funds over several investments (establish a portfolio) so that the unexpected losses from one investment may be offset to some extent by the unexpected gains from another. Therefore, the rationale for establishing a portfolio is the reduction of risk.

3.1 Security Market Line

The Security Market Line (SML) is a graph which plots the required return ($E(r_i)$) from any investment according to its systematic risk, as measured by its beta, β_i .



- The required return of an investment with $\beta = 0$ (i.e. no systematic risk) will be the risk-free return, R_f .
- The expected/required return of an investment with $\beta = 1$ will be the market return, $E(r_m)$.
- Therefore, the gradient (slope) of the line = $(E(r_m) - R_f)$

Comparing forecast returns from an investment against the required return indicates whether an investment is undervalued or overvalued.

- Any share with a return above the SML (e.g. A on the graph) is forecast to earn higher returns than predicted by CAPM for its beta. It is, therefore, underpriced and will be sought after. This will drive up its price and reduce its expected return until it earns only the required return indicated by the SML. A is a "positive alpha" investment.
- Any share with a return below the SML (e.g. B) would be sold as it appears to be temporarily overpriced. This will drive down the price and increase the expected return until it earns the required return indicated by the SML. B is referred to as a "negative alpha" investment.

In the long run, market forces should ensure that all investments give the returns predicted by the SML.

3.2 Exam Formula

Key Point

CAPM assumes that investors hold fully diversified portfolios.

If shareholders hold well-diversified portfolios, they are concerned only with systematic risk. Therefore:

- The return they require is a reward for only the systematic risk of an investment.
- Since systematic risk is measured by a beta, the required return from an investment must be related to its beta.

CAPM is the equation of the SML that relates required returns to beta values as measures of systematic risk.

The CAPM formula, as provided in the exam formulae sheet, is:

$$E(r_i) = R_f + \beta_i(E(r_m) - R_f)$$

Where:

$E(r_i)$ = expected/required return from an investment

R_f = risk-free return

$E(r_m)$ = expected return from the market portfolio

β_i = beta of the investment

The market portfolio is a portfolio containing every share on the stock market.

$(E(r_m) - R_f)$ is called the equity market *premium*, the equity risk premium, the market risk premium, or simply market premium (i.e. the extra return an investor expects for holding a diversified portfolio of shares rather than a risk-free security).

Example 1 CAPM Formula

Consider the following information:

Risk-free rate of return = 4%

Equity risk premium = 5% (i.e. return on the market is 9%)

Beta value of shares of Ram Co = 1.2

Using CAPM:

$$E(r_i) = R_f + \beta_i (E(r_m) - R_f) = 4 + (1.2 \times 5) = 10\%$$

The required return from an investment in Ram Co is 10%.

Example 2 Required Returns

Risk free rate = 5%

Market return = 14%

Calculate the required returns from investments whose beta values are:

(i) 1

(ii) 2

(iii) 0.5

Using CAPM:

$$(i) E(r_i) = 5 + 1(14 - 5) = 14\%$$

The return required from an investment with the same risk as the market is the market return.

$$(ii) E(r_i) = 5 + 2(14 - 5) = 23\%$$

The return required from an investment with more risk than the market. A higher return than that given by the market is therefore required.

$$(iii) E(r_i) = 5 + 0.5(14 - 5) = 9.5\%$$

The return required from an investment with less risk than the market. A lower return than that given by the market is therefore required.

4.1 Well-diversified Investors

If an investor already holds a well-diversified portfolio, that investor will be concerned only with systematic risk. Therefore, the CAPM is relevant.

The investor will be satisfied only if a potential investment gives a high enough return given its sensitivity to market risk as measured by its beta.

Activity 1 CAPM for Investors

[Show](#)

An investment has a forecast return over the next year of 12%. The beta of the investment is estimated at 0.9. The risk-free rate is 5% and the market return is 15%.

Required:

Determine whether a well-diversified investor should buy this investment.

**Please use the notes feature in the toolbar to help formulate your answer.*

Required return = $5 + 0.9 \times (15 - 5) = 14\%$

Expected return = 12%

Therefore the investor should *not* invest.

4.2 Companies

Companies should **not** diversify their activities to reduce the risk to their shareholders. Shareholders can diversify their shareholdings much more easily and more cheaply than a company can diversify its activities.

If shareholders are already well-diversified, the company should be concerned, on behalf of the shareholders, simply with the systematic risk of potential projects.

Therefore, the aim of a company with well-diversified shareholders should be to determine the required return from its investment projects and then compare this to the forecast return.

If the project has the same risk as the company's existing activities, the existing WACC can be used.

However, if the project has a different risk, the existing WACC will not be appropriate. In this case, a tailor-made discount rate for that project must be determined using the CAPM.

4.3 Geared and Ungeared Companies

4.3.1 Asset Beta

Suppose a company, A Co, is ungeared. As it has no debt, it has no financial risk, so it faces only business risk. Because investors can diversify away unsystematic business risk, they face only systematic business risk. The beta factor for such a company is called the asset beta (because the company has assets, no debt) β_a . As long as the business operations – and therefore the business risk – do not change, the asset beta (also called the ungeared beta) remains constant.

4.3.2 Equity Beta

Suppose B Co is identical to A Co except that its capital structure includes debt. Gearing increases and financial risk is added to business risk. This will increase the effects of the systematic risk investors in B Co face (see *Chapter 10*), so the return they require will increase to compensate for the increasing risk. The beta factor for B Co's shares (called the equity beta) increases as gearing increases. The equity beta (also called the geared beta) measures the sensitivity to market risks of the equity shareholders' returns.

- In the all-equity financed company, A Co (**ungeared** company), the asset beta and the equity beta are the **same**: $\beta_a = \beta_e$.
- However, in the **geared** company, B Co, the equity beta **exceeds** the asset beta: $\beta_e > \beta_a$.

4.4 Using the Equity Beta to Estimate the Cost of Equity

As the equity beta, β_e , measures the sensitivity to market risks of the equity shareholders' returns, it can be used in the CAPM as the required return for the equity shareholders.

Therefore, the CAPM can be used as an alternative to the Dividend Valuation Model (*Chapter 10*) for estimating a company's cost of equity.

Activity 2 Cost of Equity Using CAPM

[Show](#)

The equity beta of a company is estimated to be 1.2. The risk-free return is 7% and the return from the market is 15%.

Required:

Estimate the cost of equity of the company.

**Please use the notes feature in the toolbar to help formulate your answer.*

Activity 2 Cost of Equity Using CAPM

[Hide](#)

The equity beta of a company is estimated to be 1.2. The risk-free return is 7% and the return from the market is 15%.

Required:

Estimate the cost of equity of the company.

**Please use the notes feature in the toolbar to help formulate your answer.*

$$k_e = 7 + 1.2 \times (15 - 7)$$
$$k_e = 16.6\%$$

5.1 Different Business Risks

When the business risk of a project differs from the existing business risk of the investing company, the return required on the project is different from the average return required on the investing company's existing business operations. This means that it is **not appropriate** to use the investing company's existing cost of capital as the discount rate for the investment project. Instead, the CAPM can be used to calculate a **project-specific** discount rate that reflects the business risk of the new project.

The main reason why using the existing costs of capital is not appropriate is because projects that should be rejected might be accepted (and *vice versa*) because the present values of the project's cash flows will be mispriced. A suitable discount needs to take account of the specific risks associated with the new project (e.g. industry-specific risks, geographic risks, technological risks and regulatory risks).

5.2 Ungearing Proxy Betas

The first step in using CAPM to calculate a project-specific discount rate is to obtain information on companies with business operations similar to those of the proposed project. These companies are called "proxy companies" and their published equity betas are called "proxy betas". From a CAPM point of view, these proxy betas can be used to represent the business risk of the proposed investment project.

However, an equity beta reflects not only the business risk of a company's operations but also the company's financial risk. Therefore, removing the effect of the financial risk (gearing) from the proxy equity beta is necessary to find the asset beta that reflects business risk alone. (Section 5.4 shows this using the asset beta formula.)

When the equity betas of several proxy companies are ungeared, the resulting asset betas will usually have slightly different values (because no two companies will have precisely the same business risk).

The effect of the slight differences in business operations and business risk, as reflected in the asset betas, can be removed by averaging them. A simple arithmetic average is calculated by summing the asset betas and dividing the total by the number of asset betas.

5.3 Regearing the Asset Beta

The average asset beta represents the business risk of the proposed project. However, before a project-specific discount rate can be calculated, the financial risk of the investing company must be considered. In other words, having ungeared the proxy equity betas when calculating the asset betas, it is now necessary to "regear" the average asset beta to reflect the gearing and financial risk of the investing company.

The investing company's gearing, tax rate and average proxy asset beta are inserted into the asset beta formula (see s.5.4) to calculate the regeared equity beta. The CAPM formula can then be used to calculate a project-specific cost of equity.

The project-specific cost of equity can be used as the discount rate if the project is financed by equity (either retained earnings or new share issues). If a mix of equity and debt is used to finance a project, it would be necessary to calculate a project-specific WACC.

Exam advice

Calculation of a project-specific WACC will not be required in the exam.

5.4 Asset Beta Formula

The following exam formula can be used to convert an equity beta to an asset beta (and vice versa):

$$\beta_a = \left[\frac{V_e}{(V_e + V_d(1-T))} \beta_e \right] + \left[\frac{V_d(1-T)}{(V_e + V_d(1-T))} \beta_d \right]$$

Where β_a =asset beta

β_e =equity beta

β_d =beta of corporate debt

V_e =market value of equity

V_d =market value of debt

T=company profit tax rate

$V_e + V_d(1-T)$ =after-tax market value of the company

Exam advice

If the exam question does not give a beta for debt, assume that debt is risk free (i.e. $\beta_d = 0$).

Example 3 Asset Beta

The following information relates to RD Co:

Equity beta = 1.2

Debt beta = 0.1

Market value of shares = \$6m

Market value of debt = \$1.5m

Company profit tax rate = 25% per year

After-tax market value of company = $V_e + V_d(1-T) = 6 + 1.5 \times 0.75 = \7.125m

$\beta_a = (6/7.125 \times 1.2) + (1.5 \times 0.75/7.125 \times 0.1) = 1.024$.

5.5 Summary of Steps

The steps in calculating a project-specific discount rate can now be summarised as follows:

1. Find suitable proxy companies.
2. Find the equity betas of the proxy companies, their gearing levels and tax rates.
3. Use the asset beta formula to ungear the proxy equity betas to obtain asset betas.
4. Calculate an average asset beta.
5. Use the asset beta formula to regear the asset beta to the investing company's capital structure.
6. Use the CAPM formula to calculate a project-specific cost of equity.
7. If the project will be financed by a mixture of equity and debt, a project-specific WACC must be calculated.

Exam advice

Step 7 calculations will **not** be required in the exam.

Activity 3 Degearing

[Show](#)

Acorn Co produces electronic components but is considering venturing into computer manufacturing. Acorn Co is ungeared with an equity beta of 0.8.

The average equity beta of computer manufacturers is 1.4 and the average gearing ratio is 1:4.

The risk-free return is 5%, the market return is 12%, and the corporation tax rate is 33%.

Required:

Calculate the discount rate Acorn Co should use to appraise a computer manufacturing project, assuming it remains an equity-financed company.

**Please use the notes feature in the toolbar to help formulate your answer.*

Using the formula, calculate the asset beta of the computer industry:

$$\beta_a = \left[\frac{V_e}{(V_e + V_d(1-T))} \beta_e \right] + \left[\frac{V_d(1-T)}{(V_e + V_d(1-T))} \beta_d \right]$$

$$\beta_a = \frac{4}{4 + (1 \times 0.67)} 1.4 + 0 \text{ (as } \beta_d = 0)$$

Asset beta = 1.2

Because Acorn Co is ungeared, this asset beta is the appropriate beta for use in the CAPM to determine the discount rate that Acorn Co should use for a computer manufacturing project:

Required return = $5 + 1.2(12 - 5) = 13.4\%$

Activity 4 Regearing

[Hide](#)

As for *Activity 3* but Acorn Co has a gearing ratio of 1:2.

Required:

Calculate the project-specific cost of equity which Acorn Co should use for a computer manufacturing project.

**Please use the notes feature in the toolbar to help formulate your answer.*

Asset beta = 1.2 (as in *Solution 3*).

To find the discount rate for Acorn Co, this asset beta must be converted into an equity beta appropriate for Acorn Co:

$$1.2 = \frac{2}{2+(1 \times 0.67)} \beta_e = 0.749 \times \beta_e$$

$$\beta_e = 1.6$$

$$k_e \text{ of Acorn Co if in computer manufacturing} = 5 + 1.6 (12 - 5) = 16.2\%$$

Activity 5 Several Proxies

[Hide](#)

Puggle Co plans to invest in a new project significantly different from its existing business operations. Puggle Co financed 30% by debt and 70% by equity. It has identified three companies with business operations similar to the proposed investment:

A Co has an equity beta of 0.81 and is financed 25% by debt and 75% by equity.

B Co has an equity beta of 0.98 and is financed 40% by debt and 60% by equity.

C Co has an equity beta of 1.16 and is financed 50% by debt and 50% by equity.

The risk-free rate of return is 4% per year and the equity risk premium is 6% per year. All companies pay tax at a rate of 30%.

Required:

Calculate a project-specific cost of equity for the proposed investment.

**Please use the notes feature in the toolbar to help formulate your answer.*

Ungearing the proxy equity betas to find asset betas:

$$\text{A Co: } 75 / ((75 + 25(1 - 0.30)) \times 0.81 = 0.657$$

$$\text{B Co: } 60 / ((60 + 40(1 - 0.30)) \times 0.98 = 0.668$$

$$\text{C Co: } 50 / ((50 + 50(1 - 0.30)) \times 1.16 = 0.682$$

$$\text{Averaging the asset betas: } (0.657 + 0.668 + 0.682) / 3 = 0.669$$

Regearing the average asset beta:

$$0.669 = 70 / ((70 + 30(1 - 0.30)) \times \beta = 0.769 \times \beta_e$$

$$\text{Therefore, } \beta_e = 0.669 / 0.769 = 0.87$$

Calculating the project-specific cost of equity using CAPM:

$$= 4 + (0.87 \times 6) = 9.2\%$$

6.1 Assumptions

The assumptions of the CAPM can be summarised as follows:

- Total risk can be split between systematic risk and unsystematic risk.
- Unsystematic risk can be diversified away entirely.
- All of a company's shareholders hold well-diversified portfolios. This is fundamental to ignoring unsystematic risk. This is not an unreasonable assumption since it is quite easy and inexpensive for investors to construct portfolios that "track" the stock market.
- A risk-free security exists which provides a minimum level of return required by investors.
- Investors can borrow and lend at the risk-free rate (for which the yield on short-dated government debt is taken as a proxy).
- A standardised holding period to make the returns on different securities comparable (e.g. a return over six months cannot be compared with a return over 12 months). A holding period of one year is usually assumed.
- A perfect capital market exists, as for the dividend valuation model (see s.2.3 in *Chapter 10*), so all securities are valued correctly.

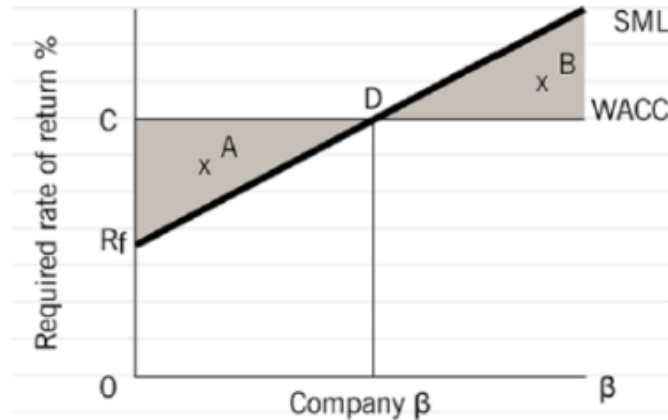
6.2 Advantages of CAPM

The advantages of the CAPM include the following:

- ✓ It considers only systematic risk (i.e. is relevant for listed companies whose institutional investors have diversified portfolios from which unsystematic risk has been eliminated).
- ✓ It generates a theoretically derived relationship between required return and systematic risk.
- ✓ CAPM has been subject to frequent empirical research and testing.
- ✓ It is a much better method of calculating the cost of equity than the dividend growth model in that it explicitly allows for a company's level of systematic risk relative to the stock market as a whole.
- ✓ It is superior to using the existing WACC as the discount rate for a project with different business risks compared to existing operations.

Example 4 CAPM v WACC

Using CAPM will lead to better investment decisions than using the WACC in the two shaded areas, represented by projects A and B.



Using WACC, Project A has $IRR < WACC$, so it will be rejected (where IRR is read from the vertical axis). This investment decision is incorrect. Using a CAPM-derived project-specific discount rate, Project A offers a return greater than needed to compensate for its level of systematic risk (because it is above the SML line), so accepting it will increase shareholder wealth.

Similarly, using WACC, as Project B has $IRR > WACC$, it would be accepted. This investment decision is also incorrect. Using CAPM, Project B offers insufficient return for its level of systematic risk (because it is below the SML line), so it should not be accepted.

6.3 Limitations of CAPM

The limitations of the CAPM include the following:

- ✗ It is a single-period model – whereas company projects are often for multiple periods.
- ✗ It is a single index model – beta is the only variable to explain different required returns on various investments.
- ✗ Lack of data for the model – particularly in developing markets.
- ✗ CAPM tends to overstate the required return on high-risk companies and understate the returns on low-risk companies.
- ✗ Assumptions do not hold in the real world. For example, the assumption that investors can borrow and lend at the risk-free rate (for which the yield on short-dated government debt is taken as a proxy) is unrealistic. The risk for individual investors is much higher than for the government.

Syllabus Coverage

This chapter covers the following Learning Outcomes.

E. Business Finance

2. Estimating the cost of capital

- a) Estimate the cost of equity including:
- ii) explanation and discussion of systematic and unsystematic risk
 - iii) relationship between portfolio theory and the capital asset pricing model (CAPM)
 - iv) application of the CAPM, its assumptions, advantages and disadvantages.

3. Sources of finance and their relative costs

- e) Impact of cost of capital on investments including:
- iii) the advantages of the CAPM over WACC in determining a project-specific cost of capital
 - iv) the application of the CAPM in calculating a project-specific discount rate.

Summary and Quiz

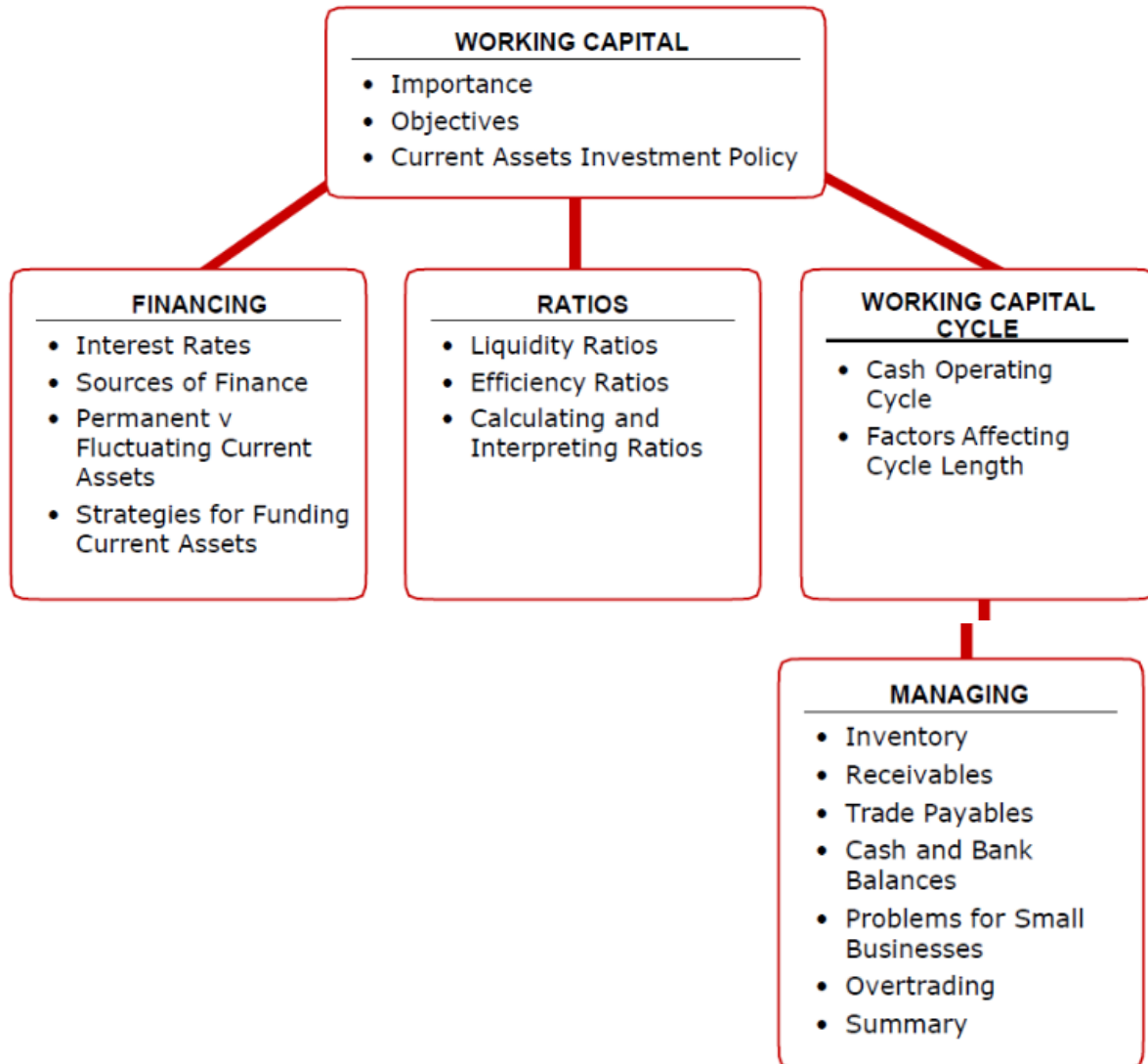
- Unsystematic risk is the risk which is unique to each company's shares. It can be diversified away.
- Systematic risk affects the market as a whole rather than a specific company's shares. It cannot be diversified away.
- A beta describes a share's degree of sensitivity to changes in the market's returns, caused by systematic risk.
- The CAPM calculates the return required on an investment, based on an assessment of its risk. It assumes that shareholders hold well-diversified portfolios and are concerned only with systematic risk.
- CAPM is also an alternative to the Dividend Valuation Model (DVM) for estimating a company's cost of equity.
- In an all-equity financed (i.e. ungeared) company, the asset beta (which measures business risk only) and the equity beta are the same: $\beta_a = \beta_e$.
- In a geared company, however, because the equity shareholders also face financial risk, the equity beta exceeds the asset beta: $\beta_e > \beta_a$.
- The asset beta is also called the ungeared beta, while the equity beta is also called the geared beta.
- The steps in calculating a project-specific discount rate are:
 - Find suitable proxy companies.
 - Find their β_e , gearing levels and tax rates.
 - Use the asset beta formula to ungear the proxy β_e to obtain β_a .
 - Calculate an average β_a .
 - Use the asset beta formula to regear β_a to the investor's capital structure.
 - Use the CAPM formula to calculate a project-specific cost of equity.
- CAPM is generally seen as a much better method of calculating the cost of equity than the dividend growth model in that it explicitly allows for a company's systematic risk relative to the stock market as a whole.
- It is also superior to using the existing WACC as the discount rate for a project with different business risks compared to existing operations.

Chapter 13: Working Capital Management

Overview	CHAPTER 13: Visual Overview
13.1 Working Capital	13.1.1 Importance
	13.1.2 Objectives of Working Capital Management
	13.1.3 Current Assets Investment Policy
13.2 Financing	13.2.1 Interest Rates
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13.3 Working Capital Ratios	13.3.1 Liquidity Ratios
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CHAPTER 13: Visual Overview

Objective: To appreciate the importance of working capital and its effective management.



1.1 Importance

i Definition

Working capital – the capital represented by net current assets which is available for day-to-day operating activities.

The definition of **working capital** is simple; it is the difference between an organisation's current assets (e.g. inventories, trade receivables and cash) and its current liabilities (e.g. trade payables). Its main function is to support day-to-day financial operations, including purchasing inventory, paying salaries, wages and other business expenses, and financing credit sales.

Many businesses which appear profitable are forced to cease trading because they cannot meet short-term obligations when they fall due. To remain in business, an organisation must successfully manage its working capital. Too often, however, this is overlooked.

Working capital management requires great care due to the potential interactions between its components.

Example 1 Working Capital's Linked Effects

A reduction in the level of inventory will reduce storage costs and the risk of obsolescence. It will also reduce the level of resources/amount of capital "tied up" in inventory.

However, such an action may damage customer relations if customers must wait for new inventory to be delivered or, worse still, it may result in lost sales if they go elsewhere.

Example 2 Working Capital Management

Extending the credit period might attract new customers and lead to an increase in revenue.

However, an organisation might require a bank overdraft to finance this new credit facility. The cost of the overdraft could exceed the profit arising from additional sales.

Management must ensure that a business has sufficient working capital. The consequences of too little will be cash flow problems highlighted by:

- Exceeding an agreed overdraft limit.
- Failing to pay suppliers (including employees) on time.
- Inability to take advantage of discounts for prompt payment.

In the long run, a business with insufficient working capital cannot meet its current obligations. It will be forced to cease trading even if it remains profitable on paper.

On the other hand, if a business ties up too much resource in working capital, it will earn a lower-than-expected rate of return on capital employed. Again, this is not a desirable situation.

Working capital management is crucial to the effective management of a business because:

- Current assets may be a high proportion of total assets.
- Failure to control working capital, and therefore liquidity, is a major cause of business failure.

Two major questions must be considered:

1. How much to invest in working capital?
2. How to finance working capital?

Example 3 Need for Working Capital

A company has a healthy liquidity position (Stage 1). Its business then doubles without investing in more non-current assets or raising more equity capital. It is a reasonable assumption that if turnover doubles, inventory, receivables and payables will also double (Stage 2).

	Stage 1			Stage 2	
	\$000	\$000		\$000	\$000
Non-current assets		1,000			1,000
Current assets					
Inventory	50		× 2	100	
Receivables	40		× 2	80	
Cash	<u>20</u>			<u>—</u>	
		<u>110</u>			<u>180</u>
		<u>1,110</u>			<u>1,180</u>
Equity		1,100			1,100
Current liabilities					
Payables		10	× 2		20
Overdraft		<u>—</u>	Balancing figure		<u>60</u>
		<u>1,110</u>			<u>1,180</u>

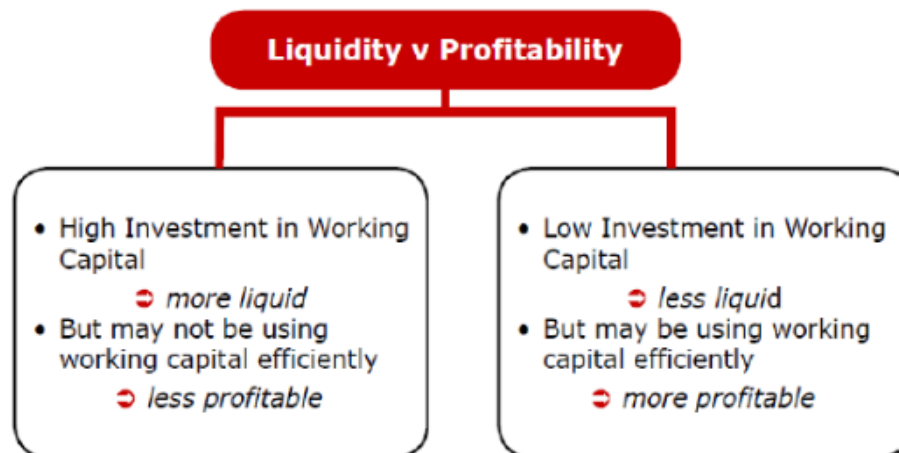
Cash will all be spent, so the company must rely on an overdraft (probably unexpected and unplanned) to finance its net current assets. As dependence on overdraft finance is risky and costly, the company should seek a more permanent and less costly form of funding.

1.2 Objectives of Working Capital Management

The two main objectives of working capital management are:

1. To ensure the organisation has sufficient liquid resources to continue in business.
2. To increase its profitability.

There is, however, a trade-off between liquidity and profitability:



As shown by *Examples 1* and *2* above, the objectives of liquidity and profitability can conflict. They are not necessarily always in conflict, however. Good warehouse management (to keep inventory low, but not so low that sales are lost if inventory is not available, say) and good credit control (to keep receivables low, but not so low that sales are lost through excessive credit control, say) could lead to higher liquidity and higher profitability.

For each company, there *will* be an optimal level of working capital. However, this can only be found by trial and error; in any case, it constantly changes.

Businesses must avoid the extremes:

- **Overtrading** arises when there is insufficient working capital to support the level of business activity. This also can be described as under-capitalisation and is characterised by an increasing proportion of short-term to long-term finance.
- **Over-capitalisation** is an excessive level of working capital, leading to inefficiency.

As a guide, many textbooks suggest that to be safe, an organisation requires a 2:1 ratio of current assets to current liabilities. For every \$1 of current liabilities, \$2 of current assets is needed to ensure that the organisation does not run into cash flow problems.

However, this is much too simplistic, and the required level of working capital will vary by industry. This is illustrated in *Example 4*, which shows a breakdown of the working capital for three UK public limited companies operating in different sectors. The figures are taken from published annual reports and the given ratios are calculated as follows:

$$\text{Current ratio} = \frac{\text{Current assets}}{\text{Current liabilities}}$$

$$\text{Quick (acid test) ratio} = \frac{\text{Quick assets}}{\text{Current liabilities}} = \frac{\text{Current assets} - \text{inventory}}{\text{Current liabilities}}$$

Example 4 Industry Comparison

Comparison of working capital by industry

- Tesco plc is a multinational retailer of general merchandise
- Airtours is a travel operator specialising in weekend trips and sun holiday charter packages
- MUFC is Manchester United Football Club

	Tesco plc	Airtours	MUFC
	\$m	\$m	\$000
Current assets			
Inventory	584	17.0	3,565
Receivables	133	413.8	10,731
Short-term investments	196	11.1	22,400
Cash at bank and in hand	<u>29</u>	<u>364.2</u>	<u>23,244</u>
	942	806.1	59,940
Current liabilities	<u>2,712</u>	<u>802.0</u>	<u>29,468</u>
Working capital	<u>(1,770)</u>	<u>4.1</u>	<u>30,472</u>
Profit before taxation	<u>832</u>	<u>140.3</u>	<u>27,577</u>
Current ratio	0.35	1.01	2.03
Acid test	0.13	0.98	1.91
Cash to current liabilities	0.01	0.45	0.79

Analysis

Each of these companies is profitable and is considered successful in its field. However, it is apparent that only MUFC meets the "suggested" current ratio of 2:1. Indeed, Tesco appears to be in real trouble with only \$0.35 of current assets and \$0.13 of "quick" assets for every \$1 of current liabilities.

Worse still, considering the ratio of cash to current liabilities, Tesco has only 1 cent of cash coverage for every \$1 of current liabilities, suggesting severe liquidity problems. Yet, Tesco is the largest supermarket chain in the UK and, in the year of this extract, has more than \$800,000 annual profit before taxation.

Airtours also falls well short of the suggested current ratio of 2:1, although its quick assets ratio of 1:1 is satisfactory. These figures illustrate that the 2:1 ratio is inappropriate, and the amount of working capital required by an organisation will vary depending on the nature of its business and the industry in which it operates.

Tesco

Although Tesco's level of working capital appears low, this must be evaluated in the context of the nature of its business. Each day, millions of customers will purchase their groceries from Tesco and pay cash before they leave the store.

Most items sold by Tesco have a shelf life of only a few days. As a market leader, Tesco can rely on regular deliveries of goods from suppliers at fairly short notice. In addition, forecasting techniques will enable managers to make reasonably reliable predictions of daily sales levels. All of these factors allow Tesco to operate with relatively low levels of inventory.

Because almost all sales are for cash, the level of receivables is also low. In addition, the company can invest surplus cash balances in short-term investments (usually money market accounts), maximising the return to its investors.

Considering current liabilities, Tesco will purchase most of its inventory on credit (resulting in a trade payables balance of \$826m included in current liabilities). Indeed, most inventory will have been sold and a profit realised before Tesco pays its suppliers. Few organisations are in such an advantageous position. Other payables will include corporation tax and dividends, amounts which Tesco will know with certainty when they are to be paid.

Due to the nature of its business, and in particular an abundance of cash sales, few receivables, low levels of inventory and most purchases being for credit, cash flow is not likely to be a problem; hence, Tesco can operate with negative working capital.

Airtours

Customers will usually pay for their holidays well before departure, ensuring that cash flow is not a problem while also minimising the incidence of irrecoverable debts. Unlike Tesco, Airtours' sales are seasonal, with most cash received from January to June. However, expenses will be incurred throughout the year, and careful planning is necessary to ensure that Airtours can meet its current liabilities as they fall due.

Because it is a tour operator, inventory levels are relatively low. Receivables mainly comprise amounts paid in advance for hotel accommodations and balances owing from customers for holidays. Payables include amounts owing for accommodation and advance payments made by customers.

Like Tesco, Airtours can operate with a lower current ratio than the suggested 2:1. However, due to the seasonal nature of its business, good budgeting and forecasting are essential to ensure that liabilities can be met even during the quiet season.

Manchester United Football Club (MUFC)

There is little evidence of any working capital problems, with the company having a 2:1 current ratio. The company also has \$0.79 of cash for every \$1 of current liabilities. This is not surprising, given the nature of MUFC's business. Cash flow is not likely to be a problem from August to May because almost every week, over 50,000 fans will crowd into Old Trafford to watch their team play. Many fans pay for their seats in advance, purchasing a season ticket before the season commences. In addition, the club receives cash from sponsors, television companies and the sale of merchandise. However, as for Airtours, "business" is seasonal and careful financial planning is necessary to ensure all liabilities are met as they fall due.

A review of the club's working capital shows that inventory and receivables are relatively small, with most of its working capital comprising short-term investments and cash, reflecting the cash received from season ticket sales.

Overall Conclusion

This analysis shows that the optimum level of working capital varies depending on the industry in which an organisation operates and the nature of its transactions.

1.3 Current Assets Investment Policy

Even within the same industry sector, companies will have different policies regarding the level of investment in current assets, depending on management's attitude to risk.

- If management is highly risk-averse, it will take a **conservative** approach to the level of investment in current assets. This is more appropriate if cash flows are erratic and unpredictable; hence, a margin of safety is needed. It is associated with:
Problems with such an approach include inventory obsolescence and a high financing cost for the high level of assets.
 - maintaining relatively high levels of inventory to ensure availability;
 - offering generous credit terms to customers to encourage demand;
 - paying suppliers promptly to ensure goodwill/minimise stockouts; and
 - holding high precautionary levels of cash.
- If management has a higher tolerance for risk, it would adopt an **aggressive** approach and drive down the levels of inventory, receivables and holdings of surplus cash. It is more appropriate if cash flows are very predictable. This policy would potentially be more profitable because of the lower level of investment in current assets. However, this would also be more risky due to, for example:
 - running out of inventory in periods of fluctuating demand;
 - losing customers to competitors who offer more generous credit; or
 - being less able to meet unexpected expenses.
- A moderate approach falls between the conservative and aggressive approaches.

2.1 Interest Rates

Whatever level of current assets the business holds, it must be matched by liabilities (i.e. current assets must be *funded*).

Management must decide whether to use *short-term* or *long-term* finance or in what proportions, if a mix is used. A key factor is the relative cost of various sources of finance.

Generally, the cost of short-term finance is below that of long-term finance. This is due to the term structure of interest rates (see *Chapter 2*).

2.2 Sources of Finance

2.2.1 Short-term

Overdraft	• Usually expensive but flexible (i.e. level of finance fluctuates to meet requirements). • Variable interest rate exposes entity to rate rises. • Repayable on demand.
Short-term loans	• Usually lower interest rate than long-term debt (unless the yield curve is inverted – see <i>Chapter 2</i>). • Renegotiation risk (i.e. the bank may refuse to refinance on maturity).
Accounts payable	• Appears cheap, but refusing settlement discounts can be expensive. • Taking excessive credit may lead to lost goodwill with suppliers and even penalties for late payment. • Trade credit can disappear (i.e. if too much credit is taken from suppliers, they may lose patience and refuse to give credit in the future).

Although short-term funding may, in some cases, be relatively cheap, there is a danger that it can quickly disappear. For example:

- The bank may ask for the overdraft to be repaid.
- Key suppliers may put a customer on a "stop list" and refuse further deliveries until all outstanding invoices are paid (and future deliveries may have to be paid for in advance).

Therefore, it may be wise to at least partly use long-term finance as, although generally more expensive, it reduces exposure to renegotiation risk.

2.2.2 Long-term

Equity	• New share issues or, to avoid issue costs, retained profits. • No legal commitment to repay (i.e. no renegotiation risk).
Debt	• Loan note issues and long-term bank loans. • If interest rates are fixed, is not affected by rising rates (until maturity).

2.3 Permanent v Fluctuating Current Assets

Some proportion of "current" assets on the statement of financial position may be more permanent. Possible reasons include:

- Holding of "buffer stock" (i.e. a minimum level of inventory held throughout the year as protection against "stock-outs");
- Holding a minimum "precautionary" balance of cash to meet any unexpected payments;
- A minimum level of trade receivables over the business cycle. Over the year, total current assets will naturally fluctuate above the minimum permanent level – only this excess is truly short-term.

2.4 Strategies for Funding Current Assets

There is an argument that any permanent portion of current assets should be matched with long-term finance, and the fluctuating portion should be matched with short-term finance. This is called a "matching policy" to the financing of working capital (i.e. finance maturity matches the realisation of the assets). A matching policy is consistent with management accepting a moderate level of risk.

An "aggressive" financing strategy would be to use short-term finance not only for the fluctuating balance of current assets but also for some, if not all, of the permanent balance. This is potentially a cheap financing strategy but indicates that management has a high and potentially dangerous tolerance for risk. Short-term finance can quickly deplete, leaving the organisation in financial distress. Lessons should be learned from cases like Northern Rock, a UK bank that relied exclusively on short-term interbank financing. In the 2008 financial crisis, Northern Rock found its only source of finance was cut off, causing the bank to collapse.

A "conservative" financing strategy would be to use long-term finance not only for the permanent level of current assets but also for part, if not all, of the fluctuating balance. This may be relatively expensive but is of lower risk for the organisation and means that management does not have to think continually about rolling over finance. Therefore, a conservative policy is consistent with highly risk-averse management (e.g. in an owner-managed business).

Funding strategies are also affected by previous funding decisions and organisation size.

3.1 Liquidity Ratios

$$\text{Current ratio} = \frac{\text{Current assets}}{\text{Current liabilities}}$$

If the current ratio falls below 1.0, this may indicate problems in meeting obligations as they fall due. Even if the current ratio is above 1.0, it does not guarantee liquidity, particularly if inventory is slow moving. On the other hand, a very high current ratio is not to be encouraged as it may indicate inefficient use of resources (e.g. excessive cash balances).

$$\text{Quick (acid test) ratio} = \frac{\text{Quick assets}}{\text{Current liabilities}} = \frac{\text{Current assets} - \text{Inventory}}{\text{Current liabilities}}$$

The quick ratio is particularly relevant when inventory is slow moving and cannot be converted into cash quickly. A company with slow-moving inventory should ideally have a quick ratio of at least 1. For a company with fast-moving inventory, the quick ratio can be less than 1 without suggesting that it is facing cash-flow problems.

3.2 Efficiency Ratios

$$\text{Inventory turnover} = \frac{\text{Cost of sales}}{\text{Average inventory}}$$

Inventory turnover shows how quickly inventory is sold, with higher turnover reflecting faster-moving inventory.

Working capital ratios are often easier to interpret if they are expressed in "days" as opposed to "turnover":

$$\text{Inventory holding period} = \frac{\text{Average inventory}}{\text{Cost of sales}} \times 365 \text{ days}$$

The inventory holding period estimates the time taken for inventory to be sold. Everything else being equal, management would prefer a shorter period.

Exam advice

When calculating ratios, use year-end figures in the statement of financial position if averages are not available.

$$\text{Receivables collection period} = \frac{\text{Average accounts receivable}}{\text{Annual credit sales}} \times 365$$

The receivables collection period estimates the time taken for a customer to pay. Everything else being equal, management would prefer a shorter period.

$$\text{Payables payment period} = \frac{\text{Average accounts payable}}{\text{Annual credit purchases}} \times 365$$

The payables payment period estimates the time taken to pay suppliers. Management would prefer to increase the credit period unless this proves expensive in terms of lost discounts or leads to other problems such as reduced reliability or quality of supplies.

$$\text{Sales/working capital} = \frac{\text{Annual sales}}{\text{Average working capital}}$$

Sales/working capital indicates how efficiently a business uses its working capital to generate sales. Everything else being equal, management would prefer sales/working capital to rise.

Exam advice

This ratio is also called the sales/net working capital ratio.

3.3 Calculating and Interpreting Ratios

Points to bear in mind when calculating and interpreting ratios:

- Seasonal and other factors may mean that the amounts in the statement of financial position are atypical.
- Ratios may be distorted by "window dressing" (e.g. large payment to suppliers at the year end to reduce the reported payables payment period).
- Ratios concern the past and do not predict the future.
- Ratios are of little value unless compared with industry averages.
- Where limited data is available (e.g. only two statements of financial position), it is more useful to calculate two sets of ratios based on year-end figures rather than a single set based on averages.

Ratio analysis is covered in more detail in *Chapter 19*.

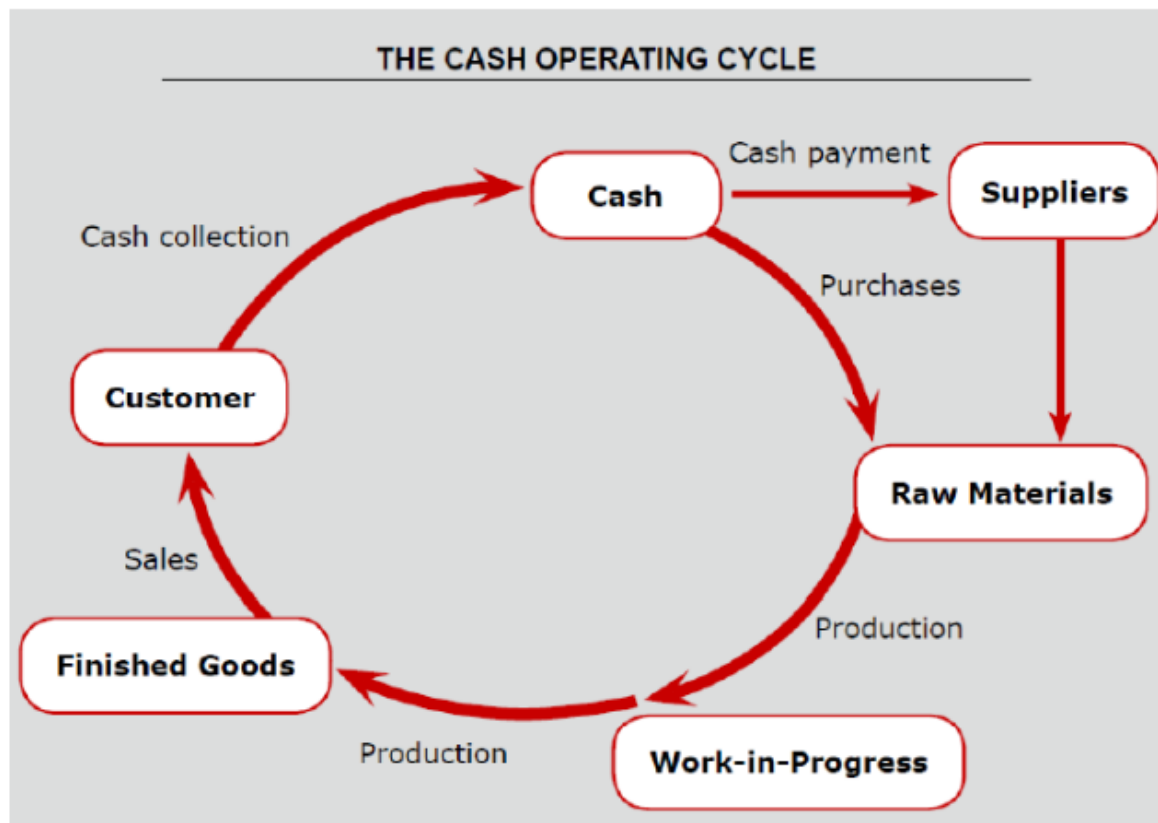
4.1 Cash Operating Cycle

The working capital cycle (also called the cash conversion cycle or the cash operating cycle) is the period (in days, weeks or months) between paying suppliers and receiving cash from customers. It can be found from standard ratios as inventory holding period + receivables collection period – payables payment period.

Inventory is purchased on credit from suppliers and is sold for cash and on credit. Cash received from customers is used to pay suppliers, wages and any other expenses. In general, a business will want to minimise the length of its working capital cycle, thereby reducing its exposure to liquidity problems.

The longer a business holds inventory (inventory turnover period), the longer it takes to collect cash from credit sales (accounts receivable payment period), and the more quickly it pays suppliers (accounts payable payment period), the longer the working capital cycle and the more adverse cash flow difficulties an organisation will face.

The longer the cycle, the greater the resources invested in working capital. Although having as short a cycle as possible is desirable, it is often challenging to differ significantly from competitors in the same trade.



4.2 Factors Affecting Cycle Length

The length of the cash operating cycle is affected by various factors, for example:

- **Type of industry** – a supermarket chain may have a low inventory holding period (fresh food), a low receivables collection period (perhaps just the time to receive settlement from credit card companies) and a significant payables payment period (taking credit from small farmers). In this case, the operating cycle could be negative (i.e. cash is received from sales before suppliers are paid). On the other hand, a construction company may have a very long operating cycle due to the investment in work-in-progress.
- **Industry norms** – if competitors offer long credit periods, reducing the receivables collection period may be challenging without losing business.
- **Power of suppliers** – an attempt to delay payments could lead to a supplier demanding “cash on delivery” (i.e. causing the payables payment period to fall to zero rather than rising).
- **Efficiency** of working capital management (e.g. weak credit control and holding of excess inventory will both lead to a longer working capital cycle).
- **Terms of trade**, such as the credit period offered to customers.

For a manufacturer, the calculation of the cash operating cycle can require a detailed analysis of the three types of inventory (raw materials, work in progress and finished goods). In this case, the cycle can be calculated as:

Receivables collection period	=	$\frac{\text{Accounts receivable}}{\text{Annual credit sales}} \times 365$	=	x
Payables payment period	=	$\frac{\text{Accounts payable}}{\text{Annual credit purchases}} \times 365$	=	(x)
Finished goods holding period	=	$\frac{\text{Finished goods inventory}}{\text{Annual cost of goods sold}} \times 365$	=	x
WIP production period (length of production process)	=	$\frac{\text{Work in progress}}{\text{Annual cost of goods sold}} \times 365$	=	x
Raw materials holding period	=	$\frac{\text{Raw materials inventory}}{\text{Annual raw materials purchases}} \times 365$	=	<u>x</u>
Length of cycle				<u>x</u>

WIP production period estimates the length of the production cycle (i.e. the number of days to convert raw materials into finished goods).

Exam advice

Use year-end figures from the statement of financial position if averages are not available.

Activity 1 Cash Operating Cycle

[Show](#)

Tipple Co has the following estimated figures for the coming year:

Sales	\$3,600,000
Accounts receivable	\$306,000
Gross profit margin	25%
Finished goods inventory	\$200,000
Work-in-progress inventory	\$350,000
Raw materials inventory	\$150,000
Accounts payable	\$130,000

WIP is 80% complete. Purchases represent 60% of production cost.

Required:

Calculate the length of the cash operating cycle.

**Please use the notes feature in the toolbar to help formulate your answer.*

Cost of sales = 75% × \$3.6m = \$2.7m

	WORKINGS	Days
Raw materials holding period	$\frac{\$150,000}{\$2.7m \times 60\%} \times 365$	34
Credit taken from suppliers	$\frac{\$130,000}{\$2.7m \times 60\%} \times 365$	<u>(29)</u> 5
WIP (production process)	$\frac{\$350,000}{\$2.7m \times 80\%} \times 365$	59
Finished goods holding period	$\frac{\$200,000}{\$2.7m} \times 365$	27
Credit given to customers	$\frac{\$306,000}{\$3.6m} \times 365$	<u>31</u>
Number of days between payment and receipt (i.e. length of cash operating cycle)		<u>122</u>

5.0 Introduction

Working capital management is a crucial factor in an organisation's long-term success. Clear and effective policies for each component of working capital are therefore vital and are the direct responsibility of the financial manager.

5.1 Inventory

The cost of holding inventory is relatively easy to measure and includes:

- storage;
- security; and
- losses due to theft, obsolescence and goods perishing.

Therefore, it is important not to hold excessive levels of inventory.

What is less easy to quantify is the cost of not holding sufficient inventory to meet customer demand. For example, if an organisation has insufficient inventory to meet demand, it will initially result in lost sales. In the longer term, it may also damage a business's goodwill as long-standing customers may turn to other, more reliable suppliers.

Inventory management is explored in *Chapter 14*.

5.2 Receivables

Too often, especially during startup, businesses concentrate on generating sales and pay little attention to collecting money from receivables. As a result, although sales exist on paper, the associated cash receipts take too long to materialise, and cash flow problems occur. Additionally, the longer a debt is outstanding, the greater the likelihood it will become bad (uncollectable).

The key areas of accounts receivable management are assessing creditworthiness, setting credit terms, monitoring and effective cash collection. These are all considered in detail in *Chapter 16*.

5.3 Trade Payables

The practice of businesses extending trade credit to one another is probably the most important source of short-term funding available to most organisations. At first glance, trade credit appears to represent a short-term, interest-free loan, which enables a higher level of trade than if everything were paid for immediately in cash.

However, there are costs associated with trade credit.

One cost of trade credit, which is often ignored, is its potential effect on the creditworthiness of a business. If a business consistently exceeds the credit period imposed by suppliers, its credit rating will be damaged in the long term. In the worst-case scenario, suppliers will be forced to take legal action and may even withdraw their credit facility, requiring cash on delivery.

Another cost associated with trade payables is the cost of forgoing early settlement discounts (see *Chapter 16*).

5.4 Cash and Bank Balances

Liquidity problems often arise because inflows and outflows of cash do not coincide. For example, a small tour operator will likely be "flush" with cash in peak holiday booking periods but have lower sales and cash balances in non-peak periods. However, business expenses such as wages and salaries, heat and light, rent and loan interest will remain more or less the same throughout the year. Therefore, businesses must plan to ensure that sufficient cash is available to meet expenses in the off-peak period.

Preparing a cash budget will indicate the flow of receipts and payments and forecast periods of surplus and deficit cash balances, thereby reducing uncertainty. If a large surplus is forecast, cash can be invested in an interest-earning account until required. If a deficit is forecast, the business can arrange a bank overdraft or loan. However, overdrafts and loans should be avoided where possible due to their high cost.

5.5 Problems for Small Businesses

Although businesses of any size can experience working capital problems, small businesses usually have the most issues, especially during startup.

For example, as sales increase, a small manufacturer must acquire more raw materials to produce enough goods to meet the demand. At the same time, workers needed to work longer hours, must be paid overtime. However, the cash from credit sales may not be received for several weeks, while suppliers and employees require immediate payment. In this situation, a business will likely depend heavily on a bank overdraft or loans. In the long term, if the business survives, the problem will be reduced through negotiating better credit terms with customers and suppliers.

5.6 Overtrading

Overtrading occurs when a company tries to support a large volume of trade from a small working capital base. Also called undercapitalisation, it often occurs when a business grows rapidly without an increase in long-term finance.

The result can be a liquidity crisis. This often can happen at the start of a new business, because:

- there is no reputation to attract customers, so a long credit period is likely to be extended to break into the market;
- if the business has found a "niche market", rapid sales expansion may occur; or
- fast-growing smaller companies often lack the management skills to maintain adequate control over debt collection and the production period.

For the above reasons, the amount of cash required will increase. However, companies in this position often will find it hard to raise long-term finance; hence, overtrading and business failure may result.

The indicators of overtrading include:

- ✗ declining liquidity;
- ✗ rapidly increasing revenue;
- ✗ increasing inventory holding period;
- ✗ increasing receivable collection period;
- ✗ increasing short-term borrowing and declining cash holdings;
- ✗ large and rising overdraft;
- ✗ declining profit margin; and
- ✗ an increasing ratio of sales to non-current assets.

If a business suffers from liquidity problems due to overtrading, the aim will be to reduce the length of the cash operating cycle. Possibilities to consider include:

- ✓ reducing the inventory holding period for both finished goods and raw materials;
- ✓ reducing the production period (this is not easy to do, but it might be worth investigating different machinery or working methods);
- ✓ reducing the credit period extended to accounts receivable and tightening up on cash collection;
- ✓ increasing the period of credit taken from suppliers;
- ✓ increasing the level of long-term finance (i.e. an equity or debt issue, with a new share issue being probably preferable to increasing debt in a risky company); and
- ✓ reducing the level of sales growth to a more sustainable level.

5.7 Summary

This chapter has examined the components of working capital and considered how organisations could improve their working capital management. Although an ideal level of working capital is difficult to calculate and will vary from one organisation to another, depending on the industry in which they operate, a business must avoid having too little or too much working capital.

Too little working capital ("overtrading") is common when a business is starting up or experiencing rapid growth. The level of sales might grow very quickly, but inadequate working capital is available to support this growth. The situation will then arise where a business may be profitable on paper but has insufficient funds to pay debts as they become due. In the short term, this situation can be solved through a combination of measures, including:

- obtaining an increased overdraft facility;
- negotiating a longer credit period with suppliers; and
- encouraging receivables to pay faster.

However, in the long term, a business is unlikely to survive without a combination of the following:

- new capital from shareholders/proprietor;
- better control of working capital; and
- the building up of an adequate capital base through retained profits.

Almost as bad is too much working capital or overcapitalisation. Poor working capital management will result in excessive resources used to fund current assets. Such a scenario will lead to a business earning a lower-than-expected return.

The shorter the working capital cycle, the faster cash, and hence profits from credit sales, will be realised. To achieve this, an organisation must regularly review its working capital, taking action where necessary.

For a meaningful evaluation of working capital management, it is necessary to compare metrics to trends (i.e. the change in a ratio over time) and external benchmarks (e.g. industry average or sector average ratios).

Activity 2 Assessing Working Capital

[Show](#)

The following are summary financial statements for Stalla Co:

	20X0	20X4
	\$000	\$000
Non-current assets	115	410
Current assets	<u>650</u>	<u>1,000</u>
Total assets	<u>765</u>	<u>1,410</u>
Capital and reserves	210	270
Current liabilities	513	982
Long-term liabilities	<u>42</u>	<u>158</u>
Total equity and liabilities	<u>765</u>	<u>1,410</u>

	20X0	20X4
	\$000	\$000
Sales	1,200	3,010
Cost of sales, expenses and interest	<u>1,102</u>	<u>2,860</u>
Profit before tax (PBT)	98	150
Tax and distributions	<u>33</u>	<u>133</u>
Retained earnings	65	17

Notes:

Cost of sales was \$488,000 for 20X0 and \$1,330,000 for 20X4.

Receivables are 50% of current assets and trade payables are 25% of current liabilities for both years.

Required:

Using suitable financial ratios and paying particular attention to growth and liquidity, discuss the significant changes faced by the company since 20X0. Comment on the capacity of the company to continue trading.

**Please use the notes feature in the toolbar to help formulate your answer.*

Discussion:

From 20X0 to 20X4, sales for Stalla have grown by 150% (all calculations are shown after these comments). The pressures of such growth in terms of supporting the business through adequate working capital can be substantial. Therefore, current assets have expanded by 54% in the same period and current liabilities by 91%. Although this aspect of the business will be dealt with in more detail below, it is worthwhile questioning at this stage whether sufficient funding for working capital is available to support the growth in sales.

Although there has been significant growth in sales during the period, PBT as a percentage of sales has declined from 8.2% to about 5.0%. This must seriously call into question either the management of costs (operational or financial) or whether there is an inability to force price increases onto customers. Given the information available, interest costs are a possible source of some of this problem. Both current and long-term liabilities have increased substantially (91% and 276%, respectively) against a background of barely increased equity funding. Debt funding (long- and short-term) looks to have increased (see following details) and will have an associated interest burden. This has an importance to the sustainability of the business.

Earnings retained do not appear sufficient to fund business growth. Hence, it is clear that borrowings have been increased to deal with this problem. However, a balance must be kept in the business between its earnings capability and its capacity to service its debt commitments.

Although the dollar amount of PBT has increased by 53% over the period, retained earnings have declined by about 74%. This may be partly explained by an increased tax burden but is due mainly to excessive distributions. In other words, the business is not retaining enough funds to support its growth. It also appears that funding is not coming from increased equity issues.

The effect of growth exceeding its funding base potentially will severely affect liquidity. Net current assets are not seriously out of line if a ratio of current assets to current liabilities of 1:1 is considered acceptable. However, a different picture emerges when current assets are viewed against sales. The ratio was 54% in 20X0 and only 33% in 20X4. This suggests, together with the other information, that inventory, receivables and cash resources might be insufficient to support sales. This may reflect

inventory, receivables and cash resources might be insufficient to support sales. This may reflect greater efficiency in current asset management if the receivables collection period is compared over the period (it declined from 99 to 61 days). However, the receivables collection period appears less favourable when compared with the payables payment period, which also declined (from 96 to 67 days during the period).

Sales to working capital increased dramatically from 9:1 in 20X0 to 167:1 in 20X4. This reflects the increase in sales combined with reduced current asset investment and increased current liabilities (e.g. overdraft).

The business faces a potentially critical situation because debt rather than equity funding has grown. Current assets mainly comprise inventory and receivables (because the business has substantial borrowings, it is unlikely to have ample cash balances simultaneously), funded by borrowings rather than retained earnings. The business generates inadequate profits and distributes too much of its earnings. The outlook is for greater borrowing.

The poor profit figures suggest that a critical point in liquidity and solvency has been reached. This is reflected in debt/equity ratios, which have increased from 2.6 in 20X0 to 4.2 in 20X4 (current and long-term liabilities are used as debt, and capital and reserves are used as equity). Unless a capital reorganisation can take place quickly, either through injected funds or conversion of debt into equity, the business will likely become insolvent.

Ratios:

	WORKINGS (in \$000)	
Sales growth:	$(3,010 - 1,200)/1,200 =$	150%
Current asset growth:	$(1,000 - 650)/650 =$	54%
Current liability growth:	$(982 - 513)/513 =$	91%
Long-term liabilities growth:	$(158 - 42)/42 =$	276%

PBT growth:	$(150 - 98)/98 =$	53%
Retained earnings decline:	$(65 - 17)/65 =$	74%

	20X0	20X4
PBT/Sales	$98/1,200 = 8.2\%$	$150/3,010 = 5.0\%$
Current assets/Current liabilities	$650/513 = 1.3$	$1,000/982 = 1.0$
Current assets/Sales	$650/1,200 = 54\%$	$1,000/3,010 = 33\%$
Sales/Working capital	$1,200/(650 - 513) = 9$	$3,010/(1,000 - 982) = 167$
Debt/Equity	$(513+42)/210 = 2.6$	$(982+158)/270 = 4.2$
Receivables at 50% of current assets	325,000	500,000
Sales per day (365 days)	3,287	8,246
Receivables collection period	$325,000/3,287 = 99$	$500,000/8,246 = 61$
Payables at 25% of current liabilities	128,250	245,500
Cost of sales per day (365 days)	1,337	3,644
Payables payment period (days)	$128,250/1,337 = 96$	$245,500/3,644 = 67$

Syllabus Coverage

This chapter covers the following Learning Outcomes.

C. Working Capital Management

1. The nature, elements and importance of working capital

- a) Describe the nature of working capital and identify its elements.
- b) Identify the objectives of working capital management in terms of liquidity and profitability, and discuss the conflict between them.
- c) Discuss the central role of working capital management in financial management.

2. Management of inventories, accounts receivable, accounts payable and cash

- a) Explain the cash operating cycle and the role of accounts payable and accounts receivable.
- b) Explain and apply relevant accounting ratios, including:
 - i) current ratio and quick ratio
 - ii) inventory turnover ratio, average collection period and average payable period
 - iii) sales revenue/net working capital ratio.

3. Determine working capital needs and funding strategies

- a) Calculate the level of working capital investment in current assets and discuss the key factors determining this level including:
 - i) the length of the working capital cycle and terms of trade
 - ii) an organisation's policy on the level of investment in current assets
 - iii) the industry in which the organisation operates.
 - b) Evaluate and discuss the key factors in determining working capital funding strategies including:
 - i) the distinction between permanent and fluctuating current assets
 - ii) the relative cost and risk of short-term and long-term finance
 - iii) the matching principle
 - iv) the relative costs and benefits of aggressive, conservative and matching funding policies
 - v) management attitudes to risk, previous funding decisions and organisation size.
-

Summary and Quiz

- Working capital's primary function is to support the day-to-day financial operations of an organisation, including the purchase of inventory, the payment of salaries, wages and other business expenses, and the financing of credit sales.
- The key issues in managing working capital are:
 - What level of current assets should a business hold?
 - How should current assets be financed?
- Even within an industry sector, companies will have different policies on the level of investment in current assets, depending on management's attitude to risk.
- The cost of short-term finance is typically below that of long-term finance, due to the term structure of interest rates.
- Managers who follow the matching principle may attempt to "match":
 - any permanent portion of current assets with long-term finance; and
 - the fluctuating portion of current assets with short-term finance.
- An aggressive funding policy will fund some or all permanent current assets with short-term finance. A conservative funding policy will fund permanent and fluctuating current assets with long-term finance.
- Good knowledge of ratio analysis is essential in many exam questions on working capital management. Liquidity ratios; include the current ratio and the acid test ratio, efficiency ratios include inventory turnover and inventory holding period, receivables collection period, payables payment period and the sales to net working capital ratio.
- The cash operating cycle is the period between paying suppliers and receiving cash from customers. The longer the cycle, the greater the resources invested in working capital. It is affected by the type of industry and the efficiency of working capital management.
- Overtrading occurs when a company tries to support a large volume of trade from a small working capital base. Also called undercapitalisation, it often occurs when a business grows rapidly without increasing its level of long-term finance.

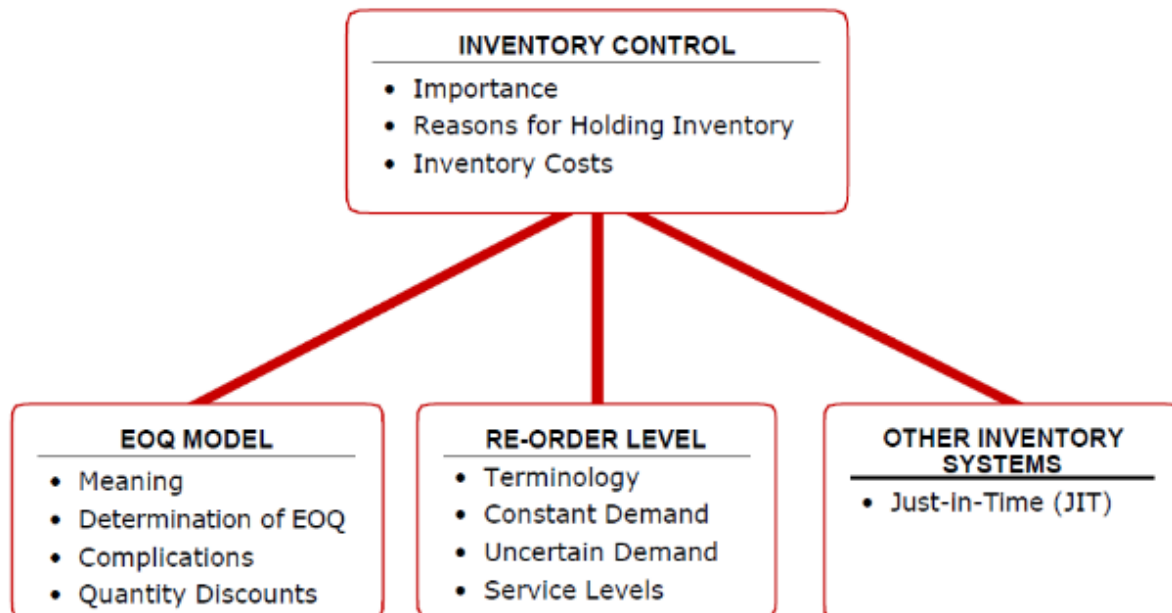
Chapter 13 Quiz

Chapter 14: Inventory Management

Overview	CHAPTER 14: Visual Overview
14.1 Inventory Control	14.1.1 Importance
	14.1.2 Reasons for Holding Inventory
	14.1.3 Inventory Costs
14.2 EOQ Model	14.2.1 Meaning
	14.2.2 Determination of EOQ
	14.2.3 Complications
	14.2.4 Quantity Discounts
14.3 Re-order Level	14.3.1 Terminology
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CHAPTER 14: Visual Overview

Objective: To understand the costs and benefits of holding inventory, calculate the Economic Order Quantity, which minimises costs, and evaluate Just-in-Time techniques.



1.1 Importance

1 Definition

Inventory control – the systematic regulation of inventory levels.

Inventory control is important because:

- Too high inventory levels incur additional costs and profits will be reduced.
- Too low inventory levels are insufficient to satisfy customers and profit will be reduced (e.g. Due to lost sales or additional costs).

For most organisations, the difficulty is determining the optimum level of inventory. This will depend on various factors, including:

- the average level of daily sales (adjusted for seasonal variations);
- the lead time between ordering goods and their delivery;
- the reliability of suppliers;
- the type of good and the danger of it perishing or becoming obsolete;
- the cost of reordering inventory;
- storage and security costs; and
- other factors such as rumours of a shortage or a price increase.

Systems must be in place to ensure that inventory levels are reviewed regularly, and appropriate action is taken when needed.

1.2 Reasons for Holding Inventory

Inventory may be raw materials, WIP, finished goods, goods for resale or consumables (i.e. supplies such as machine lubricants used in production processes). There are many reasons for holding inventory:

- ✓ To meet demand by acting as a *buffer* when consumption is unusually high. This reduces the risk of "stock-outs".
- ✓ To ensure continuous production.
- ✓ To take advantage of quantity discounts.
- ✓ To buy in ahead of an expected shortage or ahead of an expected price rise.
- ✓ For technical reasons (e.g. maturing whiskey in casks or keeping oil in pipelines).
- ✓ To reduce ordering costs.

1.3 Inventory Costs

As well as the obvious purchase price, costs associated with inventory include:

- Holding costs:
 - cost of capital tied up;
 - insurance;
 - deterioration, obsolescence and theft;
 - warehousing; and
 - stores administration.
- Re-order costs:
 - transport costs;
 - clerical and administrative expenses; and
 - batch set-up costs for goods produced internally.
- Shortage costs:
 - production stoppages caused by lack of raw materials;
 - stock-out costs for finished goods (e.g. lost sale); and
 - emergency re-order costs.
- Systems costs – people and computers.

Key Point

The benefits of holding inventory must outweigh the costs.

2.1 Meaning

The *Economic Order Quantity* (EOQ) is the quantity of inventory that should be ordered each time a purchase order is placed.

Definition

Economic order quantity (EOQ) – the **optimal** order quantity that minimises the total costs relevant to ordering and holding inventory.

2.2 Determination of EOQ

2.2.1 Formula

x	=	order quantity
C_h	=	cost of holding one unit for one year
D	=	annual demand
C_o	=	cost of placing an order

Exam advice

You will **not** be asked to derive the model in the FM exam, only to apply it. The formula is included on the formula sheet.

The total annual relevant cost to be minimised:

= annual holding cost

+ annual order cost

= the cost of holding one unit in inventory for one year $\times$ the average number of units held

+ the cost of an order $\times$ the number of orders in a year

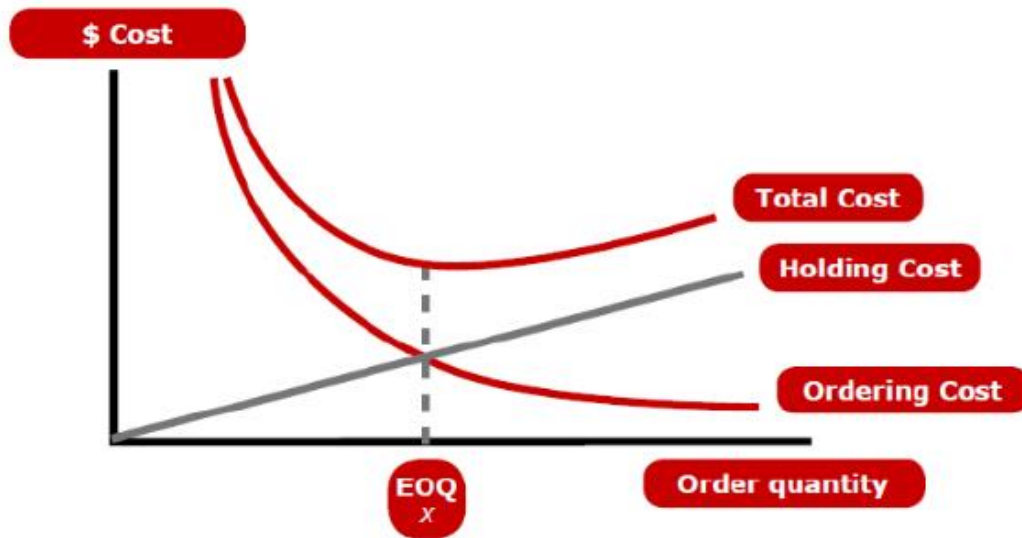
$$= \frac{x}{2} C_h$$

$$+ D \frac{C_o}{x}$$

Total cost is minimised when:

$$x = \sqrt{\frac{2C_o D}{C_h}} \quad \text{Exam formula}$$

2.2.2 EOQ graph



2.2.3 Assumption of EOQ

- Constant purchase price per unit.
- Constant demand.
- No risk of stock-outs.
- Order costs are independent of order quantity.
- Holding cost depends on the average level of inventory.

Activity 1 EOQ

[Show](#)

Using the following data, calculate the EOQ:

$$D = 40,000 \text{ units}$$

$$C_o = \$2$$

$$C_h = \$1$$

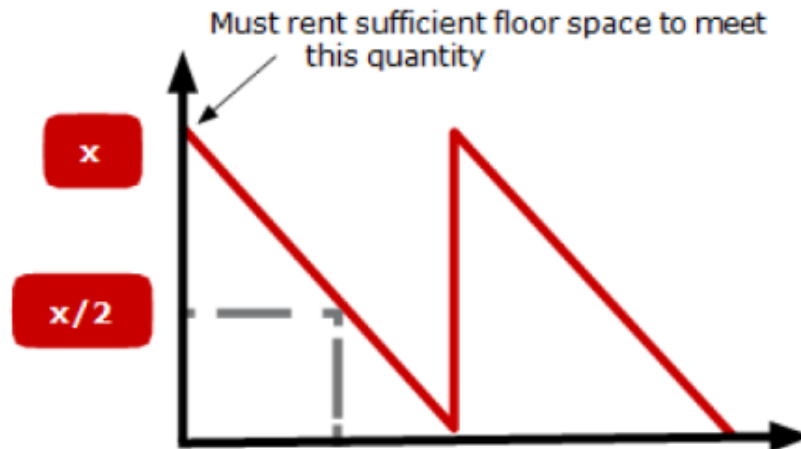
**Please use the notes feature in the toolbar to help formulate your answer*

$$EOQ = x = \sqrt{\frac{2C_oD}{C_h}} = \sqrt{\frac{2 \times \$2 \times 40,000}{\$1}} = 400 \text{ units}$$

2.3 Complications

2.3.1 Warehouse Rental

The EOQ model is based on the assumption that holding costs vary with the **average** inventory level. However, if a warehouse is rented on a long-term contract (rather than daily), it needs to be large enough to hold the **maximum** level of inventory rather than the average.



This is dealt with in the EOQ formula by **doubling the floor space** required for each unit when calculating the holding cost.

Activity 2 Floor Space

[Show](#)

Annual demand = 3,000 units

Reorder cost = \$5

Holding cost = \$3.33 per unit + rental of warehouse

Each unit occupies 3.0 square meters. Annual rental of the warehouse costs \$5 per square meter.

Required:

Calculate the EOQ.

**Please use the notes feature in the toolbar to help formulate your answer*

$$D = 3,000$$

$$C_O = 5$$

$$C_H = \$3.33 + (2 \times 3 \times 5) = \$33.33$$

$$EOQ = \sqrt{\frac{2 \times 5 \times 3,000}{33.33}} = 30 \text{ units}$$

2.3.2 Cost of Capital

Inventory, like any other asset, must be matched by a liability. Therefore, there must be a cost of financing inventory, which is a holding cost.

Example 1 Cost of capital
Cost of capital = 10%
Price per unit = \$100
Therefore, holding cost = \$100 × 0.1 = \$10
This is in addition to any other holding costs.

2.4 Quantity Discounts

When a supplier offers a quantity ("bulk-buying") discount on **each unit purchased** for specified quantities above the EOQ, the purchase price becomes relevant to the decision.

- To deal with this, calculate the **total annual cost** for the EOQ and for order quantities above EOQ for which discounts are available.
- The total annual cost is calculated as:

$$\text{Total annual cost} = \text{Annual holding cost} + \text{Annual order cost} + \text{Annual purchase cost}$$

- Choose the order quantity with the lowest total cost.

Activity 3 Quantity Discount

[Show](#)

Annual demand = 5,000

Holding cost = \$7.50

Reorder cost = \$30

Purchase price = \$1.10

A discount of 3% is available on orders of 300 units or more.

Required:

Determine whether the discount is worthwhile.

**Please use the notes feature in the toolbar to help formulate your answer*

$$EOQ = \sqrt{\frac{2 \times \$30 \times 5,000}{\$7.50}} = 200 \text{ units}$$

Total cost at EOQ	\$
Holding $\frac{x}{2} C_h = \frac{200}{2} \times \7.50	750
Reorder $\frac{D}{x} C_o = \frac{5,000}{200} \times \30	750
Purchase cost $5,000 \times \$1.10$	<u>5,500</u>
Total	<u>7,000</u>

Total cost at order quantity = 300 units

Holding $\frac{x}{2} C_h = \frac{300}{2} \times \7.50	1,125
Reorder $\frac{D}{x} C_o = \frac{5,000}{300} \times \30	500
Purchase cost $5,000 \times \$1.10 \times 0.97$	<u>5,335</u>
	<u>6,960</u>

Conclusion: The quantity to order is 300 units as the discount is worthwhile.

3.1 Terminology

i Definitions

Re-order level (ROL) – the level to which inventory should fall before placing a purchase order for replenishment.

Lead time – the time between placing and receiving an order.

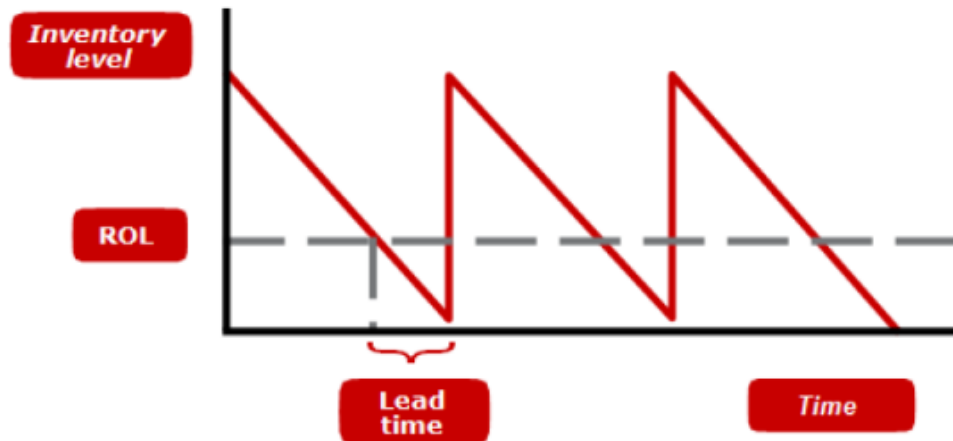
There are two possible situations to be dealt with:

1. **Constant** demand in lead time; and
2. **Uncertain** demand in lead time

3.2 Constant Demand

When demand is known and constant:

- Re-order level (ROL) = lead time (days) × demand per day
- For example, if demand is 40 units per day and lead time is two days, inventory should be reordered when inventory falls to 80 units. This can be shown graphically:



3.3 Uncertain Demand

If the level of demand is unknown, there can be only an expected level of demand. A **buffer** or safety level of inventory will then be needed to reduce the risk of a stock-out.

Method

1. Calculate **expected** demand in the lead time.

$$\text{Expected lead time demand} = \sum x_i p(x_i)$$

Where:

$$x_i = \text{level of demand} \quad p(x_i) = \text{probability of level of demand}$$

2. Take each level of demand $\geq$ expected lead time demand as a possible reorder level and calculate the expected annual stock-out cost.
3. Calculate the expected annual buffer holding cost for each possible ROL.
4. Choose the ROL with the lowest sum of stock-out cost and holding cost.

Activity 4 Re-order Level

[Show](#)

The following information relates to inventory levels of component XL5:

Holding cost = \$8 a unit

Stock-out cost = \$3 a unit

Lead time = One week

EOQ = 150 units

Weekly demand is given by:

x_i	$p(x_i)$
Demand	Probability
40	0.1
50	0.2
60	0.4
70	0.2
80	0.1

Assume 50 working weeks a year.

Required:

Calculate the optimum reorder level.

**Please use the notes feature in the toolbar to help formulate your answer*

Average weekly demand = $(40 \times 0.1) + (50 \times 0.2) + (60 \times 0.4) + (70 \times 0.2) + (80 \times 0.1) = 60$ units

Average annual demand = $60 \times 50 = 3,000$ units

Since the EOQ = 150, there will be $\frac{3,000}{150} = 20$ orders per year.

ROL	Buffer	Demand	Units short	Probability	Ave Units short	Annual stock-out cost*	Annual buffer holding cost	Total annual cost \$
60	0	70	10	0.2	2	$2 \times \$3 \times 20$	0	
		80	20	0.1	2	$2 \times \$3 \times 20$		
		Average			4	240	0	240
70	10	80	10	0.1	1	$1 \times \$3 \times 20$	$10 \times \$8$	
		Average			1	60	80	140
80	20	80	—	—	—	0	$20 \times \$8$	
						0	160	160

*Average units short $\times$ stock-out cost per unit $\times$ number of orders per year

Therefore, the optimum ROL is 70 units.

4.0 Introduction

Many other inventory systems are widely used in practice, including perpetual ("continuous") inventory systems relevant to the Audit and Assurance syllabus. However, Just-in-Time (JIT) is the only system other than EOQ in the Financial Management syllabus.

4.1 Just-in-Time (JIT)

A just-in-time *purchasing* system aims to minimise the cost of holding inventory by placing orders so that delivery is timed for when the goods are needed. For a retailer, this is when customers demand the goods; for a manufacturer, this is when components or raw materials are required to meet production schedules.

A JIT *production* system aims to minimise the cost of manufacturing by only producing goods as they are needed. Such a system should have only minimal inventory of raw materials and minimal wait time in production.

JIT inventory management aligns purchasing and production to weekly, daily or even hourly sales demand. It aims to create a continuous flow of raw materials inventory into work-in-progress, which becomes finished goods to go immediately to the customer. This means that negligible inventory needs to be held.

To operate a JIT inventory system, there are certain necessary conditions to be met:

- Sufficiently flexible suppliers and internal workforce to expand and contract output at short notice.
- The quality of raw material inventory must be guaranteed – indeed, quality must be maintained at every stage – as there is no buffer stock.
- Close working relationships with suppliers and, if possible, geographical proximity for immediate deliveries.
- A low inventory level typically requires short production runs. This is only appropriate, where set-up costs are low. High-technology production methods have made this easier to achieve.
- The workforce must be organised to accommodate flexible working hours (e.g. a core workforce combined with part-time or freelance employees).
- The factory design and layout must facilitate JIT deliveries to all areas.
- Total reliance on suppliers for quality and delivery requires contracts with strict penalty clauses.
- Significant investment by suppliers, who will require long-term agreements.

Syllabus Coverage

This chapter covers the following Learning Outcomes.

C. Working Capital Management

2. Management of inventories, accounts receivable, accounts payable and cash

- c) Discuss, apply and evaluate the use of relevant techniques in managing inventory, including the Economic Order Quantity model and Just-in-Time techniques.
- e) Discuss and apply the use of relevant techniques in managing accounts payable, including:
 - ii) evaluating the benefits of bulk purchase discounts

Summary and Quiz

- Economic order quantity (EOQ) is the optimal order quantity that minimises the total costs relevant to ordering and holding inventory.
- The EOQ formula is provided in the exam. Take care in applying it (e.g. if a holding cost varies with maximum inventory, rather than the average).
- If quantity discounts apply, choose the order quantity with the lowest total annual cost. Total annual cost includes annual purchase cost and must be calculated for the EOQ and all order quantities above EOQ for which discounts are available.
- Do not confuse the EOQ with the Re-order Level (ROL). EOQ specifies **how large** each order should be; ROL specifies **when** an order should be placed.
- Just-in-Time (JIT) is based on holding minimal inventory levels of raw materials, WIP and finished goods. It relies on close relationships with suppliers. Quality is essential to all types of inventory held.

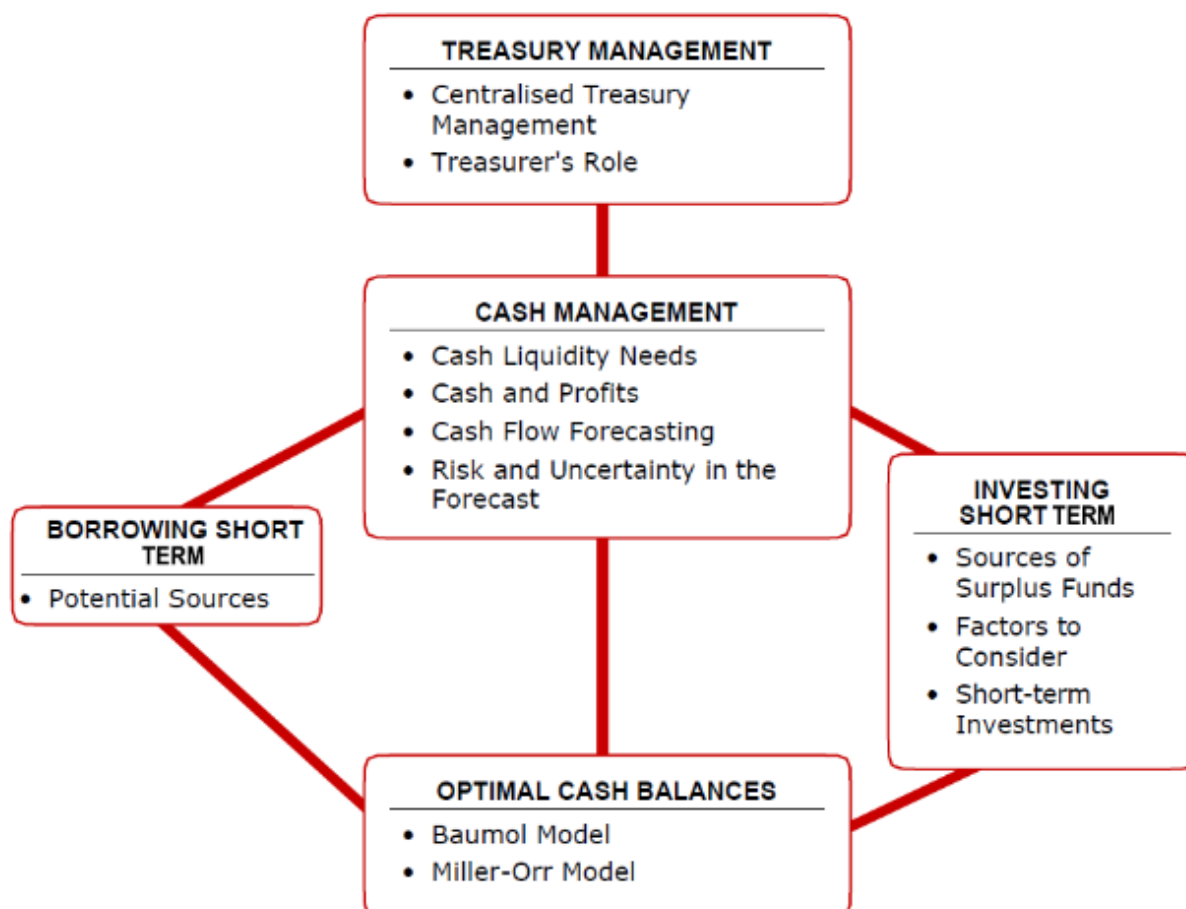
Chapter 14 Quiz

Chapter 15: Cash Management

Overview	CHAPTER 15: Visual Overview
15.1 Treasury Management	15.1.0 Introduction
	15.1.1 Centralised Treasury Management
	15.1.2 Treasurer's Role
15.2 Cash Management	15.2.1 Cash Liquidity Needs
	15.2.2 Cash and Profits
	15.2.3 Cash Flow Budgeting
	15.2.4 Risk and Uncertainty in the Forecast
15.3 Borrowing Short Term	15.3.1 Potential Sources
15.4 Investing Short Term	15.4.1 Sources of Surplus Funds
	15.4.2 Investing Surplus Funds – Factors to Consider
	15.4.3 Short-term Investments
15.5 Optimal Cash Balances	15.5.1 Baumol Model
	15.5.2 Miller-Orr Model
15 Conclusion	15 Syllabus Coverage
	15 Summary and Quiz
	15 Technical Articles

CHAPTER 15: Visual Overview

Objective: To understand the importance of cash flow management, apply theoretical models to optimise cash balances, and assess the benefits of centralised treasury management.



1.0 Introduction

Definition

Treasury management – the efficient management of liquidity and risk in a business including the management of funds (generated from internal and external sources), currencies and cash flow.

As companies and financial markets have become larger, more sophisticated and increasingly international, there has been a trend towards establishing separate treasury departments where cash control is centralised to ensure efficient use. For example, it will ensure surplus cash is invested in appropriate funds and realised when cash is required.

1.1 Centralised Treasury Management

Many organisations use centralised treasury management, which has several advantages.

- ✓ Management by specialised staff with appropriate qualifications, expertise and experience.
- ✓ Economies of scale (e.g. less staff required), as specialists are employed centrally, reducing duplication and maximising the use of skilled human resources and financial management systems.
- ✓ Ability to use "pooling", the netting of cash deficits against surpluses, to save interest expense from short-term financing.
- ✓ Increased negotiating power with banks, as the amounts borrowed or deposited would be more substantial as a group.
- ✓ More efficient foreign exchange risk management because the treasury department at the head office can find the group's net position on each currency and then consider an external hedge on this balance.

The treasury department of a large company may still have a degree of decentralisation to ensure that decisions taken are appropriate to local circumstances.

1.2 Treasurer's Role

Good cash management aims to have the right amount available at the right time. The treasurer will be involved in:

- accurate cash flow forecasting, so that shortfalls and surpluses can be anticipated;
- planning short-term borrowing when necessary;
- planning investments of surpluses when necessary;
- cost-efficient cash transmission;
- dealing with foreign currency issues;
- optimising banking arrangements;
- planning major finance-raising exercises; and
- accounts receivable/accounts payable policies.

In addition, the treasurer is often involved in risk assessment and insurance.

2.1 Cash Liquidity Needs

There are several reasons for holding cash:

- Transactions motive, where cash is held to provide sufficient liquidity to meet current day-to-day financial obligations (e.g. payroll, the purchase of raw materials, etc).
- Precautionary motive, where a cash reserve is held as a "cushion" against unplanned expenditure, like buffer inventory. This reserve may consist of "cash equivalents", which are short-term, low-risk, highly liquid investments (e.g. Treasury Bills).
- Speculative motive, where cash is held to take advantage of potential investment opportunities quickly (e.g. a "war chest" ready to use if a suitable takeover target appears).

However, cash holding must not be excessive as this leads to inefficiency. Cash balances belong to the shareholders expecting to receive significant investment returns.

Therefore, any long-term surplus of cash should be either reinvested into projects with a positive net present value or returned to shareholders via:

- dividends (regular or special dividend); or
- share buy-back programmes.

2.2 Cash and Profits

A company must be profitable to continue in existence. However, profitability is not enough; companies must also have enough cash to meet all their day-to-day payments and longer-term commitments.

2.3 Cash Flow Budgeting

A major task of the treasurer is cash flow budgeting. A simple pro forma would typically include forecasts for:

- Sales volume
- Revenue
- Costs
- One-off expenses (e.g. asset expenditure).

Typical format

Typical format

	Q1	Q2	Q3	Q4	Total
	\$	\$	\$	\$	\$
Cash inflows	x	x	x	x	x
Cash sales	x	x	x	x	x
Cash from receivables	x	x	x	x	x
Non-current asset disposals	x	x	x	x	x
Share/debt issues	<u>x</u>	<u>x</u>	<u>x</u>	<u>x</u>	<u>x</u>
Total inflow	<u>x</u>	<u>x</u>	<u>x</u>	<u>x</u>	<u>x</u>
Materials	x	x	x	x	x
Labour	x	x	x	x	x
Variable overhead	x	x	x	x	x
Fixed overhead	x	x	x	x	x
Dividends		x			x
Asset expenditure/leases			x		x
Interest/principal on debt	<u>x</u>	<u>x</u>	<u>x</u>	<u>x</u>	<u>x</u>
	<u>x</u>	<u>x</u>	<u>x</u>	<u>x</u>	<u>x</u>
Net cash flow	x	x	x	x	x
Opening balance	<u>x</u>	<u>x</u>	<u>x</u>	<u>x</u>	<u>x</u>
Closing balance	<u>x</u>	<u>x</u>	<u>x</u>	<u>x</u>	<u>x</u>

2.4 Risk and Uncertainty in the Forecast

Sensitivity analysis assesses the effect of change in a key variable.

- It is a "what if" analysis.
- It can deal with uncertainty in cash budgeting by finding the effect of a change in:
 - payment patterns by credit customers (consider the worst-case scenario);
 - timing of other receipts (e.g. sale of non-current assets, rights issues, debt issues, etc);
 - materials costs (if prices are uncertain, consider a worst-case scenario);
 - other costs (e.g. labour, overheads) or timings of outflows (e.g. fixed overhead payments, dividends, asset expenditure);
 - interest rates where borrowings are at variable rates (forecast a worst-case scenario).

An alternative to sensitivity analysis is simulation models.

- Simulation models can perform more dynamic analysis by incorporating interrelationships between variables (e.g. if interest rates rise, sales may fall).
- Unlike sensitivity analysis, simulation models (e.g. Monte Carlo) can simulate future economic scenarios to estimate the *probability* that cash flows will be higher/lower than expected. Through generating probabilities, such models provide a better analysis of cash flow risk.

Example 1 Cash Flow Forecasting

Zombie Co is worried about exceeding its overdraft limit of \$5m.

The company has used Monte Carlo analysis to produce the following forecasts of net cash flows for the next two periods and their associated probabilities.

Period 1	Probability	Period 2	Probability
Cash flow		Cash flow	
\$000		\$000	
6,000	20%	8,000	35%
3,000	50%	4,000	40%
(2,500)	30%	(8,000)	25%

Zombie Co expects to be overdrawn at the start of period 1 by \$1.5m.

Required:

a) Calculate:

- the expected value of the period 2 closing balance;
- the probability of a negative cash balance at the end of period 2; and
- the probability of exceeding the overdraft limit at the end of period 2.

b) Comment on how this analysis can assist Zombie Co in managing its cash flows.

Solution

(i) Expected value of period 2 closing balance

Period 1	Period 2	Combined probability	Closing balance (1,500) + 6,000 + 8,000 =	Expected value (EV)
6,000	8,000	$0.2 \times 0.35 = 0.07$	12,500	$0.07 \times 12,500 = 875$
6,000	4,000	0.08	8,500	680
6,000	(8,000)	0.05	(3,500)	(175)
3,000	8,000	0.175	9,500	1,662.5
3,000	4,000	0.2	5,500	1,100
3,000	(8,000)	0.125	(6,500)	(812.5)
(2,500)	8,000	0.105	4,000	420
(2,500)	4,000	0.12	0	0
(2,500)	(8,000)	<u>0.075</u>	(12,000)	<u>(900)</u>
Totals		<u>1.00</u>		<u>2,850</u>

The EV of the period 2 closing balance is \$2.85m. (This could be calculated more simply using the weighted cash flow for each period; however, ii and iii require the details in the table.)

(ii) Probability of a negative cash balance

At the end of period 2 = $0.05 + 0.125 + 0.075 = 0.25 = 25\%$

(iii) Probability of exceeding the overdraft limit

At the end of period 2 = $0.125 + 0.075 = 0.2 = 20\%$

(b) Comment

The EV analysis shows that, on average, Zombie Co will have a positive cash balance at the end of period 2 of \$2.85m. However, the actual cash balance that could occur is any of the specific closing balances shown above, rather than the average of these balances.

There could be severe consequences if the company exceeds its overdraft limit (e.g. the overdraft facility could be withdrawn). There is a 20% chance of this outcome. To guard against this, management must find additional finance of up to \$7m (\$12m – \$5m).

Also, EVs are only valid for repeat decisions rather than one-off activities, as they are based on averages. Each period and its cash flows will occur only once, and the EVs of the closing balances are not forecasted as possible outcomes. For example, the EV of a \$2.85m closing balance is not a forecasted amount and has no chance of occurring; however, a closing balance of \$5.5m has a 20% chance of occurring.

3.1 Potential Sources

Having completed a cash flow forecast the treasurer may identify a requirement to borrow funds in the short term. Potential sources of short-term funding include:

- Debt factoring and invoice discounting (see *Chapter 16*).
- A bank overdraft, but this:
 - is technically repayable on demand (although the bank may offer a revolving line of credit); and
 - normally carries a flat charge for the facility and a high variable interest rate on the balance.
- Short-term loans, but these:
 - may require security; and
 - can have fixed or variable rates of interest.

4.1 Sources of Surplus Funds

Alternatively, a treasurer may discover that the company has a cash surplus for a short-term period. Surplus funds may arise due to the following:

- overfunding – proceeds which are not yet fully required may have already been received from a share/debt issue;
- disposal of surplus assets or divisions; and
- operating surpluses.

4.2 Investing Surplus Funds – Factors to Consider

Cash surpluses should not simply be held in a bank account, which provides low returns.

- Long-term surpluses should be invested into positive NPV projects or used to pay a dividend.
- For short-term surpluses, the general rule is to invest in short-term, low-risk, highly liquid investments (e.g. Treasury bills).

Most businesses will look for a variety of investments to achieve the following objectives:

- To minimise the risks involved.
- To ensure that some cash is available at short notice.
- To obtain higher returns.

Factors to take into consideration for investing surplus funds include:

- Amount of funds available.
- Required level of liquidity (i.e. how quickly the investment can be converted to cash).
- Risk tolerance, which includes considering that shareholder funds should not be gambled with (placed in investments that are too risky). Default risk (i.e. the risk that the investment will not be repaid) should be considered.
- The expected return on the proposed investment is limited because of the requirement to select low-risk investments.

4.3 Short-term Investments

Money market deposits (i.e. bank deposits): these investments may have a notice period for withdrawals; therefore, they should only be used if there is a high certainty of cash flows.

Certificates of deposit: negotiable deposits issued by banks, with maturities from 28 days to five years. The holder can sell the certificate before its maturity date, so they are more liquid than money market deposits but with lower returns.

Treasury bills: two, three, and six-month UK government debt. These are very low risk and very liquid, but offer even lower returns.

Gilt-edged government securities ("gilts"): the long-term version of Treasury bills with maturities usually over five years. Short-term cash surpluses are not recommended to be invested in newly issued gilts as their market prices are sensitive to interest rate changes. Investing in gilts close to maturity is preferable if investing for the short term.

Other government bonds (e.g. UK local authority bonds): these bonds have rates tied to money markets and have good liquidity.

Certificates of tax deposit: deposits with UK HM Revenue and Customs that earn interest and may be surrendered for cash or used to settle tax liabilities.

Commercial paper: short-term (seven days to three months) unsecured debt issued by high-quality companies, good liquidity.

Corporate loan notes: longer maturity, fixed interest securities issued by the corporate sector. Liquidity can be poor, and corporate loan notes have a risk higher than government bonds or commercial paper.

Equities: investing short-term cash surpluses in the stock market is not recommended because of the high risk associated with equity investments.

Non-sterling instruments: most of the above instruments have non-sterling counterparts (e.g. US Treasury bills, etc). Therefore, exchange risk must be considered.

5.1 Baumol Model

5.1.1 Introduction

The Baumol model is derived from the EOQ model and can be applied when there is a *constant* demand for cash. The model suggests regular transfers from interest-bearing, short-term investments (or bank deposit accounts) into a current account.

The model considers:

- the annual demand for cash;
- the cost of each transfer from short-term investments into cash; and
- the interest rate difference between the rate paid on short-term investments and the rate paid on a current account.

The model then uses the EOQ formula to calculate the optimum amount of funds to transfer each time as short-term investments are converted into cash.

By optimising the amount of funds to transfer, the model minimises the opportunity cost of holding cash in the current account, thereby reducing cash management costs.

5.1.2 Assumptions

The simplifying assumptions of the model are:

- Cash needs are steady and predictable and funded by the sale of short-term investments.
- Constant annual demand for cash.
- Constant interest rates.
- Constant cost of each transfer/conversion of securities into cash (i.e. it does not change with the transaction size).

5.1.3 Formula

$$\text{Economic transfer} = \sqrt{\frac{2C_o D}{C_h}}$$

Where:

D = annual requirement for cash

C_o = transaction costs (brokerage, commission, etc) of selling a "parcel" of short-term investments

C_h = opportunity cost of holding cash (interest rate difference between short-term investments and cash)

5.1.4 Weaknesses

- The assumption of constant demand for cash is unrealistic. A cash management model which can accommodate a variable demand for cash, such as the Miller-Orr model, is more relevant.
 - In reality, interest rates and transaction costs are not constant; interest rates, in particular, can change frequently.
 - The model assumes a *constant* use of cash financed by selling investments. However, any worthwhile business must, at some point, *generate* cash rather than "burn" it.
-

Example 2 Baumol Model

A company has large deposits which currently earn interest of 15%. It has cash needs of \$300,000 in the next year.

Transaction costs are \$120.

Required:

Calculate the economic transfer and the average cash balance.

Solution

$$D = \$300,000$$

$$C_o = \$120$$

$$C_h = 0.15$$

$$\text{Economic transfer} = \sqrt{\frac{2 \times \$120 \times \$300,000}{0.15}} = \$21,909$$

$$\text{Average balance} = \frac{\$21,909}{2} = \$10,954$$

5.2 Miller-Orr Model

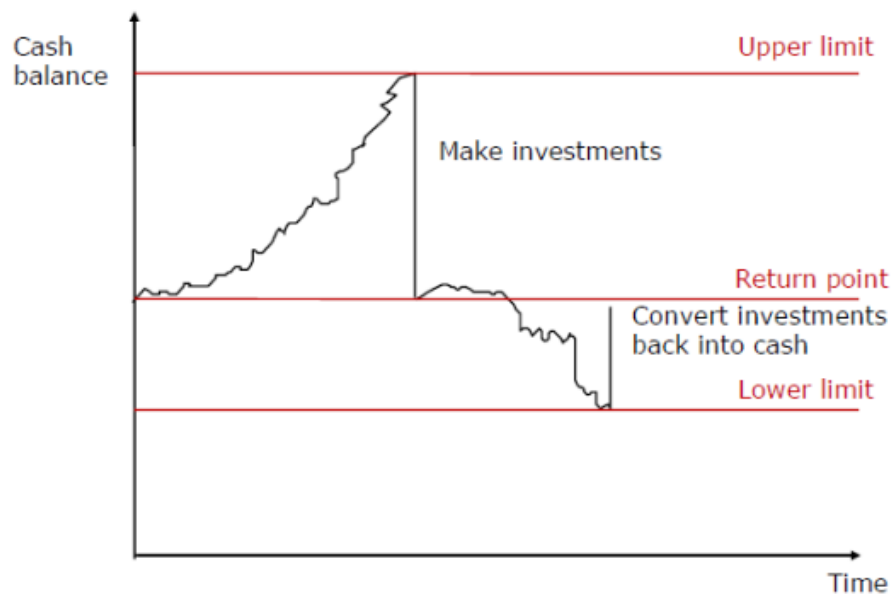
5.2.1 Introduction

The assumption of constant demand for cash in the Baumol model is unrealistic. A cash management model which can accommodate a variable demand for cash may be more relevant. This is the strength of the Miller-Orr model.

The Miller-Orr model allows for uncertainty in cash receipts and payments. The cash balance is assumed to vary between a lower limit set by management and an upper limit calculated by the model:

- If the lower limit is reached, an amount of cash equal to the difference between a default "return point" and the lower limit is raised by selling short-term investments.
- If the upper limit is reached, an amount of cash equal to the difference between the upper limit and the return point is used to buy short-term investments.

The model helps decrease the risk of running out of cash, while avoiding the loss of profit caused by unnecessarily high cash balances.



5.2.2 Assumptions

The Miller-Orr model assumes:

- A minimum acceptable cash balance, the lower limit.
- Changes in daily cash balance are random (normally distributed).
- Surplus cash can be invested in interest-earning marketable securities.
- A fixed transaction cost per sale/purchase of short-term investments.

5.2.3 Formula

The following formulae are provided in the examination:

Return point = Lower limit + ($\frac{1}{3} \times$ spread)

$$\text{Spread} = 3 \times \left[\frac{\frac{2}{4} \times \text{transaction cost} \times \text{variance of cash flows}}{\text{interest rate}} \right]^{\frac{1}{3}}$$

Where:

- | | | |
|------------------|---|--|
| Spread | = | the difference between the upper limit and lower limit |
| Transaction cost | = | the fixed cost of buying or selling marketable securities |
| Variance | = | variance (i.e. standard deviation ²) of the net daily cash flows |
| Interest rate | = | daily interest rate on marketable securities
(i.e. daily opportunity cost of holding cash) |

Note that the upper limit is the lower limit plus the spread.

Exam advice

The Miller-Orr model formulae are provided in the examination. However, you will need a scientific calculator to make the calculations.

5.2.4 Weaknesses

- ✗ Subjectivity in setting a lower limit.
- ✗ In practice, commissions for buying/selling short-term investments will likely be at least partly variable.
- ✗ The complexity of estimating future volatility of cash flows.

Activity 1 Miller-Orr Model

[Show](#)

A company requires a minimum cash balance of \$6,000, and the variance of daily cash flows is estimated at \$2,250,000. The interest rate on securities is 0.025% per day. The transaction cost for each sale or purchase of securities is \$20.

Required:

a) **Calculate:**

- the spread;
- the upper limit;
- the return point.

b) **Interpret the results.**

**Please use the notes feature in the toolbar to help formulate your answer*

$$= 3 \times \left[\frac{\frac{3}{4} \times \text{transaction cost} \times \text{variance of cash flows}}{\text{interest rate}} \right]^{\frac{1}{3}} = 3 \times \left[\frac{\frac{3}{4} \times 20 \times 2,250,000}{0,00025} \right]^{\frac{1}{3}} = \$15,390$$

Upper limit = lower limit + spread = \$6,000 + 15,390 = \$21,390

Return point = Lower limit + (1/3 × spread) = \$6,000 + (1/3 × 15,390) = \$11,130

Interpretation:

- If the cash balance rises to \$21,390, invest \$10,260 (\$21,390 – \$11,130) in securities to reduce it to \$11,130.
- If the cash balance falls to \$6,000, sell \$5,130 (\$11,130 – \$6,000) of securities to replenish it to \$11,130.

Syllabus Coverage

This chapter covers the following Learning Outcomes.

C. Working Capital Management

2. Management of inventories, accounts receivable, accounts payable and cash

- f) Explain the various reasons for holding cash, and discuss and apply the use of relevant techniques in managing cash, including:
- i) preparing cash flow forecasts to determine future cash flows and cash balances
 - ii) assessing the benefits of centralised treasury management and cash control
 - iii) cash management models, such as the Baumol model and the Miller-Orr model
 - iv) investing short term.

Summary and Quiz

- Treasury management entails the efficient management of liquidity and risk in a business and includes the management of funds, currencies and cash flow.
- There are transactions, precautionary and speculative motives for holding cash.
- A major task of the treasurer is cash flow budgeting.
- Sensitivity analysis and simulation models can deal with risk and uncertainty in the cash flow forecast.
- Short-term funding sources include debt factoring, bank overdrafts and short-term loans.
- Cash surpluses should not simply be left in a bank account, as this produces a small return.
 - Long-term surpluses should be invested into positive NPV projects or used to pay a dividend.
 - Short-term surpluses should be invested in low risk, highly liquid investments such as Treasury bills.
- The Baumol and Miller-Orr models can manage transfers between cash and short-term investments.
- The Baumol model is derived from the EOQ model and can be applied when there is a *constant* demand for cash.
- The advantage of the Miller-Orr model is that it allows for the uncertainty in cash receipts and payments. Its weaknesses include that the lower limit is subjective, estimating future volatility is complex, and transaction costs are likely variable.

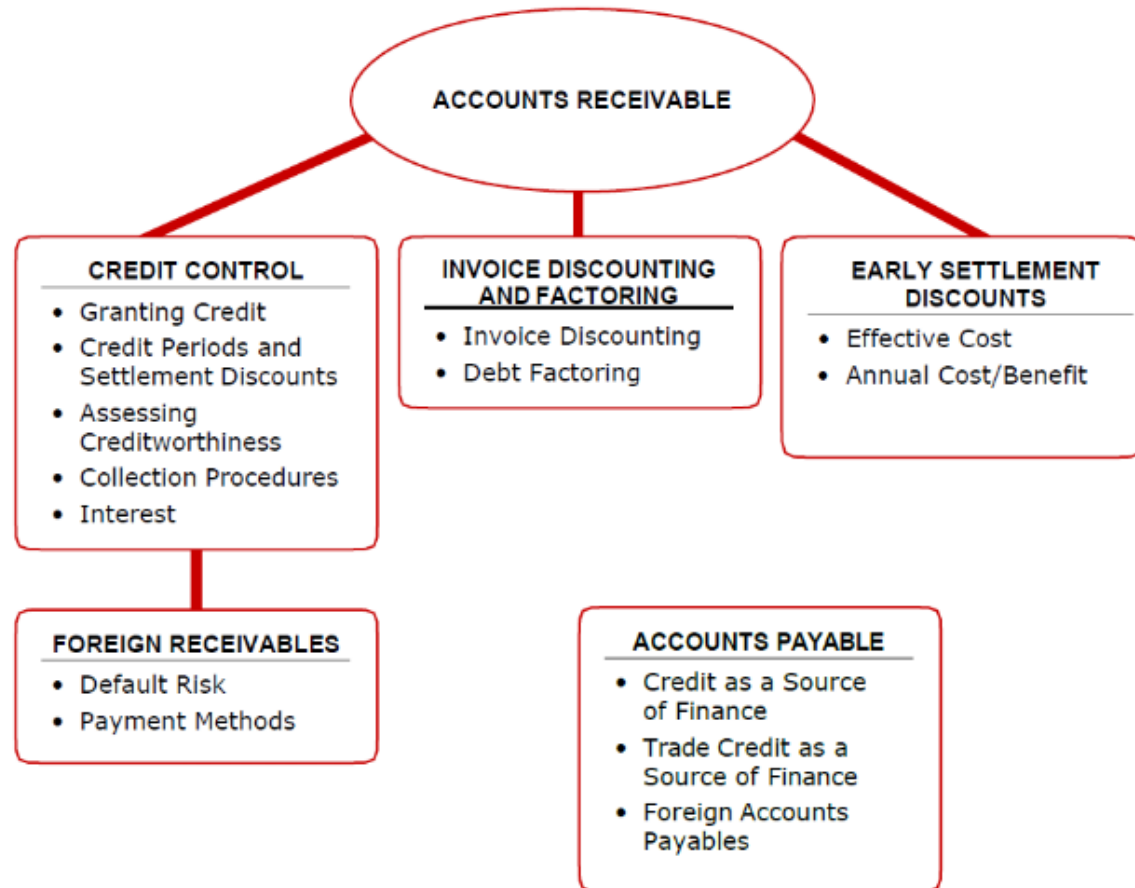
[Chapter 15 Quiz](#)

Chapter 16: Management of Accounts Receivable and Payable

Overview	CHAPTER 16: Visual Overview
16.1 Credit Control	16.1.1 Granting Credit
	16.1.2 Credit Periods and Settlement Discounts
	16.1.3 Assessing Creditworthiness
	16.1.4 Collection Procedures
	16.1.5 Charging Interest on Overdue Invoices
16.2 Foreign (Overseas) Receivables	16.2.1 Default Risk
	16.2.2 Payment Methods
16.3 Invoice Discounting and Factoring	16.3.1 Invoice Discounting
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16.4 Early Settlement Discounts	16.4.1 Effective Cost
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16.5 Accounts Payable	16.5.1 Credit as a Source of Finance
	16.5.2 Advantages of Trade Credit as a Source of Finance
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CHAPTER 16: Visual Overview

Objective: To describe and evaluate the factors involved in granting and accepting trade credit.



1.1 Granting Credit

The management of receivables was introduced in *Chapter 13*. There are two issues to be considered when deciding whether to grant credit to customers:

- Should credit be granted to all customers? Even if this is customary for the industry, a business might consider leading a change in restricting credit. Or, where credit is not the norm, offering credit may stimulate sales.
- What is the true cost to the business of customer credit? This will be influenced by the risk of bad debts and the cost of financing accounts receivable.

Exam advice

Questions on managing accounts receivable may require working capital management ratios to be applied "in reverse" (e.g. re-arrange the receivables collection period calculation to estimate receivables from a given amount of revenue).

1.2 Credit Periods and Settlement Discounts

Credit periods can be changed to respond to competition but will mainly be influenced by trade customs.

Settlement discounts (also called "cash" discounts) are influenced by accepted practice within the industry. The company must ensure that the discount's allowed expense does not exceed the benefit of reduced finance costs.

Having defined the credit periods and settlement discounts, the company must make sure that customers are aware of them by stating the terms:

- on orders forms;
- on sales invoices; and
- on monthly statements.

The settlement discount policy must be enforced since some customers will attempt to take the discount without paying on time. Therefore, the settlement discount policy must be formal, in writing, dated and quantified.

1.2.1 Setting Credit Terms

Credit limits can be set based on the results of a credit check. Once the credit limit is reached, it cannot be exceeded without the authorisation of senior management.

Credit customers should be informed in writing of the following:

- the normal credit period (e.g. 30 days after the invoice date);
- any discount for prompt payment (e.g. 2.5% of the invoice amount for payment within seven working days of the invoice date);
- any interest that will be charged on late payment.

When deciding the credit period offered to customers, an organisation must consider several factors. A longer credit period (e.g. 45 days compared to 30 days granted by competitors) may generate additional sales; however, these must be compared against the additional costs incurred by the business. These costs might include increased bad debts, higher administration costs and bank overdraft charges. The credit period should be reviewed if the profits arising from the additional credit period are less than the costs incurred.

1.2.2 Monitoring

An organisation must maintain accurate records detailing all customer transactions and the amounts owing. It should issue monthly statements and, where necessary, "chase" outstanding amounts with reminders and phone calls. Useful reports to help monitor accounts receivable include:

- An **aged receivables list** detailing the length of time outstanding amounts have been owed. It is useful because it highlights breaches of credit terms, where amounts are "past due" and debt collection procedures need to be implemented.
- A **credit utilisation report** showing the proportion of each customer's credit limit that is currently utilised. It identifies credit limits that may need to be reviewed and revised (upwards or downwards) and breaches of credit limits.

1.3 Assessing Creditworthiness

The company must balance the risk inherent in granting credit against the necessity to allow enough credit to support the level of business.

To assess the creditworthiness of a customer or potential customer, a company may consider the following:

- A bank reference, which is relatively easy to obtain. However, as the other company is the bank's customer, any reference will stick strictly to the facts and is unlikely to raise any concerns.
- A trade reference from another company that has dealings with the (potential) customer. Although another company may be reluctant to give a written reference (in case of dispute), contacts in the trade may provide an informal, oral reference.
- Credit rating/reference agencies are professional businesses which sell information about companies and individuals. They will be keen to provide the best possible information, so the company is more likely to return and use their services again.
- Latest available financial statements can be quickly and easily obtained. Although an analysis may indicate whether a company should be granted credit terms, the financial statements will be out of date to a greater or lesser extent. However, quarterly interim financial statements can generally be found on the websites of listed companies. Liquidity ratios can be calculated.
- Information from financial media, including national and local press, suitable trade journals and the Internet. For example, if a large contract has been lost or one or more directors have left recently, the company may have problems.
- A visit to discuss their exact needs and impress them with good customer service and to "get a feel" for whether the business should be given credit.

1.4 Collection Procedures

1.4.1 Cash Collection

The first step in efficient cash collection is to invoice customers quickly and accurately. Delays or inaccuracies in invoicing may signal inefficiencies that customers could seek to exploit by delaying payment.

Methods to help ensure customers pay on a timely basis include:

- Monthly statements – produced quickly and easily by any computerised sales ledger system and sent to customers. Including a remittance advice can simplify and expedite the payment process for some customers.
 - Chasing letters – directed to a specific person, preferably at a reasonably senior level. However, preparing and sending reminders and final demands is costly (if there are many debtors), and their impact may be limited.
 - Chasing phone calls – may be harder for the customer to ignore than chasing letters. A credit controller who regularly contacts a suitably senior person and politely but firmly demands payment for overdue amounts can often achieve good results.
 - Personal approach – from a senior person in the organisation to a senior person at the customer can often yield results. This is common in professional accountancy and legal firms where a partner "has a word" with a client's CEO.
 - Stopping supplies – this may be a powerful tool if, in the short term, the customer depends on the organisation's goods or services. However, the customer may source an alternative supplier in the longer term.
 - Legal action – although debt recovery through courts may not be costly in terms of fees (e.g. fees to
-

lodge a claim in a UK court are generally around 5-15%), the process can be time consuming (and likely to lead to the loss of the customer). However, just one letter from a solicitor can often be sufficient to obtain payment for fear of further legal action.

- External debt collection ("credit management") agency – fees can be high but many agencies operate on a "no win, no fee" basis. So, although the customer will invariably be lost (if not already), any recovery is preferable to write off.

Effective credit control will ensure that disputes are settled quickly without damaging the relationship with a customer while reducing the risk of bad debts. However, such a system is often expensive and time consuming; hence, many organisations, especially those which have only recently commenced trading, use debt factoring or invoice discounting to speed up cash collection.

1.5 Charging Interest on Overdue Invoices

Some powerful companies have a reputation for paying their small suppliers very slowly. In the UK, the *Late Payment of Commercial Debts (Interest) Act* allows small suppliers to charge large companies an interest rate which is 8% above the central bank interest rate on invoices unpaid from the due date, plus compensation at a fixed rate (e.g. \$100 compensation for debts of £10,000 or more).

2.1 Default Risk

The risk of default on exports may be higher than on domestic sales. Ideally, cash in advance should be requested, or at least a percentage deposit; however, such terms may not be acceptable to the customer. Credit periods on exports are often longer than for those on domestic sales. Therefore, the exporter needs to carefully consider the payment method to minimise default risk and finance the export.

2.2 Payment Methods

Organisations with foreign accounts receivable can be paid in various ways:

2.2.1 Bill of Exchange

i Definition

Bill of exchange – a document drawn by the exporter and sent to the customer, who signs to accept responsibility to pay the amount specified on the stated date.

Documents of title to the goods are not released to the customer until the customer has accepted a bill of exchange.

The exporter may choose to:

1. Hold the bill until maturity (and then receive payment from the customer); or
2. Discount the bill with a bank to receive the cash earlier.

However, if the customer does not pay the bill, the bank *will* have recourse to the exporter (i.e. default risk stays with the exporter).

2.2.2 Forfaiting

Forfaiting involves a bank discounting a series of bills of exchange *without* recourse to the exporter if the customer does not pay (i.e. default risk is transferred to the bank):

- The non-recourse aspect makes this an attractive arrangement for businesses, but as a result, the cost of forfaiting is relatively high.
- Is usually only available for large receivable amounts (over \$250,000), major convertible currencies and medium-term or longer transactions.

2.2.3 Letter of Credit

1 Definition

Documentary letters of credit – a payment guarantee backed by one or more banks.

Documents of title to the goods are released to the customer on the production of a documentary letter of credit.

- The purchaser (importer) and the seller (exporter) agree terms and conditions ("Ts & Cs").
- The purchaser's bank issues a letter of credit in the seller's favour.
- The letter of credit is issued to the seller's bank, guaranteeing payment to the seller once the conditions specified in the letter have been complied with. Typical conditions relate to presenting shipping documentation and dispatching the goods before a specific date
- The goods are dispatched to the purchaser, and the shipping documentation is sent to the purchaser's bank
 - The bank then issues a banker's acceptance
 - The seller can either hold the banker's acceptance until maturity or sell it on the money market at a discounted value

Advantages	Disadvantages
✓ Letters of credit provide a high level of payment assurance for the seller. Once Ts & Cs are met, the seller is guaranteed to receive payment. ✓ Sellers can negotiate favourable payments such as shorter payment periods to improve cash flow. ✓ The seller's exposure to credit risk is significantly reduced (because the buyer's bank, rather than the buyer, is responsible for payment).	✗ Set up can be complex and take a significant amount of time before the sale occurs. ✗ Minor discrepancies in rigid Ts & Cs can lead to delays or even non-payment. ✗ It is costly to purchasers and restricts their flexibility compared to other payment methods: payment must be made when due and buyers may be required to provide security.

2.2.4 Other Methods

Open account trading: trusting the customer to pay within the stated credit period without additional collateral or security.

Cash against documents: documents of title to the goods are not released to the customer until payment is made.

Export credit houses: organisations that give credit to the overseas customer and guarantee payment to the exporter.

Export merchants: organisations that operate as intermediaries between the exporter and the overseas customer. The merchant buys the goods (at a discount) from the exporter, sells them to the final customer and pays the exporter (usually within seven days).

Export credit insurance: protects against risks that could result in non-payment by a foreign customer. It helps reduce the impact of non-payment, but premiums are high, and insurance does not typically cover 100% of the value of the foreign sales. Examples of risks that may be insured against include:

- commercial risks (e.g. insolvency of the purchaser or slow payment);
- certain political risks (e.g. war, riots and revolution);
- currency inconvertibility; and
- changes in import or export regulations.

Export factoring: factors buy the trade receivables from the exporter and charge a commission on the transaction. Other services, such as operating the receivables ledger and credit insurance, may also be offered.

UK Export Finance (the UK Export Credits Guarantee Department (ECGD)): provides UK exporters with guarantees on bank loans to finance exports. Similar schemes operate in other countries. For example, India's Export Credit Guarantee Corporation (ECGC) provides export credit insurance and guarantees to Indian exporters and banks. Its schemes help mitigate the risks associated with international trade, allowing exporters to access financing with the backing of the government's guarantee.

Countertrade: In a countertrade arrangement, goods or services are exchanged for other goods or services instead of for cash. Countertrade can help a business enter foreign markets that it may not be able to otherwise. The main disadvantage is that the value of the goods or services received in exchange may be uncertain. Other disadvantages include complex negotiations and logistical issues.

3.1 Invoice Discounting

i Definition

Invoice discounting – the selling of selected sales invoices to a third party for a discounted cash sum, while retaining full control over the receivables ledger.

When a business enters into an **invoice discounting** arrangement, a finance company will provide a cash advance as a percentage of the outstanding sales invoices – usually in the region of 80%. As customers pay their invoices or new sales invoices are issued, the amount advanced will fall or rise to maintain the level of finance at 80% of the receivables.

- The finance company will charge a monthly fee for the service and interest on the advanced amount.
- The lender will require a floating charge over the trade receivable balances and may refuse to lend against invoices for customers with a high credit risk. Therefore, discounting is most suitable for companies selling to customers with high credit ratings and a good payment record.
- The process operates "with recourse" (i.e. the business keeps the risk of bad debts and must repay amounts advanced if a customer defaults).
- Responsibility for issuing sales invoices and credit control stays with the business. Still, the finance company will often require regular reports on the receivables ledger and the credit control process.

Advantages	Disadvantages
✓ Improved cash flow: the business receives a significant proportion of invoice amounts upfront. ✓ Flexibility: the business can choose which invoices to discount, accessing funds when needed (e.g. to cover a temporary cash shortage). ✓ Confidentiality: customers are unaware that the business borrows against sales invoices.	✗ Fees and interest charges make it an expensive form of financing compared to an overdraft or bank loan. ✗ As the finance company has a legal charge over the receivable balances, the business has fewer assets to secure other borrowings.

3.2 Debt Factoring

1 Definition

Debt factors – businesses offering a range of services in the area of sales administration and the collection of amounts due from customers.

As in invoice discounting, the business immediately receives cash representing a proportion of the total invoice value. However, unlike invoice discounting, the business does not retain responsibility for credit control. Debt factoring is an ongoing arrangement, while discounting tends to be a one-off arrangement to cover temporary cash shortages.

Debt factors may offer three closely integrated elements:

1. Accounting and collection – the company is paid by the factor as customers settle their invoices or after an agreed settlement period. The factor will maintain the sales ledger accounting function.
2. Credit control – the factor is responsible for chasing the customers and speeding up the collection of debts.
3. Finance against sales – the factor advances a percentage (e.g. typically between 50% and 85%) of the sales value immediately on invoicing.

Accounting and collection are often carried out together with credit control. The finance made available by the factor is only taken if required, as it is typically slightly more expensive than a bank overdraft.

Factoring is becoming increasingly competitive; generally, factors will act for customers with revenue in excess of \$100,000 and invoices over \$100.

The usual fees are between 0.5%–2.5% of invoice value, plus a charge for cash advances.

The main advantage of factoring is that it enables the business to concentrate on generating sales and leaves cash collection to a third party. Most importantly, it reduces the cash flow problems often experienced by new businesses and can give access to cash immediately rather than waiting 30 or more days. However, it is expensive; in the long term, it may be cheaper for a business to establish its own receivables management systems.

Advantages	Disadvantages
✓ Reduced administrative burden: outsourced debt collection saves time and effort. ✓ Improved cash flow: conversion of receivables into immediate cash. ✓ Access to expertise: the factor's expertise in credit management and collections can reduce the risk of bad debts and improve the overall efficiency of accounts receivable management.	✗ Cost: factoring fees can be relatively high. ✗ Damage to customer relationships: loss of control over customer interactions and factor's "vigour" may damage customer relationships/goodwill. ✗ Selective factoring: if the business cannot factor all receivables, managing accounts will be fragmented. ✗ Difficult to rebuild own sales ledger function at a later date.

Since outsourcing non-core business activities is now a common business practice, it is unlikely that using a factor would “signal” a cash flow problem.

Key Point

Recourse v non-recourse

- Factoring with recourse – bad debts remain the company's problem (i.e. the supplier bears the risk of non-payment).
- Non-recourse factoring – bad debts are the factor's problem. In effect, the company is insured against bad debts. Fees are higher for non-recourse factoring

Example 1 Non-recourse Factoring

Velmin Co has annual credit sales of \$700,000. The receivables collection period is currently 48 days despite the company only offering 30 days' credit, and bad debts are 3% of revenue. Velmin Co's receivables are financed by an overdraft at an annual interest rate of 8%.

Velmin Co is considering the use of a factor:

- The factor would charge 2% of revenue for a non-recourse agreement.
- The factor expects to reduce the receivables collection period to 34 days and bad debts to 2%.
- The factor would lend Velmin Co 75% of the outstanding receivables and charge Velmin 1% above their current overdraft interest cost.
- Velmin Co anticipates a \$6,000 reduction in administration costs.

Required:

Evaluate whether Velmin Co should use the factor.

Solution

Annual benefits	\$
Finance saving on reduced receivables (W)	1,659
Administration savings (as given)	6,000
Bad debts saved $((3 - 2) \times \$700,000)$	21,000
Annual costs	
Factor's fee $(2\% \times \$700,000)$	(14,000)
Net benefit	<u>14,659</u>

Exam advice

As the annual finance saving on reduced receivables is hardest to calculate, this calculation can be left to last as it is best to calculate the other costs and benefits first to obtain the easy marks.

WORKING

Situation	Existing	Revised
Receivables collection period	48 days (given)	34 days (given)
Receivables	\$92,055 ($700,000 \times 48/365$)	\$65,205 ($700,000 \times 34/365$)
Annual finance cost	\$7,364 ($92,055 \times 8\%$)	\$5,705 ($((75\% \times 65,205) \times 9\% + (25\% \times 65,205) \times 8\%)$)

Annual finance saving is \$1,659 ($7,364 - 5,705$)

4.1 Effective Cost

Exam advice

Exam questions often test calculations on whether settlement discounts should be offered to credit customers.

Offering credit customers a percentage discount on an invoice total if it is within a certain number of days is a worldwide business practice. For example, customers may be granted 30 days' credit but offered a 2.5% discount for payment within seven days, say. That would save \$250 on an invoice of \$10,000.

- To decide whether this is a good policy, the cost of the discount must be compared to the cost of financing accounts receivable (e.g. overdraft rate).
- To allow a fair comparison, the cost of the discount must be expressed as an *annual effective cost*.

Example 2 Percentage Cost of an Early Settlement Discount

A company offers its customers 30 days credit, but customers are taking 45 days credit on average. To speed up cash collection, the company is considering introducing a 1.5% discount for payment within 15 days. The company's working capital requirement is financed with an overdraft at an annual cost of 10%.

Required:

Determine whether the discount should be offered.

Solution

Assume a customer has purchased goods for \$1,000. If the customer takes the discount, the company will receive \$985 in 15 days rather than \$1,000 in 45 days. This is equivalent to the company borrowing \$985 from the customer for (45 – 15) 30 days and paying \$15 interest. Therefore, the 30-day interest rate is $15/985 \times 100\%$. Compounded to an annual rate gives:

$$(1 + 15/985)^{365/30} - 1 = 0.202$$

Therefore, the annual cost of offering the discount is 20.2%.

If the discount is not offered, the company will borrow more on its overdraft while waiting for the customer to pay. As the overdraft costs 10%, the discount should not be offered.

Exam advice

The standard 30-day credit period is a distractor and is irrelevant to answering this question.

Activity 3 Settlement Discount (1)

[Show](#)

Customers take 60 days on average to pay their credit balance. A quick payment discount of 1.5% is offered for payment within 20 days.

Required:

Calculate the annual effective cost of the discount and conclude whether the discount should be offered if the overdraft rate is 15%.

**Please use the notes feature in the toolbar to help formulate your answer*

It costs 1.5% to receive 98.5% of accounts receivable 40 days sooner.

$$40\text{-day interest rate} = \frac{1.5}{98.5} = 1.52\%$$

$$\text{Annual effective rate} = 1.0152^{365/40} - 1 = 14.8\%$$

Conclusion: The discount should be offered as 14.8% is lower than the overdraft rate of 15%.

Exam advice

In *Activity 3*, the annual effective rate is calculated using compound interest to compare to the cost of the overdraft, where interest is also charged on a compound basis. However, simple interest (i.e. $1.52\% \times 365/40 = 13.87\%$) would also be acceptable in the exam.

4.2 Annual Cost/Benefit

Example 3 Settlement Discount and Extended Credit

Melvin Co has annual revenue of \$900,000 (90% of which is on credit). The receivables collection period is 42 days despite offering only 30 days' credit. Melvin Co has a contribution margin of 30% and finances receivables using an overdraft with an annual interest rate of 8%.

Melvin Co is considering introducing an early settlement discount while extending its standard credit terms to 50 days. Under the scheme:

- Customers will be offered a 1% discount for payment within 14 days.
- It is anticipated that 40% of customers will take the discount, while those who do not will keep to the new standard credit terms.
- As a result of the extended credit terms, credit sales are expected to rise by 10%.
- Due to the extra administration involved, administration costs are expected to rise by \$10,000 a year.

Required:

Evaluate whether Melvin Co should offer the discount.

Solution

Annual benefits	\$
Finance saving on reduced receivables (W)	504
Contribution on extra 10% credit sales ($30\% \times 10\%$ ($90\% \times \$900,000$))	24,300
Annual costs	
Extra administration costs (as given)	(10,000)
Discount cost on revised credit sales ($1\% \times 40\% \times (\$810,000 + \$81,000)$)	<u>(3,564)</u>
Net benefit	<u>11,240</u>

Exam advice

Again, as the annual finance saving is the hardest to calculate, it can be done last.

WORKING

Situation	Existing	Revised
Receivable collection period	42 days	35.6 days – a weighted average i.e. $((14 \times 40\%) + (50 \times 60\%))$
Receivables	\$93,205 $(810,000 \times 42/365)$	\$86,903 $(891,000 \times 35.6/365)$
Annual finance cost	\$7,456 $(93,205 \times 8\%)$	\$6,952 $(86,903 \times 8\%)$

Annual finance saving is \$504 $(7,456 - 6,952)$. Alternatively, the finance cost (8%) could have been calculated on the reduction in receivables (i.e. $8\% (\$93,205 - \$86,903) = \$504$).

Exam advice

Revised credit sales could be reduced by the discount cost; otherwise, the finance cost is being calculated on an amount which will never be collected. However, this adjustment makes little difference, so it can be ignored.

Activity 4 Settlement Discount (2)

[Show](#)

Dodge Co has sales of \$100,000 and an accounts receivable collection period of 60 days. It pays overdraft interest at 18% a year.

It is considering a discount of 2% to customers who pay within 10 days. It is estimated that 50% of customers will take the discount.

Required:

Calculate the effect of the discount on annual profit.

**Please use the notes feature in the toolbar to help formulate your answer*

Current accounts receivable = $\$100,000 \times 60/365 = \$16,438$

New accounts receivable = $(\$100,000 \times 50\% \times 10/365 + (\$100,000 \times 50\% \times 60/365)$

= $1,370 + 8,219$

= $\$9,589$

	\$
Reduced interest expense $(\$16,438 - \$9,589) \times 18\%$	1,233
Discounts allowed expense $\$100,000 \times 50\% \times 2\%$	<u>(1,000)</u>
Change in profit	<u>233</u>

Exam advice

The solution above follows the approach shown in published answers to past exam questions. There is an argument, however, for stating the new level of accounts receivable net of discounts allowed:

$(\$100,000 \times 50\% \times 98\% \times 10/365) + (\$100,000 \times 50\% \times 60/365) = \$9,561$

This alternative approach is also acceptable.

Early settlement discounts make it more difficult to manage the sales ledger, and budgeting for customer receipts is likely to be more complex due to uncertainty about whether a customer will take a discount. In addition, some customers may abuse the discount offer – taking it without paying early.

5.1 Credit as a Source of Finance

Businesses generally use trade credit as a flexible source of short-term finance. A business may even decide to pay suppliers late. However, trade credit is not without cost.

5.1.1 Associated Costs

The business may lose any discounts for prompt payment:

- The *annual effective cost* of refusing a discount should be calculated. This should be compared to the alternative cost of financing working capital (e.g. overdraft rate).
- If the cost of refusing the discount exceeds the overdraft rate, the discount should be *accepted*.

The supplier may charge interest on late payments or increase selling prices to compensate for the finance provided.

But not all costs will be easily quantifiable:

- The loss of goodwill could be such that the supplier gives lower priority to future orders, with consequent disruption of business activities if deliveries are late.
- The supplier might eventually demand cash in advance, which the business will need to finance.

Refer back to *Example 2*. The calculations were made from the supplier's perspective. The supplier's cost of offering the discount is the same as the customer's cost of refusing the discount. If the customer in the example were to use an overdraft (say at 8%) to fund the early payment, the cost of accepting the discount (8%) is compared with the cost of refusing the discount (20.2%). Therefore, the customer would take the discount.

Example 2 illustrates an issue with settlement discounts – they might be acceptable to only the supplier or the customer, making it more difficult to decide on suitable discount terms.

Example 4 Early Payment Discount

Alanis purchases \$5,000 of goods from Celine. Celine offers all customers the option of either 30 days' credit or a 1.5% discount if cash is received in five days. If Alanis takes the cash discount, she will incur an overdraft on which interest is charged at 20% a year.

Is the cash discount beneficial to Alanis?

Solution

If Alanis takes the cash discount, she will save $\$5,000 \times 1.5\% = \75 . However, she will incur an overdraft for an additional 25 days (30 – 5 days), which will cost \$67.47 ($\$4,925$ [amount paid] $\times$ $[20\% \times 25/365]$)

Alanis will benefit by \$7.53 if she pays the invoice within 5 days.

Activity 5 Supplier Finance

[Show](#)

A supplier offers a 2% discount if the invoice is paid within 10 days of receipt but offers no discount if the payment is delayed for a further 20 days.

Required:

Calculate the annual effective cost of refusing the discount.

**Please use the notes feature in the toolbar to help formulate your answer*

Annual effective cost = 44.6% (see calculation below)

If a company receives an invoice of \$1,000 and decides to pay after 30 days, it will:

- lose the 2% discount;
- effectively have \$980 (\$1,000 – \$20) available for a further 20 days.

This is an equivalent compound rate of $\left(1 + \frac{\$20}{\$980}\right)^{365/20} - 1 = 44.6\%$

This should be compared with the cost of financing working capital. If it exceeds the cost of financing working capital, refusing the discount is expensive and the discount should be accepted.

Therefore, trade credit can be a costly form of financing when a cash discount is offered but refused.

However, the cost of trade credit decreases as the allowed payment period becomes longer relative to the discount period.

Activity 6 Discount

[Show](#)

A company currently takes 40 days' credit from its suppliers, believing this to be "free" finance. Annual purchases are \$100,000 and the company pays overdraft interest at 13%.

Payment within 15 days would attract a 1.5% quick settlement discount.

Required:

Calculate the effect on the profit and loss account of accepting the discount.

**Please use the notes feature in the toolbar to help formulate your answer*

Current accounts payable = \$100,000 × 40/365 = \$10,959

New accounts payable if discount taken = \$100,000 × 15/365 = \$4,110

	\$
Increased interest expense (((\$4,110 – \$10,959) × 13%)	(890)
Discounts received (\$100,000 × 1.5%)	<u>1,500</u>
Effect on profit	<u>610</u>

Conclusion: The discount should be accepted.

Exam advice

An exam question might ask for the “annual value of the settlement discount (in percentage terms)” rather than the annual effective cost. So, for example, if a 30-day discount is 1%, assuming 360 days in the year, the annual value is simply $1\% \times 360/30 = 12\%$.

5.2 Advantages of Trade Credit as a Source of Finance

- ✓ Convenient and informal.
- ✓ Can be used if unable to obtain credit from financial institutions.
- ✓ If settlement discounts are taken, it can result in a cheap source of financing – as there is still some time allowed before payment.
- ✓ Can be used short-term to overcome unexpected cash flow crises.

5.3 Foreign Accounts Payable

There may be specific complications (e.g. slow customs clearance, unexpected import duties or quotas) when importing. In addition, the overseas supplier may be concerned about the risk of non-payment and may demand, for example, cash against documents, bills of exchange or documentary letters of credit.

Syllabus Coverage

This chapter covers the following Learning Outcomes.

C. Working Capital Management

2. Management of inventories, accounts receivable, accounts payable and cash

- d) Discuss, apply and evaluate the use of relevant techniques in managing accounts receivable, including:
 - i) assessing creditworthiness
 - ii) managing accounts receivable
 - iii) collecting amounts owing
 - iv) offering early settlement discounts
 - v) using factoring and invoice discounting
 - vi) managing foreign accounts receivable.
- e) Discuss and apply the use of relevant techniques in managing accounts payable, including:
 - i) using trade credit effectively
 - ii) evaluating the benefits of early settlement discounts
 - iii) managing foreign accounts payable.

Summary and Quiz

- The key areas of accounts receivable management are assessing creditworthiness, setting credit terms, monitoring and effective cash collection.
- Credit terms set the payment period and state whether any discount is allowed for early payment.
- Businesses must balance the risk inherent in granting credit against the necessity to allow enough credit to support the level of business.
- Risk of default on exports may be higher than on domestic sales.
- In an invoice discounting arrangement, a finance company provides a cash advance as a percentage of the outstanding sales invoices.
- Invoice discounting improves cash flows and offers both flexibility and confidentiality. However, it is expensive and the company cannot use the receivables as security for other loans.
- Debt factors offer services in sales administration and collection of receivables through accounting and collection, credit control and finance (advances) against sales.
- Debt factoring advantages for the business are administrative savings, flexibility of finance, economies of scale and the factor's expertise.
- Disadvantages of factoring are high costs, loss of customer contact/goodwill and possible reputational damage.
- For a business offering discounts for early payment, the effective cost of the settlement discount should be compared with the cost of financing accounts receivable.
- Trade credit is often used as a flexible source of short-term finance.
- Trade credit can be a costly form of financing when a cash discount is offered but refused. The annual effective cost of refusing a discount should be calculated and compared with the alternative cost of financing working capital (e.g. overdraft rate).

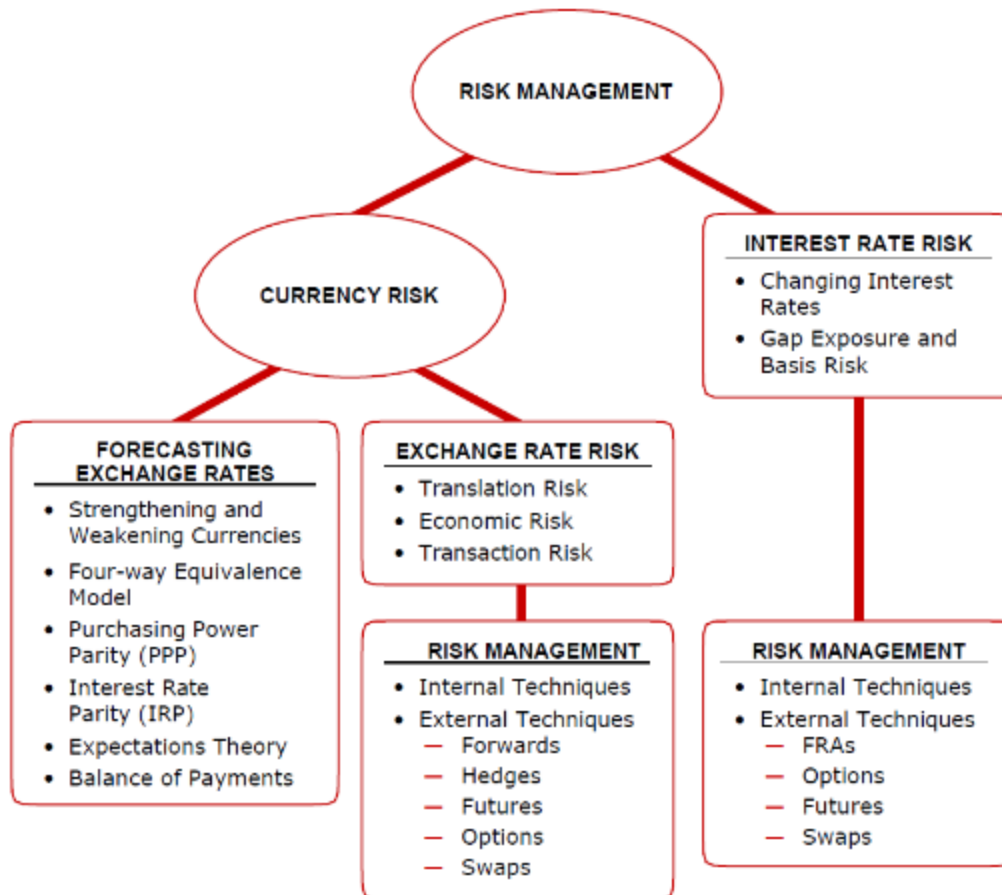
Chapter 16 Quiz

Chapter 17: Risk Management

Overview	CHAPTER 17: Visual Overview
17.1 Forecasting Exchange Rates	17.1.1 Strengthening and Weakening Currencies and Competition
	17.1.2 Four-way Equivalence Model
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CHAPTER 17: Visual Overview

Objective: To explain the causes of exchange rate fluctuations and apply traditional and basic methods of foreign currency risk management and interest rate risk management.



1.1 Strengthening and Weakening Currencies and Competition

If a company is exporting from, say, the UK to the US, and the US \$ weakens from, say, \$/£ 1.3 to \$/£ 1.4 (getting more \$ per £ means that the \$ is less valuable, so weaker), any exports from the UK will be more expensive when priced in \$. So goods priced at £1,000 will cost \$1,400 instead of \$1,300, making those goods less competitive in the UK.

Likewise, goods imported from the US will be cheaper in £ than they had been (if they cost \$1,000, their £ price will change from £769 ($\$1,000/1.3$) to £714 ($\$1,000/1.4$)).

1.2 Four-way Equivalence Model

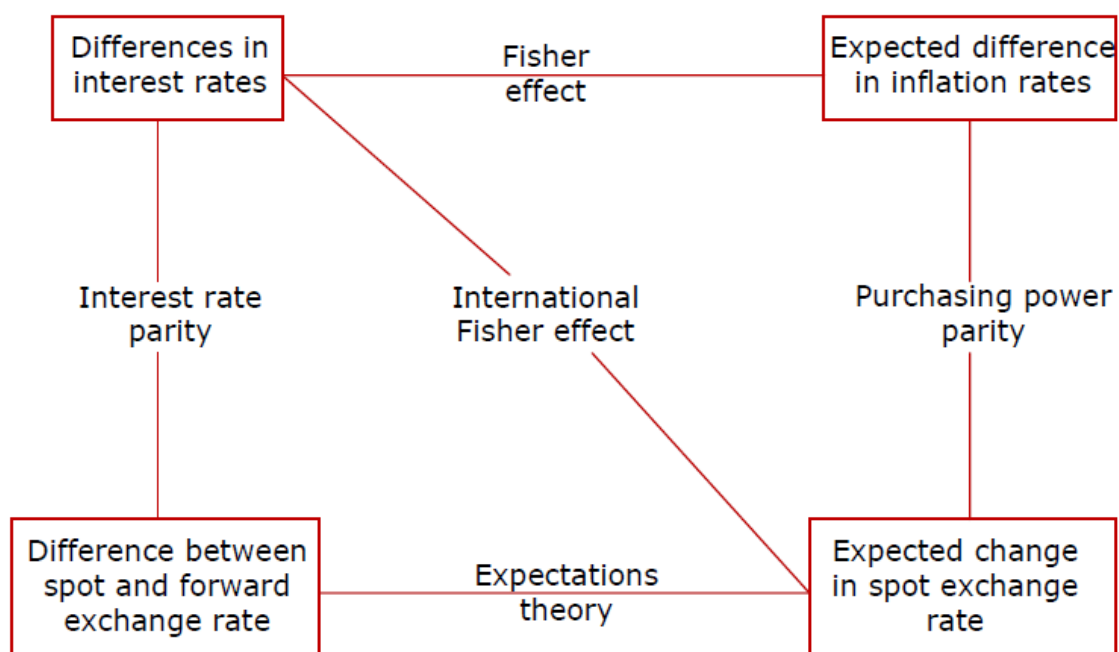
1 Definitions

Spot exchange rate – the market exchange rate for buying/selling the currency for immediate delivery.

Forward exchange rate – the exchange rate for buying or selling the currency at a specific date in the future.

The main models for forecasting future exchange rates focus on inflation or interest rate differences.

- The relationships between these macroeconomic variables can be summarised in the "four-way equivalence model" shown below:



1.3 Purchasing Power Parity (PPP)

Absolute PPP states that the exchange rate simply reflects the different cost of living in two countries.

- For example, if a representative basket of goods and services costs \$1,700 in the US and £1,000 in the UK, the exchange rate should be \$1.70 to £1.
- While absolute PPP exchange rates may represent the long-run equilibrium rate between two currencies, they are of limited practical use in financial management.
- Financial managers are more interested in market exchange rates than theoretical rates. This is where *relative* PPP is useful.

Relative PPP states that the future spot exchange rate is based on the current spot rate (the purchasing power of one currency relative to another) and the inflation rate differential between the two currencies (in other words, relative price changes).

For example, if the inflation rate is higher in the US than in the UK, relative PPP predicts that the value of the dollar will fall.

- The formula for relative PPP:

$$S_1 = S_0 \times \frac{(1+h_c)}{(1+h_b)}$$

Where:

S_1 = Expected spot exchange rate after one year

S_0 = Today's spot exchange rate

h_c = Variable currency ("foreign") inflation rate (as a decimal)

h_b = Base currency ("domestic") inflation rate

Exam advice

The relative PPP formula is provided on the Formulae Sheet.

- Spot rates should be put into the formula in the format:

Units of variable currency/Units of base currency

Activity 1 Relative PPP

[Show](#)

Spot rate 1 January 20X6 = \$1.90 per £1, so the \$ is the variable currency and £ the base currency.

Predicted inflation rates for 20X6 and 20X7:

US 2%

UK 3%

Required:

Calculate the predicted exchange rate:

- a. at 31 December 20X6; and
- b. at 31 December 20X7.

**Please use the notes feature in the toolbar to help formulate your answer*

a. Exchange rate in one year

$$S_1 = S_0 \times \frac{(1+h_e)}{(1+h_b)} = 1.90 \times \frac{(1+0.02)}{(1+0.03)} = \$1.88 \text{ per } £1$$

b. Exchange rate in two years

$$S_2 = S_1 \times \frac{(1+h_e)}{(1+h_b)} = 1.88 \times \frac{(1+0.02)}{(1+0.03)} = \$1.86 \text{ per } £1$$

This is a predicted fall in the value of sterling.

1.4 Interest Rate Parity (IRP)

IRP states that the forward exchange rate is based on the spot rate and the interest rate differential between the two currencies:

$$F_0 = S_0 \times \frac{(1+i_c)}{(1+i_b)}$$

Where:

F_0 = Forward exchange rate

S_0 = Spot exchange rate

i_c = Variable currency interest rate

i_b = Base currency interest rate

Key Point

PPP predicts the **future spot rate**, whereas IRP predicts the **forward rate**. Check the definitions in s.1.2 if necessary.

Exam advice

The IRP formula is provided on the Formulae Sheet in the exam. To find the two-year forward rate, square the differential, to find the three-year forward rate, cube the differential and so on.

Activity 2 Interest Rate Parity

[Show](#)

The spot exchange rate is \$1.78 per £ and the dollar and sterling one-year interest rates are 3.25% and 4.5%, respectively.

Required:

Calculate the one-year forward exchange rate.

**Please use the notes feature in the toolbar to help formulate your answer*

$$F_0 = S_0 \times \frac{(1+i_c)}{(1+i_b)} = 1.78 \times \frac{1+0.0325}{1+0.045} = \$1.76 \text{ per } \pounds$$

Sterling is weaker in the forward market than in the spot market.

Therefore, the theory is that the expected dollar return on dollar deposits is the same as the expected dollar return on foreign currency deposits. If this theory did not hold up, investors holding the currency with the lower interest rate would switch to the other currency for a while, ensuring that they would not make a loss on returning to the original currency by fixing the exchange rate in advance using a forward contract.

This is known as **arbitrage**. If enough investors acted this way, the forces of supply and demand would lead to a change in the forward exchange rate, making risk-free profit making impossible.

The Fisher effect (see *Chapter 5*) and international the Fisher effect underlie interest rate parity.

1.4.1 Fisher Effect

According to the Fisher effect, countries with a higher rate of inflation have higher *nominal* interest rates to offer the same *real* return as countries with low inflation:

$$(1 + i) = (1 + r) (1 + h)$$

Where:

i = nominal interest rate

r = real interest rate

h = inflation rate

1.4.2 International Fisher Effect

According to the international Fisher effect, the spot exchange rate will change to offset nominal interest rate differences between countries.

The calculations are the same as for the interest rate parity theory.

1.5 Expectations Theory

The forward rate is an **unbiased** estimate of the future spot rate. An unbiased estimate is **not** a forecast since 50% of the time, it will be too low and 50% of the time, it will be too high.

1.6 Balance of Payments Effect on Exchange Rates

Balance of payments accounts are a record of all monetary transactions between a country and the rest of the world. The two principal parts are:

1. Current account
 2. Capital account.
- The **current account** shows the net amount a country earns if it is in surplus or spends if it is in deficit. Its main component is the balance of trade, which is net earnings on exports minus payments for imports.
 - The **capital account** records the net change in ownership of foreign assets. It includes the foreign exchange market operations of a nation's central bank and loans and investments between the country and the rest of the world.

Any current account surplus (deficit) will be balanced by a capital account deficit (surplus) of equal size.

- A rise in the value of a nation's currency makes its exports less competitive and imports cheaper, with these effects tending to correct a current account surplus.
- A fall in the value of a nation's currency makes imports more expensive and increases the competitiveness of exports, and so helps to correct a deficit.

Exchange rates tend to change in the direction which will restore balance:

- When a country exports more than it imports, the demand for its currency will tend to increase (as other countries need its currency to pay for the exports). The extra demand tends to cause a rise in the currency's price relative to others.
- When a country imports more than it exports, the supply of its currency on the international market tends to increase as it tries to exchange it for foreign currency to pay for its imports. This extra supply tends to cause the price to fall.

2.0 Introduction

There are three types of exchange rate (foreign currency) risk to consider:

1. Translation risk
2. Economic risk
3. Transaction risk.

2.1 Translation Risk

Translation risk occurs when a company has foreign-denominated assets or liabilities or a foreign subsidiary or branches.

When these are presented at the closing rate in the statement of financial position of the company or parent, as exchange rates fluctuate, these amounts will change, resulting in foreign exchange profits or losses.

When converting the value of foreign currency-denominated net assets.

- If the domestic currency **appreciates** against the foreign currency, a translation **loss** arises.
- If the domestic currency **depreciates** against the foreign currency, a translation **gain** arises.

Although such gains/losses can be significant, they are not cash flows but caused by financial accounting. So, if users of financial statements understand that translation differences do not represent cash flows, they should not, **in theory**, affect the value of the company. However:

- Debt covenants often use book values, so unless they are explicitly excluded from such values, foreign currency gains and losses may have real economic consequences if a debt covenant is breached.
- Shareholders may attribute great importance to reported profits/losses, even where carefully explained.

So many companies will hedge translation risk:

- To hedge against a change in shareholders' equity, the total value of foreign currency-denominated assets should match that of foreign currency-denominated liabilities.
- To hedge against a change in the debt/equity ratio, the foreign currency debt/equity ratio should be the same as the domestic company.

2.2 Economic Risk

1 Definition

Economic risk – the risk that cash flows will be affected by long-term exchange rate movements.

As the value of a company is the present value of its future cash flows, economic risk is a significant issue for the financial manager. Unfortunately it is difficult to hedge against – diversification of the customer and supplier base across different countries will reduce exposure to this type of risk to an extent. Economic risk affects the international competitiveness of a company.

Example 1 Economic Risk of Exporting

A UK company exports to the US and invoices in dollars. Over some time, sterling becomes stronger against the dollar. Therefore, the sterling value of the export earnings will fall, reducing the cash inflow and the value of the company.

What can the company do to mitigate this risk?

- Increase the dollar price of the exports. However, this may not be practical, particularly when exporting to a competitive market.
- Diversify exports into other markets, hoping that sterling will fall against some currencies while rising against the dollar.
- Use hedging techniques such as forward contracts. Unfortunately, in the long run, this will *not* give adequate protection. As sterling rises over time in the spot markets, it also rises in the forward markets. Therefore, the hedged value of the company's exports still falls.
- Attempt to convert the cost base into dollars by importing materials from the US or setting up operations in the US. However, these may not be practical options for many companies.

Economic risk can affect a business even if it does **not** export or import. Domestic producers may face tougher competition from overseas competition if the domestic currency appreciates. Again, there is no easy method of protecting against this.

2.3 Transaction Risk

Transaction risk is the *short-term* version of economic risk. It is the risk that the *exchange rate* changes between the contracting date of a specific export/import and the related receipt/payment of foreign currency.

- Like economic risk, this affects cash flows and so affects the value of the company. Therefore, it is a significant issue for financial managers.
- Transaction risk *can* be effectively managed using internal and external hedging techniques.

3.0 Introduction

Foreign currency risk management is not concerned with maximising income or minimising expenditure but rather having certainty over receipts and payments.

3.1 Internal Techniques

Exchange rate risk may be managed internally in the following ways:

- Invoicing in the company's **domestic currency**. An exporter could denominate sales invoices in its domestic currency, effectively transferring the transaction risk to the customer. However, this may lead to lost sales.
- **Leading and lagging**, which means, for example:
 - converting domestic currency into foreign currency earlier than needed to pay suppliers ("leading"), if the domestic currency is expected to fall, which may lead to a finance cost (although early settlement discounts may be available); or
 - delaying conversion and paying suppliers late ("lagging"), if the domestic currency is expected to appreciate (although taking longer than the agreed credit period is questionable business practice).

Key Point

Only leading will reduce exposure to risk (as uncertainty about the cost of paying the supplier is eliminated). Lagging does not mitigate risk as if the domestic currency depreciates, the cost of settling the foreign currency liability will increase (and cannot be delayed indefinitely).

- **Netting** when there are both sales/receivables and purchases/payables in a foreign currency, so the net exposure is only on the difference between receivables and payables. This is only effective if there are numerous purchases and sales in the foreign currency and the transactions occur relatively closely. Still, they can result in a reduction in foreign exchange purchase costs and money transmission costs.
- **Matching of assets and liabilities** using foreign currency loans to finance overseas subsidiaries. Overseas earnings can be used to pay the loan interest and repay the principal, reducing the net foreign currency cash flow exposed to risk in the event of repatriation to the parent. This may be effective as a longer-term hedge against economic risk.
- **Matching of receipts and payments** (i.e. offsetting payments against receipts in a foreign currency) using a foreign currency bank account. There is no purchase or sale of the currency, so it does not matter if the currency strengthens or weakens. However, transactions must be relatively close together as few companies can afford to keep money in a foreign currency bank account and not use it for months.
- **Asset and liability management (ALM)**, in which overseas subsidiaries borrow locally rather than receive finance from the parent. This reduces the net assets of the subsidiary. ALM may be referred to as a "balance sheet hedge", which reduces exposure to translation risk on consolidation of the subsidiaries' net assets in the group accounts (although, as mentioned, translation risk should not, in theory, affect the value of the group).

3.2 External Techniques

Transaction risk can be managed using a variety of external hedges, including:

- Forward exchange contracts;
- Money market hedges;
- Currency options;
- Currency futures contracts;
- Currency swaps.

3.2.1 Forward Exchange Contracts

A forward contract is a legally binding agreement to buy or sell:

- a specified quantity;
- of a specified currency;
- on an agreed future date ("delivery date");
- at an exchange rate fixed today.

Forward contracts are not traded but agreements between a company and a counterparty (e.g. a bank). They are customised to match the company's exact requirements regarding quantity and delivery date.

Forward contracts are not bought; they are entered into. There is no *premium* to be paid to set up a forward hedge (unlike options).

Forward contracts do *not* require any *margin* to be posted (i.e. no cash deposit is needed when setting up a forward hedge, unlike futures contracts). However, there will usually be a small arrangement fee to set up a forward contract. Also, where the counterparty is a bank, the bank will assess the customer's creditworthiness.

- ✗ The major disadvantage of forward contracts is that *physical delivery* must occur. So, a company signing a forward contract to buy/sell foreign currency must physically exchange currency on the agreed date at the agreed rate, even if that rate has become unattractive compared to the spot rate.

Therefore, forward contracts are not a flexible method of hedging.

Example 2 How Exchange Rates are Quoted

A pair of rates is quoted for each spot and forward. For example:

Spot	€ per \$	1.2022–1.2028
Three-month forward rate	€ per \$	1.2014–1.2026

One of each pair is used to change dollars (\$) to euros (€). So \$100 would buy €120.22 now (or €120.14 in three months). It must be the rate which gives the lower amount.

The other amount in each pair is the rate for selling €s in exchange for \$s. €120.28 would buy \$100 now, and €120.26 would be \$100 in three months. In other words:

		Buy (€s for \$s)	Sell
Spot	€ per \$	1.2028	1.2022
Three-month forward rate	€ per \$	1.2026	1.2014

Example 3 Forward Exchange Contract

Lucca Co, a manufacturer in Italy, exports to the US. It has agreed to a \$500,000 sale, to be received in three months, and wants to hedge against (i.e. reduce the risk of) currency movements. (Assume the rates in *Example 2*.)

Lucca Co will want to change \$s received in three months to €. It can enter into a binding agreement with a bank to deliver \$500,000 to the bank in three months in return for €600,700 ($\$500,000 \times 1.2014$). That rate and the € amount Lucca Co will receive is now guaranteed, irrespective of the spot rate in three months.

What happens if the sale falls through after arranging the forward contract? (This does not necessarily mean a bad debt; the order might be cancelled and the goods not sent.) As the contract is binding, the bank will expect Lucca Co's commitment to be fulfilled. Lucca Co will need sufficient € to buy \$500,000 at the spot rate in three months. Lucca Co could make a loss or gain on this "closing out" of the contract, depending on the spot rate at that time.

Activity 3 Hedging with a Forward

[Show](#)

Today is 1 January 20X1. A UK-based company expects to receive dividend income of \$200,000 from its US subsidiary on 31 March 20X1.

Spot rate 1 January 20X1 (\$ per £) = 1.5123–1.5245

Three-month forward = 1.5323–1.5459

Required:

- Calculate how much sterling will be received if a forward contract is entered into.
- Calculate how much sterling would be received without a forward contract if the actual spot rate on 31 March 20X1 is 1.5247–1.5361.

**Please use the notes feature in the toolbar to help formulate your answer*

(a) Sterling received with forward contract

\$s, the variable currency, will be received, so must be sold forward. £s, the base currency, are being bought in the forward market.

Forward rate for changing \$ into £ is 1.5459. Therefore:

$$\text{£ receipt} = \frac{\$200,000}{1.5459} = \text{£}129,374$$

(b) Sterling received with no forward contract

Actual spot for changing \$ into £ is 1.5361. Therefore:

$$\text{£ receipt} = \frac{\$200,000}{1.5361} = \text{£}130,200$$

Exam advice

Calculations will only be required on forwards and money market hedges. The more complex derivatives (options, futures and swaps) can only be examined by discussion.

3.2.2 Money Market Hedges

1 Definition

Money market hedge – a technique to lock in the value of a foreign currency transaction in terms of the organisation's domestic currency using a combination of investing, borrowing and a spot currency exchange.

Suppose a UK company has dollar export earnings. A money market hedge could be set up as follows:

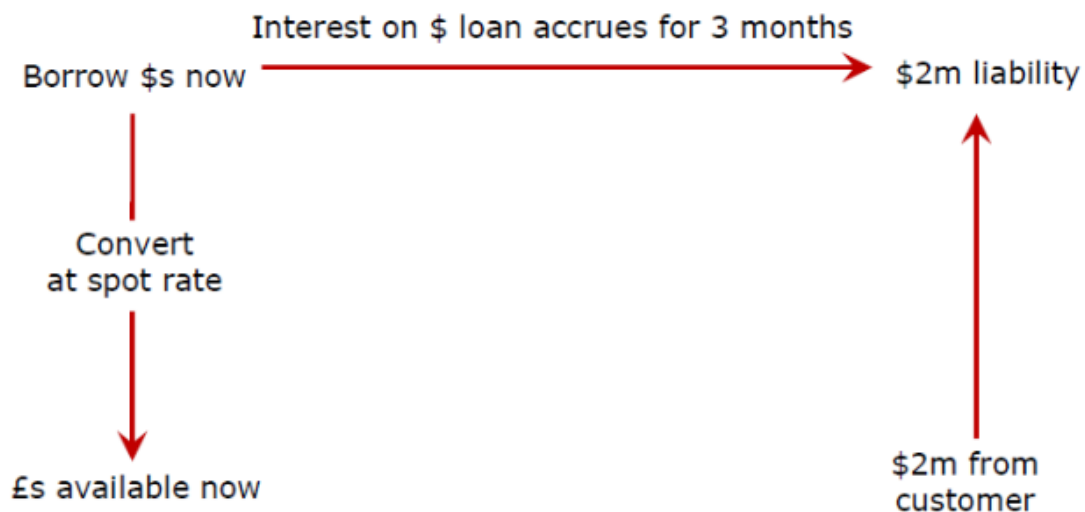
1. Borrow dollars today at the company's fixed-rate dollar borrowing rate, for repayment when the dollar earnings are due to be received. The intention is that the dollar earnings will repay this dollar loan.
2. Exchange these dollars into sterling at the current spot rate.
3. Invest the sterling received at the company's fixed rate sterling investment rate, to mature on the date the dollar earnings are due.

A money market hedge effectively produces a "homemade" forward exchange rate. The resulting exchange rate depends on the interest rate differential between the two currencies.

Example 4 Money Market Hedge

Lucky Co, a UK manufacturer, exports to the US and will receive \$2m in three months. There would be no currency risk if the \$2m could be used to settle a \$2m liability, as the currency inflow and outflow would match.

If Lucky Co does not have a \$2m liability to settle, it can create one by borrowing \$ now and repaying the loan in three months with the \$ receipt. So the plan is:



To determine how many \$s need to be borrowed now, it is necessary to know \$ interest rates. For example, the \$ three-month interest rate might be quoted as 0.54% – 0.66%. The lower rate is the rate earned for money on deposit; the higher rate is the rate charged on borrowing.

Key Point

Although described as a “three-month” rate, it is always quoted as an **annualised** rate; therefore, it must be restated as a rate applicable for the period in question.

The \$s borrowed now plus interest accrued at 0.66% (per year) for three months need to equal \$2m. So, If X is borrowed now:

$$X(1 + (0.66\% \times 3/12)) = 2,000,000$$

Therefore, $X = \$1,996,705$

This can be exchanged for £s now, at the current spot rate, say at \$ per £, 1.4701 = £1,358,210.

This amount is certain regardless of what happens to the exchange rate in the future.

Example 4 hedges a receipt of foreign currency in the future. If foreign currency has to be paid in the future, the company would change money into foreign currency now and put it on deposit to grow to the required amount by the right time. Again, because the money is changed now at the spot rate, the transaction is immune to future changes in the exchange rate.

Example 5 Money Market Hedge v Forward Contract

Why might the somewhat complicated money market hedge process (*Example 4*) be used instead of a simple forward contract (*Example 3*)?

One advantage is that Lucky Co has money now, whereas Lucca Co has to wait three months for it. Lucca Co can use the money now – or at least put it in a £ deposit account for three months. This raises an important issue – if amounts are received at different times, they cannot be compared directly because receiving money earlier is better than receiving it later.

To make a meaningful comparison, calculate the future value of the amount received now and deposited for three months. For example, if the sterling three-month deposit rate is 1.2%, £1,358,210 put on deposit for three months gives £1,362,285 ($1,358,210 \times (1 + (1.2\% \times 3/12))$).

Activity 4 Money Market Hedge

[Show](#)

A UK-based company expects to receive \$300,000 in three months.

Spot rate (\$ per £): 1.7818–1.7822

One-year sterling interest rates: 4.9%(borrowing), 4.6% (investing)

One-year dollar interest rate: 5.4% (borrowing), 5.1% (investing)

Required:

Set up a money market hedge.

**Please use the notes feature in the toolbar to help formulate your answer*

Expected receipt after 3 months = \$300,000

Dollar fixed-rate interest rate over 3 months = $5.4\% \times 3/12 = 1.35\%$

Dollars to borrow now to have \$300,000 liability after 3 months = $\$300,000 / 1.0135 = \$296,004$

Spot rate for selling dollars = \$1.7822 per £

Sterling deposit from borrowed dollars = $\$296,004 / \$1.7822 = £166,089$

Sterling interest rate over 3 months = $4.6\% \times 3/12 = 1.15\%$

Value in 3 months of sterling deposit = $£166,089 \times 1.0115 = £167,999$

3.2.3 Currency Options

A company may consider buying a currency option if it wants a more flexible hedge. Options are an example of *derivatives* (i.e. a financial instrument based on an underlying asset). In the case of currency options, the underlying asset is a currency.

The purchaser of a currency option has the right, but not the obligation, to buy or sell:

- a specified quantity;
- of a specified currency;
- on or before a specified date (*expiry date*);
- at an exchange rate agreed today (*exercise price/strike price*).

The owner of the option can either:

- exercise their right; or
- allow it to lapse (i.e. not exercise it).

However, the owner of an option must *pay* for this flexibility. The cost of an option is known as its *premium*.

Premiums are paid at the date the option is bought and are non-refundable.

A company may buy options:

- on a derivatives market; or
- directly from a bank – known as OTC ("over-the-counter").

Key Point

- A **call** option gives its owner the right to **buy** the underlying asset.
- A **put** option gives its owner the right to **sell** the underlying asset.
- **European**–style options can only be exercised **on** the expiry date.
- **American**–style options can be exercised at any time until (i.e. on or **before**) the expiry date.

Example 6 Call Option

Speck Co, a UK exporter, expects to be paid \$1m for an item of equipment delivered in 90 days. Speck Co will:

- Lose money, if the £ strengthens against the \$, as it will receive fewer £s for the \$1m;
- Make a gain if the £ weakens against the \$.

Assuming the current rate is \$1.40 per £1, Speck Co will be particularly concerned if the rate moves beyond \$1.50 per £.

It buys £ call options at an exercise price of \$1.50 per £, which give it the right to buy £s at \$1.50 per £.

- If the dollar weakens beyond \$1.50 per £1, Speck Co can exercise the option, guaranteeing at least £666,667.
- If the \$ stays stronger or even strengthens to, say, \$1.20 per £1, Speck Co can let the option lapse (i.e. ignore it) and exchange at 1.20, to give £833,333.

Therefore, Speck Co is protected against large losses but can still make gains. However, it has to pay an up-front, non-returnable premium to buy the options.

Exam advice

Calculations will **not** be required for the more complex derivatives (i.e. options, futures and swaps).
Example 6 has been provided solely to illustrate the preceding narrative.

3.2.4 Currency Futures Contracts

Definition

Futures contract – a standardised contract between buyer and seller, in which the buyer has a binding obligation to buy a fixed amount (the contract size) at a fixed price (the futures price) on a specified date (the delivery date) of some underlying asset via a recognised exchange.

Futures are *exchange-traded* forward contracts.

Currency futures contracts are *standardised* contracts for the buying or selling of a specified quantity of a currency. They are traded on a futures exchange and have various "delivery dates" (e.g. March, June, September and December).

A company can choose:

- whether to buy or sell futures; and
- which delivery date to use.

The price of a currency futures contract represents the forward exchange rate for the currencies specified in the contract.

The difference between the futures price and the spot exchange rate is known as the **basis** in the futures contract.

Key Point

The basis in a futures contract will naturally amortise to zero by the contract's delivery date.

When a currency futures contract is bought or sold, the buyer or seller must deposit a sum of money with the exchange, called **initial margin**.

- If losses are incurred (as exchange rates and, therefore, the prices of currency futures contracts change), the buyer or seller may be called on to deposit additional funds (*variation margin*) with the exchange.
- Any profits are credited to the margin account daily as the contract is "marked to market".

Although the definition of a futures contract is essentially the same as a forward contract, there is a significant practical difference between hedging with forwards and futures:

- With forward contracts there is always *physical delivery* (i.e. the underlying currency will be bought/sold when the contract reaches its delivery date).
- However, most currency futures contracts are "closed out" *before* delivery. The company executes the opposite transaction to the initial futures position (e.g. if initially buying currency futures, it is closed out by selling currency futures).
- There is no guarantee that the basis in the futures contract will be amortised at a linear rate. Therefore, the result of a futures hedge cannot be known in advance. This is known as **basis risk** (see also s.4.2.2).

If a futures hedge is correctly performed, any gain made on the futures transactions will offset, to some degree, a loss made on the spot currency markets (and vice versa).

Example 7 Currency Futures

Today is 1 February. A UK exporter expects to receive \$300,000 in three months and is considering using sterling futures to protect against transaction risk.

The company worries that sterling will appreciate, leading to a loss on the spot market sale of dollars in three months. Therefore, it needs to set up a futures position which would produce a gain on the rise in sterling.

On 1 February, it should *buy* sterling futures contracts. As it needs to hedge until 1 May, *June* contracts should be used (March contracts would only hedge until the end of March).

On 1 May the company should:

- *sell* June sterling futures; and
- sell the \$300,000 export receipts on the spot market.

If sterling has risen against the dollar, a gain on sterling futures (bought sterling low, sold sterling high) will offset the loss on the spot market.

3.2.5 Currency Swaps

1 Definition

Currency swap – an agreement in which two parties exchange the principal amount of a loan and the interest in one currency for the principal and interest in another currency.

Currency swaps can eliminate transaction risk on foreign currency loans as follows:

- On commencement of the swap – an exchange of agreed principal amounts, usually at the prevailing spot rate.
- Over the life of the swap – an exchange of interest payments.
- At the end of the swap – a re-exchange of principals, usually at the *original* spot rate (thereby removing foreign currency risk).

4.1 Changing Interest Rates

4.1.1 Exposure to Rising Interest Rates

There are two main situations in which a company may be particularly concerned about rising interest rates:

1. If it has a significant proportion of floating (i.e. variable) interest rate debt, as this leads to lower profits. However, higher interest expense also leads to higher *financial risk* (i.e. more *volatile* future profits as shown in *Example 1, Chapter 11*). An extreme interest rate rise could even cause *financial distress risk* (i.e. bankruptcy).
2. If a significant amount of surplus cash has been invested in *fixed* interest rate securities (e.g. government bonds).

4.1.2 Exposure to Falling Interest Rates

Similarly, there are two main situations in which a company may be particularly concerned about falling interest rates:

1. If it has a significant proportion of *fixed* interest rate debt and cannot benefit from falling rates (unlike its competitors, for example).
2. If it holds significant *floating* rate investments (e.g. money market investments).

4.2 Gap Exposure and Basis Risk

4.2.1 Gap Exposure

Gap exposure is the difference between the *amounts* of interest-sensitive assets and liabilities (i.e. their market prices are vulnerable to changes in interest rates).

An organisation's gap exposure can be identified through "gap analysis", grouping interest-sensitive assets and liabilities according to maturity dates. Two different types of gap may occur:

1. **Negative gap** – arises when the amount of liabilities maturing at a specific time exceeds the assets maturing simultaneously. This results in a net exposure if interest rates rise by the time of maturity;
2. **Positive gap** – arises when the amount of assets maturing at a specific time exceeds the amount of liabilities maturing simultaneously. In this situation, the organisation will suffer a loss if interest rates fall by maturity.

4.2.2 Basis Risk

Even if a company has matched its assets and liabilities with a variable interest rate, there may still be a risk if the variable interest rates are determined on different bases. For example, variable rates may be referenced to different benchmarks (e.g. one is linked to the Secured Overnight Financing Rate (SOFR) and the other to the Sterling Overnight Index Average (SONIA).

This is an example of **basis risk**.

5.0 Introduction

The primary aim of interest rate risk management is not to ensure the best possible outcome but to limit uncertainty so the business can confidently plan.

5.1 Internal Techniques

Internal techniques to hedge interest rate risk include:

- "Smoothing" – this involves maintaining an appropriate balance between fixed-rate and floating-rate borrowings or deposits; and
- "Matching" – this aims to have a common interest rate for assets and liabilities (e.g. linked to variable rates).

Example 8 Matching

A company has \$500,000 on deposit and \$520,000 borrowings.

Interest rates:

On deposits, is SOFR + 1%

On borrowings, is SOFR + 4%

SOFR is currently 3%.

Currently:

Interest expense = $7\% \times \$520,000 = \$36,400$ per year

Interest income = $4\% \times \$500,000 = \$20,000$ per year

Therefore, net expense = \$16,400

Now assume SOFR rises to 5% (a 2% increase).

Interest expense = $9\% \times \$520,000 = \$46,800$ per year

Interest income = $6\% \times \$500,000 = \$30,000$ per year

Therefore, net expense = \$16,800

The increase in interest income has almost exactly offset the increase in interest expense. The additional \$400 relates to the mismatch of the borrowing and deposit (i.e. $\$20,000 \times$ increase in SOFR of 2%).

Key Point

Matching does not guarantee the complete elimination of interest rate risk. For example, suppose the interest rate on deposits is linked to a one-month SOFR, but borrowings are based on the six-month SOFR. It is unlikely that these rates will move perfectly in line with each other. This creates a source of **basis risk** in that the internal hedge is not perfectly effective.

- **Asset and liability management (ALM)** – not only links assets and liabilities to the same interest rate, but it also attempts to match the maturity of assets and liabilities.

5.2 External Techniques

5.2.1 Forward Rate Agreements (FRAs)

Definition

Forward rate agreement (FRA) – an agreement by a bank to enter into a notional loan or accept a notional deposit from a customer for a specified period. The contract is settled based on the difference between the interest rate agreed when the contract is signed and the rate prevailing when the notional loan/deposit is deemed to start.

Forward rate agreements (FRAs) allow companies to fix, in advance, either a future borrowing rate or a future deposit rate, based on a notional principal amount over a given period.

FRAs are cash-settled, in **advance** at the start of the FRA term, based on the present value of the difference on settlement date between:

- the fixed contract rate; and
- the reference interest rate (e.g. SOFR).

The FRA can be with the same financial institution as the loan or deposit or a different institution.

Other features of FRAs include:

- The maximum maturity period for an FRA is usually about two years.
- Customised agreement is reached with a bank (i.e. OTC).
- No premium is paid for the FRA and no margin needs to be posted.

Example 9 FRA

A company plans to borrow \$20m in three months for six months and wishes to pay 7% interest no matter what happens to interest rates during the next three months.

It can enter into an FRA with the bank at an agreed rate of 7% on a notional principal amount of \$20m, starting in three months and lasting for six months. This is known as a **3-9** FRA because it begins in 3 months and ends after 9 months.

- If actual interest rates are *higher* than 7% in three months, the bank pays the company the difference between 7% and the actual rate (i.e. cash settlement is made at the *start* of the FRA period). The compensation would be calculated as the *present value* of the interest rate difference on a \$20m six-month loan (discounted at the actual interest rate).
- If actual interest rates are *lower* than 7%, the company pays the bank the difference.

No matter the actual interest rate the company will pay interest at a rate of 7% on the underlying \$20m loan.

Key Point

A borrowing for 3 months starting in 3 months would be referred to as a 3-6 FRA (because it begins in 3 months and ends after 6 months). Therefore, a 4-7 FRA would start in 4 months for 3 months, etc.

Activity 5 FRA

[Show](#)

Nero Co's cash flow forecast shows that it expects to borrow \$2m from Goodfellow's Bank in four months for three months. The company is concerned that interest rates will have risen by the time the loan is taken out. The interest rate is 5%, and Helpy Bank offers this on the required FRA.

Required:

Determine the cash flows if the interest rate when the loan is taken out is:

(a) 6.5%; and

(b) 4%.

**Please use the notes feature in the toolbar to help formulate your answer*

The FRA needed would be a 4-7 FRA at 5%.

Interest rates	(a) Rises to 6.5%	(b) Falls to 4%
Paid to Goodfellow's Bank	$3/12 \times 6.5\% \times \$2m = \$32,500$	$3/12 \times 4\% \times \$2m = \$20,000$
(Received from)/paid to Helpy Bank	$3/12 \times (5\% - 6.5\%) \times \$2m = (\$7,500)$	$3/12 \times (5\% - 4\%) \times \$2m = \$5,000$
Net cost to Nero Co	\$25,000	\$25,000

In both cases, the effective interest rate to Nero Co on the loan is 5% (i.e. the FRA-agreed rate), i.e. $(\$25,000 \times 12/3)/\$2,000,000$.

If interest rates fall, Nero Co may wish it had not entered the FRA so it would not have to pay Helpy Bank \$5,000. However, the purpose of the FRA is to provide certainty, not to guarantee the lowest possible cost of borrowing.

5.2.2 Interest Rate Futures (IRFs)

Definition

Interest rate futures (IRFs) – exchange-traded contracts whose value is determined by interest rates.

Like FRAs, IRFs specify a notional principal and a futures price.

A futures contract gives the owner the right to earn interest or obligation to pay interest:

- *Selling* a future creates an obligation to borrow money/*obligation to pay* interest;
- *Buying* a future creates an obligation to deposit money/*right to receive* interest.

The most common futures contract for interest rate hedging is a "three-month" contract. This contract is referenced to short-term interest rates.

The difference between the interest rate in the futures contract and the spot interest rate (e.g. Sterling Overnight Index Average (SONIA) for sterling transactions) is known as the *basis* in the futures contract. The basis in a futures contract will naturally amortise to zero by the contract's delivery date.

Interest rate futures are priced at 100 **minus** the implied interest rate. Therefore, if interest rates rise, the price of interest rate futures **falls**.

Key Point

Be sure you understand this. Consider, for example, a futures contract that allows lenders and borrowers to receive or pay interest at 6%, the current market interest rate. Suppose the market rate rises to 8%. Lenders will now find the futures contract less attractive because they could earn 8% at the market rate but only 6% under the futures contract. So the price of the contract must fall.

If a company wishes to hedge against rising interest rates it should use futures as follows:

- *Sell* interest rate futures today;
- Wait for interest rates to rise;
- If interest rates do rise, the *price* of futures must *fall*;
- "Close out" the futures position by *buying* the same contracts that were originally sold.

There should be a gain on futures (as the company has sold high and bought low). This gain can offset the higher interest expense the company will pay on its debts.

However, there is no guarantee that the *basis* in the futures contract will be amortised at a linear rate. Therefore, the result of a futures hedge cannot be known in advance. This is also **basis risk**.

Key Point

When hedging against rising interest rates, the futures are sold first and bought later. This "short position" is possible in futures markets because it can be closed out before contracts reach their delivery date (i.e. physical delivery does not occur).

5.2.3 Interest Rate Options

An interest rate option protects the holder against adverse rate movements and allows them to benefit from favourable movements.

The holder has the right, but not the obligation, to deal at an agreed interest rate at a future maturity date. On the expiry date, the buyer has to decide whether to exercise the right.

- Borrowers can set a maximum on the interest they have to pay by buying **put** options.
- Lenders/depositors can set a minimum on the interest they receive by buying **call** options.

Example 10 Interest Rate Options

An interest rate call option on a six-month notional loan of \$1m at an exercise price of 5% gives its holder the right, but not the obligation, to deposit/lend \$1m for six months at an annual interest rate of 5%, starting from the exercise date of the option.

If the market interest rate is 4%, the holder will exercise the option; if it is 6%, the holder will not exercise it and instead deposit/lend at the market rate.

On the other hand, an option to borrow (a put option) will not be worth exercising if market rates are below the rate specified in the option by the time the option has expired.

Various OTC interest rate options can be purchased from financial institutions and tailor-made to meet company requirements. The major types are:

Cap: if the reference interest rate rises above a predetermined level, the financial institution pays the difference to the company based on an agreed notional principal and period. This puts a cap or ceiling on the interest rate paid. The cap will not be exercised if the reference rate stays below the predetermined rate.

Floor: if the reference interest rate falls below a predetermined level, the financial institution pays the difference to the company. This would be relevant for a company with floating-rate investment income which wishes to guarantee a minimum return.

Collar: this combination of a cap and a floor keeps an interest rate between an upper and a lower limit. This is a cheaper hedge than just using a cap or a floor.

- A *borrower* would buy a cap and sell a floor, thereby offsetting the cost of buying a cap against the premium received by selling a floor.
- A *depositor/investor* would buy a floor and sell a cap.

5.2.4 Interest Rate Swaps

An interest rate swap is an exchange between two parties of interest obligations or receipts in the same currency on an agreed amount of notional principal for an agreed period.

- Interest rate swaps are a flexible method for companies to change the interest rate profile of their underlying loans or investments or to hedge against adverse interest rate movements.
- The most common interest rate swap is a *plain vanilla swap*, where fixed interest payments based on a notional principal are swapped for floating interest payments based on the same notional principal.

Exam advice

Calculations will focus on forecasting exchange rates and performing relatively simple hedges such as forward contracts, money market hedges or FRAs for interest rate management.

An appreciation of more complex derivatives such as futures, options and swaps should be sufficient.

Syllabus Coverage

This chapter covers the following Learning Outcomes.

G. Risk Management

1. The nature and types of risk and approaches to risk management

- Describe and discuss different types of foreign currency risk:
 - translation risk
 - transaction risk
 - economic risk.
- Describe and discuss different types of interest rate risk:
 - gap exposure
 - basis risk.

2. Causes of exchange rate differences and interest rate fluctuations

- Describe the causes of exchange rate fluctuations, including:
 - balance of payments
 - purchasing power parity theory
 - interest rate parity theory
 - four-way equivalence.
- Forecast exchange rates using:
 - purchasing power parity
 - interest rate parity.
- Describe the causes of interest rate fluctuations including:
 - expectations theory

3. Hedging techniques for foreign currency risk

- Discuss and apply traditional and basic methods of foreign currency risk management, including:
 - currency of invoice
 - netting and matching
 - leading and lagging
 - forward exchange contracts
 - money market hedging
 - asset and liability management.
- Compare and evaluate traditional methods of foreign currency risk management.
- Identify the main types of foreign currency derivatives used to hedge foreign currency risk and explain how they are used in hedging. (No numerical questions will be set on this topic.)

4. Hedging techniques for interest rate risk

- Discuss and apply traditional and basic methods of interest rate risk management, including:
 - matching and smoothing
 - asset and liability management
 - forward rate agreements.
- Identify the main types of interest rate derivatives used to hedge interest rate risk and explain how they are used in hedging. (No numerical questions will be set on this topic.)

Summary and Quiz

- The key models for forecasting future exchange rates focus on inflation rate fluctuations or interest rate differences. The four-way equivalence model illustrates the relationships between these macroeconomic variables.
- Absolute PPP states that the exchange rate reflects the different costs of living in two countries.
- Relative PPP states that the future spot exchange rate is based on the current spot rate and the inflation rate differential between the two currencies.
- IRP states that the forward exchange rate is based on the spot rate and the interest rate differential between the two currencies.
- A rise in the value of a nation's currency makes its exports less competitive and imports cheaper, with these effects tending to correct a current account surplus. The reverse is also true.
- The three types of foreign currency risk are translation, economic and transaction risk.
- Economic risk is the risk that cash flows will be affected by long-term exchange rate movements. It affects the international competitiveness of a company.
- Transaction risk is the risk that the exchange rate changes between the date of a specific export/import and the related receipt/payment of foreign currency.
- Internal foreign exchange hedging techniques include leading and lagging, netting and matching.
- Forward contracts can be used to hedge transaction risk.
- Money market hedges involve borrowing or investing foreign currency to protect against transaction risk.
- If a futures hedge is correctly performed, any gain on the futures transactions will offset a loss on the spot currency markets (and vice versa).
- Currency swaps can eliminate transaction risk on foreign currency loans.
- Interest rate risk can be due to rising rates, falling rates, basis risk or gap exposure.
- FRAs allow companies to fix, in advance, either a future borrowing rate or a future deposit rate based on a notional principal amount over a given period. These can be used to hedge interest rate risk.
- Interest rate options include caps, floors and collars. A cap pays off if rates rise above the predetermined level; a floor pays off if rates fall below the predetermined level.
- The main objective of interest rate and foreign currency risk management is not to guarantee the best possible outcome, but to limit uncertainty.

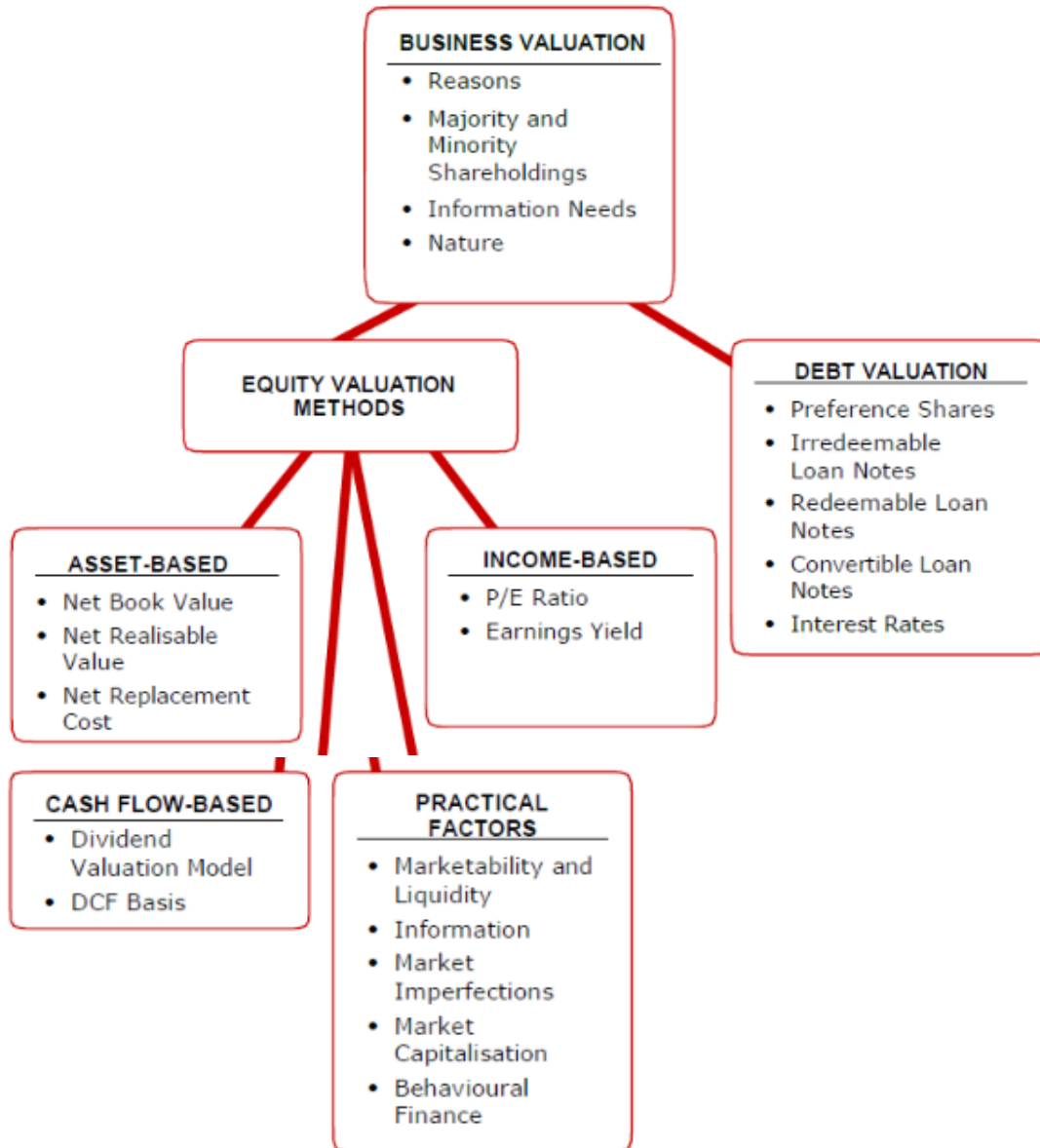
Chapter 17 Quiz

Chapter 18: Business Valuation

Overview	CHAPTER 18: Visual Overview
18.1 Business Valuation	18.1.1 Reasons for Business Valuation
	18.1.2 Majority and Minority Shareholders
	18.1.3 Information Requirements
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18.2 Asset-based Valuation Methods	18.2.1 Net Book Value (NBV)
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18.3 Income-based Valuation Methods	18.3.1 Price/Earnings Ratio
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18.6 Debt Valuation	18.6.1 Preference Shares
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18 Conclusion	18 Syllabus Coverage

CHAPTER 18: Visual Overview

Objective: To estimate the value of an organisation's shares and debt using appropriate valuation models and describe the practical considerations in the valuation of shares.



1.1 Reasons for Business Valuation

Business valuation methods may be needed:

- To determine the value of a private company (e.g. for a management buy out ("MBO")).
- To determine the maximum price to acquire a listed company (e.g. in a merger or takeover). The quoted share price is only relevant for taking a minority shareholding.
- To aid in decisions on buying/selling shares in private companies.
- To value companies entering the stock market (i.e. for an IPO).
- To value subsidiaries/divisions for possible disposal.

1.2 Majority and Minority Shareholders

Because minority shareholders have little power and no control, a 20% share of a company should be worth less than 20% of its total value. On the other hand, because of the power and control that majority shareholders have, an 80% share should be worth more than 80% of its total value. Therefore, majority shareholders should be prepared to pay a premium for control.

1.3 Information Requirements

Types of information needed for a business valuation include but are not limited to:

- Financial statements;
- Summaries of non-current assets;
- Summaries of working capital (inventory, debtors, creditors);
- Budgets and forecasts;
- Summaries of contracts such as leases;
- Details about shareholders and their shareholdings;
- Information about the industry and the broader economic environment.

There are limitations to some of this information. For example, financial statements will be out of date and budgets/forecasts might have been prepared for best-case or worst-case scenarios.

1.4 Nature of Business Valuation

Business valuation is a complex exercise. There is often no unique answer to the question of what a business is worth (e.g. the value to the existing owner may be significantly different from the value to a potential buyer).

There are various methods of valuing businesses, which may produce different overall values. These can be used to determine a relevant **range** of prices between:

- the minimum price the current owner is likely to accept; and
- the maximum price the bidder is likely to pay.

The final price will result from negotiations between the parties.

The sections that follow describe the examinable models for the valuation of shares:

- Asset-based valuations;
- Income-based valuations; and
- Cash flow-based valuations.

2.1 Net Book Value (NBV)

The NBV method uses the "balance sheet" equation (also called the "accounting" equation):

$$\text{Equity} = \text{assets} - \text{liabilities}$$

Problems and weaknesses of this method include:

- ✗ Historical cost – Balance sheet values (i.e. carrying amounts) are often based on historical cost rather than market values.
- ✗ Accounting policies – Net book values of non-current assets depend on depreciation/amortisation policies.
- ✗ Unrecorded assets – Significant assets may not be recorded in the statement of financial position (e.g. internally generated goodwill will not be recognised).
- ✗ Market conditions – There is no consideration of market conditions or the company's growth potential, which are crucial factors in share valuation.

Key Point

Because of its weaknesses/problems, a share valuation based on balance sheet net assets is unlikely to be suitable.

2.2 Net Realisable Value (NRV)

Definition

Net realisable value – Estimated selling price less estimate costs necessary to make the sale.

The NRV method estimates the liquidation value of the business:

$$\text{Equity} = \text{Estimated } \textit{net realisable value} \text{ of assets} - \text{liabilities}$$

This represents what should be left for shareholders if the assets were sold off and the liabilities settled. It may represent the *minimum* price acceptable to the present owner of the business as it ignores unrecorded assets (e.g. internally generated goodwill).

Problems and weaknesses of this method include:

- ✗ Selling prices – Estimating selling prices under forced sale or distress conditions is challenging, and not the same as “in the ordinary course of business” (e.g. for inventories).
- ✗ Market values – The NRV of some assets may fluctuate significantly if they have volatile market values (e.g. commodities and financial instruments). In addition, it is also challenging to estimate the NRV of assets for which there is no active market (e.g. a patent or specialist item of equipment).
- ✗ It ignores unrecorded assets (e.g. internally generated goodwill).

2.3 Net Replacement Cost

Net replacement cost can be viewed as the cost of setting up an identical business “from scratch”:

$$\text{Equity} = \text{Estimated depreciated replacement cost of net assets}$$

This may represent the maximum price a buyer might be prepared to pay.

Problems and weaknesses of this method include:

- ✗ Challenging to apply – There may be no comparable replacement asset (e.g. unique or specialised assets).
- ✗ Overvaluation – Valuing old and less efficient assets at current market prices of new assets.
- ✗ Unrecorded assets – It ignores unrecorded assets (e.g. goodwill).

3.1 Price/Earnings Ratio

The published P/E ratio of a quoted company takes into account the expected growth rate of that company (i.e. it reflects the market's expectations for the business).

Using published P/E ratios as a basis for valuing **unquoted** companies may indicate an acceptable price to the seller of the shares.

$$\text{Price/Earnings (P/E) ratio} = \frac{\text{Market price per ordinary share}}{\text{Earnings per share}}$$

$$\text{Earnings per Share (EPS)} = \frac{\text{Profit after tax and preference dividends}}{\text{Number of issued ordinary shares}}$$

Therefore:

$$\text{Ordinary share price} = \text{P/E ratio} \times \text{EPS}$$

$$\text{Value of company} = \text{P/E ratio} \times \text{profit after tax and preference dividends}$$

This can be used to value the shares in an unquoted company as follows:

- Step 1** Select the P/E ratio of a similar *quoted* company.
- Step 2** Adjust *downwards* to reflect the additional risk of an unquoted company and the non-marketability of unquoted shares.
- Step 3** Determine the *maintainable* earnings for the unquoted company's EPS.
- Step 4** Multiply the earnings determined in Step 3 by the adjusted P/E from Step 2 to find the value of the unquoted company's shares.

Problems and weaknesses of this method include:

- ✗ Proxy – A suitable "proxy" may not exist (i.e. no similar company listed on the stock market). Even if a proxy exists, it may be under/over-valued by the stock market.
- ✗ Accounting earnings are more subjective than cash flows – Accounting earnings are affected by non-cash items (e.g. depreciation expense) and accounting policy choices.
- ✗ Earnings manipulation – The earnings of the unquoted company may be intentionally inflated. The earnings may not represent future earnings fairly, even if they have not been deliberately distorted. Information asymmetry means the seller will know more about the company and potentially the industry than the buyer.
- ✗ Historical data – The P/E ratio is often based on historical earnings, which may not reflect earnings potential. Share prices can sometimes vary dramatically.
- ✗ Shareholder influence – A valuation based on earnings is less relevant for purchases of minority shareholdings. Minority shareholders typically have limited influence to affect earnings.
- ✗ Loss-making companies – If the unquoted company is loss-making, the P/E ratio method results in a (meaningless) negative value for its equity.

Example 1 Grocery Business

Foodie Co, a small chain of UK-based grocery shops, has just reported post-tax earnings of \$200,000, out of which it paid a dividend of \$50,000.

The P/E ratios of three large UK quoted supermarket chains are:

Morrison (Wm)	14.5
Sainsbury (J)	12.0
Tesco	17.0

Using the average P/E, 14.5 (1/3 (14.5 + 12 + 17)), the market value of Foodie Co would be estimated at \$2.9m (14.5 × \$200,000)

This might be a starting point for negotiations. However, the downward adjustment is typically a third to a half, giving a valuation range of \$1.45-\$1.93m.

Activity 1 Price/Earnings Ratio

[Show](#)

Tanglefoot Co is an unlisted company. Its most recent earnings per share (EPS) was \$0.53 ; next year's EPS is forecast to be 10% higher. Tanglefoot Co has \$50,000 of issued share capital (\$0.10 nominal value per share). The average price-earnings ratio of listed firms in the same business sector is 12 times.

Required:

Estimate the total value of Tanglefoot Co using the price/earnings ratio method.

**Please use the notes feature in the toolbar to help formulate your answer.*

Market value per share = \$0.53 × 12 times = \$6.36

Number of issued shares = 50,000/0.1 = 500,000

Total value = 500,000 × \$6.36 = \$3.18m

The sector P/E ratio should be applied to the company's most recent earnings figure rather than forecast earnings. The sector ratio would have been calculated based on the most recent published earnings – any expected earnings growth is reflected in the size of the price-earnings multiple itself.

3.2 Earnings Yield

Earnings yield is simply the reciprocal of the P/E ratio (and therefore has similar problems and weaknesses).

$$\text{Earnings yield} = \frac{\text{EPS}}{\text{Market price per share}} \times 100$$

Therefore:

$$\text{Ordinary share price} = \frac{\text{EPS}}{\text{Earnings yield}}$$

Activity 2 Income-based Valuation

[Show](#)

Accrington Co is an unquoted company.

- It has issued 400,000 ordinary shares of \$0.25 nominal value.
- Its statement of profit or loss (extract) for the year ended 31 July 20X4 shows:

	\$	\$
Profit before taxation		260,000
Less: Taxation		(120,000)
Profit after taxation		
Less: Preference dividend	20,000	140,000
Ordinary dividend	<u>36,000</u>	<u>(56,000)</u>
Retained profit for the year		<u>84,000</u>

- The P/E ratio for a similar type of business (suitable for an unquoted company) is 12.5.

Required:

Value 200,000 ordinary shares in Accrington Co on an earnings basis.

**Please use the notes feature in the toolbar to help formulate your answer.*

Note that only half of the shares are being valued.

$$\begin{aligned}\text{Valuation of 200,000 shares} &= 200,000 \text{ shares} \times \text{P/E ratio} \times \text{EPS} \\ &= 200,000 \text{ shares} \times 12.5 \times \frac{(\$140,000 - \$20,000)}{400,000} \\ &= \$750,000\end{aligned}$$

Problems and weaknesses of this method include:

- ✗ **Variability in yield** – Earnings yield can vary significantly over time; it is not a stable valuation metric and it may be a challenge to interpret. For example, a high earnings yield could be due to a low share price in a distressed company rather than indicate a good investment opportunity.
- ✗ **High interest rates** – An earnings yield may appear attractive, but it may not be competitive compared to other investments when interest rates are high.
- ✗ **Earnings quality** – Reported earnings may not be high quality and sustainable. Unsustainable earnings may lead to an inaccurate valuation.
- ✗ **Non-earnings factors** – Other factors that may be important for investors are ignored (e.g. dividends, company management, competitive advantages and industry-specific risks).

4.1 Dividend Valuation Model

4.1.1 Constant Dividend Growth

If dividends are forecast to grow at a constant annual rate to perpetuity, the valuation formula (as introduced in *Chapter 10*) is:

$$P_0 = \frac{D_0(1+g)}{(r_e - g)}$$

Where D_0 = most recent dividend

g = dividend growth rate

r_e = required return of equity investors (= k_e)

P_0 = Current share price

Note: $D_0(1+g)$ is the dividend in one year.

4.1.2 Application to Share Valuation

The model can be applied to the valuation of shares as follows:

Step 1 Identify the current dividend.

Step 2 Estimate the growth rate of dividends. Three methods may be available:

- i) economic forecasting (i.e. the exam question may give a growth rate);
- ii) assume the historical dividend growth rate will apply into the future;
- iii) Gordon's growth approximation (see *Chapter 10*)

Step 3 Determine the required return using the growth model formula or CAPM.

Step 4 Use the inputs determined above in the model to find the value per share.

4.1.3 Weaknesses

- Simplifying assumptions – The constant growth rate assumption for dividends may not hold true for many companies in the long run, leading to inaccurate valuations.
- Dependence on dividends – The DVM is less applicable to a company that does not pay dividends or pays inconsistent dividends. This limitation is particularly relevant for growth-oriented or non-dividend-paying companies.
- Sensitivity – The valuation is highly sensitive to the choice of the required rate of return.
- Shareholder influence – It has less relevance for valuing a *majority* shareholding, as an investor with control can change the dividend policy.

Example 2 Growth Model Valuation

Cross Co, an unlisted company, has just paid a dividend of \$0.12 per share. Its historical dividend growth rate of 5% is expected to be maintained

The following is available for a suitable listed company (i.e. same business and same gearing):

Share price	\$2.40
Dividend just paid	\$0.22
Historical dividend growth rate	
(also expected to be maintained)	10%

Required:

Calculate the share value of Cross Co.

Solution

Using the second growth model formula from the exam formula sheet:

$$r_e = \frac{D_0(1+g)}{P_0} + g = \frac{0.22(1+0.1)}{2.4} + 0.1 = 0.2 \text{ or } 20\%$$

Applying this to value shares in an unlisted company, Cross Co, using the first growth model formula:

$$P_0 = \frac{D_0(1+g)}{(r_e - g)} = \frac{0.12(1+0.05)}{(0.2 - 0.05)} = \$0.84$$

Key Point

This is a **theoretical** value. Although it allows for the unlisted company's lower dividend and lower growth rate, it does not consider that shares in unquoted companies are less desirable and more risky than shares in an equivalent listed company. If the shareholder's required return is 30% rather than 20%, the share price would fall to \$0.50.

Example 3 CAPM Valuation

Wave Co, an unlisted company, is entirely equity financed. It has just paid a dividend of \$0.12 a share. Its historical dividend growth rate of 5% is expected to be maintained in future.

The following information is available for a listed company in the same business:

Debt/Equity ratio	2:5
Beta	1.6

The risk-free rate is 5% and the return from the market is 15%.

The company profit tax rate is 25%

Required:

Value Wave Co's equity.

Solution

Using CAPM to find the shareholder's required rate of return in the listed company:

$$r_e = R_f + \beta(R_m - R_f) = 5\% + 1.6(15\% - 5\%) = 21\%$$

This is the return required by shareholders of a company geared in the ratio 2:5. Wave Co is ungeared, however, so 21% is inappropriate for its shareholders.

Using the asset beta formula to find the ungeared beta:

$$\beta_a = \left[\frac{V_e}{(V_e + V_d(1-T))} \beta_d \right] + \left[\frac{V_d(1-T)}{(V_e + V_d(1-T))} \beta_d \right] = \left[\frac{5}{(5+2(1-0.25))} 1.6 \right] = 1.23$$

The cost of equity of Wave Co (an ungeared company in the same business as the geared company is:

$$r_e = R_f + \beta(R_m - R_f) = 5\% + 1.23(15\% - 5\%) = 17.3\%, \text{ say } 17\%$$

The value of a share in Wave Co is therefore:

$$P_0 = \frac{D_0(1+g)}{(r_e - g)} = \frac{0.12(1+0.05)}{(0.17-0.05)} = \$1.05$$

Exam advice

Remember that the second set of brackets in the asset beta formula is nearly always assumed to be zero because β_d , the beta of debt, is assumed to be zero.

Activity 3 Growth's Effect on Valuation

[Show](#)

Claygrow Co is a company which manufactures flower pots. The following information is available:

Current dividend	\$0.25 per share
Required return on equities in this risk class	20%

Required:

Value one share in Claygrow Co under the following circumstances:

- a) No growth in dividends.
- b) Constant dividend growth of 5% per year.
- c) Constant dividends for five years and then growth of 5% per year to perpetuity.
- d) Constant dividends for five years and then sale of the share for \$2.00.

**Please use the notes feature in the toolbar to help formulate your answer.*

(a) No growth in dividends

$$\text{Constant dividend } P_0 = \frac{\$0.25}{0.20} = \$1.25$$

(b) Constant dividend growth at 5%

$$P_0 = \frac{\$0.25(1+0.05)}{0.2-0.05} = \$1.75$$

(c) Constant followed by growth

PV of five years' dividends = $(\$0.25 \times 2.991(\text{cumulative discount factor years 1-5 at 20\%})) = \0.748
plus

$$\text{PV of growing dividend from year 6 onwards} = \frac{\$0.25(1+0.05)}{0.2-0.05} \times \frac{1}{1.2^5} = \$0.703$$

$$\text{Total value} = \$0.75 + \$0.70 = \$1.45$$

(d) Constant followed by sale in 5 years

$$\text{PV five years' dividends} = (\$0.25 \times 2.991) = \$0.748$$

$$\begin{aligned} \text{PV of \$2.00 in five years' time} &= \\ & \$2.00 \times \frac{1}{1.2^5} = \$0.804 \end{aligned}$$

$$\text{Total value} = \$0.75 + \$0.80 = \$1.55$$

Activity 4 Value Using Dividend Growth Model

[Show](#)

Current EPS = \$0.30

Payout ratio = 40%

Number of shares in issue = 5m

Net assets in the statement of financial position = \$12m

Risk-free rate = 4%

Market premium = 5%

Equity beta = 1.4

Required:

Value the company's equity using the dividend growth model.

**Please use the notes feature in the toolbar to help formulate your answer.*

Current dividend per share = $\$0.30 \times 0.4 = \0.12

Retention ratio = $1 - 40\% = 60\%$

Profit after tax = $\$0.30 \times 5\text{m} = \1.5m

Return on equity = $\frac{\$1.5\text{m}}{\$12\text{m}} = 12.5\%$

Growth rate = $0.6 \times 0.125 = 0.075$ i.e. 7.5%

Required return = $0.04 + (1.4 \times 0.05) = 11\%$ (using CAPM)

Share value (P) = $\frac{\$0.12(1.075)}{0.11 - 0.075} = \3.69

Total value of equity = $\$3.69 \times 5\text{m} = \18.45m

Alternative approach (assuming the actual return on reinvested profit equals the minimum required return per CAPM):

Growth = $0.6 \times 0.11 = 0.066$ i.e. 6.6%

$P_0 = \frac{\$0.12(1.066)}{0.11 - 0.066} = \2.91

Total value of equity = $\$2.91 \times 5\text{m} = \14.55m

Example 4 Determining Growth Rate from Past Dividends

Kip Co has paid the following dividends per share in recent years:

Year	20X6	20X5	20X4	20X3
Dividend per share	\$0.65	\$0.638	\$0.628	\$0.611

The dividend for 20X6 has just been paid, Kip Co has a cost of equity of 12%.

Required:

Calculate Kip Co's share price.

Solution

To find the growth rate over n years, find the $(n-1)^{\text{th}}$ root of the difference between the latest and earliest dividend figures.

The average dividend growth rate over the four years is therefore calculated as $3\sqrt{(0.65/0.611)} - 1 = 2.1\%$

The share price = $(\$0.65 \times 1.021)/(0.12 - 0.021) = \6.7

4.2 Discounted Cash Flow Basis

A business can be viewed as a combination of its underlying projects. Just as the value of an individual project equals the present value of its future cash flows, the total value of the business should also equal the present value of the company's cash flows.

- For a business valuation, as for project valuation, the relevant cash flows are the future operating cash flows, and the appropriate discount rate is the WACC (see *Chapter 11*).
- The present value of operating cash flows provides a valuation of the company's assets. The value of equity can then be calculated as:

$$\text{Value of equity} = \text{Value of assets} - \text{Value of debt}$$

Activity 5 DCF Basis

[Show](#)

Empire Co is considering the acquisition of Juicy Co. Empire's finance director has forecast Juicy's post-acquisition operating cash flows for the next four years as follows:

Year	1	2	3	4
Operating cash flow (\$m)	15	16	18	20

Empire plans to dispose of Juicy's assets at the end of the fourth year for an estimated \$230m.

Juicy has \$15m share capital, with a nominal value of \$0.50 a share, and \$50m loan notes in issue.

Empire considers its current WACC of 10% appropriate as Juicy is in the same business and relatively small compared with Empire.

Required:

Calculate the value to Empire Co of each share of Juicy Co. Ignore taxation.

**Please use the notes feature in the toolbar to help formulate your answer.*

Year	1	2	3	4
Cash flow (\$m)	15	16	18	250
Discount factor	0.909	0.826	0.751	0.683
Present value (\$m)	13.6	13.2	13.5	170.8

Total value of assets	=	\$211.1m
Total value of equity	=	\$211.1 m – \$50m = \$161.1m
Number of shares	=	\$15m/\$0.50 = 30 million
Share price	=	\$161.1m/30m = \$5.37

DCF methods are superior to non-DCF methods in investment appraisal (see *Chapter 4*). As a model for the valuation of shares, it considers the intrinsic value of a business based on its expected future cash flows and allows for the time value of money. Although it enables a more precise assessment of a company's worth, it still has problems and weaknesses:

- ✗ Estimation challenges – Estimating future cash flows and determining an appropriate discount rate can be complex and uncertain, especially for long-term forecasts.
- ✗ Sensitivity to assumptions – Small changes in assumptions about future cash flows and discount rates can lead to significantly different valuations.
- ✗ Theory v practice – A DCF-based valuation is based on a theoretically calculated intrinsic value. In practice, market prices can be influenced significantly by other factors (e.g. investor sentiment and market trends).
- ✗ Unpredictable cash flow – A DCF-based valuation may be unsuitable for startups or high-growth companies with unpredictable cash flows.

5.1 Marketability and Liquidity

Shares in unquoted companies are not traded via the stock market, and this lack of marketability reduces their value relative to shares in quoted companies.

Furthermore, unquoted companies are not required to comply with stock market listing rules or corporate governance codes, increasing their perceived risk and further depressing their value.

Some analysts adjust the P/E ratio of a comparable quoted company downwards by as much as 40% when valuing unquoted shares.

Even shares in quoted companies can suffer from a lack of marketability during periods of "thin trading" (i.e. a lack of liquidity in the market).

5.2 Information Sources and Availability

Potential investors generally have plenty of publicly available information when deciding whether to buy/sell quoted shares (e.g. published accounts, earnings forecasts, research reports, news and analysis in the financial press and data services such as Bloomberg and Reuters).

However, much less information may be available for unquoted companies because they may not be required to publish accounts (or have exemptions from producing group accounts or showing a statement of cash flows) and are also unlikely to be watched by analysts or news agencies.

This leads to "asymmetry of information" between the managers and potential investors in unquoted companies (i.e. managers have complete information about the true value of the business, but potential investors have very little). The resulting uncertainty leads to investors potentially undervaluing the shares.

5.3 Market Imperfections and Pricing Anomalies

Although the DVM gives an idea of the fundamental fair value of a share, it is based on perfect market assumptions (i.e. many buyers/sellers of the share, zero transaction costs, freely available information and rational investors).

Even major markets such as the New York or London Stock Exchanges are not perfect markets, as evidenced by the following:

- Irrational investor behaviour (e.g. the belief that recent price rises will lead to future price rises). This over-exuberance can lead to speculative bubbles (e.g. the tech bubble in the 1990s). This is explored further in behavioural finance (see s.5.5 below).
- Pricing anomalies which appear to contradict perfect markets.
 - "The "January effect" is where investors sell shares at the end of December to utilise any tax losses on them, then buy back the shares at the start of January.
 - The "overreaction effect" is where share prices appear to overreact to unexpectedly good/bad news initially, then slowly move back toward their fundamental value.
 - The "small capitalisation discount" is caused by the behaviour of institutional investors (the primary buyers of quoted shares). Companies with market values of just a few million dollars tend to stay "under the radar" of fund managers, and in fact, some pension funds will even be restricted from buying such shares. This can lead to a "small cap" company always remaining relatively low in value.

5.4 Market Capitalisation

Market capitalisation refers to the total market value of a quoted company (i.e. number of issued ordinary shares $\times$ market price per share).

However, market capitalisation is not necessarily the amount that would have to be paid for the company's entire equity. The quoted share price is for a minority shareholding. A significant premium would usually have to be paid if control is required.

On the other hand, there may be situations where the entire capital can be acquired at *below-market* capitalisation (e.g. a small group of people holds a large proportion of the target company's shares). If the existing investors tried to dispose of their holdings via the market, they would cause a major price fall. Therefore, they may be willing to sell "off exchange" at a small discount to the current quoted price.

5.5 Behavioural Finance

Speculation by investors is a significant factor in the behaviour of share prices. Behavioural finance is an alternative view to the efficient market hypothesis and attempts to explain:

- How managers make financial decisions in real life.
- Why their decisions might not appear rational and have unpredictable consequences.

This contrasts many traditional theories, which assume investors make rational decisions.

The **market paradox** is that investors must believe they are inefficient for markets to be efficient. If they thought them to be efficient, they would not buy or sell shares, thereby slowing the dissemination of information into market prices.

Herding or "herd mentality" may be explained by:

- the desire to conform and not to act differently from others (e.g. fund managers tend to follow each other's strategies);
- individual investors, lacking confidence, believe that a large group of other investors cannot be wrong.

If many investors follow a herd instinct to buy shares in a specific sector, a significant price rise for shares in that sector can lead to a stock market **bubble**.

"Noise traders" are investors who do not base buy/sell decisions on rational analysis (i.e. without using fundamental data about the company's performance). Noise traders generally have poor timing, follow trends and overreact to good and bad news.

Loss aversion means that some investors:

- avoid investments with the risk of making losses, even though expected value analysis suggests that, in the long term, they will make significant capital gains;
- prefer to invest in companies that look likely to make stable but low profits rather than those that may generate higher profits in some years but possibly losses in others.

The **"momentum effect"** in stock markets can create optimism (e.g. investors believe a trend in price rises will continue), which increases willingness to invest in companies that show prospects for growth. If a momentum effect exists, it will likely lengthen a stock market boom (or bust) period.

An **overconfident** investor tends to hold a false or misleading assessment of their skills, intellect or ability – in other words, a belief that they are better and more able than they are. This excessive confidence can cause the investor to be reckless and make mistakes.

An investor who believes their actions will directly affect the stock market and that they have a certain degree of control of the market has an **illusion of control**.

A bias towards **optimism** often leads investors to have an unrealistically confident view of themselves and their futures.

Investors exhibit **confirmation bias** if they only seek information supporting their beliefs. This bias may lead them to focus only on information reinforcing their opinions about an investment. Selectively choosing which information to use can lead to a lack of diversification.

Therefore, individuals may not make decisions based on a rational analysis of all available information. This could lead to prices for an individual company and the market being valued either very high or very low.

6.1 Preference Shares

The market value of any share should equal the present value of the future dividend stream discounted at the investors' required return.

As preference dividends are a fixed percentage of the share's nominal value, there will be zero growth of the future cash flow, and the share price becomes the present value of a perpetuity:

$$P_0 = \frac{D}{r_e}$$

Required return, $\frac{D}{P_0}$, is the dividend yield and may be described as the cost of preference share capital.

Activity 6 Preference Shares

[Show](#)

A company has in issue 5% preference shares, each with a \$1 nominal value.

The financial press quotes the yield on these shares as 4.7%.

Required:

Calculate the market price of each preference share.

**Please use the notes feature in the toolbar to help formulate your answer.*

Annual fixed dividend = $5\% \times \$1 = \0.05

Required return = $4.7\% = 0.047$

$P_0 = \$0.05 / 0.047 = \1.06

6.2 Irredeemable Loan Notes

- The market value of any debt should equal the present value of the future payments to the investor discounted at the required return.
- In the case of irredeemable (undated) loan notes, there will be a fixed annual payment of "coupon" interest into perpetuity, with no principal repayment.

$$P_0 = \frac{I}{r_e}$$

Where:

P_0 = Ex-int market price

I = Annual interest payment

r_e = Investors' required return

- Required return, $\frac{I}{P_0}$, is the *current* yield and may be described as the company's pre-tax cost of irredeemable loan notes.

Activity 7 Irredeemable Loan Notes

[Show](#)

A company has in issue 7% undated loan notes, each with \$100 nominal value. The current yield is quoted as 7.42%.

Required:

Calculate the market price of each loan note.

**Please use the notes feature in the toolbar to help formulate your answer.*

$$\begin{aligned}\text{Annual coupon} &= 7\% \times \$100 = \$7 \\ \text{Required return} &= 7.42\% = 0.0742 \\ P_0 &= \$7 / 0.0742 = \$94.34\end{aligned}$$

6.3 Redeemable Loan Notes

The market value of a redeemable (dated) loan note should equal the present value of the coupon interest (paid each year until maturity) and the redemption price (paid at maturity), discounted at the investors' required rate of return.

The required return on a redeemable loan note is referred to as its yield to maturity, gross redemption yield or the pre-tax cost of the redeemable loan note.

Activity 8 Redeemable Loan Notes

[Show](#)

A company has in issue 8% \$100 nominal value loan notes redeemable at a 5% premium in 10 years. The bonds' yield to maturity is 10%.

Required:

Calculate the market price of each bond.

**Please use the notes feature in the toolbar to help formulate your answer.*

$$\begin{aligned}\text{Annual coupon} &= 8\% \times \$100 = \$8 \\ \text{Present value of the coupon}^{(1)} &= \$8 \times 6.145 = \$49.16 \\ \text{Redemption price} &= \$105 \\ \text{Present value of the redemption price}^{(2)} &= \$105 \times 0.386 = \$40.53 \\ \text{Market price of loan note} &= \$49.16 + \$40.53 = \$89.69\end{aligned}$$

⁽¹⁾ Applying the 10-year *annuity* factor at 10%.

⁽²⁾ Applying the 10-year *discount* factor at 10%.

6.4 Convertible Loan Notes

Definition

Convertible loan notes – allow the investor to choose between redeeming the loan notes at some future date or converting them into a predetermined number of ordinary shares.

To estimate the market value, it is first necessary to predict whether the investor will choose redemption or conversion. The redemption value will be known with certainty, but the future share price can only be estimated.

Other amounts that may be calculated for convertibles:

- **Floor value** = the value assuming redemption;
- **Conversion premium** = market value – current conversion value.

Key Point

MV (ex-interest) = PV of future interest payments and the *higher* of:

- (i) redemption value; or
- (ii) forecast conversion value,

which is then discounted at the bondholder's required rate of return.

6.5 Interest Rates and Loan Note Valuation

Market interest rates (yields) and the market value of fixed-income securities (such as loan notes) have an inverse relationship. This is because the market value of a loan note equals the present value of the fixed future payments it will make to the investor. If market interest rates (i.e. discount rates) fall, the present value of a loan note's future cash flows will rise, and with it, the loan note's market price.

Activity 9 Convertible Bonds

[Show](#)

A company has 9% loan notes in issue, redeemable at their nominal value of \$100 in five years. Alternatively, each bond may be converted into 20 ordinary shares on that date. The current ordinary share price of \$4.45 is expected to grow at an annual rate of 6.5% for the foreseeable future. The debtholders' required return is 7% a year.

Required:

Calculate the following for each \$100 convertible loan note:

- (i) market value;
- (ii) floor value; and
- (iii) conversion premium.

**Please use the notes feature in the toolbar to help formulate your answer.*

(i) Market value

Expected share price in five years' time = $4.45 \times 1.065^5 = \$6.10$ Forecast conversion value = $6.10 \times 20 = \$122$

Compared with redemption at the nominal value of \$100, conversion will be preferred.

Today's market value is the PV of future interest payments, plus the PV of the forecast conversion value:

$$= (9 \times 4.100) + (122 \times 0.713) = \$123.89$$

(ii) Floor value

Floor value is the PV of future interest payments, plus the PV of the redemption value:

$$= (9 \times 4.100) + (100 \times 0.713) = \$108.20$$

(iii) Conversion premium

$$\text{Current conversion value} = 4.45 \times 20 = \$89.00$$

$$\text{Conversion premium} = \$123.89 - 89.00 = \$34.89$$

This is often expressed on a per-share basis (i.e. $34.89/20 = \$1.75$ per share).

Activity 10 Different Conversion/Redemption Dates[Show](#)

A company has 7% loan notes in issue which are redeemable at their nominal value of \$100 per loan note in eight years. Alternatively, each loan note is convertible after seven years into 11 ordinary shares. The company's ordinary shares are currently trading at \$6.50 per share. The before-tax cost of debt of the convertible loan notes is 8%.

Required:

Calculate the current market value of each loan note, assuming the share price increases by:

- a) **4% per year;**
- b) **6% per year.**

**Please use the notes feature in the toolbar to help formulate your answer.*

(a) Share price increases by 4% per year

The share price in seven years will be \$8.55 per share ($\6.50×1.04^7), and the conversion value will be \$94.05 per loan note ($\8.55×11).

This should be compared to the redemption value of each loan note, discounted to its value as at the end of seven years. Therefore, as the redemption value is \$99.08 ($0.926 \times \107), investors will likely hold the loan notes until redemption after eight years.

The current market value of the loan notes will be \$94.23 ($(7 \times 5.747) + (100 \times 0.540)$).

This is also the "floor value" of the loan notes.

(b) Share price increases by 6% per year

The share price in seven years will be \$9.77 per share ($\6.50×1.06^7), and the conversion value will be \$107.47 per loan note ($\9.77×11). Holders of the loan notes are likely to prefer conversion, as the conversion value is now greater than the redemption value, in year seven terms (\$99.08).

The current market value of the loan notes will be \$99.10 ($(7 \times 5.206) + (107.47 \times 0.583)$).

Syllabus Coverage

This chapter covers the following Learning Outcomes.

F. Business Valuations

1. Nature and purpose of the valuation of business and financial assets

- a) Identify and discuss reasons for valuing businesses and financial assets.
- b) Identify information requirements for valuation and discuss the limitations of different types of information.

2. Models for the valuation of shares

- a) Discuss and apply asset-based valuation models, including:
 - i) net book value (statement of financial position basis).
 - ii) net realisable value basis.
 - iii) net replacement cost basis.
- b) Discuss and apply income-based valuation models, including:
 - i) price/earnings ratio method
 - ii) earnings yield method.
- c) Discuss and apply cash flow-based valuation models, including:
 - i) dividend valuation model and the dividend growth model
 - ii) discounted cash flow basis.

3. The valuation of debt and other financial assets

- a) Discuss and apply appropriate valuation methods to:
 - i) irredeemable debt
 - ii) redeemable debt
 - iii) convertible debt
 - iv) preference shares

4. Efficient Market Hypothesis (EMH) and practical considerations in the valuation of shares

- b) Discuss practical considerations in the valuation of shares and businesses including:
 - i) marketability and liquidity of shares
 - ii) availability and sources of information
 - iii) market imperfections and pricing anomalies
 - iv) market capitalisation.
- c) Describe the significance of investor speculation and the explanations of investor decisions offered by behavioural finance.

Summary and Quiz

- Asset-based valuation methods include net book value, net realisable value and net replacement cost. All three asset-based valuation methods ignore unrecorded assets.
- Income-based valuation methods include P/E and earnings yield.
- Cash flow-based valuation methods include the dividend valuation model with the dividend growth model or CAPM and the DCF basis.
- Varying levels of marketability and liquidity make it difficult to value companies.
- In completing a valuation of an unquoted company, information is often sparse.
- Practical factors to consider include marketability and liquidity, information sources and availability, market imperfections and pricing anomalies and market capitalisation.
- Issues in behavioural finance include the market paradox, herding, “noise traders”, loss aversion and the “momentum effect”.
- The market value of debt equals the present value of future interest and redemption payments discounted at the debt investors' required rate of return.

Chapter 18 Quiz

Chapter 19: Ratio Analysis

Overview	CHAPTER 19: Visual Overview
19.1 Ratio Analysis	19.1.0 Introduction
	19.1.1 Profitability Ratios
	19.1.2 Liquidity Ratios
	19.1.3 Efficiency/Activity Ratios
	19.1.4 Gearing/Risk Ratios
	19.1.5 Investor Ratios
19 Conclusion	19 Syllabus Coverage
	19 Summary and Quiz
	19 Technical Articles

CHAPTER 19: Visual Overview

Objective: To understand all ratios commonly used in business analysis.

RATIO ANALYSIS

- Profitability
- Liquidity
- Efficiency/Activity
- Gearing/Risk
- Investor

1.0 Introduction

Ratio analysis – comparing and quantifying relationships between financial variables (e.g. in the statement of financial position and the statement of profit or loss) – is often used to understand how an organisation performs and to determine whether it is achieving its objectives.

Comparison is usually over several periods for the same business or between similar businesses for the same period.

Exam advice

Because ratios are relevant to several topic areas, they are often examined. The examiner's reports often comment that candidates' performance in this area is disappointing, noting a need for improved explanation and interpretation even among those who can correctly compute them.

Ratio analysis is relevant to many topics (e.g. choice of business finance, working capital management and valuation). Some have already been covered (e.g. working capital ratios in *Chapter 13*).

1.1 Profitability Ratios

Gross profit margin	$= \frac{\text{Gross profit}}{\text{Sales}} \times 100$
Operating profit margin	$= \frac{\text{Profit before interest and tax}}{\text{Sales}} \times 100$
Return on capital employed(ROCE)	$= \frac{\text{Profit before interest and tax}}{\text{Capital employed}^*} \times 100$
Return on equity (ROE)	$= \frac{\text{Profit after tax-preference dividends}}{\text{Ordinary shareholders' funds}} \times 100$

* Capital employed = Shareholders' funds + long-term liabilities
= Total assets – current liabilities

1.1.1 ROCE

A company needs to be profitable if shareholder wealth is to be maximised. Return on capital employed (also known as the return on investment (ROI)) is useful as it relates profits to the amount of funds (the capital) employed in making the profits.

Profit margin (see below) and asset turnover (see 1.3) together explain ROCE:

Profit margin $\times$ asset turnover = ROCE

$$\frac{\text{PBIT}}{\text{Sales}} \times \frac{\text{Sales}}{\text{Capital employed}} = \frac{\text{PBIT}}{\text{Capital employed}}$$

This relationship facilitates a more detailed analysis of why ROCE is high or low, better or worse than last year.

1.1.2 ROE

ROE is another measure of an organisation's overall performance and shows the earning power of the shareholders' book investment. It is particularly useful for comparing companies in the same industry. A higher ROE could indicate:

- good cost control or investment in profitable projects; or
- that the company is more geared and has more risk associated with shareholder returns.

1.2 Liquidity Ratios

These ratios were covered in *Chapter 13*.

$$\text{Current ratio} = \frac{\text{Current assets}}{\text{Current liabilities}}$$

$$\text{Quick or acid test ratio} = \frac{\text{Current assets-inventory}}{\text{Current liabilities}}$$

1.3 Efficiency/Activity Ratios

These ratios were covered in *Chapter 13*.

$$\text{Receivables collection period} = \frac{\text{Average accounts receivable}}{\text{Annual credit sales}} \times 365 \text{ days}$$

$$\text{Payables payment period} = \frac{\text{Average accounts payable}}{\text{Annual credit purchases}} \times 365 \text{ days}$$

$$\text{Inventory holding period} = \frac{\text{Average inventory}}{\text{Annual cost of sales}} \times 365 \text{ days}$$

$$\text{Operating cycle} = \text{Inventory (days)} + \text{Receivables (days)} - \text{Payables (days)}$$

$$\text{Asset turnover} = \frac{\text{Sales}}{\text{Capital employed}}$$

$$\text{Sales/working capital} = \frac{\text{Sales}}{\text{Working capital}}$$

1.4 Gearing/Risk Ratios

1.4.1 Financial Gearing (Leverage) Ratios

Gearing was discussed in *Chapter 11*. Financial gearing, a measure of financial risk, is the degree to which a company's activities are funded by debt compared to equity. There are several gearing ratios, two of which are shown below. Note that a company would be highly geared if the first ratio was more than 100% (more debt than equity), whereas if the second ratio were more than 50%, the company would be highly geared.

$$\text{Debt to equity} = \frac{\text{Non-current liabilities}}{\text{Capital + reserves}} \times 100$$

$$\text{Debt to total capital} = \frac{\text{Non-current liabilities}}{\text{Capital employed}} \times 100$$

1.4.2 Operational Gearing Ratios

Operational gearing describes the effect of fixed costs on the relationship between sales revenue and operating profits. As operational gearing increases, profits become more volatile compared to changes in sales. If operational gearing is high, a small percentage change in sales revenue will have a much more significant percentage in operating profits.

$$\begin{aligned} \text{Operational gearing} &= \frac{\text{Fixed operating costs}}{\text{Variable operating costs}} \times 100 \quad \text{OR} \quad \frac{\text{Fixed operating costs}}{\text{Total operating costs}} \times 100 \\ &\text{OR} \quad \frac{\text{Contribution}}{\text{Profit before interest and tax}} \end{aligned}$$

1.4.3 Risk Ratios

The following ratios measure a company's ability to pay the interest on its debt. They are used by investors, lenders and creditors to determine the riskiness of their financial interests in the company. They can be used to determine both short-term financial health and company stability (a declining interest cover ratio is an indication that a company may be unable to meet its debt obligations in the future).

$$\text{Interest coverage} = \frac{\text{Profit before interest and tax}}{\text{Interest expense}}$$

$$\text{Cash flow coverage} = \frac{\text{Cash generated from operations}}{\text{Interest expense}}$$

1.5 Investor Ratios

Investors use the following ratios to determine whether the expected returns are satisfactory given the risks they will be taking.

1.5.1 Earnings per Share (EPS)

As noted in *Chapter 1*, earning per share is widely used and particularly useful when comparing results over several periods. Investors will be looking for a company to sustain its earnings, enabling it to pay dividends, reinvest for future growth and grow EPS year-on-year.

Despite its wide use, it is based on past data (whereas investors should be more concerned about future earnings) and is affected by changes in accounting policy choices or mergers and acquisitions (which affect the number of shares). Also, it is liable to manipulation when used as the basis for awarding management bonuses.

The ratio is based on the earnings due to ordinary shareholders and the number of ordinary shares only.

$$\text{Earnings per ordinary share (EPS)} = \frac{\text{Profit after tax-preference dividends}}{\text{Weighted average number of ordinary shares in issue}}$$

Diluted EPS should also be calculated where a company has a *complex* capital structure that includes potentially dilutive securities (PDSs). These are securities in issue that have an obligation to issue shares in the future (e.g. convertible debt or share options).

$$\text{Diluted EPS} = \frac{\text{Profit after tax-preference dividends} + \text{PDS adjustments}}{\text{Ordinary dividend} + \text{PDS}}$$

1.5.2 Price/earnings (P/E) ratio

$$\text{Price/earnings (P/E) ratio} = \frac{\text{Ordinary share price}}{\text{EPS}}$$

The P/E ratio was explained in *Chapter 18* as an income-based valuation model.

It is a measure of the relative worth of a share and relates two important metrics for investors – the market price of a share and the earning capacity of that share.

A company's P/E ratio tends to change only gradually. If the EPS goes up or down, the share price is also expected to go up or down, so the new share price will be the new EPS multiplied by the constant P/E ratio.

1.5.3 Other Investor Ratios

- **Dividend coverage ratio** provides an estimate for investors of the risk of not receiving dividends.
- **Dividend payout ratio** indicates how much money a company returns to its shareholders compared to how much it keeps to reinvest in growth, pay off debt and add to cash reserves.
- **Dividend yield** tells an investor the percentage of the share price paid out in dividends each year.
- **Earnings yield** was mentioned in *Chapter 18* and is used by investors to assess the rate of return on their investment.
- **Total shareholder return** was covered in *Chapter 1*.

Dividend coverage	$= \frac{\text{Profit after tax-preference dividends}}{\text{Ordinary dividend}}$
Dividend payout ratio	$= \frac{\text{Ordinary dividend}}{\text{Profit after tax-preference dividends}}$
Dividend yield	$= \frac{\text{Dividend per ordinary share}}{\text{Ordinary share price}} \times 100$
Earnings yield	$= \frac{\text{EPS}}{\text{Ordinary share price}} \times 100$
TSR*	$= \frac{\text{Year end share price} - \text{Share price at start of year} + \text{Dividends for year}}{\text{Share price at start of year}} \times 100$

* Total shareholder return

Activity 1 Ratio Analysis[Show](#)

Using the following information for Cathcart Co, calculate the ratios listed (on the next page).

Cathcart Co Statement of financial position at 31 December 200X

	\$000	\$000
Non-current assets: Cost less depreciation		2,200
Current assets		
Inventory	400	
Receivables	500	
Cash	<u>100</u>	
		<u>1,000</u>
Total asset		<u>3,200</u>
Equity		
Ordinary shares (\$1 nominal)		1,000
Retained earnings		800
Non-current liabilities		
10% loan notes		600
10% preference shares (\$1 nominal)		200
Current liabilities		
Payables	400	
Taxation	<u>200</u>	
		<u>600</u>
Total equity and liabilities		<u>3,200</u>

Statement of profit or loss for the year ended 31 December 200X

	\$000
Revenue	3,000
Cost of sales	<u>(2,400)</u>
Gross profit	600
Operating expenses	<u>(200)</u>
Profit before interest and tax	400
Interest	<u>(60)</u>
Profit before tax	340
Income tax	<u>(180)</u>
Profit after tax	<u>160</u>
Other information	
Total ordinary dividend (\$000)	125
Current market price of ordinary shares	\$1.40 per share

Solution

- (a) Gross profit margin =
- (b) Operating profit margin =
- (c) ROCE =
- (d) Return on equity =
- (e) Current ratio =
- (f) Acid test ratio =
- (g) Receivables collection period =
- (h) Asset turnover =
- (i) Proportion of debt finance =
- (j) Interest coverage =
- (k) EPS =
- (l) Dividend coverage =
- (m) Dividend yield =
- (n) Price earnings ratio =

**Please use the notes feature in the toolbar to help formulate your answer.*

(a)	Gross profit margin	$= \frac{\$600}{\$3,000} \times 100$	= 20%
(b)	Operating profit margin	$= \frac{\$400}{\$3,000} \times 100$	= 13.3%
(c)	ROCE	$= \frac{\$400}{\$3,200 - \$600} \times 100$	= 15.4%
(d)	Return on equity	$= \frac{\$160 - 20}{\$1,600} \times 100$	= 7.8%
(e)	Current ratio	$= \frac{\$1,000}{\$600}$	= 1.67 : 1
(f)	Acid test ratio	$= \frac{\$1,000 - 400}{\$600}$	= 1 : 1
(g)	Receivables collection period	$= \frac{\$600}{\$3,000} \times 365$	= 61 days
(h)	Asset turnover	$= \frac{\$3,000}{\$2,600}$	= 1.15

(i)	Proportion of debt finance	$= \frac{\$800}{\$1,800} \times 100$	= 44.4%
		OR	
		$= \frac{\$800}{\$800 + \$1,800} \times 100$	= 30.8%
(j)	Interest cover	$= \frac{\$400}{\$60}$	= 6.67
(k)	EPS	$= \frac{\$160 - 20}{\$1,000}$	= \$0.14
(l)	Dividend cover	$= \frac{\$160 - 20}{\$125}$	= 1.12
(m)	Dividend yield	$= \frac{\$0.125}{\$1.40} \times 100$	= 8.9%
(n)	Price/earnings ratio	$= \frac{\$1.40}{\$0.14}$	= 10

Syllabus Coverage

This chapter covers the following Learning Outcomes.

A. Financial Management Function

3. Stakeholders and impact on corporate objectives

- d) Describe and apply ways of measuring achievement of corporate objectives including:
- i) ratio analysis, using appropriate ratios such as return on capital employed, return on equity, earnings per share and dividend per share

C. Working Capital Management

2. Management of inventories, accounts receivable, accounts payable and cash

- b) Explain and apply relevant accounting ratios, including:
- i) current ratio and quick ratio
 - ii) inventory turnover ratio, average collection period and average payable period

E. Business Finance

3. Sources of finance and their relative costs

- d) Assess the impact of sources of finance on financial position, financial risk and shareholder wealth using appropriate measures, including:
- i) ratio analysis using statement of financial position gearing, operational and financial gearing, interest coverage ratio and other relevant ratios

Summary and Quiz

- Ratio analysis is relevant to several areas of the FM syllabus, so you must know how to calculate all ratios covered in this chapter.
- An important relationship to understand is that profit margin × asset turnover = ROCE

$$\frac{\text{PBIT}}{\text{Sales}} \times \frac{\text{Sales}}{\text{Capital employed}} = \frac{\text{PBIT}}{\text{Capital employed}}$$

- Financial gearing ratios measure the degree to which a company's activities are funded by debt compared to equity.
- Operational gearing ratios measure the effect of fixed costs on the relationship between sales revenue and operating profits.
- Interest coverage is used by lenders, creditors and investors to determine the riskiness of lending money to the company.
- Investor ratios (e.g. EPS, P/E ratio) are also used by lenders and creditors to determine the riskiness of lending money to the company.

Chapter 19 Quiz

Glossary

A

An efficient market – one in which the market price of all securities traded on it reflects all the available information.

A perfect market – one which responds immediately to the information made available to it.

Annuity – a stream of identical cash flows arising *each* year for a finite period of time.

B

Bond – a negotiable security evidencing a debt governed by a contract which specifies, for example, the coupon rate, repayment schedule, security (if any), principal value, seniority (if subordinate to other debt) and other covenants.

Business risk – the variability in the company's *operating earnings* associated with the industry in which it operates. It is determined by general business and economic conditions.

Bill of exchange – an acknowledgement of a debt to be paid on a stated date.

Bill of exchange – a document drawn by the exporter and sent to the customer, who signs to accept responsibility to pay the amount specified on the stated date.

C

Corporate governance – the system by which companies are directed and controlled.

Capital rationing – a situation in which there is not enough finance (capital) available to undertake all available positive NPV projects. Therefore, capital has to be rationed.

Crowdsourcing – using a large, evolving, relatively open group of people (usually assembled via the internet) to provide goods, services or finance.

Crowdfunding – is crowdsourcing to fund a project. Under crowdfunding, there is no financial intermediary: each individual in the crowd has a direct relationship with the entity funded.

Convertibles – loan notes or preference shares that can be converted into a pre-determined number of ordinary shares.

Commercial paper – short-term unsecured debt issued by high-quality companies. Investors can then trade the paper on the secondary market.

Coupon rate – the interest rate printed on the loan note certificate.

Currency swap – an agreement in which two parties exchange the principal amount of a loan and the interest in one currency for the principal and interest in another currency.

Convertible loan notes – allow the investor to choose between redeeming the loan notes at some future date or converting them into a predetermined number of ordinary shares.

Consumer Price Index (CPI) – a measure of the average change over time in the prices paid by consumers for a market basket of consumer goods and services.

D

Discounted payback period – the time it takes for the *discounted* returns (i.e. present values of the operating cash flows) from a project to recover the initial investment. It is also called the *adjusted* payback period.

Deep discount loan notes – loan notes issued at a substantial discount to nominal value and redeemable at their nominal value (or above) on maturity.

Documentary letters of credit – a payment guarantee backed by one or more banks.

Debt factors – businesses offering a range of services in the area of sales administration and the collection of amounts due from customers.

Derivative – a financial instrument whose value or price depends on an underlying asset.

E

Exchange rate policy – how a government manages its currency in relation to foreign currencies.

Expected value – the quantitative result of weighting uncertain events by the probability of their occurrence.

Economic order quantity (EOQ) – the **optimal** order quantity that minimises the total costs relevant to ordering and holding inventory.

Economic risk – the risk that cash flows will be affected by long-term exchange rate movements.

Equivalent annual cost – the annual cost of owning, operating and maintaining an asset over its entire life.

F

Financial management – the management of activities associated with the efficient acquisition and use of short- and long-term financial resources.

Fiscal policy – government actions to achieve economic objectives through the use of the fiscal instruments of taxation, public spending and the budget deficit or surplus. Governments can use public expenditure and taxation to regulate the level of demand within the economy.

Financial instrument – a contract for a monetary asset.

Financial intermediaries – organisations which bring together potential lenders and potential borrowers.

Financial risk – – the *additional* variability in returns due to introducing fixed-interest debt in the capital structure. As a committed fixed cost, interest on debt creates more volatile “bottom-line” profits for shareholders.

Fintech – technologically enabled innovation in financial services that could result in new business models, applications, processes or products with an associated material effect on financial markets and institutions and the provision of financial services.

Forward exchange rate – the exchange rate for buying or selling the currency at a specific date in the future.

Futures contract – a standardised contract between buyer and seller, in which the buyer has a binding obligation to buy a fixed amount (the contract size), at a fixed price (the futures price), on a fixed date (the delivery date), of some underlying asset via a recognised exchange.

Forward rate agreement (FRA) – an agreement by a bank to enter into a notional loan or accept a notional deposit from a customer for a specified period of time. The contract is settled based on the difference between the interest rate agreed when the contract is signed and the rate prevailing when the notional

loan/deposit is deemed to start.

H

Hard capital rationing – the capital markets limit the amount of finance available.

I

Inflation – the rate of increase of the general level of prices in the economy.

Islamic finance – a system of banking consistent with the principles of Islamic law (Sharia). Sharia prohibits:

- the payment or acceptance of interest (riba); and
- investing in businesses that provide goods or services considered contrary to its principles.

Internal rate of return – – the discount rate at which the present value of a project's cash flows is zero.

Inventory control – the systematic regulation of inventory levels.

Invoice discounting – the selling of selected sales invoices to a third party for a discounted cash sum while retaining full control over the receivables ledger.

Interest rate futures (IRFs) – exchange-traded contracts whose value is determined by interest rates.

L

Loan note – a security instrument (so it is tradable) that acknowledges a company's debt.

Lead time – the time between placing and receiving an order.

M

Macroeconomic policy – the setting of economic objectives by the government (e.g. full employment, economic growth, the avoidance of inflation) and the use of control instruments to achieve those objectives (e.g. fiscal policy and monetary policy).

Monetary policy – the set actions taken by the government or the central bank to achieve economic objectives using monetary instruments.

Money market hedge – a technique to lock in the value of a foreign currency transaction in terms of the organisation's domestic currency using a combination of investing, borrowing and a spot currency exchange.

Mortgage – a legal agreement to lend money, secured by a legal charge over the borrower's property.

Multi-period capital rationing – capital is rationed in two or more periods.

N

Nominal value – the value that is stated on an issued security.

Net realisable value – Estimated selling price less estimate costs necessary to make the sale.

P

Peer-to-peer (P2P) lending – a method of debt financing that enables individuals to lend money to small businesses without the use of an official financial institution as an intermediary.

Perpetuity – a stream of identical cash flows arising each year to infinity.

Preference shares – shares with a fixed rate of dividend that have a prior claim on profits available for distribution.

R

Redeemable loan note – a loan note that can be redeemed by the issuer before its maturity date. Early redemption would usually be at a premium above the loan note's nominal value.

Return on capital employed (ROCE) – the average annual operating profit expressed as a percentage of the initial (or average) investment.

Re-order level (ROL) – the level to which inventory should fall before placing a purchase order for replenishment.

Risk – a condition in which several possible outcomes exist, the probabilities of which can be quantified from historical data.

S

Security token offering (STO) – the first issue of tokens (or crypto coins) that represent an investment contract in an underlying asset.

Supply-side policies – policies which focus on creating the right conditions in which private enterprise can grow and therefore raise the capacity of the economy to provide the output demanded.

Soft capital rationing – the company sets internal limits on finance availability.

Single-period capital rationing – capital is in short supply in only one period.

Sensitivity analysis – the analysis of changes made to significant variables to determine their effect on a planned course of action.

Simulation – a technique that allows more than one variable to change simultaneously.

Supply Chain Finance (SCF) – the use of financial instruments, practices and technologies to optimise the management of the working capital and liquidity tied up in supply chain processes for collaborating business

partners.

Systematic risk – the risk which affects the market as a whole rather than a specific company's shares. (Also known as "market" risk or "undiversifiable" risk.)

Spot exchange rate – the market exchange rate for buying/selling the currency for immediate delivery.

Sustainability – meeting the needs of the present without compromising the ability of future generations to meet their own needs.

– *United Nations*

T

Treasury management – the efficient management of liquidity and risk in a business including the management of funds (generated from internal and external sources), currencies and cash flow.

Theoretical ex-rights price (TERP) – the expected share price after the rights issue.

U

Uncertainty – the inability to predict possible outcomes due to a lack of historical data (i.e. information) being available for quantification.

Unsystematic risk – the risk which is unique to each company's shares. (Also called "unique", "diversifiable" or "industry-specific" risk.)

W

Warrant – the investor's right, but not the obligation, to purchase new shares at a future date at a fixed price (the exercise or subscription price).

Working capital – the capital represented by net current assets which is available for day-to-day operating activities.

Z

Zero-coupon loan notes – loan notes issued at a discount to nominal value and which pay no coupon.